

Voting in Context: How Political and Economic Institutions Condition the Economic Vote

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Raymond M. Duch
Nuffield College
University of Oxford
Oxford OX1 1NF
44 (0)1 865 278 515
raymond.duch@nuffield.ox.ac.uk

and

Randy Stevenson
Department of Political Science
Rice University
P.O. Box 1892
Houston, Texas 77251-1892
stevenson@ruf.rice.edu

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Chapter 1

Introduction

It is virtually a universal belief among politicians, political commentators, and even voters that elections are referenda on the economy. Politicians fill their speeches with economic rhetoric; political commentators generate endless streams of economic analysis, and high paid consultants base their statistical predictions on little else. The extent and depth of this belief is revealed most starkly when it appears to have been violated. When incumbents win despite a bad economy or lose despite a good one, we observe in the post-election hand-wringing, a concerted search for *ad-hoc* explanations that preserve the more fundamental belief that the economy matters. Perhaps voters' perceptions of the "real" economy were distorted; foreign policy issues may have overshadowed its effects; or, maybe, ineffective campaign strategies undermined the expected economic vote.¹ Ultimately, however, in most elections, the economy is thought to be determinate.

Popular beliefs, of course, are not social science laws and exceptions to economic determinism are easy to find, even if they are often explained away as idiosyncratic. More generally, the systematic empirical work on economic voting, conducted in most of the advanced democracies over the last thirty years, reveals not a universal law of economic voting but rather a conditional one. Economic voting is very likely widespread and often important; but, its magnitude and nature across elections is almost certainly variable (Lewis-Beck and Stegmaier, 2000). There are limits, however, to the guidance the extant empirical literature can give us. The description of economic voting that it provides across countries and over time is at best a sketch. In some countries and in some periods there are simply no data with which to describe the economic vote. In other cases data exists but awaits a systematic analysis. In many other cases, the published estimates of the extent and nature of economic voting are simply not comparable to each other.² Overall, the vast amount of data relevant to economic voting in the world's advanced democracies has not yet been leveraged to describe variation in economic voting across countries and over time. Consequently, one of the major goals of this book is to provide a more complete and reliable description of the magnitude and nature of economic voting across a large number of countries (eighteen) and a long time period (1979-2001). With this goal in mind, our

¹ Examples in which each of these three excuses were prominent are the 1992 U.S. presidential election, the 2002 German Parliamentary elections, the 2000 U.S. presidential election, respectively.

² This murkiness is due to the fact that much of the empirical literature consists of independent studies produced by researchers working with very different data and methodologies.

study for the first time reliably compares the extent and nature of this economic vote across a large number of cases, confirming that the economic vote is both widespread and variable.

If our description of economic voting reveals a conditional law, we do not yet know its conditions. Why was the effect of the economy so pronounced in the 1980 American presidential election and so surprisingly absent twenty years later? Why is it consistently more important to the electoral fortunes of British Prime Ministers, but not their Dutch counterparts? Why, in short, does economic voting vary as it does? The second goal of this book is to answer these questions. We do this by offering a theory of how different political and economic institutions condition the economic vote and by testing hypotheses drawn from our theory. Our explanation is built on a long tradition of theoretical work that suggests economic voting is the result of instrumentally rational voters trying to use their votes to achieve the best possible economic future.

Within the family of rational choice explanations of economic voting, our contextual theory is a generalization of the competency model of rational retrospective economic voting that has been used informally by many scholars and formally by theorists studying political business cycles in the United States (e.g., Alesina and Rosenthal 1995). The competency model suggests that rational voters condition their vote on the incumbent's record of economic performance because this is the optimal way to identify and elect competent economic managers under conditions of uncertainty. Our generalization of this model explores how political and economic institutions alter the quality of the signal that the previous economy provides about the competence of candidates. Thus, both the rationality of voting in the competency model and its focus on the voter's desire for competent economic management lead to a set of theoretical propositions (and empirical hypotheses) that link variation in economic and political institutions to variability in the economic vote.

Overall, our contextual theory of rational economic voting provides a rigorous theoretical foundation for generating hypotheses about the kinds of political and economic contexts likely to condition the economic vote. Further, our empirical map of economic voting provides the raw material for testing these hypotheses. In the rest of this chapter, we preview the potential usefulness of these contributions both to the study of comparative political behavior and to the understanding of specific elections, review the theoretical and empirical work that motivated the project and that define its contribution, describe the general theoretical approach that we take to explain economic voting, and give an overview of the organization of the rest of the book.

Economic Voting in Recent Elections: The Importance of a Comparative Perspective

While the main goal of this project is to produce a set of theoretically driven empirical generalizations about the nature and sources of economic voting in different contexts, we hope that it will also help us to better understand (and predict) the role of the economy in specific elections. Our premise, however, is that a proper understanding of specific elections requires that they be viewed in a comparative perspective. As an illustration, this section briefly compares two recent German and Italian parliamentary elections (held in September 2005 and May 2006, respectively) and asks what we can learn about them (and similar cases) from the kind of comparative analysis offered in the chapters that follow?

Economic performance leading up to the elections in both Germany and Italy had been quite poor. In spite of turning in slightly improved performance over previous years, the 2005 German economy still grew by only a percentage point in real terms and unemployment rates were persistently above ten percent. Similarly, real GDP in Italy actually contracted in 2005 while the country registered the lowest level of working-age employment in Western Europe (58 percent).³ Further, as Figure 1.1 shows, voters were clearly aware of these trends. Public sentiment about the economy dropped precipitously in both countries after 2001. But German sentiment rebounded prior to the 2005 election while Italian sentiment remained at historically low levels as the 2006 election campaign began.

Before the elections, pundits argued that these records of economic performance would result in a strong rejection of the incumbents.⁴ As it turned out, however, the German SPD trailed their CDU/CSU challengers by less than a percentage point (34.3% and 35.2%, respectively) and Italian Prime Minister Berlusconi's incumbent coalition garnered 49.7 percent of the popular vote compared to the 49.8 percent obtained by Prodi's alternative center-left coalition. As a result, and in stark contrast to pre-election commentary, discussion of the economy was essentially absent from most of the post-election analyses offered by the press.

³ Just prior to the Italian elections *The Economist* noted, "But the biggest difference from the past is that the economy over which Mr. Berlusconi has presided has done so badly. Last year GDP growth was near-zero; this year's forecast has just been cut from 1.5% to 1.3%. Italy deserves its title as the new sick man of Europe. It is in far worse shape than Germany or France. And the pressing need to find a cure for that sickness means that the outcome of next week's election matters (The Economist April 18, 2006, 28)."

⁴ Some representative (English language) headlines: "Economy to dominate German election" (Deutsche Presse-Agentur, August 22, 2005); "German election boils down to jobs, economy" (Associated Press, September 18, 2005); "As Germans look toward national election, economy remains key issue" (Associated Press, September 11, 2005); German jobs the focus of tight poll race: Unemployment and welfare state dominate election campaigns as Europe's economic miracle turns sour (The Guardian, September 9, 2005).

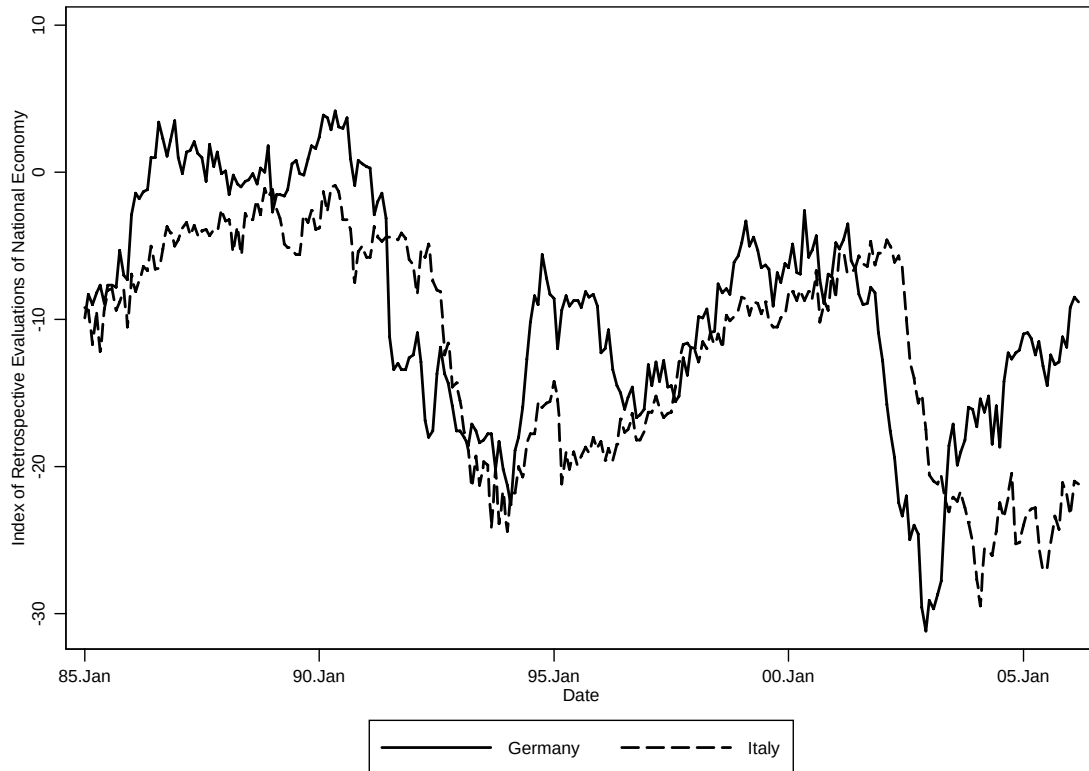


Figure 1.1. Retrospective economic evaluations for Germany and Italy

But did the economy play an important role in these elections or not? To answer this question definitively, we need a statistical analysis that estimates the impact of the economy on the election while controlling for other influences. Further, since our interest is in exploring these specific elections, we need individual level measures of economic perceptions and vote choice.⁵ We conducted surveys previous to both elections (see Chapter 6 for details) that included these data (as well as relevant controls) and were able to produce estimates of the importance of economic voting in each case.⁶ These estimates reveal that contrary to the impression that may have been created by the relatively strong performance of the incumbent parties, voter perceptions of economic performance in fact played an important (and perhaps decisive) role in both elections. In both the German and the Italian elections, our estimates indicate that the Prime

⁵ Alternatively, one could examine the role of the economy in a single election using aggregate data at the regional level.

⁶ These were produced using the methods described in Chapter 2. However, they are simply intended as illustrations here and they are not included in the data used in the rest of the book (they fall outside the time period covered in our data). Details about the data and the specific models estimated are available at www.nuffield.ox.ac.uk/economicvoting.

Minister's party (the SDP and Forza Italia, respectively) would have gained between six and seven percent of the vote had perceptions of the economy improved moderately.⁷ Further, it is likely that this increase in strength would have been enough for both incumbent parties to retain the prime ministry.

So what can we learn from estimates such as these? First, they provide a much clearer picture of whether, and to what extent, economic voting was actually important in a given election. Accordingly, they can help correct misconceptions about the role of the economy in specific elections that are typified by the journalistic accounts of the recent Italian and German elections. Of course, this is not new. Social scientists have been producing well-specified statistical models of individual level vote choice and economic voting for many years. What is different about these examples, however, is that they were produced using methods that allow us to systematically compare them to one another.⁸ Thus, based on our estimates, we can confidently assert that economic voting was about as important in the 2005 German election as it was in the 2006 Italian one.

Should we expect to see very similar levels of economic voting in these two countries? And if so, why? Perhaps the similar levels of economic voting result from some fundamental and general characteristic of individuals that applies to all voters across contexts, even those as different as Italy and Germany. Norpoth (1996a) may be correct in asserting that economic voting is “hardwired into the brains of citizens.” Alternatively, and much more likely, these cases may be similar because of corresponding similarities in context. What might these be? Well-motivated answers to this question must await the theory and analysis in the rest of this book. However, one possibility that has been suggested in the previous literature is that the distribution of policymaking power among incumbent parties was similar in the two cases: The German SPD and Italian Forza Italia were both dominant parties in their respective ruling coalition governments. Thus, voters interested in punishing an incumbent party for poor economic performance faced a roughly similar challenge in identifying whom to blame. While this may or may not in fact be the appropriate explanation, it illustrates the importance of comparable estimates of economic voting across elections and national contexts. Having estimates of the magnitude of the economic vote from multiple elections – admittedly in this example only two cases – is what leads to speculation about the interaction between the economic vote and the

⁷ We define “moderately” in Chapter 2.

⁸ Indeed, the whole point of the first part of this book is producing, for the first time, a large set of comparable estimates of economic voting at the individual level.

structure of power sharing across parties. No analysis of a single election could hope to produce a similar insight.

These two elections are even more informative when viewed comparatively. Specifically, as we show in Chapter 3, the typical size of the economic vote over the last two decades in Germany was about five percent, while in Italy it was about two and a half percent.⁹ This comparative perspective makes it clear that while economic voting in the 2005 German election was only slightly more important than usual, in the 2006 Italian election it was dramatically so. Again, a great benefit of being able to make this (somewhat richer) empirical conclusion is that it stimulates us to ask why? What has happened to cause Italians in 2006 to weight the economy as heavily in their vote as Germans usually do? It is certainly not difficult to think of ways that the electoral situation in Italy has become more similar to that in Germany over the last decade. In contrast to most of the post-war history of Italy, there is now real electoral competition between two well-defined governing alternatives. As a result, Italian votes now matter in the choice of governments at least as much as German votes do. Such changes in the electoral context may well alter the weight that voters give to retrospective economic performance in their vote choice.

This comparison of historical economic voting in these two cases leads to a more general question: Why it is that the economic vote is typically higher in Germany than in Italy? Indeed, extending the basis of our comparison again, we can see that the typical magnitude of German economic voting places it in the middle rank of the western democracies (higher than Italy, Belgium and the Netherlands, for example, but less than the US and UK). Thus, our question becomes not just why is the German economic vote typically larger than the Italian one, but why does it rank in the middle more generally? Understanding why Germany ranks as having an average amount of economic voting will provide insight into the dynamics of the 2005 elections.

Our intuition is that the German economic vote ranks as it does because it has a mix of contextual features that both raise and lower levels of economic voting. Specifically, we have already noted the potentially positive effect that real electoral competition over alternative cabinets can have on economic voting. Further, such healthy competition has been typical of the German case throughout its history and so may be one of the enduring features of the German context that contributes to a fairly large economic vote. However, other factors may well keep economic voting below that of some other countries. For example, a number of institutional features disperse policy-making authority in Germany: a long history of coalition cabinets, a

⁹ That is, a moderate worsening of economic perceptions decreases the prime ministerial party's vote by about five percent

strong federal system of government, and a moderate amount of corporatism. Again, it may be the case that the dispersion of power in Germany reduces the importance that economic performance can play in German voter's decisions – at least in comparison to the highly concentrated distribution of policy making power in more unitary and non-coalitional systems like the UK. Other explanations for Germany's mid-ranked level of economic voting may be its similarly mid-ranked level of trade dependence. It has often been suggested that voters in countries whose economies are more dependent on international trade are less willing to hold domestic politicians accountable for outcomes (that are not of their making). If this were so, we would expect German voters to be less willing than, for example, American voters to hold politicians accountable for poor performance, but more willing than voters in more trade dependent countries like the Netherlands or Belgium.

Our speculations about economic voting in the recent Italian and German elections illustrate a central theme of this book: To understand the role that the economy plays in these, or any given election, or to make predictions about its role in a future elections, we need to be able to place the voting decision in a comparative context. This does not mean describing everything that is unique about an election but rather identifying those features of the context that help explain systematic variation in the economic vote. This, of course, can only be done by building a reliable map of economic voting across countries and over time, and by identifying the kinds of contextual variables that condition the economic vote. In the rest of this book we attempt to accomplish these two tasks. Of course, both our theorizing and our empirical work builds on (and responds to) the literature on economic voting as it has developed over the last forty years. Thus, in the next sections we review the academic work on economic voting that has created both the need for this project and the opportunity for it to be accomplished.

Theoretical Foundations of Economic Voting

This book explores differences in the way voters condition their vote choice on the economy in different political and economic contexts. Consequently, our theoretical effort begins with a model of the individual voting decision that explains why the economy, or at least perceptions of the economy, matters for the vote. A number of theoretical developments are critically important to our model-building efforts. First, early in its history, the economic voting literature adopted the notion that instrumentally rational individuals make vote choices based on their utilities for competing parties. Second, efforts to specify how economic evaluations enter into the voter's utility function resulted in two modeling approaches: sanction and selection models of vote choice. Third, these sanction and selection models have been adapted to explain

how context conditions the economic vote. Finally, if economic voters are rational then developments in the literature on vote choice imply that strategic context should be one of the factors that condition the economic vote. We now briefly explore the development of these theoretical traditions and how they contribute to our rational model of retrospective economic voting.

Economic Evaluations and the Voter Utility Function

For the most part, the economic voting literature treats voters as instrumentally rational actors. Downs (1957) introduced the notion that individuals make vote choices based on their comparison of expected utilities for each of the competing parties. The notion of voters as utility maximizing political “consumers” was a significant deviation from widely accepted explanations for vote choice that borrowed from the social-psychological literature (Berelson et al 1954; Campbell et al 1960). But it was Kramer’s (1971) efforts to “test the Downsian rationality hypothesis” by exploring the link between economic outcomes and U.S. election results that inspired early economic voting research.¹⁰ Kramer in effect argued for the importance of economic well being in the voter utility function introduced by Downs. Fair (1978) took this argument a step forward by providing a formal statement of how economic performance enters the voter utility.¹¹ This was an important theoretical advance because it established a foundation for modeling vote choice from a rational utility maximization perspective that included economic well-being in the utility function. We follow in this tradition. The notion that voters rationally derive expected utilities for competing political parties and that these determine their vote choice is a central feature of our contextual model of the economic vote.

Fair’s (1978) effort though highlighted theoretical controversies as to exactly how the *economy* enters into the voter’s utility function? As Fair emphasizes, theory is a necessary guide in determining how the economy enters into the utility function. Are voters narrowly retrospective and motivated primarily by a sanctioning reflex which is suggested by the early Kramer (1971) findings? Or do voters gather more extensive information on past economic outcomes in an effort to assess how competing potential governing “teams” might perform in the future, a selection perspective hinted at by Downs (1957) and Stigler (1973). Both perspectives share a model of individual decision making in which instrumentally rational voters are

¹⁰ That is not to say that the link between the economy and elections had not been explored. In fact Kramer (1971) does a nice job of summarizing the early efforts in this regard.

¹¹ As Hibbs (2006) points out, this contribution was also important because it derived an aggregate-level vote equation from this individual-level utility function that could be estimated empirically.

maximizing a voter utility function. The sanctioning perspective has been widely adopted in much of the economic voting literature to date. Although much less widely employed in the literature, we will argue that the selection model has advantages for incorporating context into explanations of the economic vote.

The Sanctioning Model (aka the Moral Hazard Model)

The early path-breaking work of Kramer (1971) and Fair (1978) suggested that vote choice was shaped by the recent economic performance of incumbents rather than by comparative assessments of how competing parties might perform if elected. They concluded that the economy entered into the voter's utility function in a simple fashion: punish poor performance and reward good outcomes. V.O. Key's widely quoted characterization of the economic vote seemed well founded: "Voters may reject what they have known; or they may approve what they have known. They are not likely to be attracted in great numbers by promises of the novel or unknown (1966: 61)." Fiorina's (1981) classic work, *Retrospective Voting in American National Elections*, provided a more general expected utility model of voting that included economic evaluations.¹² And while both retrospective assessments and future expectations of the performance of incumbents (and challengers) figured prominently in Fiorina's models of the vote decision, Fiorina argued that future expectations are, for the most part, simple extrapolations from current trends. This provided further support for the retrospective sanctioning perspective. Hence, early in the history of economic voting this sanctioning model of voter behavior— either explicitly or implicitly — became the workhorse of models linking the economy and vote choice.

One of the attractions of this stark punishment model was that it demanded relatively little of the average voter which tended to comport with early empirical findings regarding the political sophistication of the average citizen (Converse 1964). But did this narrow focus by the voter on retrospective economic performance constitute rational behavior? Should we not expect rational voters to undertake a more comparative assessment of the likely performance of competing candidates, both incumbents and challengers, if elected? Barro (1973) and then Ferejohn (1986) showed how retrospective economic voting could be a fully rational voting strategy by building on the notion that candidates, like firms, are motivated by the maximization of rents. Voters, according to this model, are confronted with a moral hazard problem when deciding on voting for the incumbent versus opposition parties. They argue that if voters do not sanction economic performance they risk signaling to incumbents that poor economic

¹² Fiorina stopped short of providing a formal model in which retrospective economic voting emerged as rational behavior.

performance would be tolerated and hence invite rent seeking on the part of self-interested political candidates. In this model voters are not engaging in the comparative assessment of utility income streams from competing political candidates – rather they simply establish a threshold performance level and re-elect incumbents that satisfy this requirement and punish those that do not (Ferejohn 1986). This leads to the *sanctioning* feature that characterizes most accounts of the economic vote. It is the concern with re-election in the future that motivates incumbents to avoid shirking their responsibilities. They anticipate that voters will sanction them if they under-perform. And in order to maintain the credibility of this threat voters punish incumbents at the polls when retrospective economic performance is substandard.

Adopting this sanctioning perspective on rational voting had important implications for the specification of economic voting models. Pure sanctioning models of the economic vote are decidedly retrospective – voters entirely discount candidate promises (so these do not enter into the utility function) and only pay attention to economic performance over the course of the incumbent’s recent tenure in office. Expectations about future economic outcomes are not necessarily precluded from shaping vote preferences but again they simply reflect information about past economic outcomes.

The Sanctioning Model and Context. Ferejohn’s (1986) formalization of the sanctioning model provided the micro foundations for the simple vote choice rules that were articulated earlier by Kramer (1971). An interesting puzzle though is why sanctioning varies across contexts. Why are incumbents in some countries, for example, punished or rewarded with greater intensity or frequency than is the case with in others? Why might the moral hazard incentives vary from one context to another?

Sanction models of the economic vote require that voters have retrospective information about the performance of the macro-economy and that they punish or reward incumbents in a fashion that credibly signals to candidates the electoral costs of shirking or rent seeking. Context can matter in these models because signals of the incumbent’s responsibility for macro-economic outcomes may be more “noisy” in some contexts as opposed to others (Fearon 1999; Ferejohn 1986). Hence, conventional sanction models of economic voting often explain variations in sanctioning behavior in terms of variations in levels of information (e.g., Powell and Whitten 1993).¹³

¹³ There is also a coordination issue – voters need to agree on what constitutes a threshold of “bad” economic performance. This could also be the basis for contextual variations in the magnitude of economic voting in a sanctioning model (Ferejohn 1986).

Hibbs (2006) has pointed out that Kramer's (1983) famous "error-in-variables" conception of economic evaluations can be used to model contextual variations in the signal-to-noise ratio regarding incumbent economic performance. If the politically relevant portion of economic outcomes is small, then usual survey measures of economic evaluations in the voter's preference function will contain considerable error. Hence, a particularly complex governing coalition situation or a context where relatively large exogenous shocks affect the macro-economy would likely reduce the extent to which voter's economic evaluations reflect the politically relevant portion of economic outcomes. As Hibbs (2006) demonstrates, higher levels of error in the measure of economic evaluations will have a downward bias on its coefficient in a voting equation. Hence, contexts in which the politically relevant portion of macro-economic outcomes is small – reflected in relatively high measurement error when overall economic evaluations are used – will be those that have a low correlation between economic evaluations and vote choice.

The Selection Model (aka the Competency Model)

Selection models of economic voting stipulate that the vote decision entails more than a simple reward-punishment response to economic outcomes. They suggest that instrumentally rational voters are motivated by the desire to select the most competent candidates – that is, voters use information about economic outcomes to assess the future competencies of competing candidates. Selection models were not unfamiliar to the early economic voting scholars. Voters in Downs' model are future oriented and compare the platforms of contending candidates, logic which more closely resembles a selection as opposed to a sanctioning model. Likewise, Kramer (1971) viewed voters as future-oriented but unwilling to spend the resources to assess future promises. Rather, again echoing Downs, Kramer assumed that rational voters rely on the retrospective economy to inform them about the likely actions and economic success of future incumbents. Stigler's (1973) famous critique of Kramer also adopted the selection perspective, but pointed out that voters should be uncertain about how much the previous economy should tell them about the competence of incumbents: "Per capita income falls over a year or two should the voter abandon or punish the party in power? Such a reaction seems premature: the decline may be due to developments beyond the powers or responsibilities of the party (1973: 164)." However, Stigler also recognized that with knowledge of the role incumbents play in shaping policy, rational voters might be able to tease out of economic outcomes those fluctuations that are related to the actions of incumbents and those that are not. Consequently, they might use this

information in a prospective fashion to “...determine which party is on average more likely to maintain a high and steady rate of growth of income (Stigler 1973: 165).”

While this early work clearly pointed the way toward a rational model of economic voting based on the selection of competent politicians, informal models left many questions unanswered, e.g., could it be rational for voters to use the previous economy to infer competence, if politicians had incentives to manipulate the economy for electoral gain? Answers to these kinds of questions have had to await the development of the rational expectations literature in economics and its expression in the study of political business cycles.

The development of rational expectations theories provided the theoretical tools for developing formal selection models of the economic vote. Rational expectations treat forward-looking decisions in the same rational fashion as static decisions, particularly in terms of how individuals use information. This implies that individuals do not make systematic (or repeated) mistakes in forecasting the future.¹⁴ Their expectations about future outcomes should be conditioned on reasonable “models” of the “variables” and functional relationships that have historically generated such outcomes. Rational expectations theories essentially associate the same rationality assumptions with forecasts that are typically linked to static decision making. This implies, for example, that policymakers could have no permanent or predictable impact on economic outcomes because they would be anticipated by rational economic actors (Alesina, Roubini and Cohen 1997).

Building on these advances in rational expectations theory, Cukierman and Meltzer (1989) and Rogoff and Sibert (1988), independently, proposed the notion that governments differ in their competence in handling the economy and that information about their competencies plays a critical role in vote choice. By observing economic outcomes, voters become informed about the competence of an incumbent and use this information in their vote decision. Persson and Tabellini (1990) and Alesina and Rosenthal (1995) developed models that explicitly incorporated rational expectations and very nicely drew out the implications of rational expectations and competency assumptions for the economic voting model. Because voters in these models are fully informed about the incumbent’s policies and anticipate their impact on subsequent performance, only unexpected shocks to the macro-economy can have any impact on their voting decisions. And these shocks can be either exogenous shocks – unrelated to the policy action of

¹⁴ Begg (1982) presents a comprehensive description of the rational expectations revolution in macroeconomics.

incumbents – or they can be competency shocks, unexpected outcomes that can be associated with incumbent initiatives.

The Selection Model and Context. The signal extraction problem that confronts voters in selection models is to determine the extent to which observed shocks to the macro-economy are the result of incumbent competency. The usual variance (perhaps reflected in historical fluctuations) in shocks to the economy may provide the voters with information that allows them to make rationally informed assessments of the extent to which shocks to the economy in the economy are the result of incumbent competence rather than non-political exogenous factors. To the extent that historical fluctuations in the macro-economy are more likely to have resulted from competence shocks rather than non-political shocks, voters will weight current shocks to the economy – such as an unexpected change in real GDP growth – more heavily in their vote decision.

Hence, context could impact economic voting in selection models because the relative magnitudes of exogenous and politically relevant competency signals vary systematically across different economic and political contexts. This potential has certainly been recognized (e.g., Scheve 2004) but both the theoretical and empirical implications of this kind of contextual variation have yet to be fully explored. The model we offer in this book identifies the contextual implications of the selection model by assuming that the relative weight of electorally and non-electorally accountable decision makers in economic policy making directly impacts the variance of both political (e.g., competence) and non-political (exogenous) shocks on the economy. Consequently, political and economic institutions that impact this ratio should have a direct influence on the quality or strength of the competence signal that the economy provides to voters. For example, our theory suggests that voters in economies that are highly integrated into the global economy will recognize that economic policy making is weighted toward non-electorally accountable decision makers and so rationally conclude that exogenous factors play an important role in producing macro-economic shocks. Accordingly, we predict that economic voting will be higher in countries with closed economies.

Strategic Voting and Context

Rational models of economic voting should clearly reflect the strategic incentives introduced by features of the electoral system and the nature of contention amongst the competing candidates. Just as voters have rational expectations about economic outcomes, they should also have rational expectations about the relative electoral strength of competing candidates (Cox 1997) and the impact that their vote can have on the post-election distribution of policymaking

power. If rational voters believe that their vote can have no impact on who makes policy, then they should not worry too much about identifying which politicians are competent or about sanctioning underperformers. While this observation is hardly new, the fact that models of economic voting have been developed largely in the context of American two-party elections (where strategic incentives are muted) has limited work on this potential source of contextual variation in the economic vote.

Of course, the notion that voters respond strategically to electoral institutions, and to the nature of contention amongst competing parties, follows directly from our assumption that voters are instrumentally rational (Gibbard 1973; McKelvey and Ordeshook 1972; Satterthwaite 1975). One of the key insights in this literature is that rational voters will take into account whether their vote makes a difference in who wins or who loses elections and who gets into government. The usual understanding is that rational voters should not “waste” their vote on candidates that are sure to lose, but the same logic also applies to parties or candidates that are sure to win. More generally, our theoretical generalization of the strategic voting logic (to cases in which there are multiple parties and coalition cabinets) explores how characteristics of the “distribution of contention” for policymaking authority shape the incentives for strategic voting.

These strategic considerations have important implications for our contextual model of the economic vote. We anticipate there will be frequent situations in which an individual’s vote will have little impact on the post-election distribution of policy making authority (because the same parties will get into government for all plausible electoral results) and hence economic evaluations will play a minor role in voting in such elections.¹⁵ Our decision to treat voters as being instrumentally rational and our desire to explore the implications of this model in diverse political contexts dictates that we account for the impact that different strategic voting incentives, arising from different political contexts, may have on the extent and distribution of economic voting.

The Overall Message from Theoretical Developments in the Literature

Our reading of the theoretical literature on economic voting produces two clear messages about how contextual explanations of economic voting should be shaped. First, the underlying

¹⁵ A number of scholars have pointed out that contextual variation in strategic voting incentives can affect the voter’s utility function such that some factors or issues matter in vote choice in one context versus another. Myerson (1993; 1999), for example, demonstrates that under Duverger’s law for plurality voting, where parties compete on ideology, strategic voting considerations can lead voters to vote for corrupt candidates in spite of their preferences to minimize rent seeking and in spite of the fact that there are candidates with compatible ideological positions who lack the rent seeking baggage.

individual level theory of voting upon which the model is based should incorporate the assumption of instrumental rationality and thus the model should produce hypotheses about how different strategic contexts impact economic voting. Second, the existence of two quite different explanations for instrumentally rational economic voting in the literature (sanction versus selection models) makes it imperative that any contextual theory be explicit about which explanation is being invoked and what implications of the theory flow from that choice. In the theoretical chapters in Part II and Part III of this book, we propose a contextual model of rational economic voting that explicitly adopts the selection approach to understanding the motivation for rational retrospective economic voting at the individual level. In Part II we explore the empirical implications for context that follow from this choice and in Part III we generalize the model so that it reflects strategic voting incentives.

Empirical Evidence of an Economic Vote

As the above discussion indicates, theoretical efforts have for the most part focused on results that explain, in general (although with the U.S. political context in mind), why economic evaluations shape the vote decision. The early empirical work on economic voting – much of it dominated by the U.S. case – focused on establishing that in fact economic evaluations motivated vote choice. And over time there has been an impressive increase in the number of empirical studies of economic voting conducted outside of the U.S. context. A brief review of this empirical work on economic voting serves several purposes. First, the U.S. empirical evidence in particular, but also results for a number of other national contexts, provides a strong foundation for the general claim that there is an economic vote. As the time series of voting studies in the U.S. have lengthened and as the number of non-U.S. voting studies have proliferated, the empirical puzzle that motivates the book emerges: Existing empirical findings suggests that the economic vote is quite unstable both cross-nationally but also over time, even within the U.S. context. Accordingly we now review the empirical findings from the U.S. case on which much of the case for economic voting is made and then examine the cross-national findings that provide the empirical puzzle addressed in the remainder of the book.¹⁶

The United States

¹⁶ The empirical literature on economic voting has benefited from a number of excellent reviews (the most recent by Lewis-Beck and Stegmaier, 2000) and we will not attempt to replicate these efforts here. Rather, our goal in highlighting just some of the works in the vast empirical literature on economic voting is to explain how our work adds to what has already been done.

When taken together, the large literature on economic voting in the United States unambiguously confirms the existence of a robust empirical connection between the economy and vote choice in the American case. Three different strains of research, each initiated in the 1970s, contribute to this consensus.

Mueller's (1970) classic study of presidential popularity introduced one important line of inquiry: estimating the economic vote based on aggregative-level models of presidential popularity that employed, typically, monthly readings of public approval of the chief executive with objective measures of economic performance as the independent variable (typically the actual rates of unemployment, inflation, or real GDP change). Although the detailed specification of these presidential popularity functions has been the subject of some debate, almost all estimates of presidential popularity functions confirm that the economy is a significant factor driving presidential popularity (e.g., Beck 1991; Norpoth 1985). Further, the various estimates tend to agree (roughly) on the magnitude of the effects. For example, Muller's original estimate was that presidential popularity declined by about three percentage points for every percentage point rise in unemployment (Mueller's 1970: 29). Norpoth's (1985) results for inflation and unemployment are similar. He calculates that the overall change in the popularity of Nixon, Johnson, Carter, and Ford that was caused by changes in unemployment and inflation was 5.5, 2.5, 4.4, and 4 percent, respectively. While there is disagreement about the exact set of aggregate economic indicators that should appear in presidential popularity functions, there is little disagreement that *some* set of aggregate economic variables predicts changes in approval and that the overall impact of usual changes in these variables is to move popularity in the range of between 3 and 10 percentage points.¹⁷

A second avenue of empirical inquiry examines how actual election outcomes depend on objective economic outcomes. One of the first and most influential of these studies was by Tufte (1978), who provided evidence that the economy plays a predominant role in American presidential elections. Although based on only eight data points, his evidence was striking and simple: in presidential elections from 1948 to 1978, the relationship between annual real disposable income and the vote of the President's party was practically a straight line. This simple but powerful picture resonated with subsequent researchers who, using more data and refined models, confirmed Tufte's original finding (e.g., Bartels and Zaller 2001; Erikson 1990; Hibbs 2000, 2006).¹⁸ Importantly, this confirmation was not limited to the *existence* of economic voting

¹⁷ See Lanque 1986 for a review of the debate as well as estimates for disposable income that are in this range.

¹⁸ They find that the "Bread and Peace" model proposed by Hibbs (2000) performed particularly well.

in the aggregate data, but also extended to estimates of the *magnitude* of the economic vote in U.S. presidential elections. Specifically, a one percent change in annual real disposable income produces a two to four percentage point increase in support for the incumbent presidential party. These results are consistent with the magnitude of economic voting effects found by the studies discussed above, which focused on presidential popularity rather than the vote.

Early aggregate level research on the economy and elections in the United States also explored congressional contests. Kramer's (1971) influential analysis of U.S. congressional elections from 1886 to 1964 was the first to establish that change in real disposable income over the course of the election year predicts the success of the President's party in congressional elections. This result was subsequently championed by Tufte (1975; 1978) with more recent supporting evidence from Jacobson and Kernell (1983), Lewis-Beck and Rice (1992), and Kiewiet and Udell (1998). However, unlike the results for Presidential elections, this finding has been challenged by a number of scholars who failed to find a significant relationship in aggregate data (e.g., Alesina et. al. 1993; Erikson 1990; Marra and Ostrom 1989). Nevertheless, the bulk of evidence seems to argue for the existence of economic voting in congressional elections, though the relationship is almost certainly weaker than in presidential elections.

The third strain of empirical economic voting research traces its intellectual roots to Key (1966) who explored the impact of popular assessments of incumbent performance on the actual voting decision. Similar early efforts to use individual level survey data to explore economic voting specifically relied on questions that asked voters about their "personal financial situation". However, the later (and most enduring) individual level evidence for economic voting in the U.S. has come from questions that ask respondents about general business conditions or the country's "economic situation". Fiorina (1981) conducted an important early analysis of these survey data in which he demonstrated that economic perceptions of the national economy significantly impact vote choice in U.S. elections (both Presidential and Congressional). And soon thereafter, Kiewiet (1983) confirmed Fiorina's conclusions with more data and more elaborate statistical specifications. Further, he demonstrated that assessments of general economic conditions play a more important role in shaping vote choice than pocket-book evaluations (i.e., perceptions of one's personal financial condition). Fiorina's and Kiewiet's empirical conclusions quickly entered into the lexicon of American political science and with some relatively minor qualifications or extensions have withstood extensive scrutiny (some notable examples are Alvarez and Nagler 1998a and 1998b; Duch et al 2000; Markus 1988 and 1992).

Unlike aggregate studies of economic voting, which are most useful for estimating a single, average magnitude of economic voting across elections, individual level survey data, if

available for several different elections, can be used to construct (and compare) estimates of the magnitude of economic voting in each election. Kiewiet (1983) was somewhat of a pioneer in this respect because he explicitly estimated economic voting for a number of different American election studies (for the period 1958 to 1980). Given the relatively small number of election surveys in his sample (only four of the presidential election surveys included national economic evaluations), he was limited in terms of the over time comparisons he could make. Nevertheless, he was able to highlight, and propose explanations for, the average difference between congressional and presidential elections (consistent with the aggregate results described above, he found economic voting to be relatively weaker in congressional elections). As the number of these U.S. individual election studies has increased with time, scholars have become increasingly interested in using them to compare the magnitude of the U.S. economic vote from one election to the next. Fiorina et al (2003), for example, evaluate the magnitude of the presidential economic vote and speculate as to why it varies from election to election.

Despite the many individual level studies of economic voting that have been done, there is no clear consensus in the American literature on the appropriate methodology for characterizing the magnitude of the economic vote at the individual level. Kiewiet used one approach, Fiorina another, Alvarez and Nagler (1998b) still another. As a result, individual-level estimates of the magnitude of economic vote for presidents in the U.S. range broadly from election to election. For example, Kiewiet reports that a typical change in economic perceptions moves the electoral support of the incumbent by 13 percent (Kiewiet 1983: 35). In contrast, Alvarez and Nagler's (1998b, 1360) estimate is close to 38 percent.¹⁹ The lack of a uniform methodology for calculating the magnitude of the economic vote from survey data means we do not know if estimated differences reflect real variation in the size of the economic vote or whether these differences are simply artifacts of the different methods for calculating them.

Overall, the economic vote has been subject to extensive scrutiny in the U.S. context and the effort has borne some important insights.²⁰ Voting for the U.S. president is strongly influenced by economic performance. At the aggregate level we have precise estimates of the effect of various economic aggregates on the presidential vote and on presidential approval. At

¹⁹ The Kiewiet (1983) result is based on moving economic evaluations for an "average" voter from "worse than a year ago" to "better than a year ago"; the Alvarez and Nagler (1998b) result is based on moving national economic evaluations from worse to better.

²⁰ In 1990, Erikson (1990) concluded that the relationship between economic performance and the vote is probably the most widely accepted hypothesis about voting behavior and elections – here he was obviously referring to the U.S. case.

the individual-level, results reinforce the importance of national economic evaluations on the presidential vote, although estimates of its magnitude vary significantly. Nevertheless, there is no consensus on how to assess the magnitude of the economic vote, and its election-to-election variation, based on the results of individual level election studies.

Non-U.S., Single-Country Studies

Given the overall strength of the empirical results in the U.S., one might expect to find similarly strong evidence for economic voting in other developed democracies. However, an examination of the country-specific studies of aggregate and individual-level economic voting suggest considerably more variation than is the case in the U.S. In many countries we find weak empirical evidence for economic voting. This is particularly true at the aggregate level; though even with individual data the evidence is negative or mixed in a number of countries. Thus, while we do not disagree with Lewis-Beck and Stegmaier's (2000) conclusion that evidence of economic voting can be found in much of the developed world, in almost no country is there anywhere near the level of consensus or confidence that characterizes the American literature. What this means, of course, is that it is difficult to use the country-specific literature on economic voting to make anything but the grossest comparisons of economic voting across countries or (for most countries) over elections within countries.

The country-specific studies of countries other than the U.S. come in the same three flavors as those employing strictly U.S. data: aggregate studies of the economy and government popularity; aggregate studies of the economy and electoral returns; and individual level studies of economic perceptions and vote choice.

Goodhart and Bhansali (1970) provided the first aggregate study of popularity, demonstrating that the popularity of British governments was strongly influenced by movements in the aggregate economy. However, this result came under almost immediate criticism (e.g., Frey and Garbers 1971 and Miller and Mackie 1973). And although more recent aggregate work tends to support the existence of economic voting in Britain (Clark and Stewart 1995; Clark et. al. 1997; Sanders 1987; Whiteley 1986) dissenting voices have, to some extent, persisted (Crystal and Alt 1981; Norpoth 1987). Likewise, analyses of aggregate popularity and vote functions in other countries showed considerable variability. Most estimates for France exhibited strong aggregate economic effects (Hibbs and Vasilatos 1981; Lafay 1977; Lewis-Beck 1980), although there have been exceptional studies suggesting no relationship (Lecaillon 1981). The estimates for Denmark were initially negative and later mixed (e.g., Nannestad and Paldam 2000; Paldam and Schneider 1980). Kirchgassner (1991) found evidence for the impact of employment on the

popularity of German governments before 1982 but not after. Frey (1979) examined the fate of incumbent parties over 67 years in Denmark, Norway, and Sweden and found mixed results. Similarly, Madsen (1980) in a careful study of aggregate popularity in Denmark, Norway, and Sweden found no evidence for economic voting in Denmark and Norway but did find evidence for Sweden.

More generally, in the 1970's and 1980's the pattern of results from aggregate-level, single-country studies of economic voting in developed democracies (outside of the U.S.) reveals a great deal of variability. The evidence for an aggregate level relationship is stronger in some countries than in others, but in no case do we find the kind of strikingly simple aggregate level pattern of data that Tufté reported for the U.S.

Individual level analyses of economic voting in countries outside the U.S. were slow to emerge due to the relative paucity (before the mid-1980s) of individual-level surveys addressing the economic voting question. However, the few individual-level studies that were done in the 1960s and 1970s did confirm the existence of individual level economic voting beyond the U.S. context. For example, based on their study of the UK electorate, Butler and Stokes (1969: 392) concluded that "The electorate's response to the economy is one under which voters reward the Government for the conditions they welcome and punish the Government for the conditions they dislike."²¹ However, as more individual level studies were undertaken in other countries, a message similar to the aggregate level country-specific studies began to emerge – the extent and nature of economic voting at the individual level was highly variable. For example, while support for economic voting at the individual level was forthcoming in France and Britain, Miller and Listhaug (1985) found little evidence for individual level economic voting in Norway and neither did Belluci (1985) for Italy. Likewise, Nannestad and Paldam (1997), in contrast to most other results, found evidence that personal economic conditions were more important to Danish voting than evaluations of the national economy.²²

The scholars who reported negative results or that discovered evidence for unusual forms of economic voting were keenly aware of how these findings differed from expectations (stemming from the U.S. consensus) and so almost always tried to attribute them to unique features of the country's political or economic context. For example, by the early 1980s Italy had experienced almost 40 years of Christian Democratic rule (and so no true alternation of power). This distinguished it from most of the other western democracies and, according to Belluci (1985), may have depressed the Italian economic vote. Likewise, Miller and Listhaug (1985)

²¹ As cited in Lewis-Beck (1988: 34).

²² This result has, however, been challenged by Hibbs (1993) and Borre (1997).

speculated that the weak individual-level economic effects they found might be explained by the fact that Norway's economy is so heavily influenced by international economic trends. It might be the case, they argued, that voters in Norway recognize that their government has little responsibility for economic outcomes and so do not consider economic performance to be important for their vote decision. Similarly, Nannestad and Paldam (1995) suggested that Danish voters hold the government accountable for their personal economic circumstances because the extensive Danish welfare state created a connection between government policy and personal economic circumstances that did not exist in many other countries.²³

Overall, an important result of this variability in the evidence for economic voting across countries was that it encouraged scholars to speculate about the kinds of differences in domestic politics and national economies that might drive differences in economic voting across contexts. However, as Lewis-Beck and Eulau (1985) and Lewis-Beck (1988) have both emphasized, one can get relatively little leverage on the question of how political and economic context impacts economic voting by comparing the results of a relatively small number of studies that also differ dramatically in the statistical models used to produce the estimates of economic voting. Thus, the country-specific literature provides only very crude evidence of whether economic voting exists, or does not, in particular survey contexts. Instead, a better estimation of cross-national economic voting requires a more coordinated effort to produce comparable estimates – a task taken up by Lewis-Beck and a small number of others.

Cross-national Comparative Studies

By the early 1980s large-scale cross-national survey projects were becoming more common – examples included the first wave of the 22-nation World Values Survey in 1981 and the initiation of the 9-country EuroBarometer series in 1974. Taking advantage of these developments, in 1984 Lewis-Beck conducted the first cross-national individual-level study of economic voting, with evidence from five countries. His main conclusion was simply that economic voting is widespread in developed democratic contexts. However, he also demonstrated that directly comparable estimates of the magnitude of economic voting from different countries reveal considerable cross-national variance. Further, the estimated variation appeared to follow a pattern that he interpreted as consistent with the traditional reward-punishment logic for economic voting. Specifically, his evidence showed that the more parties included in the cabinet

²³ Lewis-Beck (1983) previously suggested this explanation for his results about the importance of personal financial circumstances in French economic voting. And Pacek and Radcliff (1995) provide evidence of this argument employing aggregate level data.

(e.g., Italy's five party cabinet versus Britain's single party cabinet), the smaller the impact of the economy on the voter's decision to vote for or against the cabinet parties. This pattern, he argued, might indicate that as the cabinet became more complex, voters found it difficult to attribute responsibility to specific incumbent parties and so could not reward or punish optimally.

Lewis-Beck's evidence of variability in economic voting and his tentative explanation captured the imagination of students of comparative political behavior. However, the lack of suitable comparative survey data forced these scholars to turn back to aggregate level data to pursue this comparative project further. The resulting analyses typically estimated the relationship between fluctuations in macro-economic indicators (usually, unemployment, inflation and real income) and change in vote share for the incumbent governments across a large sample of elections from multiple countries. Unfortunately, however, the aggregate approach has proven largely disappointing as a way to identify robust relationships between context and the economic vote. Specifically, some influential large-scale studies have simply found no widespread evidence for economic voting in the aggregate data. For example, Paldam (1991) analyzes electoral data from seventeen countries over the post-war period and found little evidence of economic voting in any country. Though, like Lewis-Beck, he hinted that the explanation for his weak results might lie in cross-national differences in the nature of governing institutions, his evidence did not really produce enough statistically detectable variation in economic voting for him to speculate as to the nature of these influences.

Some other cross-national aggregate level results have detected more economic voting than Paldam did (using different statistical models and, in general, smaller samples). However, the nature and magnitude of the estimated economic voting effect varies significantly from one study to the next. For example, Lewis-Beck and Mitchell (1990) examined aggregate data on 27 elections in five European countries and found that rates of unemployment and inflation do affect the vote for governing parties. However, Chappell and Veiga (2000) analyzed an expanded sample of European countries over a longer time period and found that none of the usual economic indicators had an impact on incumbent party vote share (although they did find that change in inflation rates *relative* to the European average had a significant correlation with vote share). Some of these differences in results certainly reflect differences in the elections sampled and model specification; however, one is hard pressed to have much confidence in the robustness of the economic vote in the aggregate cross-national data if the result is so sensitive to the particular sample of countries employed or the precise model specification.

Responding to the negative and inconsistent cross-national aggregate results, Powell and Whitten (1993) suggested that the impact of economic evaluations on vote choice might interact

with institutional context. Powell and Whitten (1993) developed a measure of institutional “clarity of responsibility” that measured the extent to which political institutions allowed incumbents to diffuse responsibility for economic outcomes.²⁴ The results from their analysis of 109 elections led them to conclude that, “the greater the perceived unified control of policymaking by the incumbent government, the more likely is the citizen to assign responsibility for economic and political outcomes to the incumbents” (Powell and Whitten 1993, 398).

This clarity of responsibility argument and result resonated positively with many scholars of comparative economic voting and spawned a literature consisting of efforts to identify different contextual influences on economic voting in aggregate data. Some of these studies took clarity of responsibility itself as a starting point – trying to extend the result to different samples of elections or to different dimensions of clarity. The results of these efforts, however, have been mixed. Chappell and Veiga (2000) analyze 136 European parliamentary election outcomes for the period 1960-1997 and find evidence that inflation (relative to European averages) shapes vote outcomes but find no support for the clarity of responsibility argument. Royed, Leyden and Borrelli (2000) specifically challenge the robustness of the Powell and Whitten result claiming that there is little support for the clarity of responsibility argument - in fact claiming that economic voting is higher for coalition as opposed to single party governments. Though Palmer and Whitten (2003a and 2003b) have rigorously rebutted the Royed, Leyden and Borrelli claims, it is hard to avoid the conclusion that the aggregate evidence for clarity of responsibility, like the aggregate evidence for cross-national economic voting in general, is fragile. A clue that the source for this fragility may lie in the use of aggregate electoral data (rather than in the weakness of the underlying relationship itself) comes from Anderson’s (1995) application of the clarity idea to monthly popularity data rather than electoral data. Though he worked with only five countries, his empirical analyses indicate quite conclusively “...that more responsibility in the government results in the economic variables having stronger effects on party support (Anderson 1995: 210).”

Almost all of these efforts that followed Lewis-Beck’s five nation study focused on aggregate-level electoral and economic data. One conclusion that can be drawn from this evidence is simply that the economic vote varies from election to election and from country to country. It is certainly not universally absent as Paldam’s (1991) aggregate study suggested, but neither is it pervasive (Norpoth, 1996). More importantly, given that there is variation in economic voting from election to election, the cross-national literature has begun to identify the

²⁴ Their index of clarity of responsibility combined measures of the following institutional characteristics: voting cohesion of the major parties, the extent to which legislative committee systems accommodated opposition party power sharing, bicameralism, coalition governments, and minority governments.

correlates of that variation. This search, however, has been hindered for the most part by reliance on aggregate data and less than robust evidence.

Scholars have recently though begun to heed Lewis-Beck's suggestion that a fruitful effort to explain cross-national variation in economic voting is more likely to result from individual level data on people's economic perceptions and vote choices rather than on aggregate level data about electoral returns and economic performance. Hellwig (2001) analyzes a pooled sample of nine individual-level election surveys and includes contextual measures of both economic context (economic interdependence) along with the institutional variables employed by Powell and Whitten (1993). He confirms the important role that political institutions play in mediating the magnitude of the economic vote and also provides evidence of a similar mediating role for economic variables such as the extent to which national economies are integrated into the global economy. Van der Brug et al (2006) examine voter preferences for parties competing in European elections in 15 countries over three different time periods. Accordingly, with a total of 42 different electoral contexts they are able to test institutional explanations for individual-level variations in the economic vote. They conclude that the economic vote is not very large in general and to the extent it does exist it is restricted to contexts in which there is high clarity of responsibility.

The Overall Message from Empirical Developments in the Literature

The empirical literature on economic voting that has developed over the last four decades makes it clear that economic voting is both pervasive and variable. However, the real extent of economic voting and the nature of its variability has often been obscured in the comparative literature by an over reliance on aggregate data and the incomparability of individual level studies conducted by different researchers pursuing different priorities and using different methods. What is needed, and what we will attempt to provide in this book, is a description of economic voting across countries and over time that is constructed from individual level data in a way that makes meaningful comparison possible. Fortunately, this goal can now be achieved - since over the last twenty years a large number of national voter preference surveys and cross-national survey projects have been undertaken. Further, the content of these surveys reflects a broad consensus on the basic factors motivating voters in the western democracies – so that a more or less standard battery of survey questions appear in many of these voter preference surveys. Many surveys include, for example, questions about the voter's perceptions of movements in the national economy. These developments make it possible (as we demonstrate in Chapter 3) to develop a description of economic voting at the individual level that is both comprehensive and comparable

across elections in different countries and at different times. We hope that this will serve as a foundation for identifying the elements of the political and economic context that condition the economic vote.

A Contextual Theory of Rational Retrospective Economic Voting

Our assessment of the comparative work on economic voting suggests an empirical direction that focuses on the analysis of individual-level surveys and a theoretical approach founded on rational models of individual vote choice. Since our goal is to explain contextual variation in the economic vote, we have adopted insights from the theoretical literature that we think will best explain how voters incorporate information about the political and economic context to condition their economic vote. Specifically, our contextual theory of the economic vote builds on two traditions in the theoretical literature that we reviewed above.

First, we concur with the likes of Cheibub and Przeworski (1999) and others (Fearon 1999, for example, in the same volume) that a selection model rather than a traditional sanctioning model is a promising theoretical approach for explaining cross-national variation in economic voting. While one can gain considerable insights into contextual variation in economic voting by adopting a moral hazard approach, we believe a more persuasive theoretical case can be made that the vote decision is motivated by selection (as opposed to simply a sanctioning motivation). Fearon (1999) has shown that selection incentives overwhelm sanctioning incentives when models of electoral selection and electoral sanction are combined formally. Specifically, when voters perceive any relevant differences between candidates in an election, it is rational for them to vote over those differences rather than to try to use their vote to discipline incumbents.

Second, rather than relying (as much of the current comparative literature does) on differences in voter knowledge about political and economic context (i.e., their level of “clarity”) to drive expected differences in economic voting, we assume voters are rational and have complete information about the political and economic context. Moreover, because voters are rational they use all of this information in making their vote choice. This implies that voters have rational expectations regarding the macro-economy – they use all available information about the current and retrospective performance of the economy in order to anticipate future economic outcomes under different potential incumbents. It also implies that voters have rational expectations about likely electoral outcomes and hence vote strategically. Accordingly, our contextual model of rational economic voting is a selection model in which voters have rational expectations regarding economic outcomes and also vote strategically.

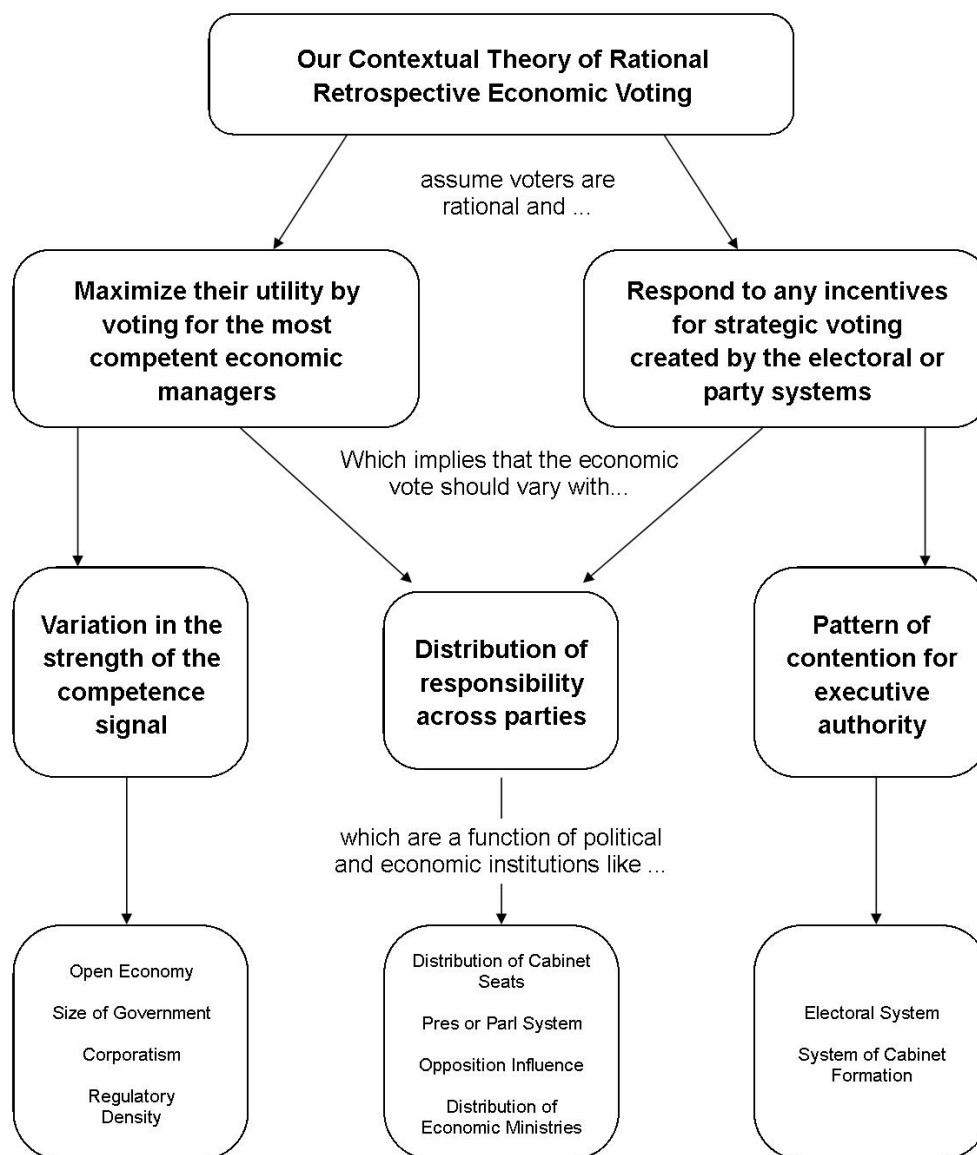
Figure 1.2 provides a simple overview of the theory as well as the empirical implications that emerge from it. The first part of our contextual theory of rational retrospective economic voting (described and tested in Part II) provides a rational selection model in which voters maximize their utility by voting for the best economic managers (ignoring strategic voting incentives). In this model, movements in the previous economy enter the rational voter's utility function for each party weighted by a "competency signal" that ranges from zero to one, depending on how much information about that party's competence economic fluctuations actually contain. The model assumes rational expectations on the part of voters so the economic outcomes associated with different incumbent parties are fully anticipated. Consequently, economic growth only depends on the natural rate of economic growth plus unanticipated shocks. These shocks, however, consist of both an exogenous component and a component that depends on the incumbent party's competence (and is persistent over time). Consequently, rational voters can use retrospectively observed economic performance to infer the competence of incumbent politicians or parties.

Specifically, in this model voters can observe shocks to the macro-economy but cannot observe the mix of exogenous and competence components that make up these shocks. Voters do, however, know the variances of the distributions of these different kinds of shocks and so are able to solve a well-defined signal extraction problem that produces a competence signal. This competence signal is simply a weighted value of economic performance (or perceptions of performance) where the weight varies between zero and one and is minimized when the variance in exogenous shocks is very large relative to the variance in the competence of politicians. In the full model, this competence signal enters the voter's utility function for each party and so the theoretical variation in the competence signal provides exactly the kind of potential contextual influence that we are trying to identify in this project. When it is near zero, there will be little economic voting because the exogenous influences on the economy are so great that the rational voter can glean little information about the competence of politicians from its usual fluctuations. Alternatively, when an economy is insulated against exogenous shocks (i.e., the variance in these shocks is quite small from period to period) observed movements in the economy would be relatively informative about the competence of politicians and so important in the vote choice of rational voters.

Two kinds of empirical implications flow from this part of the model (i.e., the selection logic that leads to the competency signal): those that say how the size of the competency signal itself should change with changes in the historical variation in the domestic economy and those

that suggest how institutions that provide more or less political control of the economy (such as the openness of the economy) should impact the size of the competency signal.

Figure 1.2
Overview of the Theory and Its Implications



The second part of the theoretical model is described and tested in Part III. It begins with the voter utility function implied by the selection model (i.e., one that includes the previous economy weighted by a competence signal) and - once we generalize the model to more complex political environments (that create strategic voting incentives) - adds an analysis of strategic voting to the utility function. That is, we ask: if the voter's preferences for parties are determined

by the utility functions implied by the selection model, which party would actually be preferred if the electoral system (and system of executive selection) create complex strategic incentives? And, how do these incentives then impact the likelihood of an economic vote? Our answers to these questions are built on McKelvey and Ordeshook's (1972) formal model of strategic voting in multiparty contexts. However, it was necessary to generalize this logic considerably to account for cases in which multiple parties compete in elections and executive authority may be shared among several parties. This generalization leads to implications about how the pattern of party competition impacts economic voting.

In addition, our analysis of strategic voting incentives interacts with the logic of the selection model to produce implications about the way that variations in the distribution of administrative responsibility among members of the incumbent government will alter the strength of the competency signal and thus the strength of economic voting. In general, the implications of the model in this regard are the same as the implications from previous arguments about the incumbent or status quo distribution of policy-making responsibility (e.g., from the clarity of responsibility literature) – that is, as responsibility is more widely shared across parties, economic voting will be weaker. Further, as any individual incumbent party has a greater share of responsibility, economic voting will be a more important influence on that party's electoral performance.

Organization of the Book

The book is divided into three parts. Part I focuses primarily on defining what we mean by an economic vote; providing a detailed discussion of how the economic vote is measured; and building a comprehensive map of the variation in economic voting across 18 countries, over 21 years, and across 655 parties. In Chapter 2 we develop a number of different conceptual definitions of economic voting that correspond to the various ways economic voting has been used in the literature (and in our theory). These include the economic vote for the chief executive, the economic vote of the government, and the volatility of the economic vote, among others. Next, we describe a four-step procedure for measuring these concepts using individual level data from 163 different electoral surveys (drawn from 18 countries from 1979-2001). In brief, our procedure generates measures of economic voting by first estimating a separate multinomial statistical model of vote choice for each of the 163 electoral surveys. The estimated parameters from these models are then used to produce measures of the general economic vote for each party in an election, which we define as the change in support for each party if the distribution of economic perceptions in the sample shifts by a given amount. Changes in predicted values calculated from the statistical model are a rather direct measure of this concept and so, for each of

the 163 cases, the procedure produces a measure of the extent of economic voting for each party. Finally, by combining these estimates in various ways, we can produce measures of the different conceptions of economic voting mentioned above.

In Chapter 3 we use these measures to construct an empirical map of the variation in economic voting across the western democracies over the last two decades. This map unambiguously confirms Lewis-Beck's assertion that economic voting in the western democracies is widespread and forcefully contradicts the most pessimistic assessments emerging from cross-national aggregate studies (e.g., Paldam 1991). More importantly, it clarifies the nature of the cross-national, over-time, and across-party variation in economic voting and suggests a number of empirical puzzles for our theory to solve.

To evaluate the quality of the map of economic voting that we have produced and to test hypotheses that come out of our theory, we build statistical models designed to explore the sources of variation in our map of economic voting. Chapter 4 reviews two strategies for doing this, both of which we use in the remainder of the book. One strategy is to pool the data from the 163 individual level surveys and then estimate hierarchical multi-level models in which we use interaction terms to look for suspected contextual influences on the extent of individual level economic voting. An alternative two-stage strategy takes the estimates that we have already used in Chapter 3 and makes them the "dependent variable" in second stage analyses exploring the sources of their variation. Both strategies typically generate similar estimates of contextual effects (Duch and Stevenson 2005). However, most of our empirical work uses the two-stage strategy, largely because it allows informative visual representations of the results, which greatly simplifies reporting and interpretation and lets the reader visualize the relationships in the data to a far greater extent than is possible with the one-stage models. Nevertheless, for the sake of robustness we also report (when appropriate) results from one-stage hierarchical models.

Chapter 4 also asks whether the variation in economic voting that is revealed in our empirical map is an artifact of differences in the 163 statistical models used to produce them. Using a variety of empirical strategies, we rule out a number of possible artificial sources of variation in our estimates, including differences in question wording across surveys, differences in the "fit" of the individual level models, differences in the timing of the surveys relative to the election, differences in the list of control variables included in the different models, differences in the applicability of the assumptions underlying the statistical model itself and endogeneity in our measure of economic evaluations. Overall, we conclude that the variation in our map of economic voting is not in any large part due to differences in the specification of the individual level models used to estimate its 163 different data points.

Part I leaves the reader with a comprehensive description of the contextual variation in the economic vote that our theory of rational economic voting will explain. Part II presents the foundations of our contextual theory; it develops the competency-signaling component of the theory; and it presents empirical tests of the hypothesized impact of political and economic contexts on the economic vote. Chapter 5 develops a selection model of the economic vote that incorporates competency-signaling arguments that, as we have seen, have significantly influenced rational expectations models in the political economy field. We build on Alesina and Rosenthal's (1995) model of rational retrospective economic voting in which rational voters try to select competent economic managers from the set of parties competing in the election. They assess competence by observing past economic performance and using this to sort out what portions of macro-economic shocks are due to political competency as opposed to exogenous factors.

Chapter 6 explores two of the assumptions of the model developed in Chapter 5 – that voters have a sense of the magnitude of normal variation in their economies compared with others and that they understand the extent to which their economy is subject to exogenous influences that domestic politicians can not control. Given the considerable skepticism that students of political sophistication have about the knowledge of the average voter, and since there was little empirical work available to give us confidence in these assumptions, we collected two data sets that can be used to explore what information about economic variation is available to voters (i.e., coding of economic information in over 10,000 newspaper front-pages) and what they actually know about the extent and sources of variation in their economies (an original survey conducted in six countries). The results indicate that a great deal of information about economic variation is available to voters and that knowledge about the comparative level of economic variation in their economies is surprisingly widespread and sensible.

Having established that voters have information about variation in economic outcomes, Chapter 6 also explores whether perceptions of differences in the extent of economic variation before elections actually lead to differences in economic voting, as the theory predicts. The results suggest that national differences in the variation in shocks to their macro-economies are sufficiently large that they could signal the economic competency of incumbents. Further, the chapter provides evidence that economic voting responds to these macro-economic signals as hypothesized by the theory.

Chapter 7 examines how institutional and structural features of the domestic macroeconomy affect the voter's signal extraction efforts. Our competency theory predicts that as the ratio of decisions affecting macro-economic outcomes by non-electorally accountable decision makers to those by electorally accountable decision makers declines, the overall

competency signal will rise and vice-a-versa. The competency signal will be lower in an open economy because the ratio of non-electorally accountable decisions (many originating with decision makers outside of the country) to elected electorally accountable decisions will be higher than in an economy less open to global influences. As a result economic voting should be lower in open economies. Paradoxically, a more expansive state sector increases this ratio of non-electorally accountable decisions to electorally accountable decisions and hence undermines the magnitude of the competency signal. Contexts with fewer statist institutions – lower government expenditures as a percent of GDP, the absence of central wage bargaining institutions, and limited government regulation – have higher levels of economic voting.

Part III explores how the economic vote may be conditioned by the structure of electoral competition, the nature of post-election bargaining over the distribution of executive authority among parties, and the distribution of policy making authority among incumbent parties. In Chapter 8 we begin by generalizing the rational selection model of individual level economic voting from Chapter 5 to contexts in which there are multiple parties competing for election and where executive authority may be shared among several parties. This generalization shows that in political contexts in which policymaking authority is widely shared, the rational voter's ability to extract an informative signal about the competence of any one party declines. Further, since multi-party competition and shared executive authority necessarily introduce incentives for strategic voting, both the extent to which different parties are competitive in the contest to obtain executive authority and the extent to which elections matter in this contest should be important contextual influences on economic voting.

Chapter 9 explores whether variations in the distribution of policy-making responsibility among parties currently in government conditions the economic vote. We find, as the theory predicts, that parties with a greater share of the status-quo distribution of administrative responsibility have a greater share of the economic vote in a given election. We also demonstrate that as the distribution of policymaking responsibility is shared more equally among the parties in an election, retrospective economic evaluations will be generally less important in the vote choices of voters.

Chapter 10 examines whether patterns of governmental contention influence the distribution of the economic vote amongst parties. We find that economic voting is more important to the support of opposition parties that are members of more, and more competitive, alternative governments than it is for other opposition parties that are not. We also find, consistent with our theory, that economic voting is smaller for incumbent parties that are members of competitive alternative cabinets than for incumbents who are not. For example, as

predicted by the theory, a prime ministerial party that is a member of each of the several different cabinets that have any real chance of forming clearly receives a weaker economic vote.

Our generalization of the selection model to political contexts, in which coalitions usually form, assumes that voters understand the likely outcomes of the coalition process and how their votes can impact this process. Chapter 10 also presents pre-election survey data from Norway, Denmark, and Sweden showing that voters in these countries have accurate perceptions of the likely outcomes of the coalition formation process (and how they are likely to differ with different electoral outcomes). We conclude that voters in these relatively complex coalitional systems have the information they need to cast strategic votes with as much confidence as any British or American voter who decides to abandon a hopeless third party candidate.

Each of the empirical chapters in Parts II and III explore how a particular set of contextual variables impacts the economic vote. An equally important task for comparativists, as Franzese (2003) has nicely pointed out, is capturing the higher-level interactions amongst contextual or institutional settings. For example, we expect the competency signal to be relatively strong in political contexts where administrative responsibility is concentrated (a single party majority government, for example). On the other hand, if domestic economic outcomes are shaped entirely by global economic trends, beyond the control or influence of incumbent governments, we would expect the competency signal in such a context to be extremely small. Thus, concentrated administrative responsibility is a necessary – although not a sufficient – condition for economic voting to occur. Given the clear interactions between different contextual variables that are implied by our theory, our final chapter begins to explore how different contextual factors interact to shape the economic vote.

Part I

Describing the Economic Vote in Western Democracies

Chapter 2

Defining and Measuring the Economic Vote

In this chapter we explain what we mean by phrases like “there is more economic voting in one election than in another” or “one party has a greater economic vote than another”. We begin this task with a discussion of our conceptual framework for understanding voting in general and then tackle economic voting specifically. Once our concepts are clear, we then move on to measurement issues and ultimately describe our empirical strategy for mapping the variation in economic voting across countries and overtime.

A Conceptual Model of Voting

We will take the act of voting to be a discrete choice. Discrete choice situations suppose that a decision maker chooses from a finite set of alternatives that are mutually exclusive and exhaustive. In democratic elections, vote choices can be represented by this structure. Voters are allowed to choose one party or candidate, to abstain, or cast a blank ballot; and these alternatives are limited to those that are on the ballot at the time of voting.²⁵ Given this structure, we will also assume that whatever cognitive processes the individual may use to make this choice, the overall process of decision can be *represented* in the following way:

1. Each alternative in the choice set has some “value” to the individual and this value is a function of variables that vary over alternatives, as well as variables that do not vary over alternatives but are characteristics of the individual making the choice. For example, a variable that would vary over alternatives is the voter’s perception of her ideological distance from each candidate. In contrast, the voter’s subjective assessment of the economy would not vary over alternatives.
2. The individual’s value for each alternative includes a random element. This randomness is usually thought of as arising from the researcher’s ignorance of all the variables that contribute to individual choice (Train 2003: 21). However, it could also arise from an inherent randomness in the value function.
3. Individuals choose the alternative that has the highest “value”.

²⁵ Any more complex electoral system (e.g., systems that allow voters to cast two votes) can also be represented as discrete choice problem - with an appropriate definition of the choice set (e.g., all the combinations of ways that a voter may cast two ballots).

Given the first two assumptions, we can represent the voter's *value function* for each alternative j as:²⁶

$$(2.1) \quad V_{nj} = g(x_{nj}, s_n) + e_{nj} \quad \forall j$$

n indexes voters, x_{nj} are characteristics of alternatives, s_n are characteristics of the voter that are constant over alternatives, and e_{nj} is a random term. In this notation, the third assumption simply requires that the voter will vote for the party or candidate, j , with the highest value function, given x_{nj} , s_n , and e_{nj} . One important implication of this framework is that the vote choice must be defined in terms of the probabilities of voting for each alternative. Accordingly, the “vote” is not a single observed ballot, but an unobserved vector of probabilities that sum to one over the alternatives. As a number of recent contributions to the literature suggest, this has important ramifications for the estimation of vote choice models in multi-party contexts (Alvarez, Nagler and Willette 2000; Blais et al 2004). Further, this has important implications for how we conceptualize economic voting that go beyond simple issues of estimation and which have not been fully articulated in the economic voting literature.

Conceptualizing Economic Voting

Given our understanding of voting as an individual discrete choice, any definition of *economic* voting at the individual level must posit some relationship between a person's perception of the economy and the probability of casting a vote for each of the available parties or candidates in an election (we will call these probabilities the voter's “support vector” or just her “support”). The task of building a conceptual definition of economic voting is to specify the nature of this relationship. The previous literature is full of possibilities. For example, much of the empirical literature conceives of economic voting as any decrease (increase) in support for the chief executive that results from worsening (improving) economic perceptions (e.g., Lewis-Beck 1981). Powell and Whitten (1993) conceive of economic voting similarly, but focus on changes in support for the governing coalition as a whole. In contrast, Stevenson's (2001) “luxury good” model conceives of economic voting as any decrease (increase) in support for leftist parties (whether incumbent or not) when economic perceptions are improving (worsening).

As these examples make clear, there is no single concept of economic voting that is appropriate for all purposes. The appropriate concept depends on one's theory or (in the absence

²⁶ This notation and the basis for much of this discussion rely on Kenneth Train's (2003) excellent treatment of the general subject of discrete choice models.

of a clear theoretical guide) how the concept is to be used. In this book, we have two empirical goals: we want to describe patterns in economic voting across countries and over time; and we want to test hypotheses about the impact of political and economic context on economic voting that are drawn from a specific theory. These different purposes sometimes call for different concepts of economic voting and so we will present and compare multiple concepts of economic voting in this chapter. We begin by describing a very general concept of economic voting and then discussing how we operationalize this concept. Next, we discuss other (more restrictive) concepts of economic voting and how they can be measured.

The Most General Concept of Economic Voting

The most general concept of economic voting that is consistent with the discrete choice framework discussed in the last section (and that maintains some relationship between economic perceptions and support) simply defines economic voting as any change in a voter's support for parties that is caused by a change in economic perceptions.²⁷ We call this *general economic voting* to distinguish it from other more narrow concepts of economic voting discussed in later sections. Notice that, consistent with the discrete choice framework, *general economic voting* is a vector of changes in the probability of voting for each party in an election. If all elements of this vector are zero, we say that there is no economic voting, while if some element is non-zero we think of this voter as an economic voter in this general sense.²⁸

This clearly differs from most conceptions of economic voting because it is neither directional nor incumbency oriented (e.g., *worse* perceptions lead to *decreases* in support for incumbents but not for opposition parties). However, because this general concept captures *any* possible connection between changing economic perceptions and electoral support, it provides a benchmark for defining other concepts of economic voting, which necessarily exclude some of these connections. Further, if we can successfully operationalize this general concept, we can use this measure to construct measures of any other concept of economic voting that we choose. Thus, our first task is to build a measure of *general economic voting* that is applicable to a diverse

²⁷ This is formally true. If we define a set of all changes in support that are caused by a given change in economic perceptions, the most general definition possible would “count” all these changes as economic voting, while every other definition necessarily excludes some elements of this set from its count.

²⁸ Indeed, since *general economic voting* is the most general concept of economic voting, if all elements of this vector are zero, there can be no economic voting under any definition of economic voting.

set of countries and time periods. Once we have achieved this, we return to the task of defining more restrictive conceptions of economic voting.²⁹

Measuring General Economic Voting

Producing a measure of *general economic voting* is complicated by the fact that the theoretical quantity we want to measure (the economic vote) is a *casual effect* (i.e., it is the impact of one variable on another) and so cannot be observed directly. Instead, a measure of this effect must be estimated from data on the two variables that make up the relationship (vote choice and economic perceptions). The information that we will use to produce measures of *general economic voting* comes from surveys of 163 different national populations in eighteen western democracies from 1979-2001.³⁰ Since each of our surveys is based on a large sample from the relevant population, we can use statistical models to produce the necessary estimates (as well as measures of our uncertainty about them). Of course, there are many different statistical models that one could use. For example, a particularly simple method would be to estimate economic voting in each population from the bivariate correlations between economic perceptions and vote choice in the corresponding survey. By comparing these correlations across populations one could characterize the strength of economic voting across countries and over time. Correlation analysis, however, is not the best way to estimate causal effects, since it does not account for the influence of other variables that can confound the relationship and bias the estimates.

More generally, in choosing an estimation method we should strive to produce estimates of economic voting that reflect its true value in the population. In statistical terms, we seek to

²⁹ A final definitional issue concerns our focus on individual vote choice as the basis of our conceptual definitions. Economic voting is an individual level phenomenon. It is only when individuals condition their votes on their evaluations of the economy, that economic voting can occur. This theoretical basis of the individual vote decision is why our language has been (sometimes laboriously) in terms of changes in support that a voter assigns to different parties, rather than changes in the aggregate vote of those parties. However, despite the impression this language may have made, we will not be interested in explaining differences in economic voting among individual voters. Rather, we will focus on a single average (or otherwise typical) voter in each population we study and try to explain variation in economic voting amongst these average voters. This has two implications for the terms we use and the inferences we make. First, if we focus on the general economic vote of a single average voter in some sample of voters, then the changes in vote probabilities for this average voter can be equated to changes in the expected vote shares of parties. Thus, instead of referring to economic voting as “changes in support of an individual voter that are the result of changing economic perceptions”, we will often drop references to the individual and instead discuss economic voting as an aggregate phenomena – i.e., the change in the vector of expected party vote shares due to changes in economic perceptions.

³⁰ Each survey randomly samples the national population of adults at the time of the survey. In some cases the sample had to be weighted to account for non-random over-sampling of some groups in the population.

produce *consistent* estimates of the economic voting in each population.³¹ In the statistical models that we will be using to produce our estimates, consistency is only achieved when one specifies the components of the statistical model correctly (i.e., they match the true process that generated the data). This means including the correct explanatory variables in the statistical model, choosing a probability distribution that correctly characterizes the random components of the process that generated the dependent variable (i.e., an individual vote choice), and specifying the functional relationship between the explanatory variables and the dependent variable correctly.

Clearly, consistent estimation is an ideal and is never achieved in practice. Indeed, since we never know the true process that generated the data, we can never know for sure if we have a correctly specified statistical model. Still, we do not come to the process of model building unarmed. We can use logical constraints to rule out many possible specifications (e.g., a variable that can only have positive values can not be generated by a random process that sometimes results in negative values). We can use our knowledge of the substantive problem (e.g., which explanatory variables are likely to be important?); and we can ask if the model produces estimates that conform to what we know (e.g., we would question a model that generated estimates implying that leftist political orientations increased the chances of a conservative vote).

In this book we produce our estimates of *general economic voting* for an average voter in each of the 163 sampled populations from carefully specified statistical models of individual voting behavior. In specifying the components of these models, we have been guided by the vast literature on voting behavior in general, the country-specific literatures on voting, and a growing body of work concerned with the particular statistical problems associated with estimating vote choice models. The 655 estimates obtained from these models (one for each party in the 163 populations) are the numbers that we use to characterize the strength of economic voting in each population. Below, we describe the four-step process we used to produce our estimates of general economic voting in each survey.

Step 1: Obtaining Comparable Vote Choice and Economic Perceptions Questions

The first step in obtaining comparable estimates from different surveys is to identify those surveys that ask respondents about their vote choice and their perceptions of the economy. Since questions are not asked in exactly the same way in each survey, we will need to make judgments about the degree of difference in question wording that is tolerable across surveys.

³¹ An estimator is consistent if it converges to the true value of the quantity being estimated as the amount of data used to produce the estimate increases.

Too high a tolerance will hurt the comparability of the resulting estimates, but too low a tolerance will unduly limit the number of surveys from which we can glean evidence. Fortunately, however, there is a remarkable degree of consistency in the question wording for vote choice and economic perception questions across surveys, and where differences do exist they are seldom idiosyncratic. Rather, there are usually only two or three ways a given question is asked and so there are almost always a group of surveys that ask questions in the same way. Hence with enough cases of each type of question wording we can do secondary analysis to look for any systematic impact that the differences have on our estimates. For example, if there are two ways in which the vote choice question is asked and this difference matters for our analysis, then we should see some systematic difference in our estimates of economic voting generated from one question type versus the other. Below, we review the questions that were used for vote choice and then turn to economic perceptions. We reviewed over 300 candidate surveys for inclusion in this study and 163 were judged to have sufficiently comparable questions on vote choice and economic perceptions.

Vote Choice: Three types of vote choice question were used in our sample of surveys. In each type, the question asks about a vote in a single election. We did not, for example, use surveys that only asked respondents to identify the party to which they felt closest (e.g., the early Italian Eurobarometers). The three kinds of questions we did use differed in two ways: (1) in their temporal relationship to the election for which the vote applied and (2) in their treatment of non-voting. With respect to the first issue, surveys conducted directly after elections simply asked respondents to report their vote choice in the preceding election. In contrast, those surveys that were conducted just before an election asked respondents for their vote intention in the upcoming election. Finally, surveys that were not proximate to an election asked the voter about a hypothetical election: “If there were a general election tomorrow which party would you support?” There is a large literature on the strengths and weaknesses of these different kinds of questions in measuring vote choice; however, the key question for our analysis, addressed in Chapter 4, is whether these differences introduce systematic biases into our estimates of the strength of economic voting.

The second difference in the vote choice variable concerns the treatment of non-voting. All the surveys we used allow the voter to express whether they did not vote or did not intend to vote. Further, most allow the voters to indicate they cast, or would cast, a blank ballot. These studies differ in how they determine whether the respondent did not intend to vote. In many surveys the option of non-voting is simply included along with the other parties in the vote choice

question. In others, however, a two-question format is used in which the respondents are first asked whether they voted (or intended to vote) and only then for whom they voted (or intended to vote). While this is a readily apparent difference in the way the vote choice question is asked in different surveys, it is unlikely to be consequential, since we decided to ignore non-voters in our analysis.³² The important point is that in all our studies voters were allowed some way to express that they either did not vote or did not intend to vote.

Economic Perceptions: Each of the 163 surveys used in the analyses in this book asks respondents a question of the following general form:

“Looking back over the last year, would you say that the economic situation in [name of country] has gotten much better, somewhat better, stayed the same, somewhat worse, or much worse.”

The key elements of this question are that it is retrospective, it refers to the national economy and it is about *change* in the economy rather than its absolute level (i.e., the economy is good or strong). For a survey to be included in our study, we required that it include an economic perception question with these three elements. However, other less fundamental deviations – like whether the retrospective evaluation was over a year or two years and whether the question had three, four or five response categories – were tolerated. We do not expect these deviations to be consequential but, as with the vote choice question, we investigate this expectation empirically in Chapter 4.

Practical constraints dictate the adoption of the retrospective version of national economic evaluations because this question by far outweighs prospective evaluations in the elections surveys conducted over the past 30 years. This, of course, is not by chance but rather reflects the theoretical importance of the retrospective model of the voter’s economic reasoning beginning with Key’s (1966) *The Responsible Electorate* and the micro-level findings in the U.S. of Fiorina (1978, 1981) and Kinder and Kiewiet (1979, 1981). Moreover, recent evidence that the economic voter in fact forms prospective expectations of economic performance in evaluating incumbents emphasizes that these prospective evaluations are dependent, although not exclusively, upon retrospective assessments of economic conditions (Duch and Stevenson 2007; Erikson MacKuen and Stimson 2000; MacKuen Erikson and Stimson 1992). Since the purpose of this book is not to determine the relative importance that voters accord retrospective versus

³² In many of the individual level models we analyzed, we had difficulty finding a parsimonious set of variables to include in the model that could consistently predict non-voting. Even when such efforts were fruitful, these factors did not often overlap with the variables that predicted which party one voted for and so including them in the models meant adding a large number of extra parameters to the equation (for each variable included we had to estimate as many parameters as parties in the election minus one).

prospective assessments in evaluating the macro-economy we focus exclusively on retrospective national evaluations. And while we do not go so far as to accept Down's (1957) suggestion that the two evaluations are identical, we believe that prospective assessments incorporate considerable retrospective information. Hence calibrating the magnitude of the economic vote based on retrospective assessments is parsimonious (in the sense that it captures a key information input into prospective evaluations), consistent with the dominant theoretical work in the field, and meets our practical data constraints. Finally, the general model of rational economic voting that we develop implies retrospective economic voting—not prospective voting—and so it is variation in this type of voting that we hope to explain with the theory.

Step 2: Identifying Control Variables

As we pointed out earlier, to improve the consistency of economic vote estimates, we specify our vote choice models with variables that control for the impact of other important factors shaping voter preferences. This increases our confidence that our estimates reflect the true relationship between economic perceptions and vote choice in the population to which the relevant survey applies. For each survey we identify the appropriate control variables from the voluminous comparative and country-specific literatures on voting behavior. There is considerable consensus in these literatures regarding what concepts to include in our models. First, the literatures in the different countries usually point to the same kinds of variables as important determinants of the vote. Second, since the scholars who have written the voting literatures in each country are usually the same people who design the surveys, measures of these basic factors are usually included in election studies.³³

In general, three theoretical traditions provide the foundations for most empirical models of voting (Alt and Chrystal 1983; Dalton 2002; Dalton and Wattenberg 2000; Franklin et al 1992; Miller and Niemi 2002). First, the sociological tradition identifies class, urban/rural residence, religion, region, language, and race as important predictors of vote choice (Alford 1963; Lijphart 1981; Lipset and Rokkan 1967; Rose 1974; Zuckerman 1982). These variables capture the voter's position within the cleavage structure of society so that in countries with different cleavage structures different subsets of these variables will be important. For example, linguistic cleavages are important in Canada and Belgium, and thus included in our Canadian and Belgium models; but they are unimportant in France and hence excluded from the French analyses. Of course a number of these socio-economic variables will be common to virtually all of the model

³³ Though the same concepts (e.g., age, religion, education, left-right self-placement) are usually included in different election studies, the specific measures used to capture these concepts are often not the same.

specifications. This is the case with age, income, religiosity or church attendance, gender, occupation, and education.

Second, expected utility theorists and students of issue voting argue that voters make choices based on the expected policy implications of electoral outcomes (Downs 1957; Kedar 2005b). This tradition emphasizes the importance to the vote decision of the spatial distance between voter issues preferences and those of contending parties and coalitions (Alvarez and Nagler 1998a; Enelow and Hinich 1984; Niemi and Weisberg 1992). In contexts with uni-dimensional political competition this can be captured by the position of the voter and the parties on a left-right scale. In contexts with multidimensional competition one must measure the positions of voters and parties on different policies.³⁴ Accordingly we included in all of our models a measure of voter placement on a left-right scale (or liberal-conservative scale, in the U.S. case).³⁵ In addition, we frequently include a measure of policy preference. For example, in many of the countries in our sample, we included respondents' attitudes toward the European Union. For the individual national election studies, or non-European studies, we included policy measures appropriate to each election (NAFTA, for example, in the case of the 1988 Canadian election).

Finally, the influence of the Michigan school, particularly in the U.S., points to the importance of the direction and strength of partisanship as an explanation for vote choice (Campbell et al 1960; Fiorina 1981; Miller and Shanks 1996). However, the distinctiveness of partisanship and vote choice has been challenged in many countries (Borre 1984; Budge et al 1976; Butler and Stokes 1969). There is a lively debate about the role of partisanship in voting in some country literatures such as Canada (Clarke, Kornberg and Wearing 2000; Clarke and Stewart 1996; Stewart and Clarke 1998) and Britain (Budge et al 1976). Accordingly, with the exception of the U.S. models, we exclude partisanship from the vote choice models. Nevertheless, because of this lack of consensus (even among scholars working in one country) we produce extensive secondary analysis to determine if our estimated variation in economic voting across countries hinges on the decision of whether to include a measure of partisanship in the vote choice model (see Chapter 5).

Most empirical analyses of voting behavior populate their models with variables derived from these three traditions and, as we noted above, most election studies include one or more measure of many of these concepts. Further, the appropriateness of variables from each of these

³⁴ There is a large literature that demonstrates voters across the western democracies use a single left-right dimension to summarize contemporary policy debates (e.g., Fuchs and Klingemann 1989; Inglehart 1984; Klingemann 1979).

³⁵ In a handful of studies this measure was excluded because it was not included in the survey instrument.

three traditions in each of the western democracies has been debated in the country specific literatures on voting behavior. Consequently, we make no effort in this book to critique or improve upon the individual level models of voting that these literatures have produced but rely on their collective wisdom to guide our selection of appropriate “control variables” for each election study we analyze. Since we have 163 individual level models, it was not possible to include the details of each specification here. However, this information, a codebook, and the estimated coefficients for each individual level model are available online at www.nuffield.ox.ac.uk/economicvoting.

It is also important to understand that nothing requires that the “control variables” appropriate to vote choice models in a given country or election be the same as in other countries or elections. While it could be that all the determinants of voting behavior are universal, this is unlikely given the diversity of influences identified in the country specific literatures.³⁶ Rather, while there does seem to be a core set of factors that are important to vote choices in all western democracies, there are also other factors that vary in importance both across countries and from election to election. Further, even when the same concepts are important influences on vote choice across our cases, there is no guarantee that the measures of these concepts used in specific surveys will be identical.

Given this variability in the likely determinants of vote choice from election to election, a practical question about the specification of the vote choice models is whether using different control variables in each model will compromise the comparability of our measure of economic voting. First, as long as including different control variables (a linguistic variable in Canada but not in France, for example) is necessary in order to ensure a properly specified model, and the rest of the model is specified correctly, then the resulting estimates of economic voting will be consistent and so comparison of the estimates amongst models from different contexts will be meaningful.

Step 3: Estimating the Statistical Models

Once we have decided on the variables to be included in the model for each individual survey, we estimate a multinomial logistic model (MNL) of vote choice. The MNL is well suited to the task of estimating a discrete choice model of voting because it allows one to estimate the probability that the voter chooses each of the available parties and to see how this distribution of support changes when perceptions of the economy change. These models have an advantage over

³⁶ For another illustration of the extent to which these variables vary in their importance for vote choice models across different national contexts see Franklin et al (1992).

binomial models because they do not force us to artificially treat multiparty elections as a contest between incumbents and opposition parties. Indeed, consistent with the concept of *general economic voting* that we use these estimates to measure, the MNL model does not even require that we define incumbency previous to the estimation.³⁷

An often used alternative to the MNL model is to estimate a set of binomial logistic regressions, one for each party in the election, in which the vote choice for each party is recorded as either being for a party or against it. The estimates from these models will not obey the constraint that estimated choice probabilities sum to one across alternatives and actually require one to estimate a larger number of coefficients than the MNL - i.e., for nine independent variables and five parties one would estimate 50 coefficients (and only 40 with MNL). Still, for purposes of comparison with the MNL estimates we have estimated these models and have found no significant differences in the general pattern of results.

In later sections we introduce estimation procedures based on pooling the data. However, these models do not allow us to produce estimates of the *general economic vote* for each survey (as defined above) because these models demand that we define the choice set similarly across the studies to be pooled (i.e., we can estimate a pooled model in which the dependent variable is a vote for or against an incumbent but not when it is for or against each of the available parties – which differ from survey to survey). Thus, pooling the data is not a useful alternative to separate estimation of the *general economic vote*. It will prove useful, however, for exploring variation in more restrictive concepts of vote choice. Thus, we defer discussion of pooled models until after we define these concepts.

Step 4: Using the Estimated Parameters to Estimate Changes in Support

Once we have obtained estimates of the coefficients of well-specified vote choice models, we still need to construct measures of the substantive impact of changes in economic

³⁷ Of course, there are disadvantages as well. The most often discussed disadvantage of the MNL is that it assumes independence of irrelevant alternatives or IIA (Alvarez and Nagler 1998a; Greene 2003). This assumption – that the addition or removal of an alternative will not change the relative selection probabilities of existing choices – seems clearly false for voting models (i.e., a new socialist party can not take a higher percentage of votes from the older socialist party than from the conservatives). However, this naïve interpretation forgets that the assumption applies *conditional* on the variables included in the model. As Train (2003) points out, ultimately a well specified choice model that accounts for the major factors driving choice will achieve conditional independence of the errors in the value functions which is equivalent to having IIA hold. In Chapter 4 we estimate choice models that do not assume IIA (i.e., the multinomial probit model) and demonstrate that in fact the estimated vote probabilities are virtually identical.

perceptions on the average voter's vector of party support.³⁸ To do this we use the estimated coefficients (and variance-covariance matrix) from the model to produce predicted changes in support for each survey respondent when economic perceptions change by a given amount. We also produce appropriate measures of uncertainty around these predicted changes.³⁹ Next we average the predicted changes in support over the sample to get an estimate of the average *general economic vote*.⁴⁰ This procedure has the advantage that one can present a single summary measure of *general economic voting* for each case; but, even more importantly, these estimates are a much better representation of the true magnitude of *general economic voting* in the population than is choosing the average individual, as is often done (see Train 2003 for a discussion of the advantages of averaging over the sample in reporting results from discrete choice models).

Finally, all our estimates of economic voting rely on the same "given change" in economic perceptions. This change is a move in each respondent's reported economic perception one unit in the direction of a worsening economy.⁴¹ If the voter's perceptions were already at the worst category, we leave them where they were.⁴² This represents a reasonable shift in the distribution of economic perceptions based on their variability in our 163 surveys. We also calculated the opposite change in which each voter's perceptions get one category better.

³⁸ Given the MNL model is non-linear, the coefficients themselves do not reflect the substantive effects. Further, since we have different (and different numbers of) parties in each of the 163 estimations the coefficients on the economic perceptions variables will not be directly comparable across cases, though the substantive effects (given a well specified model) will be.

³⁹ More specifically, we set the independent variables in the model to the values for the first individual in the sample and calculated the impact of a given change in economic perceptions on that individual's vector of vote probabilities. We then repeated this procedure 1000 times, where in each replication we drew the parameters used to calculate the prediction from a multivariate normal distribution with the mean vector and variance-covariance matrix equal to the estimates from the model. This resulted in a distribution of estimated general economic votes for each individual. We then repeated this whole exercise for each individual in the sample and obtained summary measures by averaging over the sample. All the simulations were produced using the Clarify package in STATA version 8.0 with a minor modification to produce results in a useful format.

⁴⁰ Likewise, we average over the distributions of predictions across respondents to get an average over the sample of the distribution of changes in support (and so confidence bounds on the average point predictions).

⁴¹ We always code economic perceptions into a three point scale indicating whether the economy got better, stayed the same, or got worse, over the previous year.

⁴² We have also produced estimates of economic voting when all respondents' perceptions move from the neutral perception to the worse perception.

Comparing the results from the two measures reveals no asymmetry in economic voting – the size of the economic vote is the same in both cases (though with opposite signs).⁴³

We have now described how we estimate the change in party support resulting from a unit deterioration in economic perceptions. This gives us 655 estimated economic votes, one for each of the parties in our sample of 163 voter preference surveys. For example in the 1985 Dutch survey this procedure produce estimates of the changes in support for the CDA, VVD, Left, D66, Calvin and the PvdA parties of, respectively, $-.055$, $-.029$, $-.007$, $.016$, $.015$ and $.061$. Likewise in the 1997 British survey the estimates were $-.056$ for the Conservatives, $-.053$ for Labour and $-.004$ for the SDP/Liberals. These estimates can be used to construct a variety of measures of the economic vote corresponding to particular concepts. We discuss a number of these in the next section.

Defining and Measuring Other Concepts of Economic Voting

As the British and Dutch cases illustrate, our estimated changes in support will vary in number depending on the nature of the party system and they will include parties both in and out of government. Hence, on their own these vectors of changes in party vote probabilities cannot easily be compared. For example, was there more voting in the British as opposed to Dutch case? In order to answer that question or, more generally, to compare the extent of *general economic voting* across elections in which different numbers of parties compete, we need concepts that summarize the “size” or overall amount of electoral change that is represented by the *general economic vote* vector. Further, our theory will demand a concept of economic voting that reflects an asymmetry between incumbents and opposition parties. There are a number of possibilities that have different advantages and disadvantages which we now describe.

⁴³ When one considers that the distribution of economic perceptions varies considerably in different surveys, sometimes being skewed to the right and sometimes to the left, this result means the estimation method must be choosing different coefficients for different distributions of perceptions in order to “enforce” symmetry (i.e., the best fitting models have no asymmetry in predicted values). One result of this is that alternative ways to define a “meaningful change” that are often used but that are not conditional on the distribution of economic perceptions (e.g., setting each voter’s perceptions to the neutral category and moving them one category better or worse) must create asymmetry *by definition*. Further, the result calls into question previous studies that have diagnosed asymmetry in economic voting by comparing coefficients on dummy variables for “better”, “neutral”, and “worse” perceptions. If symmetry in economic voting as we have defined it is an underlying feature of the true model, but the distribution of preferences is skewed in different ways, the estimation must produce different coefficients for these dummy variables in order to achieve symmetry in the substantive impact of economic voting.

Volatility of the Economic Vote

One possible summary measure of the economic vote is based on the familiar concept of “electoral volatility” that has often been used by scholars interested in the electoral stability of modern electorates (e.g., Bartolini and Mair 1990). This concept captures how much electoral change occurs from election to election. It is calculated from observed changes in party vote shares by summing the absolute value of the change in vote share for each party (from one election to the next) and then dividing by two. The resulting number is the amount of the electoral support which “changed hands” from one election to another. We can produce an analogous concept for *economic* voting by applying the formula for electoral volatility to the *general economic vote* (which is, after all, just a vector of changes in support over parties). The result is the *volatility of the economic vote* (or just *volatility*) and is just the amount of electoral support that “changes hands” as a result of a given change in the distribution of economic perceptions. We can operationalize this concept by applying the standard formula for electoral volatility to our estimates of the *general economic vote*. Specifically, we sum the absolute values of the estimates of the *general economic vote* for each party and divide this number by 2.

The concept of the *volatility of the economic vote* is illustrated in Table 2.1 along with our concept of *general economic voting* and some other concepts of economic voting that we will discuss below. The table presents three hypothetical cases in which three or four parties are competing (we ignore non-voting). In each case, the columns provide a set of hypothetical changes to each party’s initial support that results when the voter’s economic perceptions worsen, holding all other influences on the vote constant. Thus, these columns are the *general economic votes* defined above.

Just looking at the *general economic vote*, it is difficult to compare the overall level of economic voting in these three cases. However, when we calculate the *volatility of the economic vote*, we discover that the amount of electoral support that changed hands due to the economy is the same for each case. Thus, the magnitude of the *general economic vote* is the same in the three cases, though its distribution across parties differs.

Table 2.1. Illustrating Different Concepts of Economic Voting

	<i>Changes in support due to worsening economic perceptions</i>		
	Case 1	Case 2	Case 3
Socialist (PM)	+.05	+.025	-.05
Liberal (Cab Partner)	-.03	+.025	+.025
Christian Democratic	-.02	-.025	+.025
Conservative (Cab Partner)	--	-.025	--
<i>Volatility of the Economic Vote</i>	.05 tied	.05 tied	.05 tied
<i>Economic Vote of the Chief executive</i> (changes "in the wrong direction" set to zero)	0 tied	0 tied	-.05 most
<i>Economic Vote of the Chief executive</i> (changes "in the wrong direction" included)	+.05 least	+.025 middle	-.05 most
<i>Economic Vote of the Government</i> (changes "in the wrong direction" set to zero)	-.03 least	-.025 middle	-.05 most
<i>Economic Vote of the Government</i> (changes "in the wrong direction" included)	+.02 middle	+.025 least	-.025 most
<i>Mean Economic Vote of the Government</i> (changes "in the wrong direction" set to zero)	-.015 middle	-.0083 least	-.025 most
<i>Mean Economic Vote of the Government</i> (changes "in the wrong direction" included)	+.01 least	+.0083 middle	-.0125 most

"Most" refers to the most economic voting and similarly for other rankings. Larger negative numbers indicate more economic voting.

The *general economic vote* and its aggregate, *volatility of the economic vote*, provide an upper bound to the amount of economic voting that can be found in a given case. This occurs because these concepts "count" any change in support that results from a change in economic perceptions as part of the economic vote. However, Table 2.1 also illustrates other concepts of economic voting that differ from *volatility* because their definitions exclude some changes in support that are included in the *general economic vote*. As an example, while the *volatility of the economic vote* is the same in the three cases in Table 2.1, the other concepts would lead to a different conclusion. The point here is that when we are comparing economic voting from one case to another our conclusions regarding the relative ranking of cases may depend on the particular conceptualization we choose. Consequently we describe several alternative concepts of the economic vote and identify the one most appropriate for our theory.

Economic Vote of the Chief Executive, Cabinet Parties, and the Opposition

The concept of economic voting that is most often used in empirical work is the *economic vote of the chief executive*. We define this concept as any decrease (increase) in support for the party of the incumbent chief executive that is caused by worsening (improving)

economic perceptions.⁴⁴ Similarly we define the *economic vote of a cabinet party* as any decrease (increase) in support for a cabinet party that is caused by worsening (improving) economic perceptions. The *economic vote of an opposition party* is similarly defined as any increase (decrease) in support for an opposition party that is caused by worsening (improving) economic perceptions. Unlike our definition of *general economic voting*, these definitions are explicitly directional and so require us not to “count” as economic voting any changes in support that are not in conceptually consistent directions (even if these changes are due to changing economic perceptions). To see why this is important, consider the *economic vote of the chief executive* for the cases in Table 2.1. *Volatility* is the same in all three cases, but the *economic vote of the chief executive* differs. Furthermore, we get a different ordering depending on how we “count” changes in support for the chief executive that are inconsistent with the definition given above. For example, in cases 1 and 2, worsening economic perceptions lead to an increase in support for the prime minister; however the definition of the *economic vote of the chief executive* applies only when worsening perceptions result in declining support.

Given this, what might cases “in the wrong direction” mean? If we apply the conceptual definition of the *economic vote of the chief executive* rigorously, the only thing these examples tell us is that economic voting is absent in cases 1 and 2 – they say nothing more about its magnitude. Indeed, the information about the extent of the *economic vote of the chief executive* is exactly the same as it would be if the change in support had been zero. Consequently, in trying to sort out “how much” economic voting there is from case to case, it would be appropriate to set those that are “in the wrong direction” to zero. This accurately reflects their status with respect to the conceptual definition we are using and removes the impression that positive values of the *economic vote of the chief executive* somehow indicate less economic voting than would a value of zero. Looking again at the table, this understanding leads us to the conclusion that the *economic vote of the chief executive* in both of the first two cases is properly considered zero, while for the last case it is -.05.

While this conclusion is consistent with our conceptual definition of the *economic vote of the chief executive*, when we turn to measuring the concept we must recognize that the relationship between support and economic perceptions is not observed directly but must be estimated in each case. This raises the possibility that estimates “in the wrong direction” can provide information that is usable (and interpretable) within our conceptual definition. Specifically, such estimates may arise due to estimation error, even though the true relationship

⁴⁴ By “incumbent chief executive” we mean the party that was chief executive at the time of the survey or election to which a survey refers.

(that we cannot observe) is consistent with the concept. Indeed, we are increasingly likely to produce such estimates when the true relationship is closer to zero. Thus, conclusions like “the *economic vote of the chief executive* should be seen as larger for case 2 than for case 1” can have a sensible interpretation if one understands that such statements assume (1) that the true relationship conforms with the conceptual definition (i.e., it is zero or negative) though the estimates do not, and (2) the estimation method is likely to produce larger inconsistencies the closer the true level of economic voting is to zero.

In the end, because our empirical analysis of contextual variation in the *economic vote of the chief executive* relies on estimates, we chose to include all the estimates in our maps of variation in economic voting (and in other analyses) even if they are “in the wrong direction” and to treat those estimates that are further from zero (in the wrong direction) as indicating increasingly less economic voting (interpreted as a greater chance economic voting for the particular case is zero).⁴⁵ As it turns out we are quite confident of this interpretation since the estimates of the *economic vote of the chief executive* described in the next chapter produce very few results in the wrong direction and those that are in the wrong direction are close to zero, which suggests strongly that they are in fact due to estimation error.⁴⁶ Similar observations apply to the *economic vote of a cabinet party* and the *economic vote of an opposition party*.

Economic Voting of the Government

Another concept of economic voting that has been used in previous empirical studies is the *economic vote of the government*. This concept is similar to the *economic vote for the chief executive* but places different restrictions on the elements of the *general economic vote* that “count” as economic voting. Specifically, it is defined as any decrease (increase) in support for the incumbent party or parties that is caused by worsening (improving) economic perceptions. Clearly, this reduces to the definition of the *economic vote of the chief executive* with one

⁴⁵ An alternative strategy would be to constrain the models that produce the estimates to only produce estimates that are consistent with the conceptual definition. If the constraint is wrong because the theory that produced the concept is wrong, then the model will not fit the data well and the underlying theory and concept should be rejected. We do not pursue this approach here because, correctly or not, many other scholars view such constraints in empirical models with suspicion (did we somehow cook the results?). Of course, if this really mattered to our substantive conclusions we would spend the time necessary to convince readers that this is an appropriate strategy. However, as it turns out, there are so few estimates “in the wrong direction” it does not matter to our substantive results which of these methods we choose.

⁴⁶ Of course, future researchers examining other samples (e.g., in developing countries) may well find many more inconsistent estimates. If the number of these is large then it becomes less and less plausible that the true relationships are actually consistent with the definition provided here and so the utility of the concept itself (and the theories from which it arises) should be questioned.

incumbent party, but may produce different levels of economic voting when there is more than one incumbent.

Although this concept is often used with little discussion of its meaning and measurement, there are at least three sources of ambiguity in the concept that limit its usefulness. The first is the definition of incumbency. Few theories of economic voting are precise enough to dictate a single appropriate concept of incumbency. For example, the traditional reward/punishment explanation for economic voting has voters' attributing responsibility (and punishment) to incumbents for bad economic performance but is silent about which parties should be considered incumbent.⁴⁷ The situation only gets worse when many parties participate in government, where minority government is possible, and where opposition parties play an important role in policy making.

The second conceptual ambiguity that arises in defining the *economic vote of the government* concerns the rule for aggregating the economic vote for each party in the cabinet into an economic vote for the government as a whole. There are at least two immediately plausible options for defining the concept: the *economic vote for the government* could be the sum of the *economic vote of the cabinet parties* or the average economic vote over these parties.⁴⁸ Again, theoretical discussions in the empirical literature that have used this concept provide little guidance in making an appropriate choice, though the measurement convention seems to be to sum the economic vote for each party (e.g., Powell and Whitten 1993). While this choice does not seem to reflect a theoretical commitment to a particular conceptual definition, it could certainly matter empirically.

The final ambiguity that arises in defining the *economic vote of the government* is closely related to the one we identified for the *economic vote of the chief executive* - it asks how we should include changes in support for cabinet parties that are in the wrong direction in the summation that produces the aggregate economic vote? In this case, however, a close adherence to the conceptual definition does not give us any guidance since the definition is about an aggregate (though it would suggest counting as zero any *sum* that was in the "wrong direction"). Our discussion of how to count changes in support in the "wrong direction" for chief executives applies directly to the economic of each cabinet party that goes into the summation for the *economic vote of the government*. Again, there is no conceptual justification for including these

⁴⁷ Indeed, even in the relatively simple American case, this ambiguity has spawned an empirical literature that tries to resolve the question by testing whether economic voting is apparent when one focuses on different plausible incumbents (e.g., American Governors, Representatives, Senators, or the President – Leyden and Borrelli 1995; Lowery, Alt, and Ferree 1998; Peltzman 1987).

⁴⁸ We could also consider a weighted sum or average over the government parties (where the weight is the relative influence of the party in cabinet).

errant changes in the summation; however, because they are estimates, they may only be in the wrong direction because the true values were close to zero (but in the right direction). For chief executives, we were reassured that interpreting wrong-signed estimates as due to estimation error was reasonable because in our data there were few estimates of *economic voting for the chief executive* in the wrong direction. However, in the case of the *economic vote of a cabinet party* we have many more wrong-signed estimates and so we are correspondingly less comfortable assuming they arise due to estimation error.⁴⁹

The issues we have raised about the concept of *economic voting for the government* should give us pause. There was certainly no universally applicable resolution for any of them and taken together the various plausible ways the different ambiguities could be resolved leads to a large number of plausible definitions of *economic voting for the government*, each of which could be used as the basis of an empirical study.⁵⁰ Thankfully, however, there is no real need to tolerate such ambiguity in the study of contextual variation in economic voting. Specifically, as we show in the next chapter, focusing on the *economic vote of the chief executive* rather than the *economic vote of the government* is seldom consequential substantively.⁵¹ Further, when parties other than the chief executive are of interest, rather than aggregate the economic votes of several parties, one can simply focus on the economic vote of the parties individually (defining an appropriate concept specific to these parties). This greatly reduces the problem of defining incumbency and avoids the aggregation problem entirely. Further, it presents few additional methodological difficulties, and provides much more detailed information about the pattern of economic voting across parties. This is the strategy that we pursue in Chapter 9 where we reveal that the extent and consistency of economic voting for cabinet partners is very different from that of the chief executive. This difference underlines further the wisdom of avoiding aggregating the economic vote of multiple cabinet parties in studying contextual variation.⁵²

⁴⁹ As was the case with individual parties one could similarly define an economic vote for the opposition as a whole and which would face the same issues with aggregation.

⁵⁰ Given these problems, we avoid the use of this concept in the rest of the book. However, in the next chapter, we do compare it to the *economic vote of the chief executive*. Thus, for that purpose we follow convention and identify incumbents as those parties that hold cabinet seats, use the sum of the economic votes of the cabinet parties (rather than their average), and include “wrong-signed” estimates in the sum.

⁵¹ Clearly for a large subset of cases – one party governments – these are equivalent, but (as elaborate in chapter 11) even when several the economic votes of other governing parties are aggregated along with that of the prime minister, their economic votes tend to be very small relative to that of the prime minister and so while the two measures *could* differ considerably, in our sample they do not.

⁵² This caution applies both to the procedure of aggregating estimated economic votes for parties as we do here and the more common procedure of aggregating each individual respondents vote over government

Summary

In this chapter we have tried to delve somewhat more deeply than is usual into the conceptual definition of economic voting and to make plain the tradeoffs between different conceptual choices. This attention to definitional issues is absolutely necessary for studies, such as this one, that must compare the magnitude of economic voting across cases. As we have seen, different definitions of economic voting can easily lead to different orderings of cases when size comparisons are necessary. Of course it is an empirical question whether these definitional differences turn out to be practically important for understanding the pattern of variation in economic voting or for exploring the sources of this pattern. Nevertheless, our conceptual discussion does lead us to favor some measures over others. Specifically, the theory that we present in chapters 5 and 8 clearly calls for a definition of economic voting that is directional in the usual way (i.e., economic voting is any decrease (increase) in the vote of incumbent parties that is caused by worsening (improving) economic perceptions and vice-a-versa for non-incumbents).⁵³ Thus, when our theory calls for consideration of the economic vote of a particular party we use the *economic vote of the chief executive*, the *economic vote of a cabinet party*, or the *economic vote of an opposition party* rather than the general economic vote.⁵⁴ Likewise, when our theory calls for consideration of the general level of economic voting in an election rather than for a particular party, we should rely on directional measures. Thus, we seldom use the volatility of the economic vote since it “counts” any change in the support of parties due to the changing economic perceptions as an indication of economic voting. Likewise, the various problems with the concept of *the economic vote of the government* lead us to rely on the *economic vote of the chief executive* as our usual summary measure of overall economic voting in an election.

With some examples from our sample of surveys we can illustrate the consequences, for the ranking of cases, of adopting different concepts of the economic vote. Table 2.2 presents estimates for the *economic vote of the chief executive*, *economic vote of the government* and *volatility of the economic vote* from the 1983 German and the 1985 Dutch surveys. What is particularly interesting here is that each of the measures suggests a different conclusion regarding

parties (i.e., coding a 1 if the voter voted for a government party and 0 otherwise) and then using this individual level variable to estimate a single model of economic voting for the government as a whole.

⁵³ The various versions of the theory are quite specific in providing appropriate definitions of incumbency, so there is no ambiguity of the kind mentioned above.

⁵⁴ Though not mentioned above, the last of these is defined analogously to the others.

the ranking of the two cases. The *volatility of the economic vote* suggests that the two cases have similar levels of economic voting: .085 for German and .091 for the Netherlands. The other two measures suggest exactly the opposite ranking of the two cases: In the case of the *economic vote of the government measure*, the Netherlands has a much higher economic vote (-.084) than Germany (-.058). On the other hand, the *economic vote of the chief executive* measure ranks Germany with a much higher economic vote (-.08) than the Netherlands (-.055). While it is an empirical question (addressed in the next chapter) whether this type of reversal is a more general problem, it illustrates the need for careful theoretical justification of the particular concept of the economic vote employed for estimating contextual effects. The next chapter will report our initial description of contextual variation in the overall economic vote which is measured by the *economic vote of the chief executive*. Subsequent empirical chapters will use some of the other concepts, again depending on the particular theoretical issue being addressed.

Table 2.2. Two examples of alternative measures of the overall economic vote

	Germany 1983	Netherlands 1985
<i>Economic vote of the chief executive</i>	-.08	-.055
<i>Economic vote of the government</i>	-.058	-.084
<i>Volatility of the economic vote</i>	.085	.091

Chapter 3

Patterns of Retrospective Economic Voting in Western Democracies

The analyses of the 163 surveys described in the previous chapter generate a wealth of data: a total of 678 economic vote measures for 105 political parties in 18 countries over a 20-year period. As we pointed out in the introduction, the theoretical and empirical focus of this book is on explaining contextual variation in the economic vote, as defined in the previous chapter. Before proceeding to these efforts, we address two particularly critical empirical tasks in this chapter: Is there a significant and pervasive economic vote in advanced democracies? Secondly, is there significant contextual variation in the economic vote that is sufficiently interesting to justify a book-length treatment?

The first question we posed is not simply rhetorical. There is considerable skepticism as to the importance of economic voting (a recent statement is Cheibub and Przeworski, 1999). If, as some would argue, economic voting were apparent in only a small number of countries or elections, or if it were wildly inconsistent in direction and magnitude, then there would be little reason to try to understand how contextual factors might affect it. Part of this skepticism concerns the importance of economic evaluation relative to other influences on the vote? Again, if economic evaluations have a much weaker influence on the vote than other factors, then our time would be better spent studying contextual variation in those factors. Readers can probably anticipate the answers to these questions. The evidence presented in this chapter indicates that economic voting of the usual kind (i.e., incumbents are hurt by a poorly performing economy) is pervasive in the advanced democracies. Indeed, the size and pervasiveness of economic voting estimated here should put an end to any speculation about the importance of cross-national economic voting. Further, in the last section of this chapter, we show that economic voting is not only important in many different contexts, but is also among the most important influences on the vote in any given context.

The economic vote may be pervasive but as we establish in this chapter it varies quite significantly across contexts. An important goal of this chapter is to demonstrate that this contextual variation in our particular measure of the economic vote conforms to the patterns of variation noted in the previous literature (albeit for limited samples of cases). And indeed, the comparison of our contextual variation with those in the literature provides strong face validity for our measure of the economic vote. Our description of contextual variation in the economic vote in this chapter also serves an important theoretical purpose. Our measurement strategy and the relatively large size of our sample of voter preference surveys allows us to provide a much richer description of contextual variation in the economic vote than has been the case in the

previous literature. This allows us to highlight patterns of variations in the economic vote that have never been examined in the previous literature. Economic voting varies across countries; it also varies over time, or from one election to the next, within countries; it varies across political parties (controlling for their governing status), but also the economic vote of parties varies over time. These patterns of contextual variation in economic voting that we graphically depict in this chapter are the foundation for our theoretical and empirical efforts in the rest of the book. They provide the clues as to what features of the political and economic context voters likely use to condition the magnitude of their economic vote.

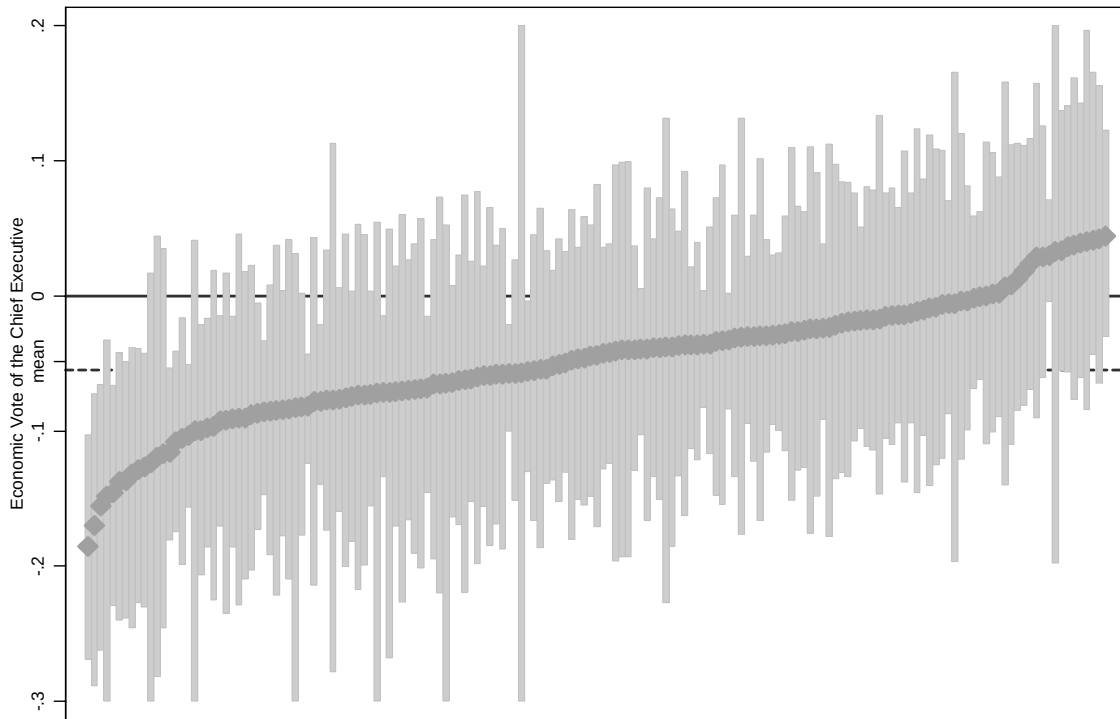
Magnitude of the Economic Vote and its Variation

In the previous chapter we pointed out that there are good conceptual reasons for focusing on the *economic vote of the chief executive* as a measure of the magnitude of overall economic voting. Most importantly, and unlike the *volatility of the economic vote*, it is appropriate for testing *directional* theories of economic voting (in which only some changes in electoral support are “counted” as economic voting). It also avoids the need to aggregate (in essentially arbitrary ways) the economic votes of incumbent parties. In this section, we demonstrate that patterns of variation in the magnitude of the *economic vote of the chief executive* are quite consistent with the existing economic voting literature – again providing assurance that our measurement strategy is sound. Furthermore, the contextual variations in the economic vote magnitudes point to interesting patterns of variation, the explanation of which is the subject of subsequent chapters.

We begin with Figure 3.1 which orders our estimates of the *economic vote of the chief executive* from highest on the left to lowest on the right (there are 163 estimates here corresponding to each of the voter preference surveys in our sample). The line rising from the lower left to upper right quadrants of the graph represents our point estimates of the *economic vote of the chief executive* while the vertical bars represent 95 percent confidence intervals around each point estimate. Recall that given our measurement strategy a larger economic vote is one in which the party *loses* more support as economic perceptions worsen. Thus, larger negative values indicate more economic voting. The *economic votes of the chief executive* range between about -20 percent to about 8 percent with a standard deviation of .04. Approximately 90 percent of the *chief executive* votes have negative values and about 50 percent have values less than -4.5 percent.⁵⁵ The average *economic vote of the chief executive* is -5 percent and is represented by

⁵⁵ The significantly larger number of “wrong-signed” values for incumbent parties suggests that voters treat junior coalition partners differently than the Chief executive party – a theme we address in later chapters. It

the horizontal dashed line. These simple distributional plots indicate that ruling parties are punished as economic perceptions worsen. A moderate decline in perceptions of economic performance typically results in a 5 percent drop in support for chief executive parties.



Confidence bounds greater than .2 and less than -.3 truncated for display

Figure 3.1. Confidence Bands for Economic Vote of the Chief Executive

The magnitude of this result is consistent with the historically modest electoral volatility found in most elections in established democracies. For example, in their analysis of 93 parliamentary elections, Powell and Whitten (1993) estimated that the average single party majority government – the equivalent of our chief executive party – lost 3.6 percent of the vote. Our estimated *economic vote of the chief executive* of 5 percent is certainly consistent with such a finding. In contrast, estimated levels of economic voting that deviate significantly from these modest levels are not really plausible. Hence reports in other studies of typical shifts in support of 30 or 40 percent of the vote due to changing economic perceptions are probably not representative of the kinds of changes that actually occur. These inflated economic vote

also suggests why treating the combined votes of incumbent parties as a single measure of the economic vote may be ill advised.

magnitudes likely result from reporting substantive effects for non-representative individuals or for massive shifts in perceptions that do not reflect typical changes in economic evaluations.⁵⁶

Clearly our estimated shifts in support due to the economy are substantively important. The median economic vote magnitude is about 5 percent and one can easily identify elections in which swings in support of this order determined winners from losers.⁵⁷ Nevertheless, Figure 3.1 indicates that if one were to base an assessment of economic voting on a single election study one might conclude that the economic vote is not significantly different than zero. The advantage of working with a large number of diverse surveys is that we can examine not just the statistical significance of each study separately but can also consider the total density of the 95 percent confidence intervals for all of 163 estimates of the economic vote (the grey area in Figure 3.1). If the density of the confidence intervals was equally distributed above and below the zero line – or if it was concentrated above the zero line – we would reject the notion of an economic vote significantly different than zero. In fact we find that over 70 percent of the density of these intervals falls *below* zero. The picture, then, leaves little doubt that there is a consistent cross-national economic vote that hurts incumbents who preside over a period of worsening economic perceptions. Further, Figure 3.1 indicates that drawing conclusions regarding the magnitude of the economic vote from one – or even a small number – of voting studies is likely to be misleading given that any one study has a good chance of generating point estimates that are not statistically different from zero in typical sample sizes.⁵⁸

This finding is the first comprehensive evidence of economic voting at the individual level across a large number of countries and over an extended period of time. These results are important because they lay to rest any reservations about the generality of economic voting in the western democracies. Cheibub and Przeworski (1999), for example, have raised questions about the economic vote, noting that, with some relative minor exceptions, “...economic performance does not affect the survival of heads of democratic government even during the years that elections are held (Cheibub and Przeworski 1999: 237).” Our findings suggest that these, and other (Paldam 1991), reservations about the prevalence of economic voting in developed

⁵⁶ Lewis-Beck (1988: 84) for example reports substantive effects in the range of 16 to 54 percent which are based on changing scores on all his economic variables from neutral to a position of discontent.

⁵⁷ The recent U.S. presidential elections electing G.W. Bush are an obvious case in point. Lewis-Beck (1988: 86) also points to the 1974 and 1981 French presidential elections where the margin of victory was significantly lower than 5 percent.

⁵⁸ To be clear, we are arguing that even in cases in which economic voting for the chief executive is small – so that a typical sample of 1000 respondents is not enough to distinguish the small effect from zero – the larger view proposed by Figure 3.1 should give us comfort that economic voting in this case really is there.

democracies are unfounded. In addition, our results provide rigorous support for comparativists who have found economic voting at the individual level, although based on survey results from a small sample of countries (Anderson 1995; Blais et al 2004; Lewis-Beck 1988). Hence, economic voting is alive and well and pervasive, at least in developed democracies.

Contextual variation

We clearly have established that there is an economic vote and that it varies significantly. Let's now explore the patterns of contextual variation in the economic vote. Again, our overall objectives here are twofold: One goal is to establish that the patterns of contextual variation in our measure of economic voting are consistent with the results of other comparative economic voting studies. A second objective of this exploratory data analysis is to flag patterns in the data that raise interesting theoretical puzzles to which we will return in subsequent chapters.

Recall that the estimates for our economic vote are derived from 163 individual voter preference surveys (from 19 countries and over a 20 year period) and that estimates from each of these specific surveys are in some sense the unit of analysis for our second stage estimations. Hence the total variation in economic vote summarized in Figure 3.1 can be broken down into three different components: country, temporal and survey-specific variation. Variation in average economic voting from country to country is simply the economic vote for each country averaged over all of the years in our sample. Variation in average economic voting from year to year is the economic vote in each year averaged over all countries in our sample. What remains is survey-specific variation which is variation not captured by the country and time specific effects.

Figure 3.2 provides some insight into these components of this variation by taking the *economic votes of the chief executive* and their confidence intervals from Figure 3.1 and organizing them according to country and vote survey date. We can, for example, identify country differences in the graph: We find an identifiable group of countries in which the *economic vote of the chief executive* is usually close to zero: Italy, The Netherlands, and U.S. congressional elections, for example. We can also identify a cluster of countries in which the *economic vote of the chief executive* is very high: the U.K., U.S. presidential elections, Ireland, New Zealand, and Spain. But what Figure 3.2 highlights, and much of the literature tends to underplay (an exception is Nadeau et al 2002), are the interesting patterns of variation within each of the countries. Ireland and Germany, for example, seem to have decreasing levels of economic voting throughout our sample period which of course could be a temporal trend (which is shared with the other countries in our sample) or the result of survey-specific effects (changes in the perceived competitiveness of the incumbent coalition parties, for example). But before we assess

the relative importance of these different components of economic vote variation, let's look more closely at each of these individual country results which are in themselves quite informative.

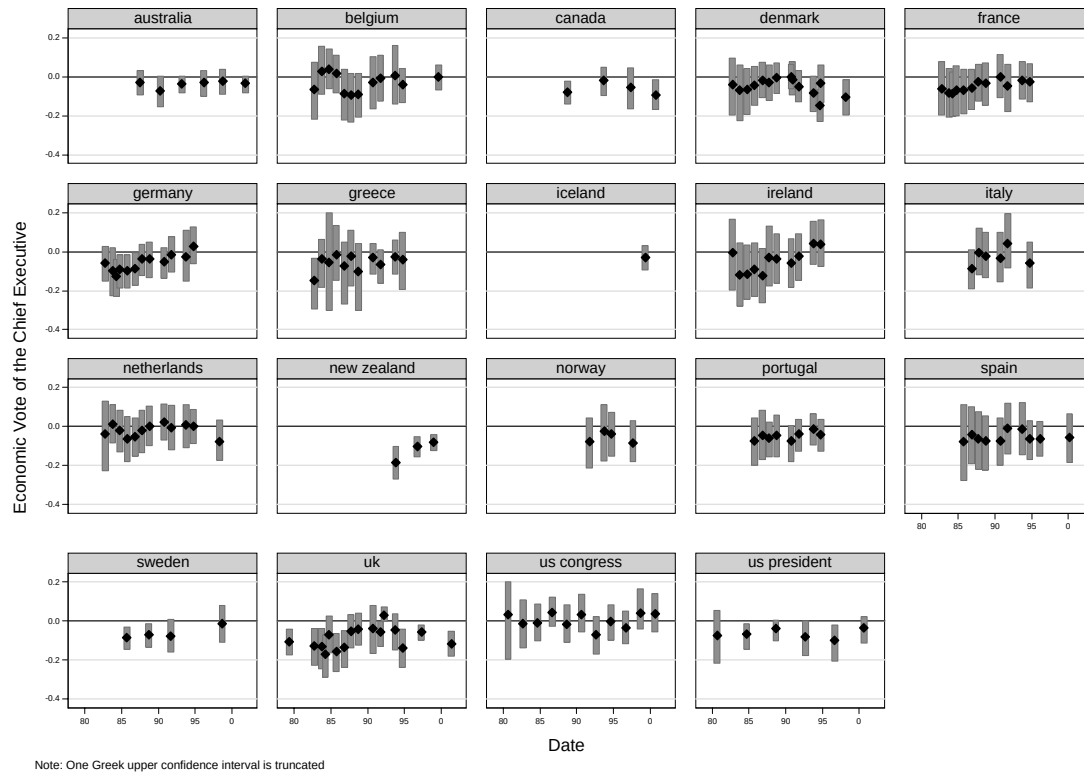


Figure 3.2. Economic Vote of the Chief Executive by Country and Voter Survey

The variations in the economic vote in each of the individual countries in Figure 3.2 are quite consistent with other findings in the comparative economic voting literature; at the same time they point to interesting puzzles. The U.S. results are a case in point of the results lining up exactly as the literature predicts. Specifically, the extensive literature on economic voting in the U.S. has consistently found a strong economic vote in presidential elections (Erikson, Mackuen, Stimson 2002; Kiewiet 1983) while indicating a much weaker impact of the economy in congressional elections (Alesina and Rosenthal 1995; Erikson 1990; Keech 1995). Our results are consistent with both these expectations: U.S. presidential elections produce high levels of economic voting, compared to the other countries in our data, while we show almost no economic voting for the party of the president in congressional elections. It is also interesting (and encouraging) that the modest negative spikes that do occur in the congressional economic vote series (i.e., years of higher economic voting) correspond with presidential elections, which is again what the U.S. economic voting literature would predict. The relatively weak levels of

economic voting in the 1988 and 2000 presidential elections are also consistent with other findings and speculations that the economy should play a less important role in elections where there is no sitting U.S. president (Nadeau and Lewis-Beck 2001; Norpoth 2002). And the relatively high levels of economic voting in the 1992 and 1996 presidential elections are consistent with other analyses of these elections (Alvarez and Nagler 1995, for example).

The U.K., as most would expect, also ranks as having a relatively high level of economic voting but this clearly varies. During much of the 1980s the economic vote for the party of the prime minister was extremely high, in the -10 to -15 percent range, while in the 1990s the economic vote was much closer to -.05 percent. This is consistent with other efforts to measure the magnitude of the British economic vote. For example, Evans (1999a) documents the relatively weak impact economic evaluations had on vote choice in the period 1992 through 1995. And Blais et al (2004) found that the 1987 election in Britain generated a particularly high level of economic voting, dropping to relatively low levels in the 1992 and 1997 elections and then perking up again in the 2001 election. Readers will note in Figure 3.2 that this is precisely the pattern of results that we obtain. One explanation for this pattern that we explore in more detail in Chapter 6 is that the economic context differed significantly between the late 1980s and 1990s which may have affected the importance of the economy for vote choice in the U.K.

We see a similar pattern in the German case with economic voting declining over the sample period. Again economic context is a plausible explanation here. This decline could be the result of a significant improvement in inflation performance during this period; this would be consistent with Anderson's (1995) argument that inflation has been a key economic variable driving the economic voting in Germany. But changes in the German political context offer an equally plausible explanation. Note that until 1986 the German economic vote is in the neighborhood of -10 percent and it drops rather significantly to -.04 in 1987. The 1987 election was a watershed election; it marked the transition of Germany from a three- to a four-party system with the Green party polling an equal number of votes as the FDP (roughly 10 percent each). By complicating the strategic voting considerations and the alternative coalitions that could replace the incumbent Kohl government this may have moderated the economic vote.⁵⁹

The Italian case is also suggestive, although based on only one observation, that dramatic changes in the party system can affect the magnitude of economic voting. Consistent with the common wisdom on post-war Italy (Bellucci 1991) we find almost no economic voting (although

⁵⁹ Our 1987 measure of the economic vote is based on a EuroBarometer survey that was conducted in November of 1987. Hence, voters are responding to a more complex political landscape that had been in effect for approximately 10 months when the EB survey was conducted (the 1987 German election occurred in January).

1986 is an exception) prior to the electoral reforms of 1993. We do though find a very large economic vote in 1994 which was the first election under the new election laws of 1993 that provided for single member districts and was the catalyst for subsequent majoritarian trends in the Italian party system.

At least superficially, the French results seem to confirm arguments regarding the impact of cohabitation on economic voting in that country (Lewis-Beck and Nadeau 2000). Note that the magnitude of the French economic vote ranges between -6 to -8 percent prior to 1987 (the first full year of cohabitation) but then falls rather precipitously to around -3 percent in 1987 and 1988. It recovers to around -5 percent in 1991 but then falls again during the next period of cohabitation, 1993 and 1994. We will revisit this variation in French economic voting in our discussion of government responsibility and the economic vote.

Spain is another interesting case where levels of economic voting are a rather constant five to seven percent over most of the time period in our sample. The exception is 1991 and 1993 when we see the economic vote drop to essentially zero. Part of the explanation may lie with the recognition by voters that Spain was feeling the effects of a European recession – which by 1991 was beginning to take a serious toll on the Spanish economy – over which the González government had little control. Our theoretical results, discussed in later chapters, suggest that voters likely discount the importance of economic outcomes in their vote decision when these outcomes are seen as resulting from exogenous factors beyond the control of incumbent governments. The 1993 economic vote could also have been influenced by the narrow election victory for the González government that forced the Socialists to govern in a minority with the parliamentary support of the Catalan nationalists. Again, we will argue in subsequent chapters that as policy making responsibility is shared more equally declines – which would be the case with a minority as opposed to majority government – economic voting will decline.

Our results for Spain are at odds with the excellent set of historical accounts of Spanish economic voting described in Maria Maravall and Przeworski (1998) and Fraile Maldonado (2001). Both accounts focus on post-Franco Spanish democracy and show the size of the economic vote (particularly the retrospective economic vote) increasing as the Socialist government aged (it was initially elected in 1982 and replaced by the People's Party in 1996). We in fact find little variation in the economic vote with the exception of 1991 and 1993, the tail end of the Socialist rein, when, contrary to these historical accounts, we find very little economic voting.

Other country results exhibit similarly interesting patterns of temporal variation. In Belgium during the early 1980s and much of the 1990s there is essentially no economic voting

but during the second part of the 1980s, the economic vote is actually quite large. The 1993 economic vote in Canada declines rather precipitously relative to the 1988 election, which is consistent with Alvarez et al (2000) although not with Blais (2004).⁶⁰ Our finding might not be entirely unexpected given that the 1993 election represented such a radical melt-down of the Canadian party system. The economic vote in Ireland has a remarkably linear and negative trajectory; moving from having a rather high economic vote in the early 1980s to having no economic voting by the mid-1990s.

Overall, the pattern of variation within countries that is revealed in Figure 3.2 is both encouraging and intriguing. It is encouraging because, with the possible exception of the Spanish case, it conforms to what we already know about the magnitude of economic voting in those few cases in which explicit temporal comparisons of economic voting are available or for which a “common wisdom” has otherwise emerged. And it is intriguing because it reveals patterns of variation that on their face look like they could have systematic causes. As we have already pointed out, some of the largest temporal changes in economic voting occur during equally dramatic periods of political and economic change. Further, much of the estimated temporal variation in economic voting within countries is “smooth”. That is, the estimates for any country are not fluctuating up and down from year to year but seem to trend up or down in somewhat longer cycles – as if they are responding to systematic forces rather than simple randomness.⁶¹ We can gain further insights into these variations by examining the individual components of the variation in the economic vote described above. Specifically, by aggregating the estimates from Figure 3.2 within countries we can get an ordering of the average economic vote for each country and compare our ordering to other work. Likewise, by aggregating within years we can examine the secular changes in economic voting that may apply across countries and benchmark them in a similar fashion.

Cross-national variation

Figure 3.2 suggests that countries vary in their overall level of economic voting. In order to put this in better perspective, Figure 3.3 provides a box-plot of each country’s *economic votes of the chief executive* over the sample period. The median country in this sample is Canada with an *economic vote of the chief executive* of about -5 percent. And the pattern of cross-national

⁶⁰ Our comparisons with Blais et al (2004) are with respect to their Table 2 (page 526) which employs a method of estimating the magnitude of the economic vote that is most similar to ours.

⁶¹ There is a significant correlation between adjacent surveys within countries (a positive regression coefficient of .35 in a bivariate regression of *the economic vote of the prime minister* and its “lag”, $t = 5.25$).

variation is consistent with what we know from previous work. A case in point is Lewis-Beck's (1988) classic study that measures magnitude of cross-national differences in the economic vote, based on individual-level surveys. Our rank ordering of countries in terms of the average magnitude of the economic vote is precisely the same as those in Lewis-Beck's analysis (from highest to lowest: Britain, Spain, Germany, France, and Italy). The fact that Britain ranks as having the highest level of economic voting (next to New Zealand) is consistent with Anderson (1995) who finds that Britain unambiguously has the highest level of economic voting in his sample of five countries. The low ranking of The Netherlands and Denmark also fits with Anderson (1995) findings. Our rankings are also roughly consistent with the ranking of the eight nation study conducted by Nadeau, Niemi and Yoshinaka (2002): the UK has the highest magnitude in their results; France and Germany rank second which is consistent with Figure 3.3; and in their estimates Italy, the Netherlands and Belgium have the lowest economic vote magnitudes, again consistent with our findings. Our rankings of Denmark and Ireland, on the other hand, do not correspond with theirs.

The U.S. has been the focus of much of the empirical work on economic voting which, for the most, part suggests relatively high levels of economic voting in the country (Alvarez and Nagler 1995; Kiewiet 1983). Our results in Figure 3.3 tend to confirm this general expectation: The median economic voting score for U.S. presidential elections is about -6 percent and falls slightly above the median level of economic voting in our sample. Note though that the variation in U.S. economic voting is significant, rising to as high as -10 percent in 1996 but falling to almost zero percent in the 2000 election study. And as we pointed out above, the U.S. results for congressional elections support the conventional wisdom – the median economic vote in U.S. congressional elections ranks as the lowest case of economic voting in our sample (but note again that there is considerable variation amongst these elections).

In short, for those cases in which previous work provides some expectation about the likely pattern of cross-national variation in economic voting, our measures are consistent with these expectations. This should, of course, increase our confidence in the overall patterns of variation described in Figure 3.3.

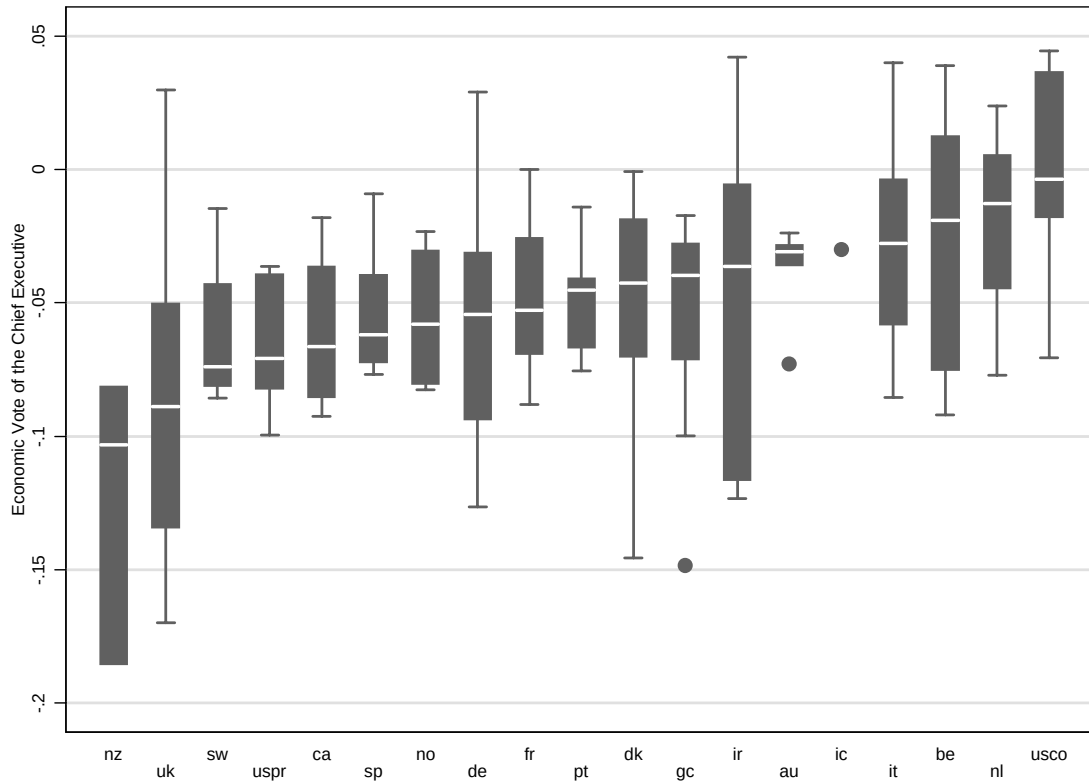


Figure 3.3. Magnitude of the Economic Vote by Country

Overtime variation

A number of the individual countries examined earlier exhibited clear temporal trends in the level of economic voting – the declining levels of economic voting in the U.K., Ireland and Germany are a case in point. Some in the literature suggest that the developed democracies share a common temporal trend in the economic vote. Anderson (1995: 35), for example, identifies a positive trend in the economic vote and suggests that it is the product of the government's increasing role in the economy which he argues raises the voters' sensitivity to economic performance. Anderson's (1995) results suggest that there might be an asymmetric period effect whereby economic voting is moderated during particularly good economic times (in his case pre-1973) and exaggerated during difficult times (again, in his case, the post-oil crisis). Others have argued that global economic convergence should reduce overall levels of economic voting because national governments have declining control over economic outcomes (Alesina, Cohen and Roubini 1997; Alvarez, Nagler and Willette 2000). Political factors have also been suggested as a possible explanation for trends in economic voting. Kayser and Wlezien (2005) argue that economic voting is negatively related to levels of partisanship and hence declining levels of partisanship in Europe over the 1976-1992 period has resulted in an increase in the economic

voting. They speculate that the opposite has occurred in the U.S. where they point out partisanship appears to be rising. Clearly there are conflicting views of the precise nature of recent trends in economic voting. Our relatively large sample of economic vote magnitudes can provide some insight into the veracity of these claims regarding trends in the economic vote.

Figure 3.4 allows us to assess visually whether there is any overall trend in the economic vote. In this figure we present box-plots of the economic vote in each of the years of our sample. By now it should not surprise readers to see considerable country variation in the magnitude of the economic vote in any particular year. Judging by the median vote in each year, there is some evidence that in the pre-1990 period the typical *economic vote of the chief executive* was larger than in the post-1990 period. In the period 1990 through to 2001, seven of the eleven years had median economic votes of less than 5 percent. In the period 1979 to 1989, three of the eight years had median economic votes of less than 5 percent. Hence to the extent that there is any temporal trend in economic voting it appears to be declining moderately.

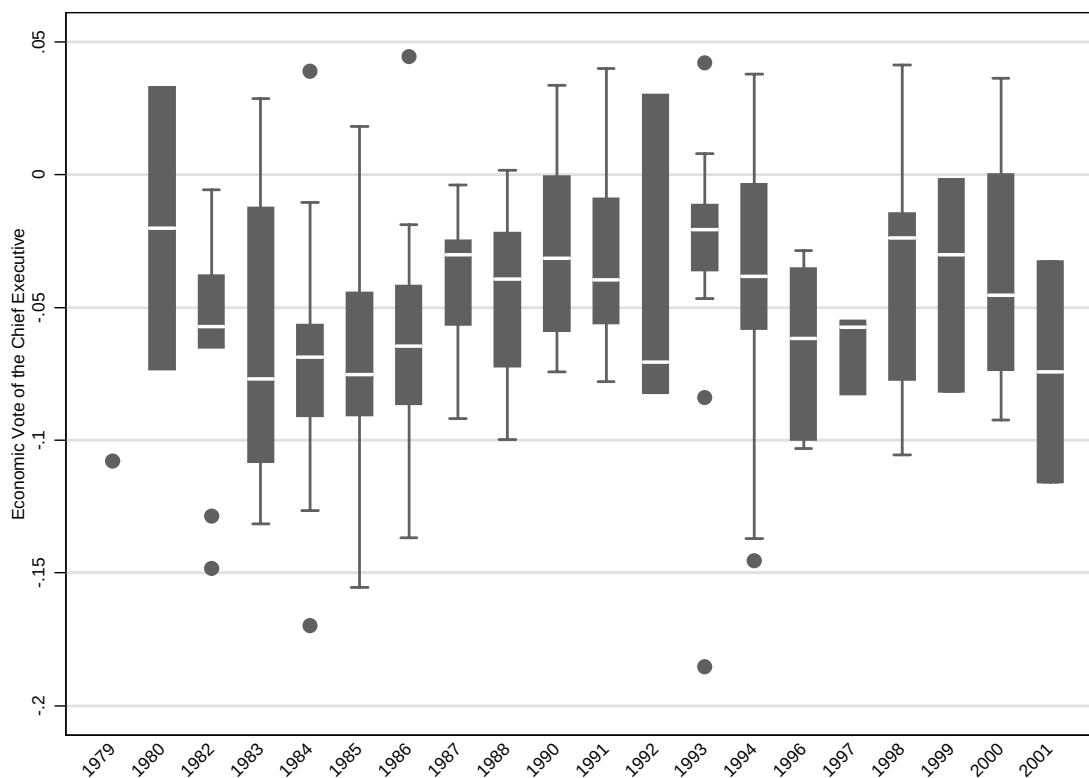


Figure 3.4. Magnitude of the Economic Vote by Year

Are these temporal variations in the economic vote consistent with the economic voting literature? Claims that the global economies are converging which should in turn reduce the

magnitude of economic voting (Alvarez, Nagler and Willette 2000) might be consistent with the fluctuations in Figure 3.4.⁶² Arguments, such as those of Kayser and Wlezien (2005) that presume rising economic voting in developed democracies, are problematic because the case for increasing economic vote magnitudes seems unlikely given the results summarized in Figure 3.4. The decline in economic voting (i.e., trending toward zero) is particularly obvious between 1984 and 1990 which is part of the period in which Kayser and Wlezien (2005) argue the economic vote should be rising.⁶³ And Anderson's (1995) claim that the economic vote tends to be exaggerated during particularly bad economic performance might be consistent with our finding that economic vote is moderately declining over the sampled period. Clearly there is no consensus in the literature regarding temporal trends in the economic vote. Accordingly, our evidence of a moderately declining economic vote would seem to be consistent with some of the arguments in the literature but obviously challenges others.

The components of the variation in economic voting

Our visual inspections of the economic vote confirm that there is contextual variation of a nature that poses some interesting theoretical puzzles. What is not entirely clear from these graphical displays is precisely how this variation is apportioned among the three different components described earlier. How large is the temporal variation relative to the cross-national variation? And is there any significant residual variation that is survey-specific? Understanding these relative magnitudes is important because they provide a guide to the theoretical variables that are likely to have the highest explanatory pay-offs. One convenient way to do this is to estimate a variance components model in which we partition the economic vote for any particular case into four additive parts, an overall mean, a country specific effect, a time specific effect, and a residual effect that is particular to the specific case. By treating the last three of these effects as random draws from appropriate zero-mean distributions, we can estimate the respective variances of these distributions and so get an indication of how much of the total variation in the economic vote is accounted for by differences in country means, differences in temporal means, and factors

⁶² Although the year 2001 seems to call into question this argument, it should be noted that there were only two voter surveys in our sample from this year – the UK and Australia – and hence this particular time point is not particularly informative when assessing temporal trends in the data.

⁶³ The Kayser and Wlezien (2005) analysis includes EuroBarometers extending back to 1972 while our analysis begins in 1979. Nevertheless, this trend is quite unlikely given the declining pattern of economic voting that we observe in the 1980-1990. Kayser and Wlezien (2005) analyze only a subset of the countries in our sample but most of them – the UK, Germany, Ireland and France – exhibit declining economic voting in our analysis.

associated with specific surveys.⁶⁴ This is useful because it will immediately tell us whether explanatory variables that are constant within countries, constant across countries, or variable over time and across countries are most likely to account for the total variation in economic voting that we observe.

Our estimate of the total variation in the *economic vote of the chief executive* across all cases is .066.⁶⁵ 23% is accounted for by variation in country means, 4% by variation in yearly means, and 73 percent by survey specific variation. To put it another way, if we could identify variables to account for 100 percent of the variance in economic voting, then we would expect almost three-quarters of the explanatory power to come from variables that vary from survey to survey (e.g., the extent of electoral competition, clarity of responsibility, and many others), slightly less than a quarter of the explanatory power to come from variables that vary across countries but are relatively constant over time - at least for the time period covered in our data (e.g., trade openness, size of the government sector, and number of parties in competition), and about 5 percent from variables that are constant across countries but vary over time (e.g., secular trend variables).

This section demonstrated that the *economic vote of the chief executive* is a reasonable measure of our concept of economic voting – at least in so far as its magnitude and variation are quite consistent with existing findings in the literature. We pointed to a number of interesting patterns in this variation that we hope to explain in the subsequent chapters. And finally, by estimating how the total variation in the *economic vote of the chief executive* gets parsed into its three different components, we provided an indication of the type of theoretical variables that might be most powerful in explaining this variation. Recall from Chapter 2 that there are alternative plausible measures of the economic vote. This raises the possibility at least that the contextual variation we’ve just described in this section might be very dependent on our choice of this particular measure of the economic vote, i.e., the *economic vote of the chief executive*. In the next section, we examine whether these descriptions of variation in economic voting are independent of the particular measure of the economic vote employed.

⁶⁴ These numbers come from a variance components model (estimated using Stata 9.0’s xtmixed function) in which the dependent variable is the *economic vote of the chief executive*, there are no covariates in the model, and the grouping of observations is by country and by year. Specifically, we assume $y_{it} = \alpha + \mu_i + \tau_t + v_{it}$, where α is the mean of y_{it} , μ_i is a random effect for country i , τ_t is a random effect for year t , and v_{it} is the remaining survey specific effect. The estimates reported in the text are simply the estimated variances of the three different error components.

⁶⁵ This is simply the sum of the estimated variances from the model in the last note.

Alternative Measures of the Overall Economic Vote

In Chapter 2 we discussed two alternative ways of measuring the amount of overall economic voting in an election – the *volatility of the economic vote* and the *economic vote of the government*. As we pointed out in that discussion, there are good theoretical reasons to focus on the *economic vote of the chief executive* as our summary measure of overall economic voting (rather than these other measures). Still, it is certainly worth examining how closely related these different measures are, if only to know whether our results are likely to be dependent on the choice of measure.

The measure of the *volatility of the economic vote* discussed in Chapter 2 tells us how much party support would “change hands” as a result of worsening economic perceptions.⁶⁶ Figure 3.5 presents a histogram of overall *volatility of the economic vote* for our 163 voting studies. Worsening economic perceptions produce, on average, about 7 percentage points of electoral volatility (the median is also 7 percentage points). This suggests that in a typical election in our sample of democracies, variations in economic evaluations will generate total shifts in support for political parties that amount to approximately 7 percentage points. This is consistent with Blais et al (2004) who generate a similar estimate of economic vote volatility, although for a much smaller sample of countries – 11 surveys conducted in Britain, Canada and the U.S. They estimate a mean overall economic vote volatility of 6 percent (a median of 5 percent).

⁶⁶ For example in the 1984 Danish election study we estimated economic vote effects of 13.4 percent for the Socialists; -6.5 for the Conservatives; -5.8 for the Liberals; -.8 for the Center Party; and -.3 for the Radicals. The sum of the absolute value of these is 26.9. This sum divided by 2 gives a *Volatility* score of 13.5 for the 1984 Danish election study.

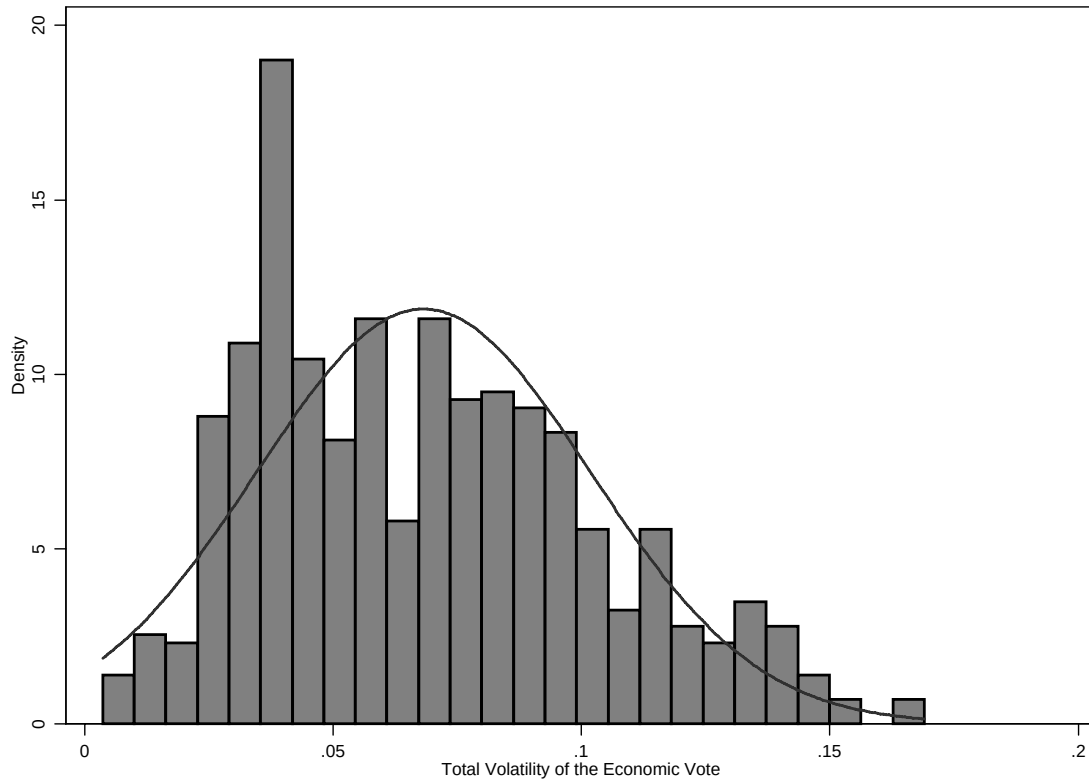


Figure 3.5. Volatility of the Economic Vote

In the empirical analyses in subsequent chapters we will use the *economic vote of the chief executive* when we need a measure of the overall level of economic voting in an election, rather than the *volatility of the economic vote*. The main disadvantage of the *volatility* measure, as we pointed out earlier, is that it treats all changes in support due to changes in economic perceptions as equal evidence *for* economic voting – including changes in support that are “wrong-signed” and would (in our other measures) count *against* economic voting (e.g., improvements in the vote of a chief executive when economic perceptions worsen). Nevertheless, both of these variables could be used as a summary measure of the overall magnitude of the economic vote and so we would like to know how closely they track one another.

Figure 3.6 summarizes the extent to which our measures of the economic vote track one another. We begin with *volatility* and the *economic vote of the chief executive*. The upper-left panel of Figure 3.6 shows a scatter plot of the *economic vote of the chief executive* against the *volatility of the economic vote* and it confirms that there is indeed a close relationship between these two measures. By construction the *economic vote of the chief executive* can never be larger than *volatility*, which accounts for the scatter plot’s V-shape. The left boundary of the V-shape represents two-party cases in which *economic vote of the chief executive* necessarily equals

volatility. Although there are relatively few cases, occurrences of positive values of the *economic vote of the chief executive* attenuate the correlation between *volatility* and the *economic vote of the chief executive* because these cases get treated differently in the two measures. In the case of the *economic vote of the chief executive*, they count as evidence of little or no economic voting. However, in the case of the *volatility* measure, positive estimates of the *economic vote of the chief executive* are treated the same (in terms of their weight in the overall measure) as negative estimates of the same size. Consequently these estimates increase the overall *volatility* score. For the arguments we test in subsequent chapters, it makes more sense to treat positive values of the *economic vote of the chief executive* as evidence against economic voting and hence we favor the *economic vote of the chief executive* as our dependent variable.⁶⁷

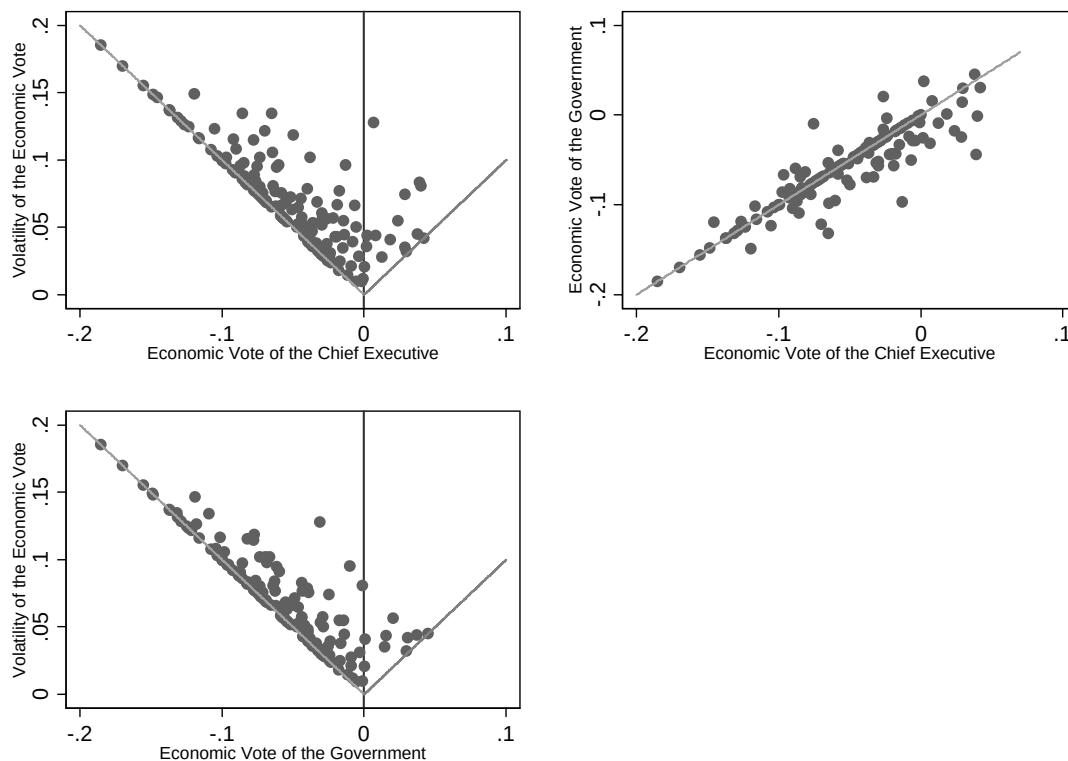


Figure 3.6. Alternative Measures of the Overall Economic Vote

Another alternative measure of overall economic voting is the *economic vote of the government*. Recall from the previous chapter that the most common measure of the *economic vote of the government* that is used in the literature is the sum of the economic votes for each of

⁶⁷ The R^2 from a simple bi-variate regression of *economic vote of the chief executive* against the *volatility of the economic vote* is .5 with a slope coefficient of -.88.

the cabinet parties.⁶⁸ For those countries with single-party governments this measure will not vary from the *economic vote of the chief executive*. However, differences will occur for those studies in which the incumbent government consists of multiple parties. In these cases we would naturally expect the sum of the economic vote for each party in the government to be larger (i.e., more negative) than the *economic vote of the chief executive*. However, because the estimates for many cabinet partner parties can be “wrong signed” this need not be the case. The upper-right panel in Figure 3.6 plots the *economic vote of the government* against the *economic vote of the chief executive*. The graph reveals that (due to many “wrong-signed” estimates for cabinet partner parties) variation around the 45 percent line is more symmetric than one might otherwise guess and so there seems to be a fairly good fit between the measures for our estimates. Likewise, the corresponding comparison between the *economic vote of the government* and *volatility* in the bottom-left panel is quite similar to the one directly above it. Further, the similarity between the *economic vote of the chief executive* and the *economic vote of the government* in these data leads to similar maps of economic voting using either concept. For example, the median (and mean) magnitude of the economic vote employing the *economic vote of the government* measure is -5 percent which is precisely what we obtained using the *chief executive* measure (this only happens because of the surprisingly muted asymmetry in the scatter plot in the upper-right panel of Figure 3.6).

The Distribution of the Economic Vote over Individual Parties

Many theoretically-motivated hypotheses regarding the economic vote concern the distribution of the economic vote across political parties and hence are addressed by none of the three separate measures of the economic vote discussed to this point. The most basic notion of retrospective economic voting suggests that the parties in power will be affected differently by the economy than those parties in opposition. And as we describe in later chapters, theories of economic voting suggest more refined distinctions regarding the distribution of the economic vote across incumbent political parties or across opposition parties. Arguments, for example, that single out particular parties in a coalition government as being held responsible for economic outcomes (Anderson 1995) imply that within a specific context certain governing parties have a higher (or lower) economic vote than others. An example might be that the Prime Ministerial party or the party holding the Finance portfolio receives a greater economic vote than other

⁶⁸ Returning to the 1984 Danish election study example, the Conservative, Liberals and Center parties made up the incumbent coalition government at the time. Recall we estimated their economic vote as -6.5 for the Conservatives; -5.8 for the Liberals; -8 for the Center Party. The sum of these three values gives the measure of the *economic vote of the government* which is -13.

parties in a governing coalition. Accordingly, we now describe how the economic vote varies across political parties in our sample. Again, our goal here is to establish that these patterns generate results consistent with the current economic voting literature but also to highlight potentially interesting theoretical insights that might result from the analysis of these distributions.

A histogram of the economic vote for each of the 678 parties in our national surveys is presented in Figure 3.7. The parties' economic votes have a normal distribution centered on zero and range from a loss of 15 percent to a gain of 15 percent support. Approximately half of the party economic votes are small, falling between plus or minus 2.5 percent. On the other hand, the remaining 50 percent are greater than plus or minus 2.5 percent. And approximately 25 percent of the estimates are either greater than plus or minus 5 percent. As we would expect, the tails of this distribution are dominated by cases from countries that tend to have single party governments. Over half of the cases falling below -10 percent, in fact, are from the U.K. and over half of those falling above 10 percent are either in the U.K., Ireland or the U.S.

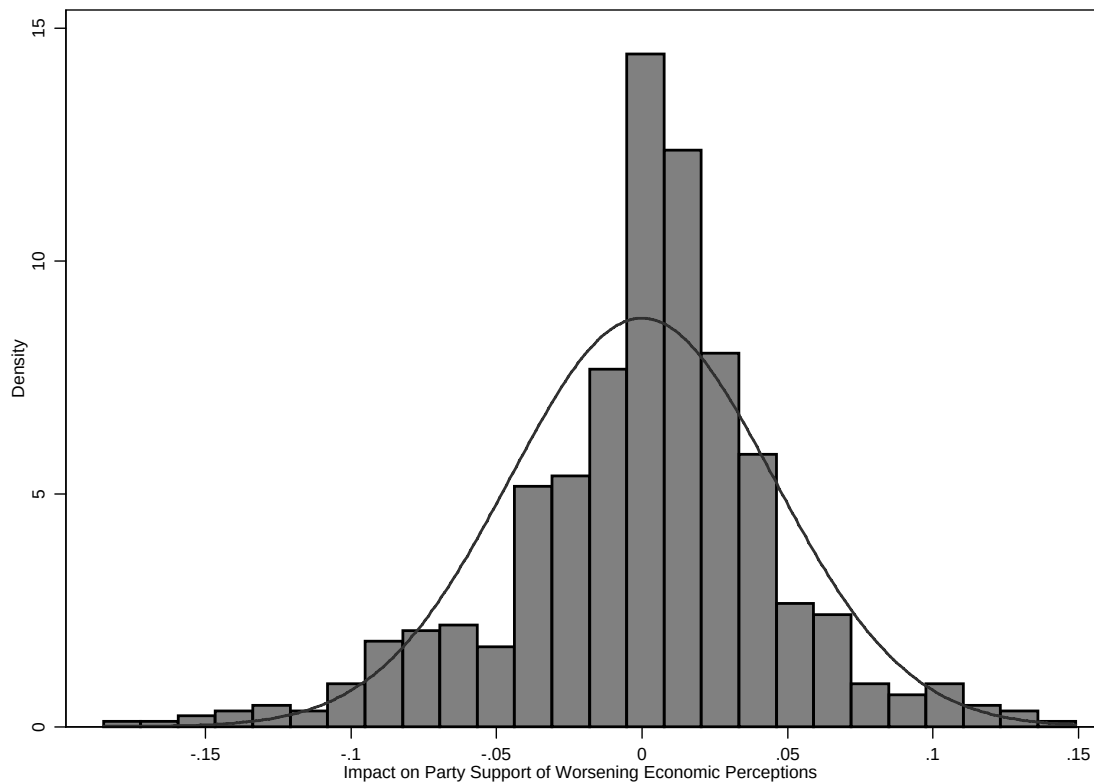


Figure 3.7. Economic Vote of Political Parties

As we noted above, and will develop in much more detail in later chapters, arguments regarding the economic vote of political parties draw a basic distinction between the economic vote of incumbent and opposition parties. Our expectations of course are that the economic vote of incumbents should be in some sense the mirror image of those of the opposition parties. The box-plots in Figure 3.8 summarize the distribution of economic voting across our sample of 105 parties according to whether each party was an incumbent or in opposition at the time of the survey. Each bar summarizes the impact of a declining economy on a particular party's electoral support. Most importantly, the overall result is consistent with the literature: incumbents tend to get punished by a deteriorating economy while opposition parties are rewarded when the economy does poorly. This is confirmed by the fact that the median vote for the incumbent parties falls below zero while it falls above zero for the opposition parties. But Figure 3.8 makes quite clear that the economic vote of both types of parties varies considerably. This is evidenced by the fact that the median vote for incumbent parties varies between about -.12 to approximately .02. And for opposition parties the median vote varies between .12 to -.04.

Not only does the economic effect vary across these parties, but there is also considerable variation in the effect for any particular party over the period 1979-2001, and this variation holds up even when we control for incumbent status. We can illustrate this with two extreme cases identified in Figure 3.8. The economic vote of the UK Conservatives, while they were the Prime Ministerial party, ranged between a high of -.17 and a low of .03. And while its economic vote exhibited quite a broad range of values, it was essentially negative (with only one exception) resulting in a median economic vote of -.06. This ranks the party as having one of the strongest incumbent economic votes. The Belgium Christian Democrats, on the other hand, have a much lower median incumbent economic vote of -.015 but their vote is more equally spread over both positive and negative values (its high value is -.09 and its low value is .04). In spite of being an incumbent party, in a number of voter preference surveys the Belgium Christian Democrats vote probabilities increase in response to a negative evaluation of economic performance. The French UDF is even more of an anomaly in the sense that while in the governing coalition three of its four economic votes were positive (its vote probabilities increasing in response to a negative economic evaluation); in fact the party registered one of the most positive economic votes for an incumbent party (.05).

We see similar variation in parties' economic votes while they are in opposition. The Irish Fianna Fael has the highest economic vote of all opposition parties – its median economic vote is .11 and it ranges between .08 and .15. On the other extreme we find, for example, the Dutch Liberal Party. The Dutch Liberal Party has one of the most negative economic vote of all

opposition parties measured either by its median value of -.015 or its extreme positive value of .04. Hence, for some parties their experiences in opposition produced both electoral rewards and punishments for perceptions of poor economic outcomes.

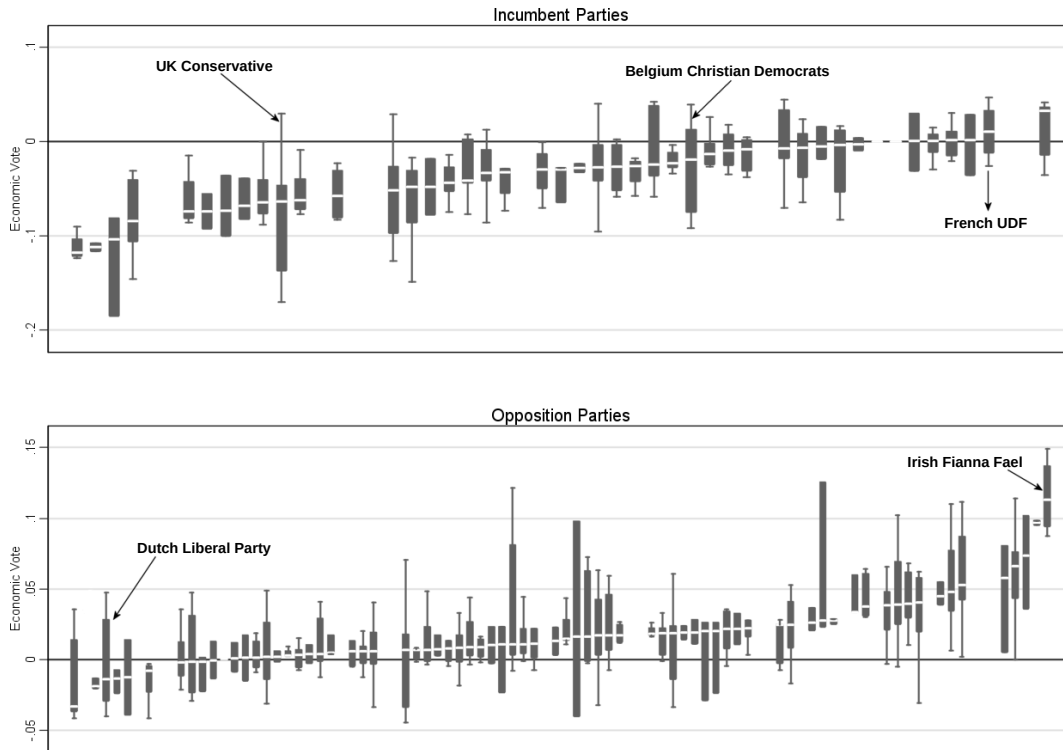


Figure 3.8. Economic Vote of Incumbent and Opposition Political Parties

There are more general patterns in the plots in Figure 3.8 that speak to interesting theoretical issues. A rather significant degree of asymmetry in the incumbent versus opposition economic vote is evident from the slope of the median votes in the two figures: the median economic vote for incumbents is -3 percent which, in absolute value terms, is almost twice as large as the 1.6 median economic vote for opposition parties. The incumbent vote change is order-of-magnitude consistent with Powell (2000) who reports that the average vote loss for government parties in his sample of 153 elections is approximately 2 percent.⁶⁹ Recall from Figure 3.5 that the modal economic vote in our sample of 678 simulated party economic votes is zero. Figure 3.8 gives some insight into which parties have no economic vote. Note that the median economic vote for opposition parties is much more concentrated around the zero line than

⁶⁹ Rose and Mackie (1983) report an average government party vote loss of 1 percent over the entire post-War period; Paldam (1986) reports an average vote loss for governing parties of 1.7 percent.

is the case for incumbents. In a typical election a much large number of opposition parties are unaffected by changes in economic evaluations than is the case for incumbent parties. So, for example, when the economy deteriorates and voters abandon the incumbent parties, there are a large number of opposition parties to whom voters are not inclined to switch their vote. Why is this the case? One possible explanation is that voters who are responding to bad economic performance are searching for opposition parties that are credible governing parties. There are in fact a number of opposition parties that are simply not serious potential governing partners and these parties may make up the bulk of those parties that receive no economic votes. We will explore these, and similar, explanations in subsequent chapters.

While it is true that in general incumbent parties tend to be rewarded by an improving economy and the opposition reaps the rewards of a worsening economy, there is considerable variation around this central tendency. The magnitude of this effect appears to be more pronounced amongst incumbent as opposed to opposition parties. And these effects are more likely to be the case for some parties than others. Moreover, controlling for their incumbent-opposition status, individual parties exhibit considerable variation in their economic vote over our twenty year period – for many parties it varies over both positive and negative values. Our contextual theory of the economic vote will propose an explanation for this variation in the economic vote of political parties.

We have now examined a number different measures of economic voting – *the economic vote of the chief executive, the volatility of the economic vote, the economic vote of the government*, and various measures of the economic vote of individual political parties. They all indicate that economic voting is pervasive in western democracies, although our estimates clearly vary across parties, over time and cross-nationally. While economic voting might be pervasive, does its importance compare favorably with other factors that shape the vote decision? We turn to this issue in the next section.

Economic versus Non-economic Voting

The attention accorded the economy in much of the voting literature is predicated on the widely accepted notion that economic evaluations are one of the more important factors in the vote decision. Thus whatever the magnitude of the economic vote is in absolute terms, many would argue that it outweighs other factors in the voter's decision (Alvarez, Nagler and Willette 2000; Lewis-Beck 1988; Lewis-Beck and Stegmaier 2000). In their list of stylized facts about the economic vote, Lewis-Beck and Paldam (2000) suggest that the economy accounts for about one-third of the change in the vote. Much of the evidence to this effect comes from U.S. voting

studies. Alvarez and Nagler (1995) find that the economy was the dominant issue shaping the 1992 vote, as does Lockerbie (1992). And Erikson (1989) is quite categorical in arguing that in aggregate-level models economic factors – specifically income change – outweighed other variables in predicting presidential vote. For the U.K. case, Sanders (2003b) argues, based on the analysis of Conservative party vote data from 1974 to 1985, that economic evaluations outweigh the impact of partisanship by at least a factor of 3 to 1. Similar claims have been made in the comparative voting literature. In his classic treatise on economic voting, Lewis-Beck (1988: 84) concludes that “Changing economic conditions exert a force on Western European voters that approaches and sometimes exceeds the force of more traditional factors.” More recent comparative data from Alvarez, Nagler and Willette (2000) also suggested that the economy tends to weigh more heavily than other factors in the vote decision.

But does it? As we and others have pointed out, the absolute impact of the economy can be small. Indeed many of the estimates of the economic vote in Figure 3.1 are less than 1 percent suggesting the economy has a relatively small impact on vote choice probabilities. For example, we estimate that the economic vote in the 1993 Canadian election was essentially zero which comports with most accounts of the election. The failed constitutional initiatives of the Mulroney government, the introduction of federal sales tax and opposition to NAFTA were issues that seemed to have overwhelmed evaluations of the performance of the national economy (not to mention the unpopularity of Mulroney and the disastrous election campaign undertaken by Campbell). Foreign policy concerns can easily trump the economy as has been pointed out by Nickelsburg and Norpoth (2000), Mueller (1970), Zaller (2004) and Bueno de Mesquita and Siverson (1995). This may well have been the case with the 2002 German Parliamentary elections. Incumbent fatigue (Zaller 2004) can also overwhelm the importance of economic evaluations – which may have been the case with the 1997 elections in the UK where the incumbent was defeated in spite of a strong national economy.⁷⁰ It also may have contributed to the very small economic vote in the 2000 U.S. presidential elections. Analyzing 11 elections from Canada, the U.K. and the U.S., Blais et al (2004) suggest in fact that issues weight more heavily than the economy in the vote decision.

Clearly, then, issues matter in elections and can sometimes overwhelm economic voting in the vote decision. Still, the examples of this are largely anecdotal or based on small samples.

⁷⁰ Our 1997 estimate of the magnitude of the economic vote in the UK is the second lowest reading over the 1979-2001 sample period which is consistent with the notion that the economy had relatively little impact on vote choice in this election.

More problematic from the perspective of the economic voting literature would be evidence that economic evaluations systematically play a minor role in vote choice compared to other factors.

There have been efforts using comparative data to assess the importance of economic voting relative to other factors but they have been based on small samples of countries. Michael Lewis-Beck's (1988) assessment of the importance of economic evaluations on vote is based on five national surveys, all conducted in 1984. Alvarez, Nagler and Willette (2000) base their assessments on seven surveys (two each in the U.S., The Netherlands, and Canada and one in the U.K.). And the Blais et al (2004) conclusions are founded on 11 election surveys from three countries. We address this particular shortcoming here by basing our comparative assessment on 163 surveys, conducted over a 20 year time period and covering 18 nations. A variety of factors typically compete for the voter's attention in any particular election campaign. We are simply interested in those competing factors that are most prevalent in the vote calculus and have persisted over time. Accordingly, we explore two factors that meet these criteria: ideology and issues.

Ideology

There is wide-spread agreement that voters in most developed democracies employ the Left-Right continuum to organize the political landscape (Fuchs and Klingemann 1989; Huber and Gabel 2000; Laver and Budge 1992). And there is extensive evidence supporting spatial models of vote choice (Enelow and Hinich 1984; Hinich and Munger 1997). Hence, movements along the Left-Right continuum either by voters or by the parties will affect the spatial distance between voters and candidates and in turn the voters' support of political parties. In fact, the comparative voting literature offers considerable support for the notion that ideology represents one of the most powerful predictors of vote choice (Dalton 2002; Franklin et al 1992).

Accordingly, we compare the impact of Left-Right self-placement on party support with our estimate of the magnitude economic voting, employing an estimation strategy similar to the one described in the previous chapter. Most of the studies in our sample employ a 10 unit scale for Left-Right self-placement that ranges from 1 for most Left to 10 for most Right. However, in the U.S. national election survey, ideology is typically calibrated with a liberal-conservative self-identification measure. For comparative purposes we have mapped the U.S. liberal-conservative scale onto a 10-point Left-Right scale. Fluctuations in the ideological self-placement of the mass public are quite modest. The overall mean Left-Right score of the 145 studies including this variable is 5.4 with a standard deviation of study means of about 0.4. Within each country, the standard deviation for these series ranges between .11 and .32 and the mean ranges between 4.6

and 5.9. The maximum range of the study mean scores within any country is approximately 1 unit (Belgium, Denmark, Great Britain and Spain experience a range of values close to 1) although the typical range is about a half a unit.⁷¹ Hence, from one election to the next, the maximum swing in ideological preferences of the electorate typically would not exceed a half a unit.

To assess the relative importance of Left-Right self-identification in shaping voting intention we estimate a model of the vote for the chief executive party that includes both Left-Right self-placement and retrospective economic evaluations. We estimated a separate model for each of the 145 surveys (from our overall sample of 163) that included a measure of Left-Right self-identification. The dependent variable in the model is the dichotomous *chief executive* party vote measure (coded 1 for those supporting the incumbent *chief executive* party) which was described in the previous chapter.⁷² *Ideology* is measure employing the demeaned (employing the survey mean) Left-Right self-placement of the voter (coded so higher numbers indicate a more right leaning voter). We employ the same dummy variables measuring retrospective perceptions of the national economy that was utilized for generating economic voting effects: *Worse*, which equals one if voter thought the economy had gotten worse in the last year; *Better*, which equals one if they thought the economy had gotten better; and *Same*, which equals one if they thought the economy had stayed the same. Two of these, *Better* and *Worse*, were included in the individual level model. Clearly we expect these two dummy variables to have opposite signs with *Worse* being negative and *Better* being positive.

To estimate the impact of ideology on vote choice we employ the coefficients estimated from each of these 145 individual samples. We begin by estimating a predicted vote for the incumbent with Left-Right self placement set to 0 and the *Better* and *Worse* dummies set to zero. We then shift the value of Left-Right self placement by 0.5 (the two economic evaluation variables are set to 0 in this simulation implying a neutral evaluation of the economy) to generate an estimate of the predicted incumbent vote when the sample is made 0.5 units more

⁷¹ The other component of spatial models, candidates for political office, also exhibit very stable positioning on the Left-Right scale. As Poole and Rosenthal (1997) point out ideological shifts in the Republican and Democratic parties tend to be very gradual. And as Alvarez and Nagler (1995) suggest in their analysis of the 1992 presidential elections, candidates did not have much of an electoral incentive to shift their ideological positioning. European political parties also exhibit similar stability in their left-right placements (Budge 1994).

⁷² The model estimations are similar to those described in Chapter 2, except that instead of estimating the full MNL we simply use a logit estimation and we employ a dichotomous dependent variable for whether or not the party choice is the chief executive party or not. As we show in Chapter 5, this simplification gives results that are virtually identical to the MNL estimation.

conservative.⁷³ For each sample we then calculate the difference in the predicted incumbent vote by subtracting the base prediction from that of the more conservative sample. This resulted in 145 simulated effects of a .5 unit shift in the Left-Right self-identification variable.⁷⁴

Figure 3.9 presents a histogram of the 145 predicted changes in the probability of voting for the chief executive party that results from a change in ideology. This is the absolute value of change in the vote probabilities since of course the direction of the predicted change would be a function of the incumbent chief executive party's ideology. The Ideology result is very similar to the estimated *economic vote of the chief executive*. In the case of a 0.5 unit change in Left-Right identification, chief executive party vote shifts on average about 4 percentage points. This compares to our estimate of 5 percent for the *economic vote of the chief executive*.

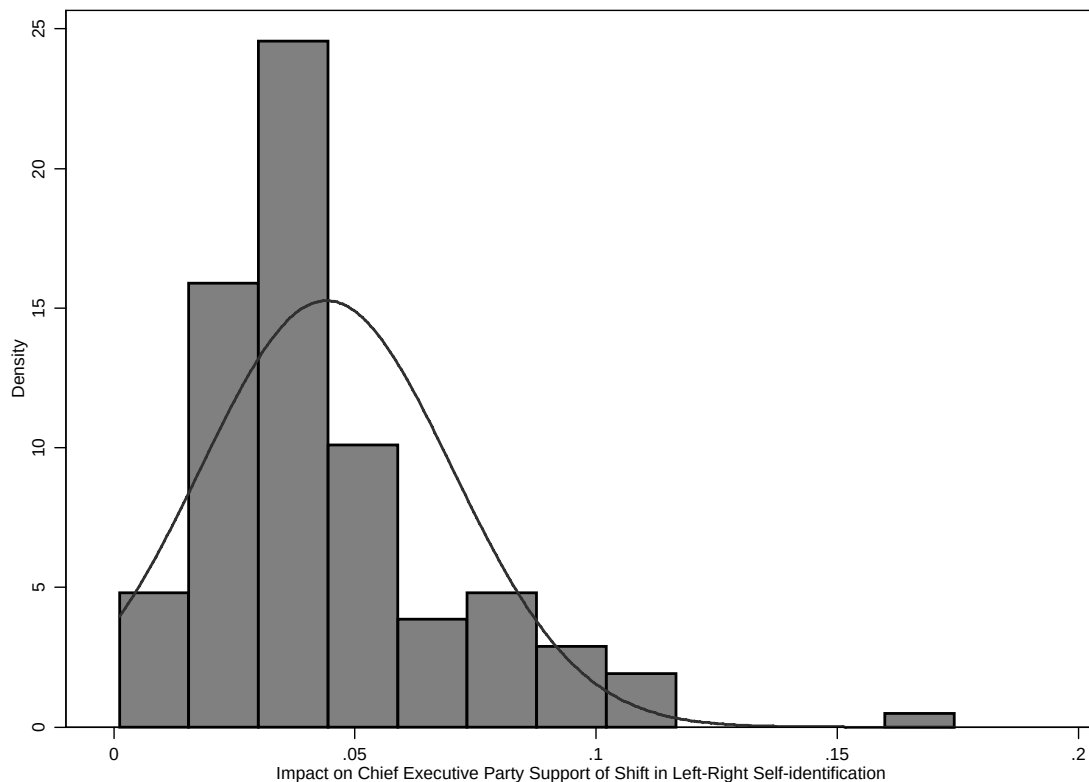


Figure 3.9. Ideological Vote for Chief Executive Party

⁷³ As a benchmark, Lewis-Beck (1988: 84) in comparing the ideology effect to the economic vote effect on vote shifts the ideology score 3 units (from 9 to 6) which is six times the magnitude of our simulation and, reassuringly, obtains results approximately seven times the magnitude of ours (an average shift of about 30 percentage points).

⁷⁴ The simulated effects and their standard errors were generated using Clarify as described in King, Tomz, and Jason Wittenberg (2000).

Ideology is generally considered to be one of the most important factors shaping the voting decisions. As a result, the impact of ideology on party vote probabilities serves as an ideal metric against which to compare our estimate of the economic vote. Our assessments are quite favorable to the notion that economic performance is one of the most important factors shaping voting decisions. The typical shift in voter Left-Right sentiment results in a median change in vote probabilities on the order of 4 percent which is very similar to the 5 percent effect we estimate for shifts in economic evaluations.

Issues

To test the relative importance of issues we draw upon the Post-Materialism items developed by Ron Inglehart (1977). Although hardly a perfect measure (Duch and Taylor 1993), the concept has the virtue of being included in large numbers of voter surveys. The questionnaire items making up this construct are asked over the entire time period of our study and in 105 of the European studies in our sample.⁷⁵ These items are designed to measure the extent to which respondents prioritize different political goals and in this respect they are a plausible proxy for the issue preferences of voters. The standard battery of questions asks respondent to indicate which of the following four political goals they would rank as a first and second priority: 1) maintaining order in the nation; 2) giving the people more say in important political decisions; 3) fighting rising prices; 4) protecting freedom of speech. Respondents selecting two “Post-Materialist” items (items 2 and 4) are coded 3; those selecting two materialist items (items 1 and 3) are coded 1; respondents selecting a mix of materialist and Post-Materialist items are coded 2. This construct is widely used in comparative voting studies – particularly in Western Europe – and has been shown to be strongly correlated with vote choice (see Franklin et al 1992; Kitschelt 1994). Post-Materialism, like the Left-Right measure, does not vary significantly over time – the overall standard deviation for country means is about .21 and the maximum variation across studies within any country is on the order of .75 units. Again, given this relatively modest dynamic variation, we do not expect that typical fluctuations in this particular indicator of policy preferences to have a significantly higher impact on vote choice than economic evaluations.

We estimated a model of vote choice that included the same variables as the ideology equations described above with the addition of the Post-Materialism indicator. Following the same strategy we employed in the case of ideology, we simulated the impact of Post-Materialism

⁷⁵ The Post-materialism question was for the most part not included in vote choices studies conducted in non-European countries.

on vote probabilities by varying the Post-Materialism measure in each country from a value of 1.75 to a value of 2.5. Again, this strikes us as a very conservative strategy for assessing the relative impact of Post-Materialist issues given the over-time stability of this measure. Figure 3.10 reports the results for these simulations across our sample of European countries. The effect over the 105 samples ranges from between 0 and 12 percent with the distribution clearly skewed towards a very small effect (the simulated effect in about 15 percent of the studies was less than 1 percent). The median vote probability shift across the surveys we sampled here is slightly less than 4 percent (the mean is 4 percent). It is quite clear that fluctuations in Post-Materialism certainly are no more important than the economy in vote choice and likely have relatively less weight in the voter choice function.

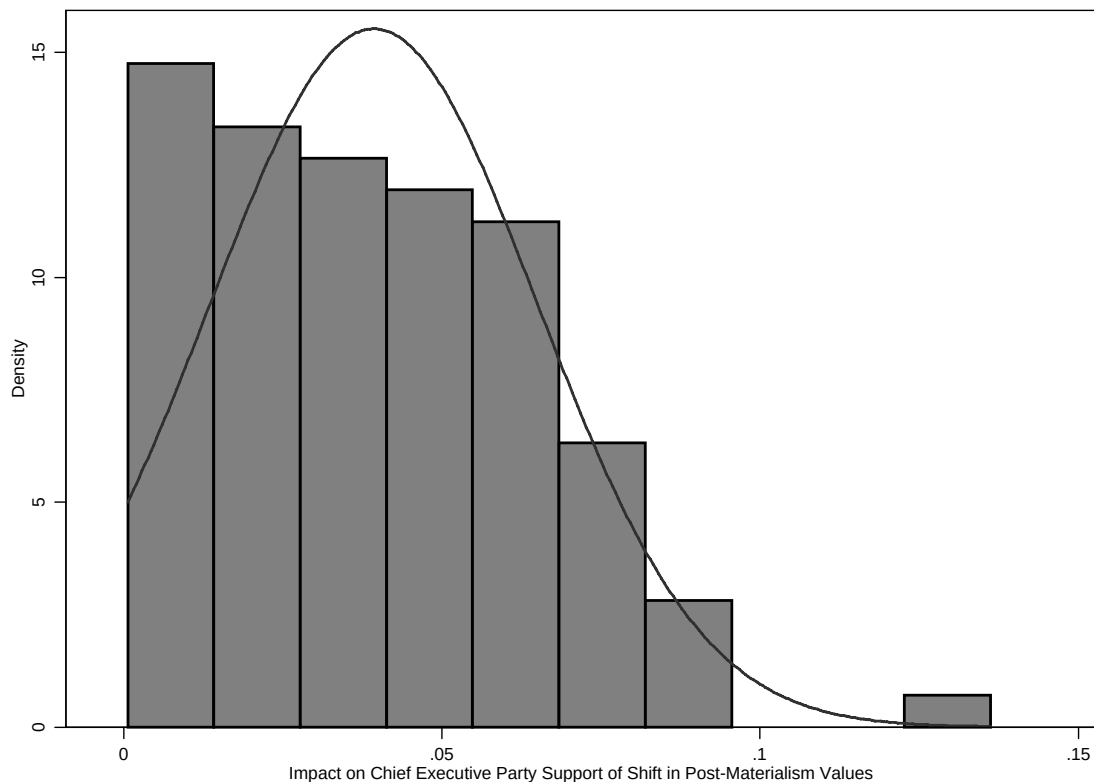


Figure 3.10. Post-Materialism Vote for Chief Executive Party

There are conflicting claims regarding the importance of the economy, relative to other issue concerns, in the typical vote decision. In this section we provide some insight into the relative importance of the economic vote. Working within the constraints of the common questions included in our 163 voter preference surveys, we examine the relative impact of two

factors that likely compete with the economy for consideration by the voter: ideology and Post-Materialism values. While both of these variables have an important impact on the vote choice, in neither case is their impact more important than the economy. At best, these two other considerations have an impact that is similar in magnitude to the economic vote.

Summary

Yes, the conventional wisdom amongst politicians, political pundits, media personalities and even academics is that the economy matters when it comes to voting decisions. And even the most casual students of politics would seem to believe that this phenomenon is ubiquitous across the developed democracies. But, surprisingly, the cross-national evidence of the wide-spread impact of the economy on the party preferences of voters is rather limited. We have assembled a body of evidence that spans a relatively large number of countries (eighteen) and an extensive time period (1979-2001) to provide what we believe is definitive evidence in favor of the conventional wisdom. Yes, there is economic voting in developed democracies. How much economic voting? It varies, of course, but we situate the median impact of economic evaluations on the vote probabilities of parties in an election between 5 and 7 percent depending on the way we measure it. Moreover this impact stacks up quite well against other plausible factors that might sway voter preferences for competing political parties. We estimate that a typical change in Left-Right sentiment (.5 units) or in Post-Materialist values (.75 units) both shift support for incumbent parties by about 4 percent, slightly less than the mean size of our estimated economic vote.

An equally important goal of this chapter is to demonstrate that there are interesting patterns of variation in this economic vote. The overall picture of contextual variation in the economic vote confirms the general patterns that others have observed – for example relative rankings of countries (the UK tends to have more economic voting than the Netherlands) and temporal trends (a recent moderation in the magnitude of the economic vote). This provides face validity for the particular measure of the economic vote that we have adopted in this book.

The extent to which the economic vote varies becomes even more evident once we dig a bit deeper and examine over time variation within countries. Here we see examples of a large economic vote in the early 1980s dwindling to virtually no economic voting in the year 2000 (Ireland and Germany, for example) or just the opposite (Denmark, for example, where we see economic voting increasing in recent years). We also see examples of economic voting rising and falling over time, or vice versa, such as in the UK, Spain and the Netherlands. Unlike most other treatments, we drill down even further and examine the economic vote of each of the political

parties. At this level there are a myriad of interesting variations. The Dutch Liberal Party, for example, that, when in opposition, frequently gets punished for perceptions of poor economic performance. Or the French UDF party that, in spite of being a member of the governing coalition, rarely is punished for poor economic performance.

Our expectation is that these contextual variations in the overall economic vote and in economic voting for political parties are not just random fluctuations. Rather we suggest they are the result of instrumentally rational voters responding to features of the political and economic context that also vary over time. The contextual theory we develop in later chapters identifies those contextual features that voters respond to and hence that condition the magnitude of their economic vote.

Chapter 4

Estimation, Measurement and Specification

The evidence presented in the previous chapter established that economic voting in the world's advanced democracies is both pervasive and variable. The goal of the rest of this book is to identify the political and economic contexts that explain that variation. To do so, we will develop a contextual theory of rational retrospective economic voting that points to specific kinds of political and economic institutions as the likely sources of cross-national and temporal variation in economic voting. However, before we make the effort to develop that theory and test its implications, we need first to establish that the variation in our estimates of economic voting described in the last chapter cannot be explained away as an artifact of the empirical methods we have employed to generate them. Three general methodological issues are of concern here: the choice of statistical models, measurement decisions, and model specification. We begin with an explanation of the two main statistical methods we will use to conduct the empirical analysis in this, and subsequent, chapters. We then proceed to analyze our data to determine whether our estimated variation in the economic vote is an artifact of our particular statistical models, measurement choices, or specification decisions.

Two Methods for Exploring the Sources of Contextual Variation in the Economic Vote

In this section we introduce two general empirical methodologies for testing hypotheses about the impact of contextual variables on individual level behavioral relationships: The first is a one-stage approach in which the individual-level data from multiple contexts (in our case voter preference surveys) are pooled and the impact of context is estimated simultaneously with the individual behavioral relationships of interest. The other method is a two-stage approach in which the strength of a behavioral relationship is first estimated in each context and then these estimates (in our case the estimates of economic voting that were presented in the previous chapter) become the dependent variable in second-stage analyses in which the independent variables are measured features of context. The methods have different strengths and weaknesses but as Bowers and Drake (2005) have argued, they complement one another well and so are best used together.⁷⁶ In the empirical chapters that follow we do just that.

We begin with the observation that our data – multiple individual level surveys on vote choice and retrospective economic perceptions – can be represented as a hierarchical, multi-level structure with two levels, where level-1 units are individuals (i.e., the respondents in each of the

⁷⁶ For a more detailed description and comparison of these two approaches see Duch and Stevenson (2005).

election studies) and level-2 units are the surveys themselves. Since each survey corresponds to a particular country and time point (i.e., the date the survey was administered), this structure allows for variation in the individual behavioral relationship between economic perceptions and vote choice across countries and over time. For illustrative purposes, we will focus on how one can model contextual influences on the *economic vote of the chief executive*, but the discussion below applies equally well to all the other concepts of economic voting discussed in Chapter 2, except *general economic voting*.⁷⁷ A general representation of the multilevel structure of the data could be written using a logit link function as follows:

$$(4.1) \quad \begin{aligned} v_{ik} &\sim \text{Bin}(\pi_{1ik}) \\ \text{logit}(\pi_{1ik}) &= \beta_{0k} + \beta_{1k}X_{ik} + \phi_{1k}Z_{ik} \end{aligned}$$

In this notation, *Bin* is the binomial distribution and v_{ik} indicates a vote for the chief executive by voter i in survey k . X_{ik} are retrospective economic evaluations measured at the individual level and Z_{ik} are other characteristics of individuals that shape vote choice. The coefficients that describe economic voting in any particular survey is β_{1k} and this (as well as the other coefficients) is allowed to vary from survey to survey. If we think that economic voting is conditioned on known contextual variables that vary over the surveys (i.e., the level-2 units), then we can include measures of these contexts in this general model as interactions.

$$(4.2) \quad \begin{aligned} v_{ik} &\sim \text{Bin}(\pi_{1ik}) \\ \text{logit}(\pi_{1ik}) &= \beta_{0k} + \beta_1 C_k + \beta_{2k}X_{ik} + \beta_3 X_{ik} C_k + \phi_{1k}Z_{ik} \end{aligned}$$

where C_k are contextual variables measured at the survey level. By making different assumptions about the sources and nature of variation in the coefficients in this equation we produce different specifications that are best estimated using different methods. We will use two different strategies for filling out the specification of the model in equation (4.2) and estimating its parameters. Both allow the coefficients describing the economic vote at the individual level to vary systematically as a function of C_k , but do not assume this is the only source of variation in

⁷⁷ As we noted there, since the *general economic vote* vector has different (and different numbers of) parties for each case, these can not be combined into a sensible pooled analysis without first defining a different concept of economic voting (e.g., economic vote for the prime minister or government) that makes the different cases conceptually comparable.

the coefficients.⁷⁸ One of these produces estimates of the individual-level relationships and the contextual effects in a single step while the other strategy uses two steps to estimate the contextual effects.

One-stage Strategy

The one-stage strategy pools the data from all the surveys and assumes that, conditional on C_k , variation in the parameters across surveys can be described by a normal distribution. Consequently, estimating variation in the parameters reduces to estimating the parameters of this normal distribution. Specifically, the model assumes that:

$$(4.3) \quad \begin{aligned} v_{ik} &\sim \text{Bin}(\pi_{ik}) \\ \text{logit}(\pi_{ik}) &= \beta_{0k} + \beta_{1k}X_{ik} + \phi_1Z_{ik} \end{aligned}$$

$$(4.4) \quad \beta_{0k} = \gamma_{00} + \gamma_{01}C_k + \omega_{0k}$$

$$(4.5) \quad \beta_{1k} = \gamma_{10} + \gamma_{11}C_k + \omega_{1k}$$

$$(4.6) \quad \begin{bmatrix} \omega_{0k} \\ \omega_{1k} \end{bmatrix} \sim N(0, \Omega), \quad \Omega = \begin{bmatrix} \sigma_{\omega 0}^2 & \sigma_{\omega 0, \omega 1} \\ \sigma_{\omega 0, \omega 1} & \sigma_{\omega 1}^2 \end{bmatrix}$$

This specification only allows for variation in the intercept and the coefficients on economic perceptions, but variation in the coefficients on Z could also be included.⁷⁹ Models such as this one are quickly becoming standard in political science (see Snijders and Bosker 1999 for a good introduction) and are usually estimated using either Bayesian simulation or a quasi-likelihood method developed by Goldstein (1995).⁸⁰ The most important feature of the model, for the purposes of this project, is that the estimates of γ_{01} and γ_{11} and their associated standard errors provide direct tests of the impact of measured contextual factors on economic voting. The model does not assume, however, that the measured contextual factors represented by C_k are the only

⁷⁸ An approach we do not discuss explicitly in this section is the “interactive” model in which the coefficients in equation (4.2) are considered fixed parameters that do not vary randomly over contexts. Essentially, this model assumes that C_k captures all the relevant contextual variation in the parameters. Since interactive models are nested in the alternative multi-level models we discuss, we do not deal with them separately.

⁷⁹ Over or under-dispersion in the individual level binomial variation can be included at the cost of estimating one additional parameter. In the models we estimate we include this parameter and test for over or under-dispersion.

⁸⁰ We use Goldstein’s method as implemented in the program MLwin 2.0.

source of variation in the economic vote parameter, β_{1k} . Instead, additional random variation in the parameters is allowed but is assumed to follow a normal distribution.

These distributional assumptions are most helpful in describing individual level relationships when one has a small number of cases for some of the surveys. In these cases, the estimates of the β_k 's, for these low-observation surveys “borrows strength” from the other surveys, which (by assumption) arise from the same distribution. Specifically, when information in a survey is sparse, the estimated coefficients for that unit “shrink” toward the estimate one would get from the pooled sample. In the most extreme case, in which some survey had no data at all, this shrinkage would be complete and the estimated coefficient for the case with no data would just be the pooled estimate. In contrast, when there is an abundance of individual-level data in each survey, little information from other surveys may be necessary for precise estimation and so the individual survey estimates will borrow little from others. Our case resembles the latter one in which there are relatively numerous individual-level data in each survey and hence we would not expect much reliance on the pooled sample.

Because the data are pooled in the one-stage strategy, we need to include the same variables (measured in the same way) in the voting model for each survey. With respect to the core variables in the economic vote equation -- vote choice (v_{ik}) and economic perceptions (X_{ik}) -- this is not a constraint: we could not estimate the economic voting effect without these two core variables and, as described in Chapter 2, we only included surveys in our study that had comparable vote choice and perceptions questions. However, the requirement that the control variables, Z_{ik} , be measured in exactly the same fashion in all the individual-level studies is more constraining.⁸¹ In our case, this means the specification of the individual-level vote choice model can include only a very limited number of control variables and may necessarily exclude some variables known to be important in specific cases.⁸² Later in this chapter we explore whether our

⁸¹ One can imagine pooling data that have different variables in the voting model for each survey by including a large number of interaction terms that activate a particular variable only for the case or cases in which it is appropriate. With five control variables and fifteen countries this would mean including seventy-five such dummy variables in the model. The practical difficulty of estimating such a model with the thousands of individual cases in the pooled data led us to use alternative methods.

⁸² An alternative option is to restrict the pooled data set to those studies that have similarly measured v_{ik} and X_{ik} and have a full set of similarly measured control variables Z_{ik} . This effectively restricts the estimation of contextual effects to those individual surveys that were designed explicitly to have the same set of variables. An example is Hellwig (2001) who estimates pooled economic vote model employing the CSES studies. This enabled the author to estimate fully specified vote choice models although at the cost of including only nine surveys.

estimates of the influence of contextual variables on economic voting are impacted by estimating a pooled model that excludes possible important confounding variables.

Two-stage Strategy

An alternative strategy for estimating contextual effects begins with the estimates of the economic vote described in Chapter 3. Recall, these were based on 163 individual-level vote choice models that took the general form of Equation (4.3).⁸³ We can use these estimates in a second stage statistical analysis to test hypotheses about the impact of specific measured contextual variables on economic voting.⁸⁴ We do this using “second-stage” regressions of the following general form:

$$(4.7) \quad EV_k = \alpha_0 + \alpha_1 C_k + v_k$$

where EV_k is, for example, our estimate of the *economic vote of the Chief Executive* from Chapter 3. Similar equations could be written for our estimates of the other concepts of economic voting discussed in Chapter 2.⁸⁵

The “data” that we use for the dependent variable in the second-stage model are the estimates of economic voting obtained in the first stage and so this analysis must account for the uncertainty of these estimates. The common procedure for doing this is to weight the second stage analysis by the inverse of the standard error of the estimates of the dependent variable. However, as Lewis and Linzer (2005) show, this procedure can produce incorrect estimates of uncertainty because it treats all of the second-stage error as if it came from estimation uncertainty, when in fact some of it is inherent to the stochastic process governing the second stage. Lewis provides a weighting procedure that produces appropriate standard errors but he suggests (and his Monte Carlo results show) that using standard errors that are robust to unspecified forms of heteroskedasticity is almost always as good. Given this (and because we must combine this correction with others discussed below), we adopt the later procedure.

⁸³ For ease of explanation Equation 4.3 is a logit but our estimates generated here were generated from a Multinomial Logit but the multi-level structure would be the same.

⁸⁴ Jusko and Shively (2005) show that in the case in which the first stage estimates are coefficients from a linear regression, the coefficients on the contextual variables are consistent.

⁸⁵ This equation is intended to be general. The actual specification of the second stage models could take a variety of forms.

Another challenge for our second-stage analyses is that our data often includes the same party in different surveys. Since we might expect the size of the economic vote for the same party across studies to be correlated (even conditional on a well-specified set of covariates), we need to allow for this in the second-stage estimation. We do so by grouping the data (by party in this case) and calculating standard errors that allow for an unspecified pattern of correlation within groups, while assuming that there is no correlation across groups. The resulting standard errors, when combined with the correction in the last paragraph, simply allow for unspecified forms of heteroskedasticity across surveys, correlation between surveys when the party included is the same, and no correlation otherwise. These corrections are now commonly used in political science and can be implemented easily with standard statistical packages.⁸⁶ Once these corrections are made, we test hypotheses about context by examining the estimates of α_i from equations of the general form given in (4.7).⁸⁷

In Chapter 3 and the chapters that follow we rely primarily on the two-stage estimation method for presenting our empirical results (although, for comparative purposes, each of the following empirical chapter also includes a one-stage pooled hierarchical model for all of the main results). An obvious question is whether the distribution of contextual variation described in Chapter 3 is sensitive to employing a one- versus a two-stage estimation strategy. Given our relatively large and diverse sample, we can assess whether there are any significant differences associated with employing one or the other of these approaches. We can also examine whether the variation we described in Chapter 3 is sensitive to the measurement and model specification decision we have adopted. We now turn to these empirical analyses.

Estimation Effects

Within the comparative economic voting literature the strategies adopted for estimating the economic vote, at the individual-level, are quite diverse. The previous section describes one of the more important estimation decisions that need to be made in comparative efforts of this nature. But there are a number of other important estimation choices that we had to make and these may not necessarily be the choices others would entertain. Accordingly, in this section we

⁸⁶ We implement this strategy using Stata 9.0's robust and cluster options.

⁸⁷ This discussion is intended to be general. The model specifications we employ are all of this general form but vary according to the hypothesis being tested.

review the most important estimation decisions we made and assess whether our choice of specific strategies has implications for the results we described in Chapter 3.

One- versus Two-stage Strategy

The previous section described two different approaches for measuring the economic vote. The patterns of contextual variation in the economic vote described in Chapter 3 are based on the two-stage approach that involves separate estimations from each of our 163 surveys. One could also generate similar descriptions of the economic vote estimating a one-stage hierarchical multi-level model in which the individual level data from the various surveys are pooled. In order to compare the two methods we will estimate 1) the economic vote obtained from each survey separately and 2) the survey-level (or level-2) residuals of a pooled multi-level model. In order to produce the later estimates we combined individual level data from our 163 electoral surveys to estimate a single logistic model of the individual level vote for, or against, the chief executive party. This self-reported vote was modeled as a function of economic perceptions and a much reduced set of control variables that were available in most of our surveys. However, even with this reduced set of controls, the requirement that we have a set of common control variables eliminated 14 studies, so that 146 studies remain in the pooled data set. The hierarchical model we estimate is as follows:

$$(4.8) \quad \text{logit}(\pi_{ik}) = \beta_{0k} + \beta_{1k} \text{Worse}_{ik} + \beta_{2k} \text{Better}_{ik} + \phi_{1k} \text{Ideology}_{ik} + \phi_{2k} (\text{Ideology}_{ik} * \text{CE_Ideology}_k)$$

$$(4.9) \quad \beta_{0k} = \gamma_0 + \omega_{0k}$$

$$(4.10) \quad \beta_{1k} = \gamma_1 + \omega_{1k}$$

$$(4.11) \quad \beta_{2k} = \gamma_2 + \omega_{2k}$$

The usual assumption of multivariate normality of the second level errors is also imposed. The variables included in the model are *Ideology*, which is the demeaned (employing the survey mean) left-right self-placement of the voter (coded so higher numbers indicate a more right leaning voter) and its interaction with a dummy variable (*CE_Ideology*) indicating whether the Chief Executive's party is a leftist party.⁸⁸ We expect then that the coefficient on left-right self-placement (*Ideology*) will be positive and its interaction with the ideology of the Chief Executive

⁸⁸ In order to include as many studies as possible, we had to combine some left-right self-placement questions that gave the responded different response options. This was done by normalizing the various scales and then using the normalized scales.

(*Ideology*CE_Ideology*) to be negative. The variable measuring retrospective perceptions of the national economy has been broken into three dummy variables: *Worse*, which equals one if the voters thought the economy had gotten worse in the last year; *Better*, which equals one if they thought the economy had gotten better; and *Same*, which equals one if they thought the economy had stayed the same. Two of these, *Better* and *Worse*, were included in the individual level model. Clearly we expect these two dummy variables to have opposite signs with worse being negative and better being positive. The pooled model is thus limited to one main control variable – the ideology of the voter relative to that of the Chief Executive. This sparseness in specification is almost always the cost one must pay for pooling data from disparate surveys.

Having pooled the data, we estimated the hierarchical model in Equation (4.8), obtaining a single set of coefficients that describe the average relationships across all the studies in the sample. All these estimates were in the expected direction and statistically significant.⁸⁹ In Equation (4.8) the level-1 units are individuals while the level-2 units are election studies.⁹⁰ In this section we are not really interested in the estimates of the average level of economic voting across surveys (γ_1 or γ_2). Rather, we are interested in the survey-level residuals for each of the k election studies (estimates of ω_{1k} or ω_{2k}) that indicate how the impact of economic evaluations in different surveys deviates from its overall average effect. Specifically, we want to compare these to the economic vote estimates obtained from the separate, survey by survey estimates.

In order to isolate the effect of the different estimation procedures we estimate separate logistic regressions for each survey that mirror the pooled, multi-level specification in Equation (4.8) – the interaction term is dropped because *CE_Ideology* doesn't vary within election studies:

$$(4.12) \quad \text{logit}(\pi_{ik}) = \beta_0 + \beta_{1k} \text{Worse}_{ik} + \beta_{2k} \text{Better}_{ik} + \phi_{1k} \text{Ideology}_{ik} + \varepsilon_{ik}$$

We estimate this logistic model for each of the same 146 election studies included in the pooled model from Equation (4.8). This generates 146 estimates of each of the β and ϕ coefficients in

⁸⁹ We do not show the coefficients from these models here since they are not needed for this discussion. However, they are provided in Duch and Stevenson (2005).

⁹⁰ One could designate a third level for country, but fixed country effects are easier to estimate simply by including country dummies and may make more sense, since we probably do not think of these countries as a sample out of a large set of similar units, which is really the motivation for treating the second level coefficients as random and estimating features of the distribution of these coefficients.

the model.⁹¹ In order to compare these estimates to the estimates of the level-2 residuals from the pooled, multi-level model, we generate “survey-level residuals” for the estimated coefficient on the *worse* dummy variable:⁹²

$$(4.13) \quad \psi_{1k} = \hat{\beta}_{1k} - \frac{1}{N} \sum_{k=1}^K \hat{\beta}_{1k} n_k$$

where n_k is the number of observations in survey k and N is the total number of observations in all the surveys. Figure 4.1 plots these survey-level residuals (on the y-axis) against the level-2 residuals from the pooled multi-level model (ω_{1k} in Equation (4.10)). Figure 4.1 reveals a tight fit between the two sets of estimates (the correlation is .81). The slight differences in estimates that we do see in Figure 4.1 are in the direction we would expect. In Figure 4.1 we have included both a regression line for the two sets of residuals and a 45 degree line (the expectation if residuals from both models were exactly the same). The manner in which the regression line cuts the 45 degree line illustrates that when information in a survey is sparse the multi-level estimated coefficients for that unit “shrink” toward the estimate one would get from the pooled sample. Hence at the extremes the residuals are smaller for the multi-level estimates than they are from each of the individual surveys. Overall, though, we can conclude that there is not much systematic difference between the estimates that is due to the estimation method.⁹³

⁹¹ We focus on the coefficient on the *worse* dummy variable. An examination of the better variable generates the same conclusions.

⁹² To be comparable to the multi-level result, this must be the weighted mean (where sample size provides the weight).

⁹³ One additional difference in the specification results from the fact that in the pooled model, we did not allow random coefficients for ideology and its interaction, while in the separate models these coefficients are clearly unconstrained. To be sure this was not critical, we estimated this model as well but the picture is essentially the same. We keep this presentation because the choice is inconsequential and it requires less notation.

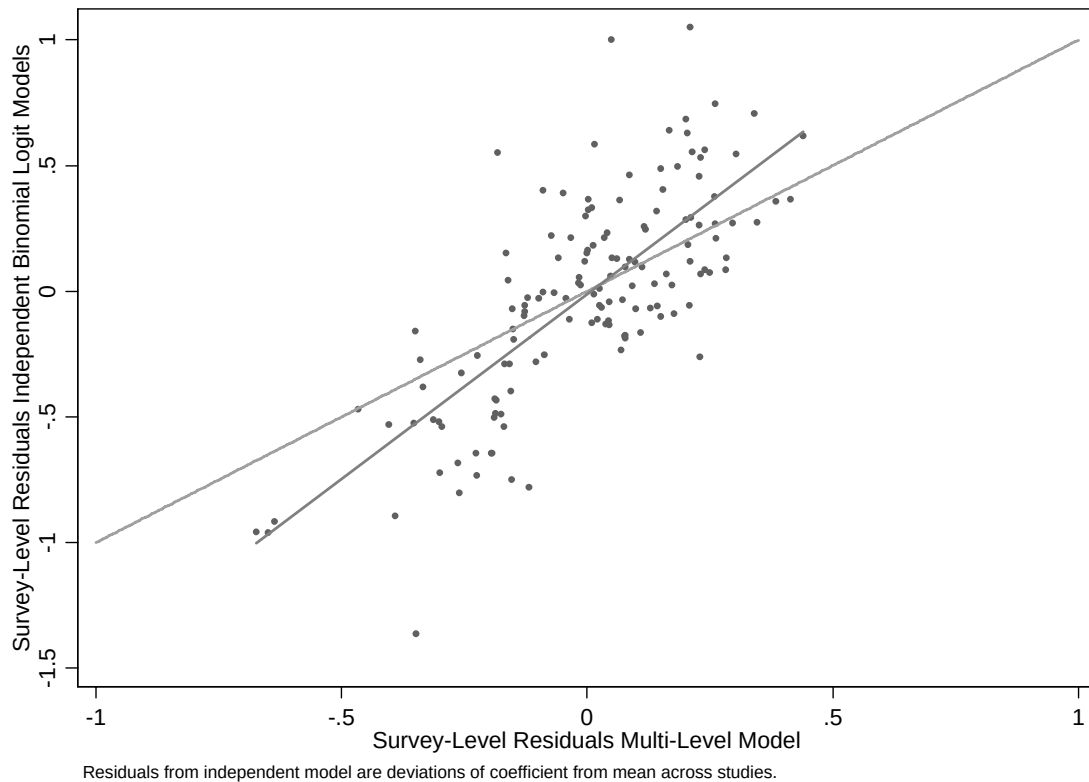


Figure 4.1. Correspondence between estimated coefficients from separate logistic models and pooled multi-level model (with identical control variables in both models)

Multinomial Estimation

We use multinomial logistic models to estimate the coefficients of the individual-level vote choice models from which we produce our estimates of the economic vote. As we pointed out in Chapter 2 multinomial logistic estimates of well specified models will give us consistent estimates of the economic voting effects. Of course this is not the only strategy available for estimating vote choice models. And over the past decade comparative scholars have become increasingly concerned with identifying which of a number of different available estimation techniques should be employed to properly estimate multi-party vote choice models (examples are Alvarez and Nagler 1998a; Quinn, Martin and Whitford 1999). Broadly speaking there are three approaches that have been employed to estimate discrete vote choice models: binomial logit models; multinomial logit models and multinomial probit models. The debate regarding the appropriateness of these different techniques raises the possibility that the variation in our estimates of the economic vote reported in previous chapters is an artifact of the multinomial logistic estimation strategy we have adopted. Because we are in the enviable position of having a

very large number of individual-level surveys, we can carefully assess in fact whether our estimates of economic voting vary as a function of adopting different estimation strategies.

Until relatively recently, the norm in the economic voting literature has been to employ dichotomous measures of vote choice, estimated with binomial logit. One strategy (used, for example, by Lewis-Beck, 1988) is to dichotomize the vote choice into a vote for any party in the governing coalition versus a vote for any challenging party. A similar strategy that does not focus only on the governing parties is to estimate separate logistic models for each party, where the dependent variable is coded as a vote for or against each party in turn. Anderson (1995) employs this strategy in his comparative evaluation of economic voting in European democracies, as do Clarke et al (2004) in their recent analysis of vote choice in the 2001 British elections. Our concern here is to understand whether adopting this binomial logistic estimation strategy results in patterns of economic voting that deviated significantly from the multinomial logit estimates summarized in Chapter 3. Accordingly, in Figure 4.2 we generated estimates from a series of binomial logit estimates for each party's vote (each dependent variable assumes a 1 for votes for the party and a 0 for votes for all other parties) and compare them to the estimates reported in Chapter 3.

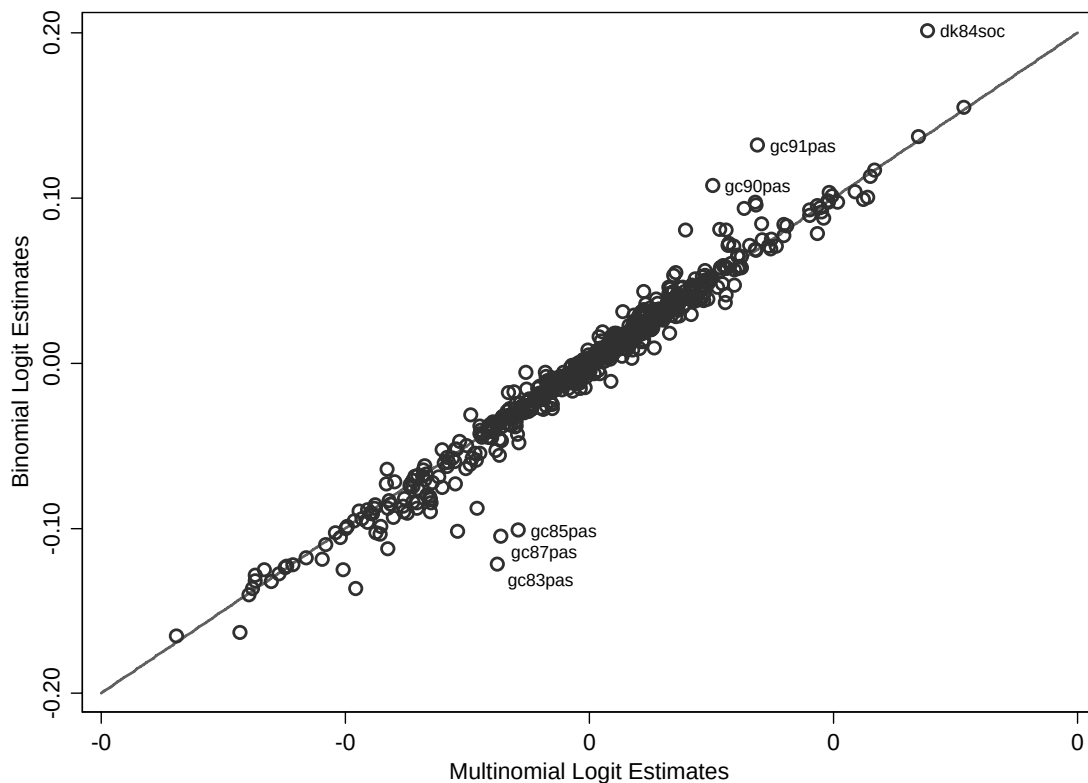


Figure 4.2. Binomial and Multinomial Logit Estimates of Economic Voting for Each Party

Clearly, the choice of a binomial or multinomial logistic estimation method is not consequential for our estimates of economic voting. Both estimation methods produce essentially identical results. The only consistent deviation is for the Greek cases: the multinomial estimates for these cases are attenuated toward zero compared to the binomial ones. The similarity between the two estimation methods likely helps explain why our rank orderings of national average economic vote magnitudes are similar to those of Lewis-Beck (1988) and Anderson (1995) who employ binomial logit for estimating the economic vote.

Multinomial logistic estimations of discrete vote choice models for political contexts with multiple competing parties have been criticized because the estimator assumes the independence of irrelevant alternatives (IIA). This assumption – that the addition or removal of an alternative will not change the relative selection probabilities of existing choices – seems clearly false for voting models.⁹⁴ Multinomial probit allows for a relaxation of the IIA assumption and has, accordingly, been championed by some as a preferred estimation strategy for discrete vote choice models with multiple parties (Alvarez and Nagler 1998a; Rudolph 2003).

It is important to note though that the IIA assumption for multinomial logit applies *conditional* on the explanatory variables included in the model. As Train (2003) points out, ultimately a well specified choice model that accounts for the major factors driving choice will achieve conditional independence of the errors in the value functions which is equivalent to having IIA hold. To the extent that our multinomial logit models are well specified we would not expect that their predicted vote probabilities would deviate significantly from those resulting from multinomial probit estimates of the same models.⁹⁵ We explore whether this is in fact the case by re-estimating party vote probabilities for a subset of the surveys in our sample. Specifically, since estimation of the multinomial probit requires a set of alternative specific covariates, we only compare those cases in which we could use the survey data to construct a set of variables measuring the ideological distance between the respondent and each of the parties for whom they could vote.⁹⁶ In order to construct this variable we need a measure of both respondent self-

⁹⁴ The assumption implies that the introduction to the choice set of a new socialist party will not have a disproportionate impact on the votes for the existing socialist and conservative parties, i.e., the new socialist party will not take a higher percentage of votes from the older socialist party than from the conservatives.

⁹⁵ Our concern is only with the robustness of the predicted vote probabilities to differences in specification. We make no claims about the robustness of other quantities that might be of interest for other purposes, such as substitution patterns.

⁹⁶ We chose to use ideological distance variables rather than other alternative-specific measures because we had included a similar, non-alternative specific, measure (left-right self placement) in each of our MNL models.

placement on the ideology scale and the placement of parties on the ideology scale. Since these ideological placement questions are infrequently included in election studies, we have a much more restricted sample size here – thirteen surveys.⁹⁷ The other variables in the model resembled those described for the fully specified economic voting models described in Chapter 2.

Figure 4.3 provides two plots. First, we compare estimated vote probabilities from multinomial probit and the multinomial logit models that both contain ideological distance variables. Next, we show these same multinomial probit estimates (again with the ideological distance variable) plotted against estimates from a MNL model that includes left-right self-placement instead of ideological distance variables. It is clear that for these cases the fit between the predictions is quite close, especially for those generated with MNL with left-right self-placement (our usual specification) and those estimated with MNP.⁹⁸ These results suggest that in the case of well-specified multinomial vote choice models – such as the ones we employ to generate the economic votes in Chapter 3 – the predicted vote choice probabilities generated by MNL and MNP will be very similar.

⁹⁷ For a detailed discussion of these analyses and results see Duch, Stevenson and May (2007).

⁹⁸ A result confirmed by Dow and Endersby (2004) in their comparison of Multinomial Probit and Multinomial Logit.

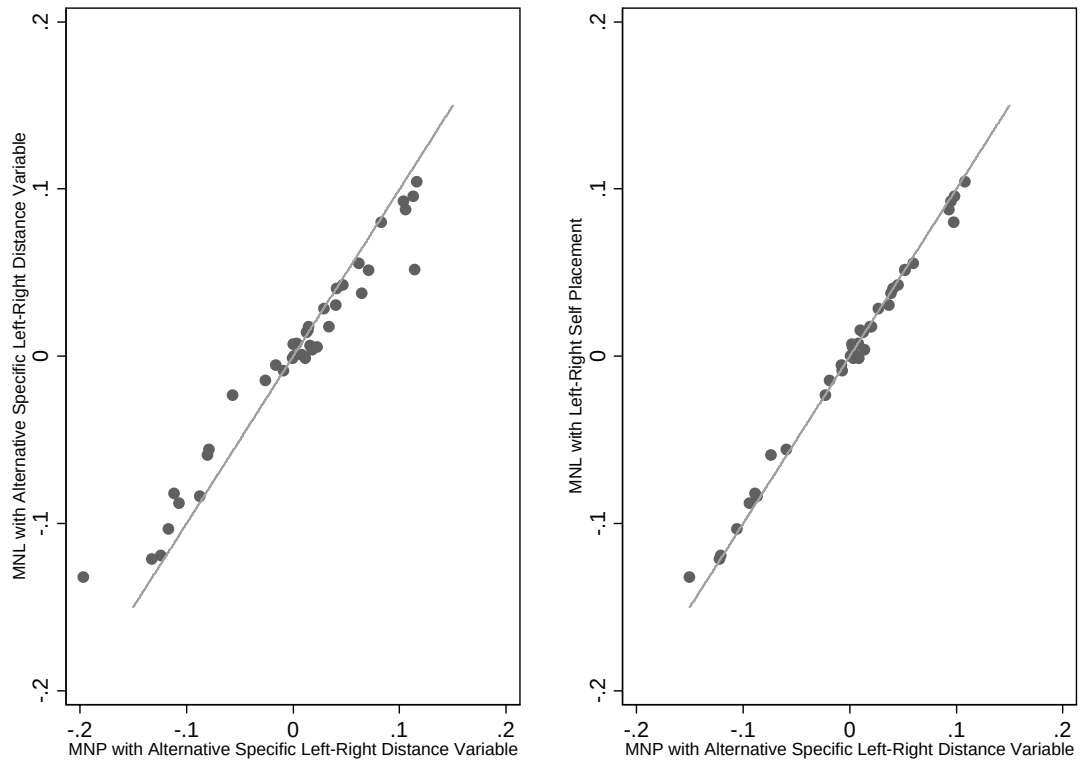


Figure 4.3. MNP and MNL Estimates of Economic Voting for 39 Parties in 13 Surveys

Asymmetric Economic Vote?

We arrive at our measure of economic voting by taking the coefficients from the multinomial logit estimates and calculating predicted changes in party support for a unit *decrease* in economic perceptions for each individual in the 163 surveys. These predicted changes were then averaged across each sample to generate an economic vote for each party in the country. The decision to estimate economic vote magnitudes based on worsening perceptions rather than improving perceptions was an essentially arbitrary decision on our part. Some have argued that the economic vote is asymmetric in the sense that negative economic outcomes receive more weight in the voter's value function than positive outcomes (Bloom and Price 1975; Haller and Norpoth 1994; Nannestad and Paldam 1997a, 2000; Price and Sanders 1995). This raises the possibility that our decision to base our estimated economic vote on a negative shift in economic perceptions might inflate our estimates of economic voting. Since we used a set of dummy variables to capture perceptions in the individual models, there was nothing in the specification itself that forced our estimates of the economic vote to be the same when economic perceptions worsened or improved by the same amount. Thus, if there is asymmetry we should see it by

comparing the estimated economic vote based on a worsening perception with those based on an improving assessment.

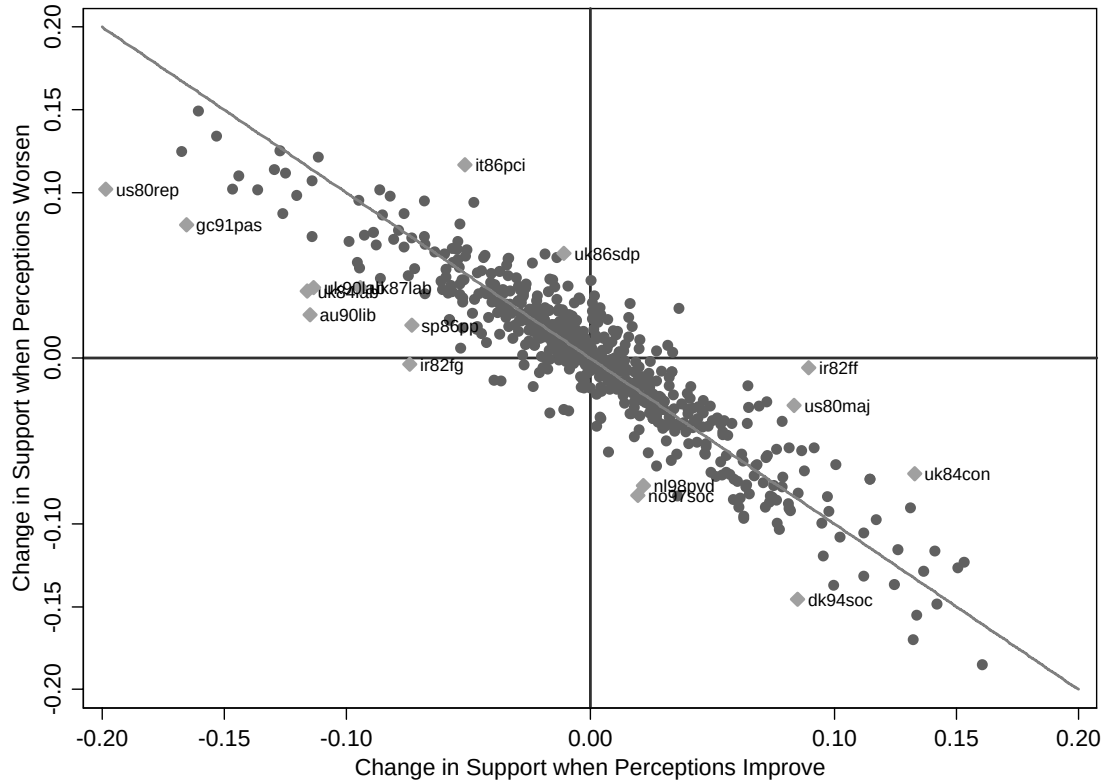


Figure 4.4. Symmetry of Economic Voting for Each Party

Figure 4.4 compares economic voting for each party when economic perceptions worsen or improve by the same amount. Clearly, there is little evidence for any significant asymmetry in economic voting. This echo's the early U.S. results reported by Kiewiet (1983) and the Western European results of Lewis Beck (1988) that fail to find asymmetry in economic voting. Our results though represent the first challenge to this asymmetry argument at the individual level that is based on such a large set of cases.

The estimation issues treated in this section have implications that are much broader than our comparative economic voting study. Because our project assembles a relatively large sample of individual-level surveys we can provide a number of unique insights into these debates. First, we provide insights into strategies for estimating contextual effects in multi-level models where there are large numbers of observations for each of the second level units (i.e., surveys). We

compare a two-stage approach with a one-stage pooled strategy and find that there are almost no differences in the results that are due to the estimation approach. This result is likely due to the large number of observations we have for each of our second level units, and so *should not* be taken as evidence that the two methods would produce the same results when the number of observations for some second-level units is limited.

A prominent methodological debate amongst students of comparative voting behavior concerns the appropriate estimation strategies for vote choice models in which there are more than two competing parties – hence a multinomial dependent variable. Again, given the nature of our sample we can provide some insight as to whether the adoption of different limited-dependent variable estimation techniques affects the estimated vote probabilities obtained from these different methods. We estimate party vote probabilities (based on 146 different voter preference surveys) employing logit estimation (where each party’s vote is estimated in a separate logit equation), multinomial logit, and multinomial probit. We then use the resulting coefficients to generate the economic vote of each party. Our comparison of the economic vote of each party, across estimation methods, demonstrates that these three different estimation strategies generate very similar results. At a minimum this suggests that, at least for the kinds of vote choice models we employ in this study, the results one can expect from employing any one of these three estimation strategies are likely to be quite similar.

Finally, we address a third estimation issue that is very much particular to the economic voting literature: the claim that the economic vote is asymmetric, and in particular that it tends to be stronger in the face of bad, as opposed to positive, economic evaluations. Again, our large sample size allows us to provide a unique and quite definitive assessment of this claim. Our evidence clearly suggests that there is no asymmetry in the economic vote – we obtain the same results by estimating the economic vote employing the coefficients for both our positive and negative (dummy) evaluation variables.

Differences in the Measurement of the Dependent Variable

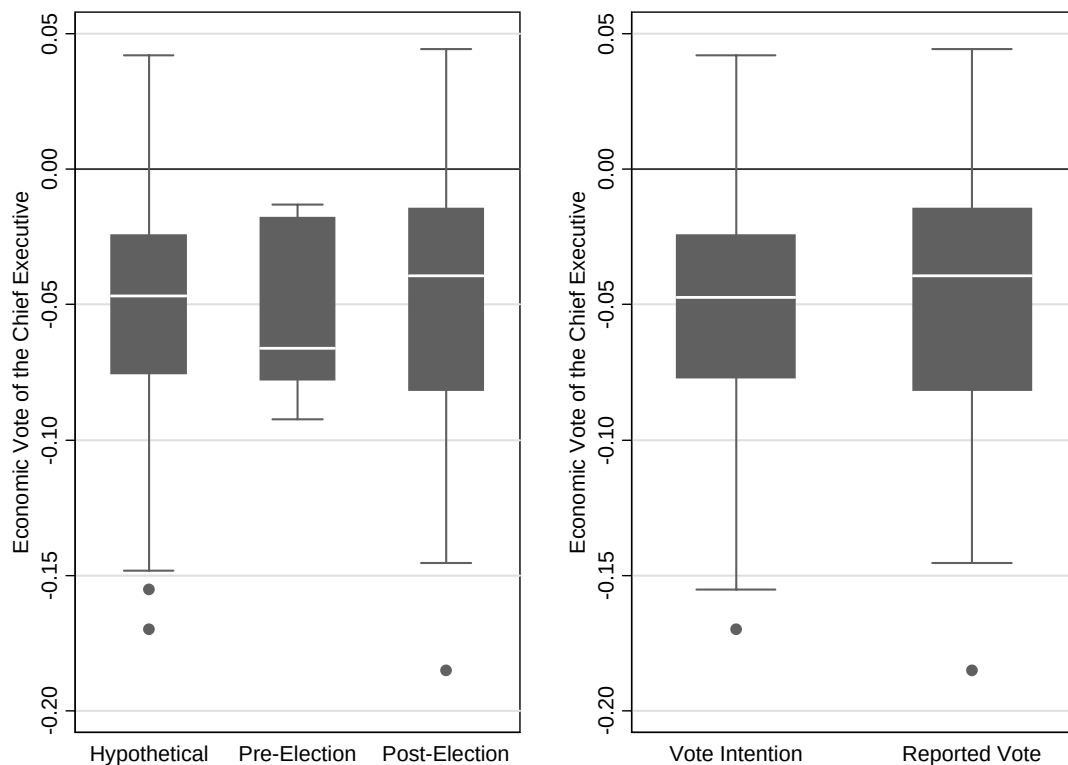
Another potential source of artifactual variation in our measure of the economic vote is differences in the measurement of vote preference. Recall that the 163 voter preference surveys, from which we estimate the economic vote, asked three types of vote choice questions. In most of the voter preference surveys (70 percent), voters were asked a “hypothetical” vote choice question: Which party they would vote for “if an election were held today”. The next most frequent question (25 percent of our surveys) was not hypothetical at all but occurred in post-election surveys usually conducted in the months directly after an election. These questions

asked voters to report which party they had voted for in the last election. Finally, 5 percent of our surveys occurred just prior to an election and asked voters to indicate which party they would vote for in the upcoming election. Survey researchers have certainly established that there are, in general, differences in party support when these different questions are asked (Gelman and King 1993; Lau 1994). The best-established result is that post-election vote reports over-state the true support for the winner of the election and that this may come from mostly non-voters who subsequently report both that they voted and that they voted for the winner. There is also strong evidence of over-reporting of voting in post-election surveys (Silver, Anderson and Abramson 1986; Wolfinger and Rosenstone 1980). It is possible that this kind of effect (or others) induces a systematically higher or lower economic vote in our individual level models.

If there is a systematic effect on our measure of economic vote magnitudes associated with vote choice question wording then it should be apparent in comparisons of the magnitudes we obtain using the different wordings. Figure 4.5 compares the economic vote of the prime minister by question type. In the left hand graph we distinguish hypothetical votes, from vote intention that refers to an upcoming election, and reported vote after the election.⁹⁹ The right hand graph compares the economic vote for cases in which vote intention (whether hypothetical or not) is employed versus those in which reported vote is employed as the dependent variable. It is immediately apparent that if there is over reporting for the winner of the election it is not reflected in a systematically higher or lower estimate of the economic vote for prime ministers. None of the means of the three types of questions are significantly different from one another.¹⁰⁰

⁹⁹ There are only six surveys that ask pre-electoral vote intention (the middle category in the left hand graph).

¹⁰⁰ The p-values for significant differences are all greater than 0.5.



Shaded areas of boxplots include the 25-75 percentiles and the “whiskers” extend to 1.5 the interquartile range. The center line is the median.

Figure 4.5. Vote Choice Question and Economic Vote of the Chief Executive

Question type is not the only measurement issue that might impact the vote choice question. Another is the proximity of the question to an election (Lau 1994). For reported vote and pre-election vote questions, these generally occur quite close to the election, but for hypothetical vote questions, these can sometimes be asked in surveys that are far from an upcoming election or far from the previous one. Various scholars have suggested that economic voting should decline as a function of the time separating the election and when voters are surveyed (either after or before an election); perhaps because media cues attaching incumbents to economic outcomes are less frequent or because voters are paying less attention (Palmer and Whitten 2003). It could also simply be that when the hypothetical vote choice question is asked closer to an election it provides a better measure of what the voter would do in an actual election. And there is evidence to suggest that responses to vote choice questions reflect more information about strategic incentives and issues as the timing of the survey gets closer to the election and hence are more strongly correlated to their actual vote choice (Gelman and King 1993).

We can assess empirically whether the proximity of our sample of surveys to elections had an impact on economic vote magnitudes. Figure 4.6 graphs the economic vote of the prime minister against the time since the previous election or until the next one, whichever is closer. There is no trace in this graph (as well as the corresponding regression) of any systematic relationship between the proximity of the election to the time the survey was conducted and the *economic vote of the chief executive*.¹⁰¹

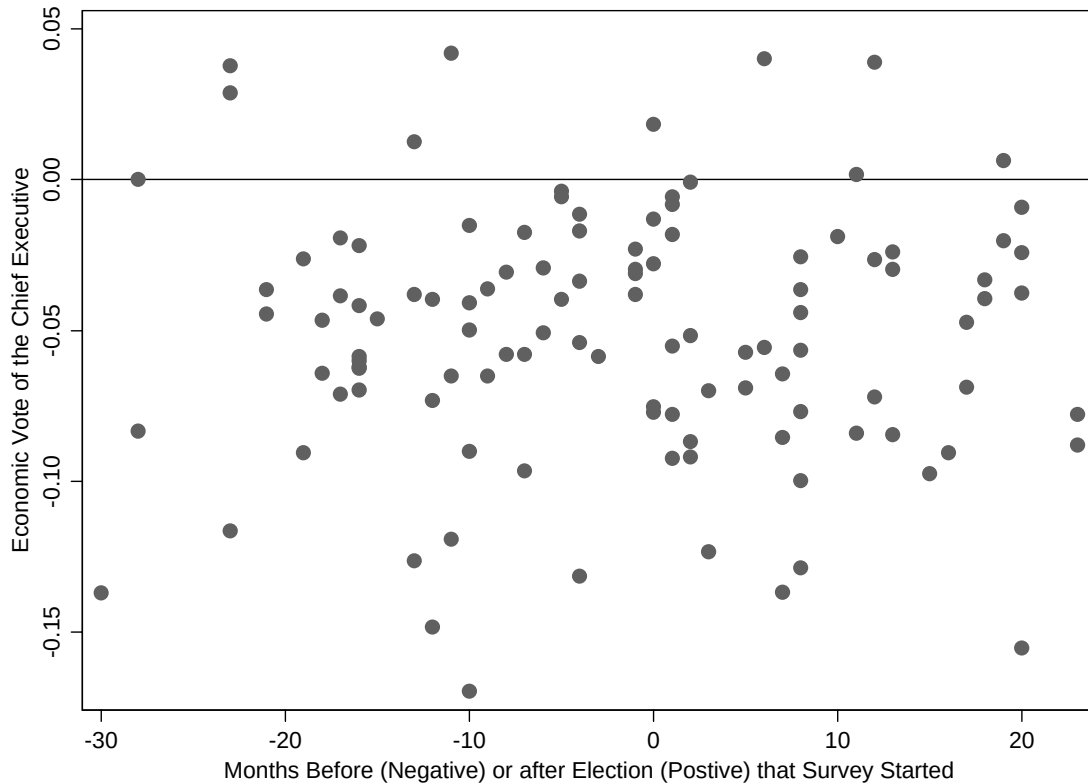


Figure 4.6. Proximity of Survey to an Election and Economic Vote of the Chief Executive

Our dependent variables in the 163 vote choice models is based on vote preference questions that varied in terms of whether they referred to an hypothetical as opposed to actual election and whether the questions were asked prior to, or after, an election occurred. These variations do not result in any systematic differences in our reported empirical results. These vote preference questions were asked at different time points relative to actual national elections.

¹⁰¹ The p-value for a linear regression of proximity to election and economic vote of the chief executive is 0.984.

Again, we demonstrate that the period in which the surveys were conducted has no systematic effect on the magnitude of the economic vote.

Model Specification

Recall from our discussion in Chapter 2 that the specification of the 163 economic vote models was guided by the prevailing literature on vote choice models for the sampled country. There are nonetheless controversies regarding the appropriate specification of vote choice models, and economic voting models in particular. This raises the possibility that model specification may have systematically inflated or deflated our estimates of economic vote magnitudes. We explore in this section two model specification issues. First we examine whether our choice of control variables for these 163 models had any systematic effect on the economic vote magnitudes summarized in Chapter 3. Second, we assess whether our measure of economic evaluations is sufficiently exogenous of vote preference to provide us with a meaningful estimate of economic voting.

Variation in control variables

As we pointed out in Chapter 2, the specification of control variables varies from study to study – the linguistic variable, for example, is, appropriately, included in Canada but excluded from France. This raises the possibility that there is a systematic relationship between the control variables included in our different models and our estimates of the economic vote. So, for example, models that include a measure of social class might generate systematically different estimates of economic vote magnitudes than those that do not include this variable. If the individual level models were properly specified so that, for example, those equations that should include social class do include it and those that should not include it do not, we would not expect a systematic relationship between its presence and the impact of economic perceptions on the vote.

Our strategy for exploring the possibility of these kinds of effects was to create measures for each of our 163 studies that indicated which concepts were controlled for in each individual level vote choice model and then to use these variables to search for systematic impacts of differences in the sets of control variables on our estimates of economic voting. Importantly, our coding was of the *concepts* controlled for in the models not the measures used. As we emphasized in Chapter 2, while many surveys include the same concepts, they often measure these concepts differently – so, for example, occupation is often solicited in surveys but the coding into occupation categories can be quite different. For the purpose of this analysis what

matters is whether or not an occupation variable was included in the equation regardless of how it was measured.

Table 4.1 provides the list of concepts that are controlled for in our models and the percentage of our 163 cases that include a measure controlling for the concept. The reported test is simply whether the mean of the *economic vote of the chief executive* differs in models with and without the concept.¹⁰²

Table 4.1 Control Variables and Economic Voting of the Chief Executive

<i>Was Some Measure of the Concept Included in Model? (1 = "yes", 0 = "no")</i>	<i>Frequency of Inclusion and Statistical Significance of Impact on Economic Voting</i>
Age	100%
Gender	97%
Education	97%
Left/Right Self Placement	95%
Religiosity	92%
Occupation or Self Identified Class	88%
Satisfaction with democracy	83%
Lives in an Urban Area	78%
Support for EU	65%
Household Income	45%
Union Membership	35%**
Language or Ethnicity	27%
Home ownership or Lives in public Housing	24%
Region of Residence	16%
Policy Opinions (various policies)	15%
Party Attachment (US only)	11%*
Feelings or Evaluations about PM Candidates (Australia only)	9%*

* and ** indicate that the means between groups defined by the dummy variable for the *economic vote of the chief executive* is significant at $p < .10$ and $P < .05$, respectively. No test was possible for Age since it was included in all studies.

For the most part, whether a particular concept was or was not controlled for is unrelated to our estimate of the *economic vote of the chief executive*. For three cases, however, there is a statistically significant relationship. We discuss one of these cases, party attachment, in the next section, where we address the issue of the use of party attachment in vote choice models and defend our decision to include party attachment only in the US case. Another case, candidate evaluations, is also discussed in a later section where we show that this variable appears to be necessary to get sensible estimates for the Australian case. However, for the third concept, union membership, a closer examination of the relationship reveals no group of cases that is clearly

¹⁰² These tests are from dummy variable regressions that account for correlations between observations of the same party across elections and the fact that the dependent variable is estimated.

driving the result, nor does it provide any clear intuition as to why it occurs. We are left then with the possibility that whether or not we control for union membership in different empirical models accounts for some of the systematic variation in our estimates of economic vote magnitudes. We lack, however a clear theoretical rationale for understanding why this might be the case. Nevertheless, for the vast majority of the cases, the data seems to support the idea that the list of control variables we include in the model is not driving most of the cross-national and over time variation in economic voting that we will explore in subsequent chapters.

Alternative specifications

A second issue concerning the set of control variables included in different studies does not concern the presence of systematic differences across studies, but instead focuses on our confidence in the specification (and resulting estimates) in a particular study. Specifically, the estimates we provide for each case are not the only ones that could have been obtained. In each case some alternative set of control variables could be found that would lead to different estimates of economic voting in that case. Our strategy in this book has been to use a set of controls in each case that are well accepted in the literature on economic voting for that case and which will get us as close to the “true” model as possible. Thus, the only alternative sets of control variables that should concern us are those that are as plausible or as well supported in the literature as the ones we choose. If there are alternative specifications that are theoretically credible and that result in very different estimates of economic voting, then we need to qualify our assessment of economic voting magnitudes with the warning that different reasonable specifications of cases would produce different results.¹⁰³

Of course, our review of the various country specific literatures on voting behavior did reveal a number of controversies about exactly which variables should be included in vote choice models. In most cases, however, these were relatively minor controversies and we usually found that the inclusion or exclusion of specific variables did not change our estimates of economic voting significantly. One controversy in the literature on voting behavior, however, is both

¹⁰³ To be clear, removing control variables from a vote choice model that are widely recognized as important will very likely have some effect on our estimates of economic voting. Indeed, we should not expect them to be insensitive to these kinds of changes (hence the need for control variables in the first place). Thus, when we (and the literature) have confidence that a particular variable belongs in the equation, there is no reason to remove it and the fact that doing so would change our estimates does not create uncertainty about the magnitude of economic voting for the case, since the estimate without the variable is clearly (according to the collected wisdom of the literature) wrong. However, when the literature is uncertain about a whether a variable is appropriately included in the vote choice model and its inclusion or exclusion has an impact on economic voting, then we should be more uncertain that the estimates of the economic voting for the case that are based on an uncertain specification are correct.

widespread and consequential for our analysis – whether party identification (or attachment) should be included or not.

The case of the United States is the least controversial. Indeed, our review of the literature on individual level vote choice in the U.S. indicates an overwhelming consensus in favor of including party identification as an explanatory variable. The variable has been a pillar of the Michigan School's approach to voting behavior, but is also important in the arguments and empirical models of scholars coming from very different theoretical approaches. Fiorina (1981), for example, makes a strong case for including party identification in models of rational retrospective voting.

Kiewiet (1983) questioned the appropriateness of including party identification in models of economic voting on the grounds that party identification might itself be highly correlated with economic evaluations and hence bias the estimated effects of these variables. Kiewiet (1983) explored this possibility by estimating two different sets of voting models – one with partisanship included and a second set that substituted a battery of socio-economic variables for partisanship. The substantive effects of the economic variables in the two different models did not vary significantly, suggesting that, at least in the U.S. context, inclusion of the party identification variable doesn't seem to either inflate or underestimate the economic voting effect. And much of the recent work on economic voting in the U.S., such as Erikson, Mackuen and Stimson (2002), reflect the importance of including this “running tally” as a control variable in economic voting models. Following this consensus, then, we include party identification in our U.S. equations.

When one goes beyond the United States, one sometimes finds a controversy regarding the appropriateness of including party identification as an explanatory variable in voting models or, more usually, a wide-spread consensus that it is inappropriate. The principal critique, popularized by Budge, Crewe and Farlie (1976) but taken up by many other European students of voting behavior, is that outside the U.S. the concept of party identification and vote choice cannot really be distinguished empirically (at least how it is usually measured) because voters simply report their vote choice when asked about party attachment. There has been considerable empirical evidence in support of this contention (Holmberg 1994; Thomassen 1976) and so the concept has not often been used in studies of voting behavior in many of the European democracies.¹⁰⁴

In the Anglo-democracies (including Australia, Canada, New Zealand and Britain), Party Identification has been frequently used in voting studies, but its use has often spurred controversy

¹⁰⁴ Moreover there has been a widely documented trend of declining partisanship in many advanced democracies that has added to controversies regarding this variable (Dalton and Wattenberg 2000).

as students of voting behavior have also questioned whether it in fact it is very distinctive from the vote. The Canadian literature is a case in point. The “Michigan model” has had an important influence on the design of election studies in Canada – hence considerable attention is given to the inclusion of partisanship questions in Canadian national election studies. Nevertheless, students of Canadian voting behavior seem pretty uniformly convinced that party identification historically has been very low (see the early studies of John Meisel 1975 and Jane Jenson 1976). And the evidence seems to suggest that partisanship has been declining in Canada and is becoming increasingly unstable (Bowler 1990; Clarke, Kornberg and Wearing 2000; Stewart and Clarke 1998).¹⁰⁵ The virtual disappearance of two major Canadian parties in 1993 attests to the relative weakness of party attachments. One of the contributing factors to the weakness or instability of partisanship in Canada is the existence of two distinct party systems at the national and provincial level which contributes significantly to partisan inconsistency and hence instability (Stewart and Clarke 1998).

The use of party identification in UK voting models evokes similar equivocations as in Canada. In their classic study of the British voter, Butler and Stokes (1976) make the case that party identification is much more unstable than is the case in the U.S. and that it tends to shift with changes in vote choice. Budge, Crewe, and Farlie (1976) make an even more aggressive case against employing party identification in British vote models on the basis that partisanship is highly correlated with—and likely not causally prior to – vote choice. Subsequent research seems to support this conclusion that conventional measures of party identification in Britain are essentially tapping voting intention (Bartle 2001; Sanders et al 2003). In addition, Clarke, Stewart and Whitely (1997) have documented the extent to which partisanship in Britain is highly sensitive to movements in short term forces such as economic conditions and issue controversies.

Based on our review of the voting literature there certainly is some question as to whether party identification ought to be included in the British and Canadian models. And there clearly are strong grounds for excluding party identification from voting models estimated with European data.¹⁰⁶ Our general rule then was to exclude party identification from our economic voting models with the exception of the U.S. models where there is overwhelming consensus that it should be included. But, does this decision to include or exclude party identification from economic voting models have implications for the magnitude of the economic effect? The

¹⁰⁵ Although see Schickler and Green (1997) who argue that once measurement error is taken into account party identification has been relatively stable in Canada and other advanced democracies.

¹⁰⁶ In our data from the European cases, the correlation between the party attachment question and vote choice was so high that they were essentially identical so could not be included in the same model.

answer to this question is important in those cases in which the use of the variable is controversial. If the range of estimates of economic voting across reasonable models is large (for example, models including and excluding party identification), we must qualify the conclusions drawn from any one set of economic voting estimates with the caution that other reasonable but very different estimates are possible. Given this, we explore the impact of including or excluding partisan identification on economic voting in Canada, the UK, and Australia, the three cases in our sample in which the literature provides little consensus on the issue. Our strategy is simply to estimate models of economic voting with and without partisanship and examine the differences in our estimates of economic voting.

Table 4.2 reports the difference in the *chief executive* vote between the estimates we used in Chapter 3 (that did not include party identification) and those we get when we add party identification to these models. With only one exception, all of the models, whether they contained party attachment or not, showed evidence of economic voting in the expected direction – i.e., a negative change in support for the prime minister’s party when economic perceptions worsened.¹⁰⁷ Further, again with only one exception (the 1993 Canadian election), all the differences between the two estimations suggest that including partisanship tends to attenuate the economic voting effect. However, only two of the estimates are large enough to really change the substantive picture of the amount of economic voting for the prime minister in the election: the British 2001 and the Canadian 2000 cases.

Table 4.2. Change in chief executive vote when Party Identification dropped from Model

<i>Survey Year</i>	<i>Britain</i>	<i>Canada</i>	<i>Australia</i>
1979	.006		
1987			.009
1988		.03	
1990			.012
1992	0.00		
1993		-.005	.001
1996			.004
1997	.012	.032	
1998			.01
2000		.046	
2001	.06		.005

The economic vote of the prime minister is larger when it is more negative, so a positive number here indicates that adding party identification to the model decreases the economic vote of the prime minister.

Sparse versus rich specification

¹⁰⁷ The exception was the Canadian 1997 case when party identification was included.

Concerns about the sensitivity of variations in the economic vote to model specifications can be addressed in a different way by adopting a uniform specification for all 163 vote choice models in which all of the explanatory variables were identical. When the explanatory variables in each of the 163 equations are identical then variations in the magnitude of the economic vote from one study to the next cannot be attributed to differences in the list of control variables in the models (since it does not vary). We believe our richer specification of each of the 163 vote choice models provides a better estimate of economic vote magnitudes in each national context. Nevertheless, if our estimates of the economic vote deviate significantly from those obtained from a uniform specification one might, reasonably, conclude that a significant amount of the contextual variation in the economic vote is driven by model specification decisions. We can look at the issue of how sensitive the estimates of individual models are to the set of controls included by comparing the estimates from the rich and varied sets of controls that were used to generate the economic vote distributions in Chapter 3 to corresponding estimates based on a more limited, but uniform specification. The sparse but uniform specification includes only the economic perceptions question and left-right self placement, while the rich specifications include the various controls discussed in Chapter 2 (details of the full-specified 163 models can be found at www.nuffield.ox.ac.uk/economicvoting).¹⁰⁸

¹⁰⁸ Only cases in which we could measure left right self placement are included.

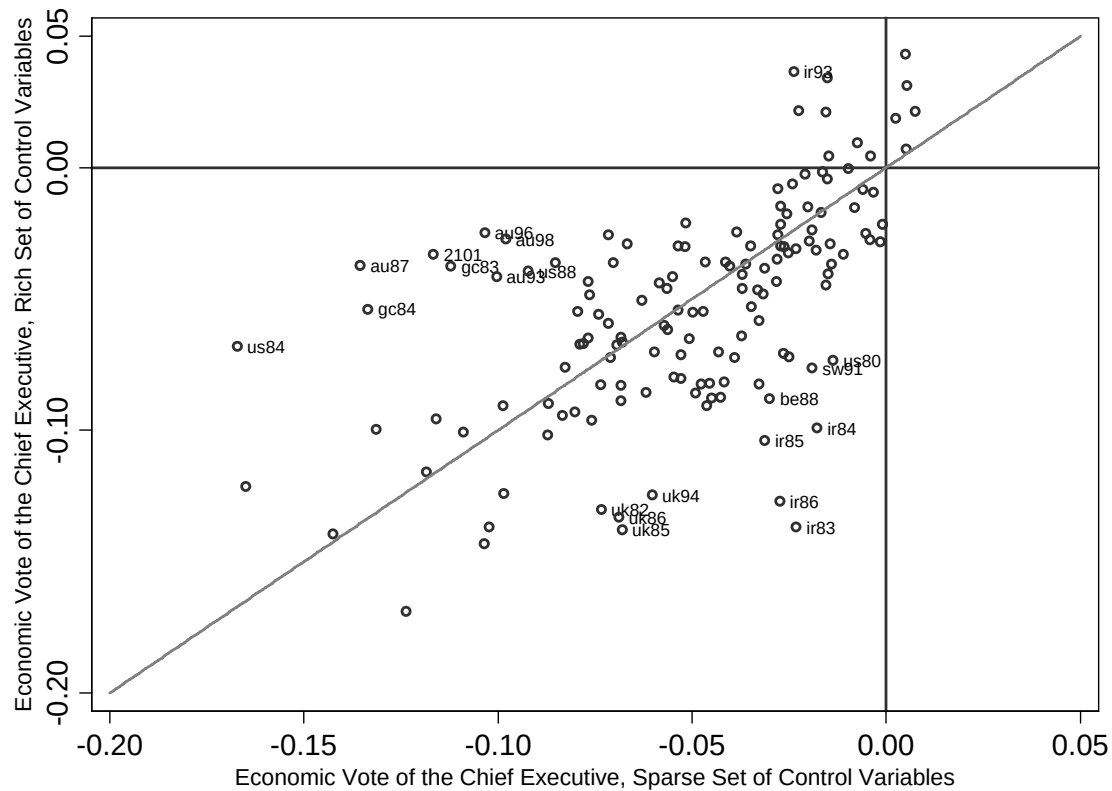


Figure 4.7. Comparison of Estimates with Rich and Sparse Sets of Control Variables

The comparison in the chart is between sets of controls that are about as different as one could find. The inclusion of left/right self-placement in the sparse model, however, was not only due to the fact that it could be measured similarly in many models. It was also included because our exploration of different specifications (as we worked with each of the 163 individual level models) revealed to us an important fact that is reflected in Figure 4.7. If one includes left-right self placement in a vote choice model (or we suspect some other variable that “nails down” the voter in a general ideological space) then the estimate of economic voting is fairly stable to even drastic revisions in the list of other controls. The outliers labeled in the figure are exceptions, but instructive ones. Outliers below the 45 degree line represent cases in which the magnitude of the estimated economic vote is significantly lower in the sparse specification. Outliers above this line are cases in which the estimated vote is higher in the sparse specification. First, the Irish and British cases make up the majority of the outliers below the 45 degree line and in both of these cases the role of left-right self placement in vote choice has been frequently questioned in the literature (e.g., Evans 1999a). In the case of our voting models it appears that in the UK and Ireland left-right self placement is not a sufficient control for generating reasonable estimates of the economic vote – a richer specification seems to be necessary. Indeed, in the British cases the

question is often not even asked in the national elections studies (the cases here come from Eurobarometers). In Ireland, it has long been argued that although a left-right dimension of politics may be emerging, the principal organization of political conflict has traditionally not been the left-right but rather the policies and ideological profiles that surrounded the pro and anti-treaty traditions of the major parties.

Above the 45 degree line the Australian cases are prominent exceptions. Our investigation of this case suggests that predicted vote is quite sensitive to the inclusion of a prime minister evaluation question.¹⁰⁹ Hence in this case adding a control for prime minister evaluation dampens our estimate of the economic vote compared to the sparse model. Finally, we again see that the Greek cases are exceptions – reinforcing a general pattern that is emerging in this chapter; that it is hard to nail down the magnitude of economic voting in Greece, the estimates of which depend more than we would like on various modeling decisions.

Fit of the Individual Level Models

The “fit” of each of the individual-level models varies from one study to the next. At the low end some of the models have “pseudo- R^2 s” in the neighborhood of .07 while the outlying Greek cases reach values as high as .8. The typical pseudo- R^2 is about .35. One possibility here is that our estimate of the size of the economic vote might vary with the fit of each model. It might be the case, for example, that models with a poor fit tend to inflate the economic vote effect because the economic evaluation variable is capturing some of the effect of excluded independent variables. The final question we raise in this chapter is whether the estimates that make up our map of economic voting vary systematically with the fit of the individual level models from which they come. To examine this, we graph, in Figure 4.7, the pseudo- R^2 from the multinomial logit estimation of the model against the economic vote of the prime minister.

¹⁰⁹ On the importance of leadership variables in voting models in the Australian case see Bean (1996); Bean and Mughan (1989); and Bean (2001).

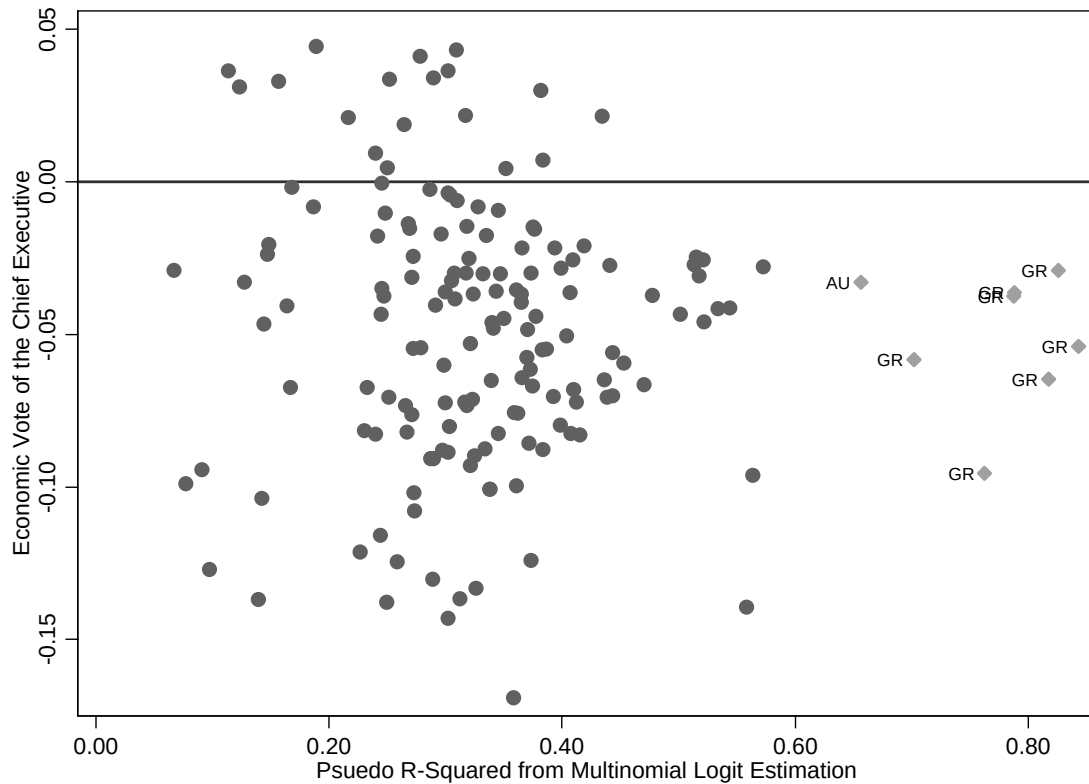


Figure 4.8. Fit of the Individual Level Model and Economic Voting for the Prime Minister

The graph reveals no clear relationship between this measure of fit and our estimates of the *economic vote of the chief executive*. It does however again single out the Greek cases unusual. The pseudo- R^2 's for the Greek cases are much larger than other countries. Our investigation as to why this should be the case suggests that this is due to a very strong relationship between left-right self-placement and the vote.

Endogenous Economic Perceptions

Typically economic voting models – such as the ones generating the economic vote magnitudes described in Chapter 3 – treat evaluations of the economy as exogenous with the implicit assumption that they reflect some aspect of objective economic performance. But the literature on attitude formation – economic evaluations are attitudes after all – suggests that voter characteristics will affect the extent to which this “objective” information shapes their economic evaluations. One of the pioneers of the economic voting literature, Gerald Kramer (1983), suggested that individual level analyses of economic voting are potentially problematic because individual reports of changes in economic outcomes exaggerate net changes that are politically accountable. And he pointed out that when individual level perceptual errors correlate with

partisanship or vote preference this can inflate the magnitude of the economic vote. More recently, Zaller (2004) echoed this concern, arguing that political sophisticates and partisans resist economic information from the media that is at odds with their partisanship while the economic evaluations of the less partisan are more receptive to media messages regarding economic performance. This suggests, of course, that certainly for some elements of the population, economic evaluations will be tainted by partisanship. And depending on the distribution of sophisticates and partisans in the population this could inflate or dampen the magnitude of economic voting in some contexts.

Evidence from the analysis of individual-level surveys suggests this might be a serious problem for economic voting studies. Duch et al (2000) consider the case of economic voting in U.S. presidential elections and conclude that national economic evaluations are strongly shaped by partisan pre-dispositions. Erikson examines individual-level ANES survey data and concludes that "... cross-sectional variation in respondent's reported perceptions of national economic conditions are largely random noise that has no bearing on political evaluations (Erikson 2004, 5)." And he argues that any observed relationship between economic evaluations and vote choice is an artifact of vote preference shaping individuals' economic perceptions. Based on their analysis of individual-level UK panel data Evans and Anderson (2006) conclude that the causal impact of vote preference on economic evaluations is stronger than the impact of these evaluations on vote choice. Finally, in their study of economic voting in four nations, Wlezien et al (1997) find that economic evaluations are strongly influenced by vote preference.

If economic perceptions are in fact endogenous, the theoretical implications of endogeneity can not be denied, however, the practical implications of endogeneity for the quality of the estimates depends on a number of factors that may well vary from study to study. To put it another way, the extent to which endogenous economic perceptions will move estimates of the magnitude of the economic vote away from their true values in a population may well vary from survey to survey depending on the specifics of the empirical model that was estimated. Thus, it is possible that variation in the practical implications of endogeneity across our 163 surveys is driving some of the variation in the map we have drawn. Given this, the only real way to know if this is happening is to re-estimate our 163 models using methods that can account for the endogeneity of economic evaluations and then compare these estimates to those in Chapter 3. In this section, then, we report estimates of the *economic vote for the chief executive* that "correct" for endogeneity using a method proposed by Duch and Palmer (2002a) that first purges economic evaluations of the systematic influences of variables like partisanship and political knowledge (as

well as others) and then uses these purged evaluations in vote choice models. The details of the Duch and Palmer (2002a) method are described in Appendix 4.1.

Figure 4.9 compares our estimates of the *economic vote of the chief executive* from models using purged and un-purged measures of economic evaluations respectively. The result clearly suggests that, despite persistent worries in the literature about the endogeneity of economic perceptions, the practical impact of this problem is relatively small – at least for our goal of making comparative assessments.

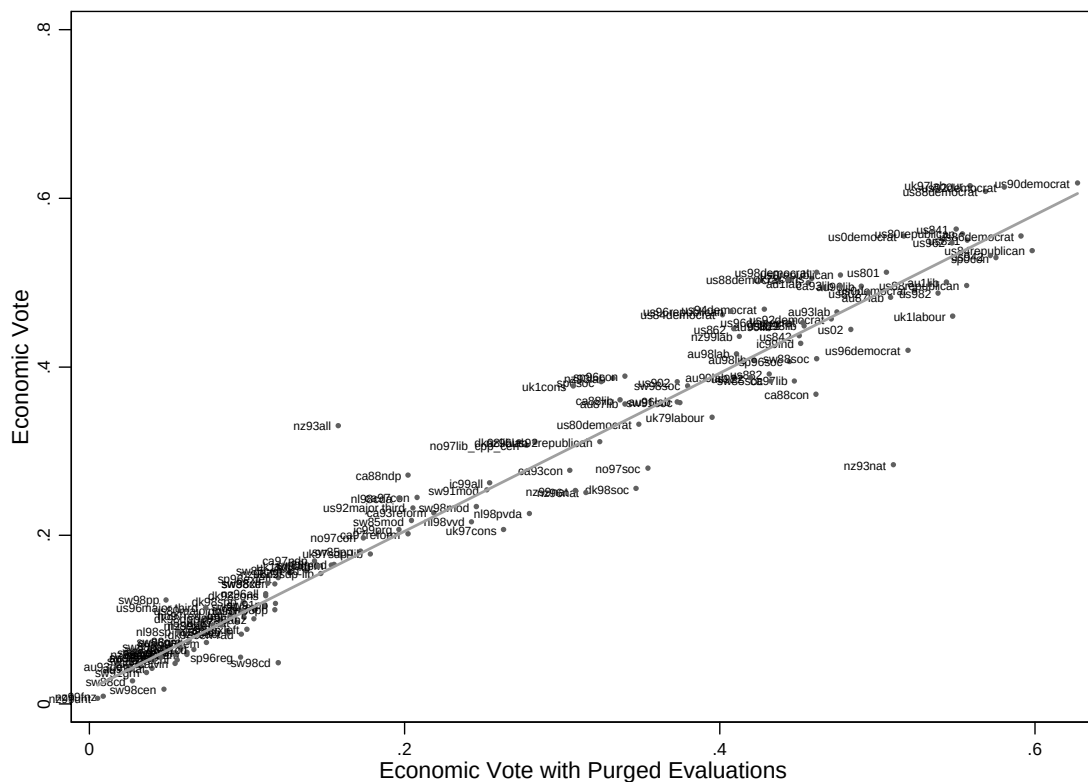


Figure 4.9. Economic Vote with and without Purged Economic Evaluations

Is the contextual variation in the economic vote, such as that which was described in Chapter 3, an artifact of model specification? Again with the advantage of a relatively large number of individual-level surveys we are able to provide quite conclusive answers to this question. First, with respect to the control variables included in economic voting models, so long as left-right self placement is amongst the control variables then the estimate of economic voting is fairly stable to even drastic revisions in the other controls included in the model. Second, while it certainly is the case that subjective evaluations of the economy are influenced by vote

preference, this endogeneity has surprisingly little influence on the patterns of contextual variation in the economic vote described in Chapter 3.

Summary

Students of comparative economic voting are primarily interested in the contextual variation in the economic vote that we described in Chapter 3. And these estimates will be the subject of both the theoretical and empirical efforts we undertake in the rest of the book. But in order to generate these estimates of the economic vote from our 163 voter preference surveys, we make a number of methodological choices, some of which are the subject of current debates in the literature or some that readers might simply question. There are three broad types of methodological issues associated with estimating contextual variation in the economic vote. This chapter assesses whether our choices in any one of these areas likely affected in any systematic fashion the empirical results we report. We introduce two general empirical methodologies for testing hypotheses about the impact of contextual variables on individual level behavioral relationships – a one-stage and a two-stage estimation strategy – both of which give very similar estimates of contextual variation in the economic vote. Our comparison of the party vote probabilities employing logit, multinomial logit, and multinomial probit suggests that these estimation strategies generate very similar results. The measurement of our dependent variable varies somewhat across voter preference surveys and we demonstrate this variation has no significant impact on our results. And with respect to model specification, we demonstrate that our results are very robust to the control variables included in the model, with the provision that left-right self placement (or some other variable capturing ideology) is amongst them. We also demonstrate that endogeneity in our measure of economic evaluations has surprisingly little influence on the patterns of contextual variation in the economic vote described in Chapter 3.

This first part of the book has been devoted to measuring the economic vote with the ultimate goal of explaining any cross-national or over-time variation in its magnitude. We provide quite conclusive evidence, based on 163 individual-level surveys, that economic evaluations have a very important impact on vote choice in the developed democracies but that the effect does vary significantly across countries and over time. Part I also outlines the second-stage strategy we will employ for modeling contextual variation in the economic vote. We now move to Part II in which we propose, and empirically test the first part of our contextual theory of economic voting.

Appendix 4.1

Following Duch and Palmer (2002) we can think about economic evaluations (X_i) in the following terms,

$$(10.1) \quad \begin{aligned} X_i &= \lambda_i X^O + X_i^S + \varepsilon_i \\ X_i^S &= f(W) \end{aligned}$$

where X^O is the objective economic economy, X_i^S captures systematic differences due to information and subjective factors (i.e., W), and ε_i is the stochastic component. In this formal definition, individual-level evaluations contain two forms of “noise”: subjective considerations and random fluctuations. Both forms of noise constitute sources of non-attitudes. If we include national economic evaluations in an economic voting model without controlling for this systematic measurement error (X_i^S), the evaluations will “pick up” the direct effect of partisan pre-dispositions, thereby producing an inflated estimate of the relationship between economic perceptions and vote choice.

Hence it is important to understand to what extent this endogeneity affects our estimate of the magnitude of the economic vote. By controlling for systematic and random measurement error in economic evaluations, we can estimate whether this error seriously affects estimates of the magnitude of economic voting at the individual level. One approach is to purge X_i , reported economic evaluations, of systematic measurement error (X_i^S) which would result in a measure of economic evaluations that included only factors contributing to meaningful fluctuations in economic evaluations (X^O) and random measurement error (ε_i). This would entail

estimating X_i^{Purged} , which is economic evaluations purged of systematic measurement error.

Duch and Palmer (2002) suggest that the systematic component of the measurement error in economic evaluations consists of three broad factors. First, economic evaluations are shaped by partisanship. Duch and Palmer (2002) have demonstrated that national economic evaluations are to a large extent shaped by partisan pre-dispositions – this is the case in both the American and European contexts. In addition to political partisanship, respondents may rely on personal experiences and regional economic circumstances to formulate an evaluation of the national economy. Citizens who infer national conditions from personal and local experiences are effectively evaluating the economy in a subjective rather than objective manner. Similarly, national economic evaluations may vary across individuals due to differences in levels of

information and sophistication about government policy and economic outcomes. Third, social class differences may systematically shape economic evaluations. Individuals in different socioeconomic circumstances might view the same economy in a very different light (MacKuen and Mouw 1995). Similarly, citizens of different sexes and races may perceive the economy differently due to biases in their general attitudes toward the economic and political systems (on gender and economic voting, see Welch and Hibbing 1992).

Note that the coefficient (λ_i) in Equation (10.1) varies across individuals. This variation results from groups of individuals placing different weight on different aspects of economic performance. These differences are captured by λ_i . This “Policy Related Variation” represents differences in economic evaluations that are grounded in self-interested differences in opinion regarding measures of economic performance and hence capture variation in economic evaluations that are grounded in meaningful policy differences, or emphases, across individuals. For example, individuals who value reducing inflation more than maintaining low unemployment (e.g., retired respondents on fixed incomes) might emphasize price stability more than job creation when evaluating the national economy.

Hence we can identify three factors contributing to systematic measurement error: party identification (PID), Information and socio-economic status (SES). Accordingly X_i^{Purged} is estimated as follows:

$$\begin{aligned}
 X_i &= \alpha_1(PID) + \alpha_2(\text{Information}) + \alpha_3(\text{SES}) + \alpha_4(\text{Policy}) + \varepsilon_i \\
 \hat{X}_i^S &= \hat{\alpha}_1(PID) + \hat{\alpha}_2(\text{Information}) + \hat{\alpha}_3(\text{SES}) \\
 X_i^{\text{Purged}} &= X_i - \hat{X}_i^S
 \end{aligned}
 \tag{10.2}$$

The estimate of vote probabilities is then,

$$\begin{aligned}
 v_{ik} &\sim \text{Bin}(\pi_{ik}) \\
 \text{logit}(\pi'_{ik}) &= \beta_{0k} + \beta_1 C_k + \beta_{2k} X_{ik}^{\text{Purged}} + \beta_3 X_{ik}^{\text{Purged}} C_k + \phi_{1k} Z_{ik}
 \end{aligned}
 \tag{10.3}$$

Part II

A Contextual Theory of Rational Retrospective Economic Voting: Competency Signals

Part II begins by developing a rational model of retrospective voting in which voters extract signals about the competency of incumbent politicians from movements in the retrospective economy. Given this model of individual level economic voting we can explore how the quality of the competency signal that voters extract from the observed economy may vary across different contexts and hence induce different amounts of economic voting. Chapter 5 develops a selection model of the economic vote that first describes the competency signals that voters extract from macro-economic outcomes. We then explain how the quality of these competency signals condition the economic vote. Contexts in which the quality of these competency signals are high should generate more economic voting.

We argue in this chapter that political contexts can be distinguished in terms of the number of elected versus non-elected decision makers that are shaping political and economic policymaking. And our theory suggests that the ratio of these elected to non-elected decision makers can condition the magnitude of these competency signals. As the ratio of decisions by non-elected decision makers to those by elected decision makers rises the overall competency signal will decline and vice-a-versa.

Chapter 6 provides empirical evidence that voters indeed are attentive to fluctuations in the macro-economy and are equipped to undertake the competency signal extraction task implied by our theory. The chapter also presents evidence that these competency signals vary over time and cross-nationally in an informative manner and that these signals condition the economic vote consistent with our theory. Chapter 7 presents empirical evidence suggesting that this ratio of non-elected to elected decisions impact the economic vote consistent with our theory.

Chapter 5

Competency Signals and Rational Retrospective Economic Voting

Introduction

In Chapter 3 we highlighted the extent to which economic voting varies from one context to the next. For example, Figure 3.2 from Chapter 3 suggested that in 1985 economic evaluations in The Netherlands had a rather significant impact on vote choice but 10 years later economic assessments had no effect on the Dutch vote decision. In this chapter we begin building our explanation for this kind of variation in economic voting. As we discussed in Chapter 4, modeling vote choice is equivalent to properly specifying the voter's value function for each alternative in her choice set (i.e., each party or candidate). The question then becomes: What should these value functions that produce economic voting look like? And how will their parameters vary in different political and economic contexts to produce variation in the economic vote?

The Competency Model with Two-Party Competition for a Unified Executive

We begin to answer these questions with a theoretical model of economic voting in which the previous economy enters the voter's value function for parties modified by a competency signal. Alesina and Rosenthal (1995) and Persson and Tabellini (1990) have suggested two closely related formal models in which rational, expected utility maximizing voters maximize their utility by conditioning their electoral choice on the retrospective performance of the economy. The models differ in their assumptions about the information available to voters when they form their expectations about what economic policies politicians will pursue, but this difference only affects implications about the behavior of politicians, not about voters. Specifically, the Persson and Tabellini model implies that politicians will create political business cycles by distorting the pre-electoral economy for political purposes, while under the (arguably more realistic) information conditions in Alesina and Rosenthal's model, this does not occur. This result is important because it shows that rational retrospective economic voting is not dependent on the existence of pre-electoral business cycles or on the existence within the model of an "adverse selection" problem in which politicians have incentives to try to hide their type from voters by manipulating economic policy and outcomes. Retrospective economic voting can occur in such models, but it also occurs in selection models in which politicians have no ability (or incentive) to deceive voters about their competence as economic managers. This result simplifies our analysis of context in these models because it allows us to focus our generalization

of the model on the voter's *decision theoretic* problem of choice under uncertainty, rather than on the technically more complicated problem of a strategic interaction between voters and politicians.

We begin, then, by examining Alesina and Rosenthal's model of rational retrospective voting as it was originally conceived and then suggest an extension and reinterpretation of one key component of the model that makes it easier to use the model to generate predictions about the impact of specific changes in political and economic context on the extent of economic voting.¹¹⁰ The model assumes that two parties compete for unified control of the executive and that this executive can choose an inflation rate directly (their economic policy) that determines growth in the following expectations augmented Phillips curve:

$$(5.1) \quad y_{it} = \bar{y} + \pi_{it} - \pi_{it}^e + \eta_{it}$$

Where y_{it} is the rate of economic growth in period t , under incumbent party i ; π_{it} is the inflation rate, the level of which is chosen directly by the incumbent party; π_{it}^e is the rate of inflation that voters expect the incumbent to choose, $\bar{y}$ is the natural rate of growth, and η_{it} is a random shock to the economy.

The economic shock consists of two parts as follows:

$$(5.2) \quad \eta_{it} = \varepsilon_{it} + \xi_t$$

One part, ε_{it} , is simply an increment to growth that depends on the identity of the incumbent but not on her economic policy (which is captured in π_{it}). This increment to economic performance is meant to capture the economic impact of the incumbent administration's managerial competence. More specifically, this shock includes any unobserved economic impact of the behavior of the incumbent administration that is not constant over time or administration.¹¹² We refer to this impact as a "competency shock".

¹¹⁰ We generally follow the notational conventions used in Alesina and Roubini (1997).

¹¹² If voters observe the impact of this behavior it cannot be part of the shock but is part of the observed policy represented by π_{it} . Likewise, any unobserved impact of behavior on growth that is constant is subsumed in the natural rate of economic growth.

The other part of the total shock to economic growth, ξ_t , though also unobserved and not constant over time, does not depend on the identity of the administration. We refer to it as an “exogenous” shock or sometimes as a “non-political” shock.

Voters cannot observe competence shocks or non-political shocks directly but can glean information about incumbent competence from the fact that the observed economy is partially dependent on it. Of course, to be useful in forecasting the future economy, the level of competence inferred from observed economic performance must provide some guide to the incumbent’s future level of competence. Consequently, we assume that competence is persistent over time in the following way:¹¹³

$$(5.3) \quad \varepsilon_{it} = \mu_{it} + \mu_{it-1}$$

Thus, current competence is just a first-order moving average from a sequence of competency shocks. We assume that each of these competency shocks is drawn from identical distributions, with mean zero and finite variance σ_μ^2 . Likewise, we assume that the non-political shocks, ξ_t , are drawn from identical distributions each with zero mean and finite variance σ_ξ^2 . We assume voters know the expected values and variances of these distributions.

All voters in the model are identical and care about achieving the highest possible economic growth and lowest possible inflation in the next period. Specifically, we will write the utility of a typical voter in period $t+1$ as a function of which party is elected, what policy that party pursues, and what the resulting level of economic growth will be. Given some governing party, i , that pursues a particular economic policy (a choice of π_{it+1}), the voter’s utility in period $t+1$ is:¹¹⁴

$$(5.4) \quad u(\pi_{it+1}, y_{it+1}) = -\frac{1}{2} \pi_{it+1}^2 + b y_{it+1}, \quad b > 0$$

¹¹³ One can also discount the impact of past competence as long as it is at least partially persistent.

¹¹⁴ It is possible to state the voter’s preferences more generally to include the whole sequence of time

discounted future periods, e.g. $U = E \left\{ \sum_{t=0}^{\infty} \delta_t u(\pi_{j,t}, y_{j,t}) \right\}; 0 < \delta < 1$. However, the usual assumption

restricts the discount factors to make this equivalent to a voter who looks into the future only so far as the next period of incumbency. For our purposes, then, the simpler formulation given in the text is adequate.

where b indexes the voter's preference for growth relative to inflation. The particular functional form of utility for inflation and growth is quite flexible: as we will see any choice that has the voter's utility increasing in growth will produce the same substantive implications for rational economic voting given the other assumptions in the model. The one provided above is a common formulation in the literature and so was chosen for its familiarity. Since utility is increasing in y and is maximized, for a given y , when inflation equals zero, this expression says that voters prefer more growth and price stability and that they would be willing to trade price increases for growth at a rate governed by the size of b .

Since a voter's future utility will depend on choices made today, he must forecast the likely economic future under different possible incumbents. Our assumption is that these expectations are formed rationally based on all the information available at the time of the election. Politicians in the model all care only about being in office and understand that voters will vote to maximize their expected utility.¹¹⁵

Since voters form expectations about inflation and growth rationally they know that incumbent politicians will pick the level of inflation (and correspondingly growth) that will maximize the incumbent's expected utility. Voters are assumed to know current inflation and are never surprised by the government's inflation policy. Consequently, politicians have nothing to gain from doing anything but choosing the voter's optimal inflation rate (zero). Thus, in this simple version of the model, all politicians, no matter how competent, will choose the same economic policy and differences in growth associated with different politicians can only result from differences in their types (which are exogenous to the model). Clearly, then, the decisions of the politicians play no real role in the model and so it is equivalent to a reduced form, decision-theoretic version of the usual formulation that has been used to explore political business cycles. Since our focus is on the decision of voters given the observed economy and not on the decisions of politicians about policy, this seems an appropriate simplification.

With this, the growth rate from Equation (5.1) is just the natural rate plus any shock. Further, voters can actually observe the total shock, since they can calculate it via Equation (5.1). However, they cannot use that equation to parse out how much of the observed shock is due to the incumbent's competence, since they do not observe the two shock terms separately, but only overall growth.

¹¹⁵ We could allow politicians to differ in their policy preferences, for example leftist politicians might prefer a non-zero inflation rate. As we will see, however, economic voting in the model does not in any way depend on the policy choices of politicians and so we ignore this complication.

The voters in the model form their expectations about the competence of the incumbent rationally and because of the moving average structure of the competence shock in Equation (5.3), growth rates at time t that differ from $\bar{y}$ will provide voters with information regarding the competence of an incumbent re-elected for period $t + 1$. This follows from taking expectations in Equation (5.3) (recall that the unconditional expectation of μ_{it+1} is zero).

$$(5.5) \quad E[\varepsilon_{it+1}] = E[\mu_{it+1}] + E[\mu_{it} | y_{it}] = E[\mu_{it} | y_{it}]$$

Voters form their expectations about the competence of an incumbent re-elected in period $t+1$ by evaluating μ_{it} or, more precisely, the noisy signal provided by y_{it} . A key assumption of Alesina and Rosenthal's (1995) model is that voters learn the value of competency with a one-period delay – that is, in period t they know μ_{it-1} but not μ_{it} . Hence voters base their forecast of the economic competence of the incumbent on both y_{it} and μ_{it-1} . Specifically, in the current period voters know the competency of the incumbent in the last period, the natural rate of growth, the current realization of growth, and the current economic shock (which is composed of some unknown mix of the current competence of the incumbent and the non-political shock). Growth in the current period is thus:

$$(5.6) \quad \begin{aligned} y_{it} &= \bar{y} + \eta_{it} \\ &= \bar{y} + \mu_{it} + \mu_{it-1} + \xi_t \end{aligned}$$

Rearranging this gives:

$$(5.7) \quad \mu_{it} + \xi_t = y_{it} - \bar{y} - \mu_{it-1}$$

Where everything on the right hand side of this equality is observed and so the sum of the terms on the left is also observed, though not the individual components. Denote the sum on the left hand side as $k_{it} = \mu_{it} + \xi_t$. Since k_{it} is observed, the voter can compute her expectation about μ_{it} given k_{it} (i.e., her expectation about the incumbent's current level of competence, given the observed level of growth and the incumbent's competence in the last period). To calculate this conditional expectation, we need to know the distribution of both k_{it} and μ_{it} . k_{it} is the sum of two

normally distributed random variables, both with zero means and variances σ_μ^2 and σ_ξ^2 , respectively. The distribution of k_{it} is thus:

$$(5.8) \quad k_{it} = (\mu_{it} + \xi_t) \sim N(0, \sigma_\mu^2 + \sigma_\xi^2)$$

Given that both, k_{it} and μ_{it} are distributed normally, their joint distribution is bivariate normal and the optimal forecast of μ_{it} given k_{it} is just the conditional expectation, which is computed from the appropriate conditional distribution of the bivariate normal. Using standard results, this conditional expectation is (Greene 2003):¹¹⁶

$$(5.9) \quad \begin{aligned} E[\mu_{it} | k_{it}] &= E[\mu_{it}] + \frac{\sigma_{\mu,k}}{\sigma_k^2} (y_{it} - \bar{y} - \mu_{it-1}) - E[k_{it}] \\ &= \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) \end{aligned}$$

Since $E[\mu_{it} | k_{it}] = E[\mu_{it} | y_{it}]$, this expression is the rational voter's assessment of the current competence of the incumbent given the observed economy.¹¹⁷ Further, from Equation (5.5), we have $E[\mu_{it} | y_{it}] = E[\varepsilon_{it+1}]$, so we now have what we need to explore the implications of the model for economic voting by comparing the voter's expected utility for voting for the incumbent in this model to her expected utility for the challenger.

Since there are no incentives to vote strategically in a two party contest, voters will vote for the party that they expect to deliver the most utility in the next period. So we can write their expected utility for voting for incumbent party i as equal to the expected utility voters will accrue in the next period if party i is in office.

¹¹⁶ In general, $E[x | y] = \rho_{xy} \frac{y - E[y]}{\sigma_y} \sigma_x + E[x]$, where ρ_{xy} is the correlation between x and y .

¹¹⁷ $E[\mu_{it} | y_{it}] = \frac{\sigma_{\mu,y}}{\sigma_\mu^2 + \sigma_\xi^2} (y_{it} - y - \mu_{it-1})$ by applying the same signal extraction solution as above. Further, it is easy to show that $\sigma_{\mu,y} = \sigma_{\mu,k}$, so the claim in the text follows.

$$\begin{aligned}
E[u_{t+1} | v_i] &= E[u(\pi_{it+1}, y_{it+1})] \\
&= \frac{1}{2} E[\pi_{it+1}^2] + bE[y_{it+1}] \\
&= 0 + b(\bar{y} + E[\eta_{it+1}])_i \\
(5.10) \quad &= 0 + b(\bar{y} + 0 + E[\varepsilon_{it+1}]) \\
&= b\left(\bar{y} + \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}(y_{it} - \bar{y} - \mu_{it-1})\right) \\
&= b\bar{y} + b\left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}\right)(y_{it} - \bar{y} - \mu_{it-1})
\end{aligned}$$

Lacking any information about the challenger's level of competence, the voter's expected utility for voting for the challenger, k , is just:¹¹⁸

$$\begin{aligned}
E[u_{t+1} | v_k] &= E[u(\pi_{kt+1}, y_{kt+1})] \\
&= \frac{1}{2} E[\pi_{kt+1}^2] + bE[y_{kt+1}] \\
(5.11) \quad &= 0 + b(\bar{y} + E[\eta_{kt+1}]) \\
&= b(\bar{y} + E[\xi_{t+1}] + E[\varepsilon_{kt+1}]) \\
&= b\bar{y}
\end{aligned}$$

Thus, the voter will vote for the incumbent when the expected utility in Equation (5.10) is larger than that in Equation (5.11). The difference is:

$$\begin{aligned}
E[u_{t+1} | v_i] - E[u_{t+1} | v_k] &= b\bar{y} + b\left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}\right)(y_{it} - \bar{y} - \mu_{it-1})_i - b\bar{y} \\
(5.12) \quad &= b\left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}\right)(y_{it} - \bar{y} - \mu_{it-1})
\end{aligned}$$

This result makes it clear when voters can and cannot extract information from fluctuations in the previous economy in order to access the current competence of an incumbent

¹¹⁸ Alesina and Rosenthal assume that the expected competence of the challenger is always zero. We discuss what happens when we change this assumption (and why we might need to) below.

and cast an economic vote. The term $y_{it} - \bar{y} - \mu_{it-1}$ is simply observed economic performance less the parts of economic growth whose sources are known to the voter. The term captures what the incumbent has “done for the voter lately” (i.e., how the current period differs from the natural level of growth, discounted by the impact of the incumbent’s known level of competence in the previous period). We interpret the coefficient on this term, i.e. $\frac{\sigma_{\mu}^2}{\sigma_{\mu}^2 + \sigma_{\xi}^2}$, as the “competency signal” that controls how much information about the competence of incumbents voters can extract from observed movements in the economy. This competency signal will always be positive and will approach one as the variance in the random (non-political) shocks to the economy, σ_{ξ}^2 , goes to zero. In that case, the voter knows that growth above or below the natural rate is completely due to competency shocks – consequently, deviations from the natural rate of growth can be completely attributed to the competence of the administration. More generally, if σ_{μ}^2 , the variance of the competence term μ_{it} , is large relative to the variance in the non-political component of growth, σ_{ξ}^2 , then changes in the economy will provide a strong signal about the competency of the incumbent and the voter will weight the retrospective economy more heavily in her utility function. Alternatively, growth that is above or below the natural rate is a poor signal of the incumbent’s competence if observed growth is more likely to result from non-political shocks than from competency shocks – i.e. if σ_{ξ}^2 is high relative to σ_{μ}^2 .

We can conclude, then, that economic voting in its traditional form obtains in this model – voters will vote against the incumbent when economic performance is sufficiently weak and for the incumbent when it is sufficiently strong. In addition, the economic vote is weighted by a competency signal that captures the extent to which shocks to the economy are due to the competence of governments.

Competency Shocks, Economic Policy, and Political and Economic Context

Scheve (2004) and others used this formulation of the competency signal to speculate about the way that economic contexts like the extent of trade openness might impact economic voting. For example, Scheve (2004) suggests that increased openness will decrease the size of the variance of exogenous shocks and so increase the extent of economic voting. We could certainly proceed similarly – i.e., by speculating about how different political and economic contexts impact the two variance terms in the competency ratio. However, the problem with this strategy is that the theory in its current form provides relatively little guidance about the

substantive meaning of these variance terms and so these sorts of speculations are relatively unconstrained. This is especially true for the concept of the competency shock. What exactly is the distribution of competency shocks and what would account for differences in its variance across contexts? Certainly we could think of the competency distribution as the pool of political talent available in a country and suppose that this pool differs from country to country in a way that might account for different competency signals (though differences like this seem unlikely across the western democracies). Alternatively, we could think of the competence distribution as describing the variation within a country of bureaucratic skill, political corruption, or the efficiency of markets that translate policy to outcomes. The point here is that one is relatively free to posit a substantive interpretation for what the competency distribution means and then to suggest what kinds of political and economic contexts would impact the variance of this distribution (and so economic voting).

In addition to ambiguities in the substantive meaning of the variance of the competence distribution, one must also remember that there are two different variance terms in the competency ratio. Thus, it is important that any speculation about the impact of some particular contextual variable on the size of the competency signal address how that variable impacts both the variance of competence shocks and the variance of exogenous shocks. In the small literature that has taken this approach, however, it is rare to find discussion of both kinds of shocks.

Given this, in this section we suggest an extension and reinterpretation of Alesina and Rosenthal's model that, we believe, gives the model a clearer substantive interpretation and so allows for more substantively grounded speculations about how different contexts are likely to alter the competence signal and thus economic voting. Further, our extension leads us to identify ways in which variables like trade dependence and the size of government can impact the competency signal that have not been emphasized in the previous literature. For example, while previous theoretical speculation about the impact of expanded government on economic voting has emphasized the larger number of functions that might well be associated with such an increase, our theoretical formulation makes it plain that other aspects of expanded government – like the further political insulation of entrenched bureaucracies from electoral control – may also play a role in the ability of voters to extract a clear signal of incumbent competence from observed changes in the economy.

Our modification of the original Alesina and Rosenthal (1995) formulation begins by distinguishing between two types of decision makers, which we will call “electorally-dependent decision makers” (EDDs) and “non-electorally dependent decision makers” (NEDDs). The first of these labels (EDD) is just shorthand for referring to the elected officials that make up the

national government and the bureaucracy that is responsible to them. The second label (NEDD) refers to everyone else whose decisions might impact the economy including individuals, firms, interest groups, non-electorally dependent (entrenched) bureaucrats, foreign leaders, the WTO, and many more. The reason this distinction is important is that we assume that competency shocks are only associated with the decisions of the EDDs (electorally-dependent decision makers), while the exogenous shocks are associated with the decisions of everyone else. Note that included amongst the NEDDs are government officials from sub-national levels of government that contribute to exogenous shocks in the macro-economy. Hence, federalist systems that multiply the number of non-national government entities that contribute to exogenous shocks would have a lower overall competency signal and hence less economic voting.¹¹⁹

In the model developed above we assumed a single decision – the setting of interest rates, for example. In fact, EDDs make many economically consequential decisions; as do NEDDs. Just as in the model above, each of these decisions has some systematic effect on the economy and some random effect, and the competence and exogenous shocks discussed above are nothing but these many random shocks, summed over the sets of decisions made by elected and non-elected decision makers respectively.

Formally, let the number of decisions made by EDDs be α and the number made by NEDDs be β and assume, for notational simplicity, that there is a single elected decision maker and a single non-elected decision maker making all of these decisions.¹²⁰ With this, we can write the growth equation from above as:

$$(5.13) \quad y_{it} = \bar{y} + \sum_{l=1}^{\alpha} \omega_{ilt} + \sum_{l=1}^{\beta} \psi_{lt}$$

¹¹⁹ The effect of federalism on the magnitude of the economic vote in national elections is typically explained as the result of an increased level of confusion on the part of voters (Anderson 2006; Cutler 2004). In our theory voters are not confused but rather are fully informed and simply discount the national incumbent's competency signal when there are a larger number of "non-elected" actors affecting shocks to the macro-economy. For similar reasons we would expect to see relatively low levels of economic voting in state or provincial elections where the importance of "non-elected" actors (i.e., the federal government) on economic outcomes is considerable (Stein 1990). We are unable to actually test this hypothesized relationship with our sample of countries because they have very little variation on the widely accepted measures of federalism. For example, on the Franzese (2002) measure of federalism, twelve of the seventeen countries in our sample (for which there is a federalism measure) have the lowest score of 1 leaving only five countries that register as federalist (and one, the U.S., is a very large outlier compared to the other federalist countries).

¹²⁰ This can be extended to represent both the number of decision makers and their volume of decisions although the notation becomes much more complex; the results nevertheless are exactly the same.

where ω_{ilt} is the growth shock associated with the l th decision of i , the EDD. Likewise, ψ_{lt} is the growth shock associated with the l th decision of the NEDD. Any known or systematic impacts of these decisions on the economy are subsumed in the natural rate of growth or are anticipated by voters and so have no effect on growth.¹²¹

We assume that for the EED, the l 'th shock is persistent in the same way as in our earlier discussion (i.e. $\omega_{ilt} = \mu_{ilt} + \mu_{ilt-1}$).¹²² And we assume that μ_{ilt} and ψ_{lt} are independent normally distributed random variables with zero means and variances σ_μ^2 and σ_ψ^2 , respectively.

The growth equation can then be expressed as,

$$(5.14) \quad y_{it} = \bar{y} + \sum_{l=1}^{\alpha} (\mu_{ilt} + \mu_{ilt-1}) + \sum_{l=1}^{\beta} \psi_{lt}$$

Rearranging this gives:

$$(5.15) \quad \sum_{l=1}^{\alpha} \mu_{ilt} + \sum_{l=1}^{\beta} \psi_{lt} = \bar{y} - y_{it} + \sum_{l=1}^{\alpha} \mu_{ilt-1}$$

Where everything on the right hand side of this equation is observed and so the sum of the terms on the left is also observed, though not the individual components. Denote the sum on the left hand side as $k_{it} = \sum_{l=1}^{\alpha} \mu_{ilt} + \sum_{l=1}^{\beta} \psi_{lt}$. Since k_{it} is observed, the voter can compute her

¹²¹ Specifically, if $\sum_{l=1}^{\alpha} (\phi_{ilt} + \omega_{ilt})$ is a generic term capturing the total impact of all decisions relevant to the economy that are made by the elected decision maker, rather than simply an inflation choice, we can think of $\pi_{it} = \sum_{l=1}^{\alpha} \phi_{ilt}$ as the “policy” part of this impact over which voters have rational expectations. Thus, as before, π_{it} is fully anticipated by the rational electorate (so voters anticipate the many choices the decision maker will make) and can have no real effect on growth (i.e., given rational expectations and the information assumptions from the earlier model, π_{it} will equal π_{it}^e). Consequently, the decision maker will always choose the mix of policies that are most preferred by the voter.

¹²² This persistence means we need to think of the l th decision at time t and the l th decision at time $t+1$ as being members of the same “category” of decisions so that the shock to decision l at time $t-1$ tells us something about the shock to decision l at time t . One can simplify matters considerably by assuming that the shocks for all decisions made in a single period by the same decisionmaker are the same. In that case, we would think of the shock as a sort of characteristic of the decision maker so that all his or her decisions about the economy “worked” a little better or a little worse. An equivalent version of the model would allow for many decisionmakers each making many decisions. However, this kind of generalization has no impact on our conclusions, though it does change the substantive interpretation of the persistence assumption as noted above.

expectation about $\sum_{l=1}^{\alpha} \mu_{ilt}$ given k_{it} (i.e., her expectation about the EDD's current overall competence shock, given the observed level of growth and the decision maker's overall competence shock in the last period). To calculate this conditional expectation, we need to know the distribution of both k_{it} and $\sum_{l=1}^{\alpha} \mu_{ilt}$. $\sum_{l=1}^{\alpha} \mu_{ilt}$ is the sum of α normally distributed random variables each with zero mean and variance σ_{μ}^2 , so $\sum_{l=1}^{\alpha} \mu_{ilt} \sim N(0, \sigma_{\mu}^2 \alpha)$. Likewise, $\sum_{l=1}^{\beta} \psi_{lt} \sim N(0, \sigma_{\psi}^2 \beta)$. Thus, k_{it} is the sum of two normally distributed random variables, both with zero means and variances $\sigma_{\mu}^2 \alpha$ and $\sigma_{\psi}^2 \beta$, respectively. The distribution of k_{it} is thus:

$$(5.16) \quad k_{it} = \sum_{l=1}^{\alpha} \mu_{ilt} + \sum_{l=1}^{\beta} \psi_{lt} \sim N(0, \sigma_{\mu}^2 \alpha + \sigma_{\psi}^2 \beta)$$

Given that both, k_{it} and $\sum_{l=1}^{\alpha} \mu_{ilt}$ are distributed normally, their joint distribution is bivariate normal and the optimal forecast of $\sum_{l=1}^{\alpha} \mu_{ilt}$ given k_{it} is just the conditional expectation, which is computed from the appropriate conditional distribution of the bivariate normal. Using standard results (Greene 2003), this conditional expectation is:

$$(5.17) \quad \begin{aligned} E\left[\sum_{l=1}^{\alpha} \mu_{ilt} \mid k_{it}\right] &= E\left[\sum_{l=1}^{\alpha} \mu_{ilt}\right] + \frac{\sigma_{\mu,k}}{\sigma_k^2} \left(y_{it} - \bar{y} - \sum_{l=1}^{\alpha} \mu_{ilt-1}\right) - E[k_{it}] \\ &= \left(\frac{\alpha \sigma_{\mu}^2}{\alpha \sigma_{\mu}^2 + \beta \sigma_{\psi}^2}\right) \left(y_{it} - \bar{y} - \sum_{l=1}^{\alpha} \mu_{ilt-1}\right) \end{aligned}$$

The numerator of the competence signal is now the variance of the overall competence shock, which is the product of the variance of the distribution of competence shocks associated with a single decision and the number of decisions made by the EDD ($\alpha \sigma_{\mu}^2$). This is important because it implies that expanding the number of economically consequential choices that the electorally-dependent decision maker makes will increase this product. More substantively, this variance should be larger in countries in which EDDs make more of the economic decisions that determine the country's growth path.

The principal advantage of this formulation of the model is that it alleviates the need to speculate about how differences across cases in particular political and economic institutions lead

to differences in the variance of the distribution of individual competence shocks (σ_μ^2) or individual exogenous shocks (σ_ψ^2) across cases. Instead, since these are the variances governing the shocks associated with individual decisions, each of which is likely to have only a very small effect on growth, we can simply assume that these variances are both small (implying that the maximum impacts of any individual decision on growth is small) and constant across cases. So the positive or negative shock associated with any one decision in any one place and time may range in size similarly to any other decision. What differs across context, however, is the number of decisions over which we take the sum. Thus, the task in connecting political and economic institutions to the strength of the competence signal is shifted in our revised model from speculation about the impact of these institutions on the variances of the distributions of competence and exogenous shocks to speculation about how they impact the number of economically consequential decisions subject to electoral control.

Turning back to Equation (5.17), since $E\left[\sum_{l=1}^{\alpha} \mu_{ilt} \mid k_{it}\right] = E\left[\sum_{l=1}^{\alpha} \mu_{ilt} \mid y_{it}\right]$, this expression is the rational voter's assessment of the current competence of the incumbent given the observed economy.¹²³ Further, from Equation (5.5), we have $E\left[\sum_{l=1}^{\alpha} \mu_{ilt} \mid y_{it}\right] = E\left[\sum_{l=1}^{\alpha} \omega_{ilt+1}\right]$, so we now have what we need to explore the implications of the model for economic voting by comparing the voter's expected utility for voting for the incumbent in this model to her expected utility for the challenger.

The voter's expected utility for voting for incumbent party i is just the expected utility the voter will accrue in the next period if party i is in office.¹²⁴

¹²³ $E[\mu_{it} \mid y_{it}] = \frac{\sigma_{\mu,y}}{\sigma_\mu^2 + \sigma_\xi^2} (y_{it} - y - \mu_{it-1})$ by applying the same signal extraction solution as above. Further, it is easy to show that $\sigma_{\mu,y} = \sigma_{\mu,k}$, so the claim in the text follows.

¹²⁴ All voters in the model are identical and care about achieving the highest possible economic growth and lowest possible inflation in the next period. Specifically, we will write the utility of a typical voter in period $t+1$ as a function of which party is elected, what policy that party pursues, and what the resulting level of economic growth will be. Given some governing party, i , that pursues a particular economic policy (a choice of π_{it+1}), the voter's utility in period $t+1$ is in part:

$$u(\pi_{it+1}, y_{it+1}) = -\frac{1}{2}u(\pi_{it+1}^2) + by_{it+1}, \quad b > 0$$

$$\begin{aligned}
(5.18) \quad E \left[\sum_{l=1}^{\alpha} \mu_{ilt+1} \mid v_i \right] &= E \left[u(\pi_{it+1}, y_{it+1}) \right] \\
&= \frac{1}{2} E[\pi_{it+1}^2] + b E[y_{it+1}] \\
&= 0 + b(\bar{y} + E[\eta_{it+1}])_i \\
&= 0 + b \left(\bar{y} + 0 + E \left[\sum_{l=1}^{\alpha} \omega_{ilt+1} \right] \right) \\
&= b \left(\bar{y} + \left(\frac{\alpha \sigma_{\mu}^2}{\alpha \sigma_{\mu}^2 + \beta \sigma_{\psi}^2} \right) \left(y_{it} - \bar{y} - \sum_{l=1}^{\alpha} \mu_{ilt-1} \right) \right) \\
&= b\bar{y} + b \left(\frac{\alpha \sigma_{\mu}^2}{\alpha \sigma_{\mu}^2 + \beta \sigma_{\psi}^2} \right) \left(y_{it} - \bar{y} - \sum_{l=1}^{\alpha} \mu_{ilt-1} \right)
\end{aligned}$$

Lacking any information about the challenger's level of competence, the voter's expected utility for voting for the challenger, k , is just:¹²⁵

$$\begin{aligned}
(5.19) \quad E \left[\sum_{l=1}^{\alpha} \mu_{klt+1} \mid v_k \right] &= E \left[u(\pi_{kt+1}, y_{kt+1}) \right] \\
&= \frac{1}{2} E[\pi_{kt+1}^2] + b E[y_{kt+1}] \\
&= 0 + b(\bar{y} + E[\eta_{kt+1}]) \\
&= b \left(\bar{y} + E \left[\sum_{l=1}^{\beta} \psi_{lt} \right] + E \left[\sum_{l=1}^{\alpha} \omega_{klt+1} \right] \right) \\
&= b\bar{y}
\end{aligned}$$

Thus, the voter is more likely to vote for the incumbent when the expected utility in Equation (5.18) is larger than that in Equation (5.19). The difference is:

where in Alesina and Rosenthal's formulation π is just inflation and b indexes the voter's preference for growth relative to inflation. However, in our formulation π is a package of policies and b indexes the voter's preference for growth relative to her (non-growth related) utility for the overall policy package offered by the government. The particular functional form of the voter's utility is quite flexible: any choice that has the voter's utility increasing in growth will produce the same substantive implications for rational economic voting given the other assumptions in the model. The one provided above is a common formulation in the literature and so was chosen for its familiarity. In our application, it requires that we normalize the utility of the policy bundle so that the voters preferred policy is achieved at a level of zero, but this can be done without loss of generality.

¹²⁵ Alesina and Rosenthal (1995) assume that the expected competence of the challenger is always zero. We discuss what happens when we change this assumption (and why we might need to) in Chapter 8.

(5.20)

$$\begin{aligned} E\left[\sum_{l=1}^{\alpha}\mu_{ilt+1} \mid v_i\right] - E\left[\sum_{l=1}^{\alpha}\mu_{klt+1} \mid v_k\right] &= b\bar{y} + b\left(\bar{y} + E\left[\sum_{l=1}^{\beta}\psi_{lt}\right] + E\left[\sum_{l=1}^{\alpha}\omega_{klt+1}\right]\right) - b\bar{y} \\ &= b\left(\frac{\alpha\sigma_{\mu}^2}{\alpha\sigma_{\mu}^2 + \beta\sigma_{\psi}^2}\right)\left(y_{it} - \bar{y} - \sum_{l=1}^{\alpha}\mu_{ilt-1}\right) \end{aligned}$$

This result makes it clear when voters can and cannot extract information from fluctuations in the previous economy in order to access the current competence of an incumbent and cast an economic vote. The term $y_{it} - \bar{y} - \sum_{l=1}^{\alpha}\mu_{ilt-1}$ is simply observed economic performance less the parts of economic growth whose sources are known to the voter. The term captures how the current period differs from the natural level of growth, discounted by the impact of the incumbent's known level of competence in the previous period. We can interpret the coefficient on this term, i.e. $\left(\frac{\alpha\sigma_{\mu}^2}{\alpha\sigma_{\mu}^2 + \beta\sigma_{\psi}^2}\right)$, as the “competency signal” that controls how much information about the competence of incumbents voters can extract from observed movements in the economy. This competency signal will always be positive and will approach one as the variance in the random (non-political) shocks to the economy, $\beta\sigma_{\psi}^2$, goes to zero. In that case, the voter knows that growth above or below the natural rate is completely due to competency shocks – consequently, deviations from the natural rate of growth will perfectly identify competent and incompetent administrations. More generally, if $\alpha\sigma_{\mu}^2$, the variation in the competence term μ_{it} , is large relative to variation in the non-political component of growth, $\beta\sigma_{\psi}^2$, then changes in the economy will provide a strong signal about the competency of the incumbent and the voter will weight the retrospective economy more heavily in her utility function. Alternatively, growth that is above or below the natural rate is a poor signal of the incumbent's competence if $\beta\sigma_{\psi}^2$ is high relative to $\alpha\sigma_{\mu}^2$.

Given our assumption that σ_{ψ}^2 and σ_{μ}^2 are constant over all contexts, the impact of political and economic institutions on the strength of the competence signal (and ultimately on economic voting) must come through differences in the α and β terms in the above equations. The numerator of the competency signal is the variance of the overall competence shock ($\alpha\sigma_{\mu}^2$), which implies that expanding the number of economically consequential choices that the EDDs

makes will increase the value of this term. More substantively, the variance in the overall competence shock should be larger in countries in which EDDs make more of the economic decisions that determine the country's growth path.

When comparing the overall competency signal in different contexts the ratio of non-electorally-dependent to electorally-dependent decisions in each context can shape the relative size of their overall competency signals. Consider the case where the competency signal is larger in one context than another:

$$(5.21) \quad \frac{\alpha\sigma_{\mu}^2}{\alpha\sigma_{\mu}^2 + \beta\sigma_{\psi}^2} > \frac{\alpha'\sigma_{\mu}^2}{\alpha'\sigma_{\mu}^2 + \beta'\sigma_{\psi}^2}$$

$$\frac{\beta\sigma_{\psi}}{\alpha\sigma_{\mu}^2} < \frac{\beta'\sigma_{\psi}}{\alpha'\sigma_{\mu}^2}$$

$$\frac{\beta}{\alpha} < \frac{\beta'}{\alpha'}$$

where α and β are from a large competency signal context and α' and β' are from a small signal context. The resulting inequality in Equation (5.21) suggests that the ratio of NEDDs to EDDs in the large signal context must be smaller than this ratio in the smaller signal context. The relative impact of political and economic contexts on the overall competency signal is the magnitudes of these ratios.

Contextual effects that will have the most unambiguous impact on this ratio are ones that simultaneously increase α and decrease β or vice-a-versa. These are cases where the rising number of NEDDs affecting variations in shocks to the macro-economy is displacing the number of EDDs affecting these shocks. This will generate the inequality in Equation (5.21) and an unambiguous predicted effect. Contexts where the number of NEDDs affecting macro-economic shocks is high will tend to have fewer EDDs impacting macro-economic shocks – hence an overall signal that is smaller. An example here might be the adoption of monetary unions or common tariff regimes which typically result in a smaller number of EDDs affecting macro-economic shocks (thereby reducing the magnitude of the competency shock term) and a larger number of NEDDs having an impact (thereby increasing the exogenous shock term). The opposite case that results in a strong competency signal is contexts with relatively larger numbers of EDDs and relatively fewer NEDDs. Large states with unitary constitutions and relatively closed economies compared to small states with federal constitutions and that have extensive exposure to international trade might be an example of this case.

There are also contextual differences that only influence the numerator or denominator in these terms. Take the example of holding the numerator constant by setting $\beta = \beta'$ – two such contexts would not differ in terms of the number of NEDDs affecting macro-economic shocks. The inequality in Equation (5.21) then simply reduces to $\alpha > \alpha'$; contexts with a higher number of decisions by EDDs affecting economic shocks will have a higher competency signal. An example here might be presidential versus parliamentary constitutions. One might think of presidential regimes as having larger numbers of elected decisions that affect the macro-economy than is the case with parliamentary contexts (on the grounds that there is a fusion of executive and legislative officials in a parliamentary regime while they are elected separately in a Presidential regime). Nevertheless, all things being equal, parliamentary and presidential constitutional contexts may have similar numbers of NEDDs impacting a nation's macro-economic shocks. Hence, the result of holding β and β' constant and increasing the number of decisions by EDDs (α) will be a higher competency signal.

Contextual differences will not always generate different competency signals. When the two ratios in Equation (5.21) are equal it is less likely that competency signals will vary across contexts. Obviously, this can happen when $\alpha = \alpha'$ and $\beta = \beta'$. These are simply cases in which context does not matter. A somewhat more problematic case is when the two ratios in Equation (5.21) are equal but $\alpha \neq \alpha'$ and $\beta \neq \beta'$. We expect this to be relatively rare since it presumes that contextual differences reduce or increase both the number of EDDs and of NEDDs in similar ratios to each other.

Our theoretical results in this section provide a foundation for understanding how economic and political context shapes the economic vote. As the ratio of NEDDs to EDDs rises the overall competency signal will decline and vice-a-versa which, we argue, should impact the economic vote. Our theory does not indicate which specific contexts are associated with particular ratios of NEDDs to EDDs. Hence, in order to generate predictions regarding the impact of economic and political context on economic voting we need to understand how these contexts impact the ratio of NEDDs to EDDs. Below, we examine a number of such hypotheses by providing arguments about how specific institutional differences across political and economic contexts should impact the ratio of NEDDs to EDDs.

This chapter has developed a selection model of the economic vote that explains contextual variations in economic voting in terms of the quality of the competency signal that voters are able to extract from shocks in macro-economic outcomes. In addition, we have

described how the ratio of electorally dependent to non-electorally dependent decision makers can condition the magnitude of these competency signals. We now turn to examining empirically the nature of economic competency signals, whether voters are attentive to these signals, whether these signals condition economic vote magnitudes and whether economic and political institutions, through their impact on the competency signal, also affect economic voting in a fashion consistent with our theory.

Chapter 6

What Do Voters Know about Economic Variation and its Sources?

Our explanation for contextual variation in economic voting assumes that voters know how much of the variation in random shocks to the economy is due to the competence of the government rather than the influences of non-electorally dependent exogenous factors.¹²⁶ By making this assumption, we were able to derive hypotheses about the way in which variation in the competency signal (and therefore voters' beliefs about this signal) likely impacts economic voting across countries and over time. Further, in elaborating the meaning of the competency signal and thinking about what drives variation in it, we will suggest a broad set of political and economic contexts that might systematically condition the magnitude of the signal (e.g., those that impact the relative number or importance of electorally versus non-electorally dependent decision makers in economic policy making).

This chapter considers whether it is even reasonable to assume that voters have sensible beliefs about the variance in competence shocks to the economy. Or even about the total variation in shocks to the economy. Skepticism abounds in this respect and with good reason. Early efforts by Alesina and Rosenthal (1995) to demonstrate that fluctuations in the U.S. macro-economy might plausibly inform voters about incumbent competency suggested they did not. And even if they did, should we expect the typical voter to pay attention to, and have informed beliefs about, these signals? Given the often-touted ignorance of voters when it comes to political and economic matters, some skepticism is surely warranted (on levels of political information see Delli Carpini and Keeter 1996; on levels of information about the economy see Paldam and Nannestad 2000). Given this skepticism and given the relatively limited research that has been done on these issues in the existing literature, this chapter explores a variety of original data that we have collected that provides insight into these issues.

How do we do this? Well, our theory is quite precise about the information demands made of the voter. This chapter describes these assumptions, lays out their empirical implications and provides some initial empirical tests. First we assume there is a competency signal that is composed of variations in the competence and exogenous shocks described in the previous chapter. Hence, this chapter begins by demonstrating that there is information in economic fluctuations that voters can plausibly use to inform themselves about competency: most importantly that there is variation and that the contextual differences (either over time or across

¹²⁶ That is, they know $\sigma_{\mu}^2 / (\sigma_{\mu}^2 + \sigma_{\epsilon}^2)$.

countries) can inform voters about the two components of the competency signal. Second, media reporting of economic news is an important – although obviously not the only – source of information regarding fluctuations in the macro-economy. Hence we expect that the content of economic news will be consistent with our model of how voters extract competency signals from fluctuations in the macro-economy. Third, voters have sensible beliefs about these fluctuations in economic outcomes that are consistent with the signal extraction efforts we attribute to them. Finally, they use this information to condition the economic vote. We do not pretend to provide the definitive evidence for all four of these claims. Nevertheless, we think the evidence is suggestive enough to invite further inquiry into the sources and nature of voters' information about the economy and its variability.

Competency Signal Extraction Illustrated

Our theory of the economic vote assumes that the voter observes fluctuations in economic outcomes that inform them about incumbent competency. What are informative macro-economic fluctuations? Take, as an example, the typical voter in the first round of the 2002 French Presidential elections. Figure 6.1 (focus on the dots initially) presents the quarterly real GDP growth rates that occurred in the 7 years preceding Jacques Chirac's 2002 re-election bid for the French Presidency. The annual real GDP growth rate was about 2.4 percent over the course of Chirac's 7-year term (represented by the horizontal line drawn in the figure). But these quarterly readings clearly varied – in the early part of his term the growth rate almost dropped to zero, rising quite a bit higher than the mean in 1998 and in the year 2000, and then dropping below one percent in the first quarter of the election year. Our theory presumes that French voters know $\sigma_{\mu}^2 + \sigma_{\xi}^2$; we contend that one source of this knowledge is exactly the kinds of historical variations in economic outcomes that are illustrated in Figure 6.1.

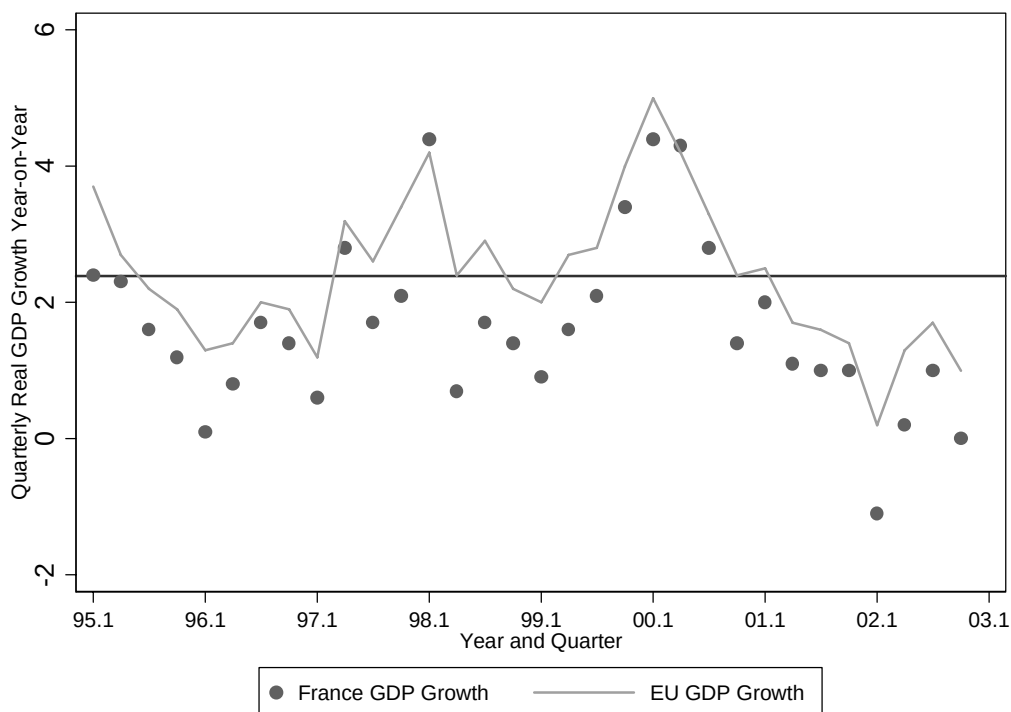


Figure 6.1 French Real GDP Growth during President Chirac's First Term

Besides the total variation in macro-economic shocks our theory assumes voters can separate shocks to the macro-economy that are associated with incumbent competence (σ_{μ}^2) as opposed to non-political factors (σ_{ξ}^2). So, for example, would the average French voter conclude that elected officials are primarily responsible for the spike in GDP growth to almost five percent in the first quarter of 2001 and then its rapid decline to less than one percent in the first quarter of 2002? In our theory they would draw this conclusion, if they believed that σ_{μ}^2 was large relative to the total variation. In this chapter we argue that voters who are attentive to global economic outcomes have information that allows them to distinguish the relative sizes of σ_{μ}^2 and σ_{ξ}^2 .

It is unlikely that all of the variation in Figure 6.1 is the responsibility of elected French officials. Note for example, that the spike in GDP growth in the first quarter of 2000 and the steep subsequent decline in GDP growth conformed very closely to overall fluctuations in EU growth – in fact much of the volatility in French economic growth over the period seems to resemble that of the overall EU economies. Since fluctuations in the two GDP growth rates are so similar, voters are likely to conclude that competency shocks historically have been small relative to exogenous shocks. Consequently they can infer that the unobserved variance of the

competence shocks (σ_{μ}^2) must be larger than that of the exogenous shocks (σ_{ξ}^2) . If these French voters focused exclusively on GDP growth, we would conclude that the overall competency signal is quite small which, all things being equal, would imply low levels of economic voting. And we believe that voters recognize this is the case; moreover we hope to demonstrate in this chapter that some voters attribute significant amounts of this variation to global economic influences that are outside of the control of elected officials. The solid line plot in Figure 6.1 is the quarterly GDP growth figures for the, then, 15 members of the EU and could assist voters in their signal extraction task. In this case our expectation is that French voters will, for the most part, perceive small variation in the competency shock term (σ_{μ}^2) and relatively high variation in the exogenous shock term (σ_{ξ}^2) .

Hence we expect that the typical voter can a) compare fluctuations in the country's macro-economy with that of the overall global economy; and b) draw conclusions about the impact of domestic politicians on economic outcomes based on whether variations in the national economy diverges from the global norm? The next section establishes that national economic outcomes in general fluctuate in a fashion that likely informs voters of incumbent competency.

Contextual Variation in Macro-economic Outcomes

We begin by generalizing the French example from Figure 6.1 and demonstrating that in fact cross-national variations in macro-economic outcomes can be pronounced compared to within-country variations and so may figure in the voter's efforts to distinguish incumbent competence from random shocks. Figure 6.2 makes a convincing case that cross-national variations in macro-economic outcomes can be large (and persistent) and hence they may provide voters with the information necessary to infer the competence of incumbent governments. The first panel in Figure 6.2 presents change in the CPI index for the fourteen countries in our sample over the 1979-2002 period – the solid line in the graph represents the mean change in CPI for this sample of countries (weighted for size of the economy). In the early 1980s we see very significant cross-national variation with countries experiencing changes in their CPI index that were as much as 5 percent lower than the mean and as much as 10 percentage points higher. By the end of our sampled period, the range had narrowed radically to between about -2 and +3 percentage points. This is consistent with findings in the political business cycle literature that the increased credibility of the European Monetary System (EMS) and the reduced number of currency realignments lead to a greater convergence of inflation rates in Europe and a reduction,

if not elimination, of a partisan impact on inflation rates in the OECD countries (Alesina and Roubini 1997, 161).

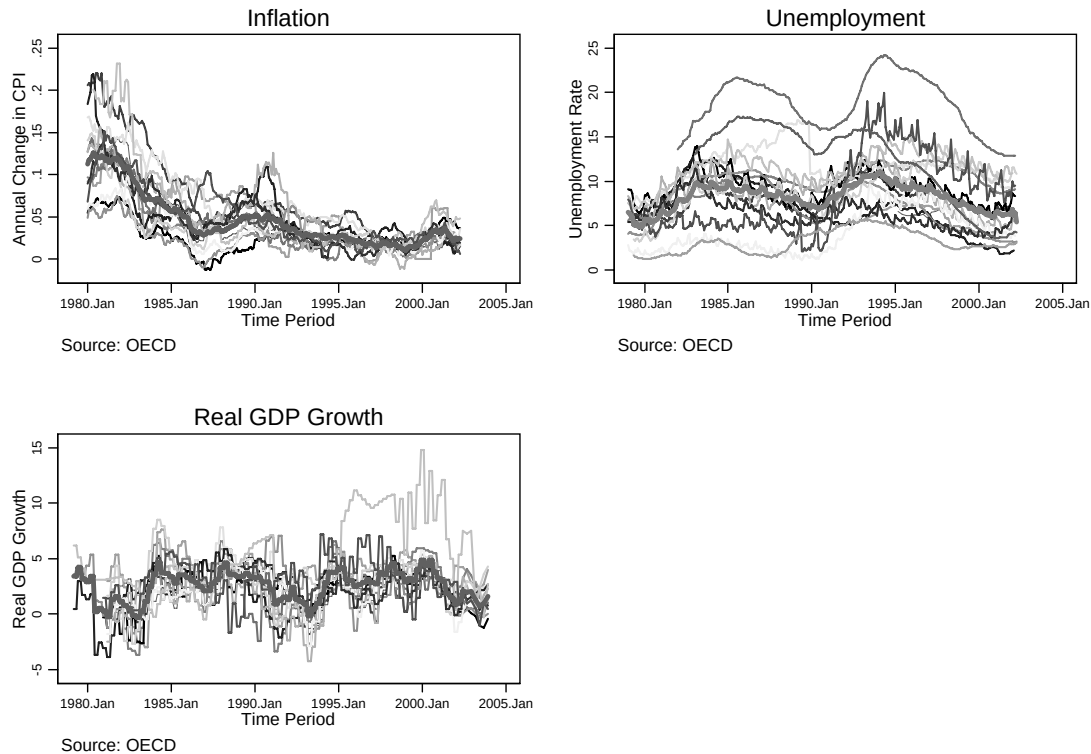


Figure 6.2. Macro-economic Outcomes: 1979-2002

The second panel in Figure 6.2 presents the raw unemployment rates along with the overall sample mean (again, weighted for the size of each economy) that is represented by the bold black line. Unlike inflation where there was a clear dynamic trend converging on a low rate of inflation amongst all the countries, average unemployment for the sample has fluctuated significantly around a mean of about 7 percent but rising to over 10 percent in 1985 and 1995 and dropping to as low as 5 percent early in the sampled period. There are single countries in the sample that experience radically different rates of unemployment – Ireland registered unemployment rates of 17 percent in the 1980s and early 1990s but we see these rates drop to less than 5 percent in the early 2000s. Belgium begins the sample period with around 7 percent unemployment but experiences levels reaching 12 percent by the end of the 2000s.

In a fashion similar to unemployment, real GDP per capita growth rates vary considerably over the sampled period. The overall sample mean drops to about 0 percent real

growth in the early 1980s, the early 1990s and the early 2000s and almost hits 5 percent during the high growth periods. The equilibrium level appears to be around 3 percent during the period we sampled. It appears that over this sampled period most countries experienced this full range of fluctuation in real GDP growth (zero to five percent) – some even witnessed serious economic contractions in the order of negative five percent and a couple of countries exceed ten percent growth for short periods of time. But there is also considerable cross-country variation in the series although more muted than was the case with unemployment.

These descriptive plots suggest the existence of significant cross-national variations in macro-economic shocks. But we can summarize the plots of Figure 6.2 with a measure of the relative magnitudes of cross-national to within-country variations in macro-economic shocks. We do this in two steps. First, we estimate a ratio, where the numerator is between-country variance and the denominator is the sum of within-country and between country variance in macro-economic outcomes. To estimate these quantities from data on economic outcomes, we specify a simple random effects model for each of our three macro-economic variables for country j in period t ,

$$(6.1) \quad \begin{aligned} Y_{ij} &= \beta_0 + \mu_j + \varepsilon_{ij} \\ \beta_0 &= \text{grand mean of } Y_{ij} \text{ all countries and periods} \\ \mu_j &= \text{deviation from grand mean for country } j \\ \varepsilon_{ij} &= \text{deviation from mean for country } j \text{ for period } t \end{aligned}$$

The total variance of Y_{ij} is the sum of the variance of μ_j which we label τ^2 and the variance of ε_{ij} which we label σ^2 . The intra-class correlation coefficient (ρ) captures the extent to which overall variation is due to differences between countries (Goldstein 1995). It is defined as

$$(6.2) \quad \rho = \frac{\text{variance between countries}}{\text{total variance}} = \frac{\tau^2}{\tau^2 + \sigma^2}$$

As the proportion of the total variance accounted for at the cross-national levels rises, ρ gets closer to 1. As the intra-class coefficient (ρ) gets larger cross-national variance in macro-economic outcomes dominates total variation in macro-economic outcomes and voters can infer national macro-economic shocks are the result of incumbent competence rather than exogenous shocks from the global economy (i.e., $\sigma_\mu^2 > \sigma_\varepsilon^2$).

Our expectation is that voters establish the competence of incumbents over macro-economic outcomes based on a sampling of economic outcomes that span a reasonable period of time – this does not result from viewing a single monthly economic outcome. At the same time we can see from the inflation series in Figure 6.2 that the proportion of total variance in economic shocks accounted for by the competence term may vary over time. Accordingly, we estimate the random effects model over approximately five-year periods (1979-85; 1986-1990; 1991-1995; 1996-2002). The resulting intra-class coefficients for these periods are summarized in Figure 6.3. Cross-national variations in unemployment clearly dominate total variation in this macro-economic indicator, suggesting that it should play an important role in signaling incumbent competence. Cross-national variations in inflation seem to account for less of the total variation in this series than was the case for unemployment, but appear to be relatively significant throughout our sample period. Even though, as we saw in Figure 6.2, total variation in inflation has declined significantly, the portion accounted for by cross-national variations continues to be quite significant. Finally, the results for cross-national variations in real GDP confirm the French example in Figure 6.1: For the most part between-nation variations in real GDP have been small relative to total overall variation in this indicator. Given the magnitude of cross-national variations relative to within country fluctuations for both unemployment and inflation we would expect these two series to have a greater impact on economic voting than variations in real GDP growth.



Figure 6.3. Intra-class Correlation Coefficients for Macro-economic Indicators

The point of presenting these macro-economic statistics is simply to establish that the real economy fluctuates over time and cross-nationally in a fashion that at least potentially provides voters with the information necessary for undertaking the signal extraction task described in our theory. The fact that the necessary information is available to voters is obviously no assurance that they pay any attention to it and that voters are ultimately knowledgeable about it. One factor that would certainly contribute to the voter's ability to learn about these economic fluctuations is the reporting of economic news by the media. If the media accords little attention to variations in macro-economic shocks this would compromise the voter's ability to learn about them. This is an empirical question that we turn to in the next section.

Information about Variance in the Macro-Economy

Do the voters get much information about the variance in economic outcomes? There is considerable evidence that the media plays an integral role in shaping economic evaluations, which of course are critical to economic voting models (De Boef and Kellstedt 2004; Duch and

Stevenson 2007; Erikson, Mackuen and Stimson 2002; Nadeau et al 1999).¹²⁷ We believe the media transmits information that helps voters assess overall variance in shocks to the macro-economy. Specifically, media reports of economic performance include extensive references to how the economy has changed. Our analysis of 2,700 front-page newspaper stories reporting on economic outcomes from six countries suggests that about half of these stories include some reference to how the economy changed, either in the current period or over the recent past.¹²⁸ Moreover what typically captures the attention of media outlets are unexpected changes in macro-economic outcomes. Again, a careful analysis of our sample of media reports suggests that these stories tend to focus on the unexpected. Hence, to the extent that the media primes voters to think about the economy at all, it seems to emphasize changes or fluctuations in macro-economic outcomes.

This suggests that fluctuations in the economy trigger media attention. We evaluate the extent to which this is the case by examining media coverage in the U.S. For each month over the 1980-2004 period we conducted a count of *New York Times* and *Washington Post* stories mentioning inflation and a similar count for unemployment. These counts were then averaged over each year in our sample to give us two variables measuring annual average monthly counts of inflation and unemployment stories. We also estimated the standard deviation of inflation and unemployment for each of the years in our sample. Our argument is that greater variation in economic outcomes stimulates greater coverage. This suggests that the count of media stories concerning inflation or unemployment should be higher during periods in which the standard deviations of actual inflation or unemployment outcomes are high. Figure 6.4 presents the plots of inflation stories against the standard deviation of inflation rates and unemployment stories against the standard deviation of unemployment rates. The results support the notion that variance in economic outcomes tends to trigger media attention. Note that in both cases, high standard deviations are associated with a high story count. This lends credence to our argument that voters are quite likely to be informed about variance in economic outcomes.

¹²⁷ The precise manner in which the media affect economic evaluations is the subject of debate – for example, De Boef and Kellsted (2004) suggest the effect is indirect via popular assessments of the president's economic management while Nadeau et al (1999) argue for a direct effect between the media and economic evaluations.

¹²⁸ This project samples newspaper front pages the day of and the day after the release of national economic statistics by the relevant country's statistical agency (from 1980-2001). All headlines and stories that mention the economy were coded for content, including the extent to which the information was presented in changes or levels or was compared to previous experiences.

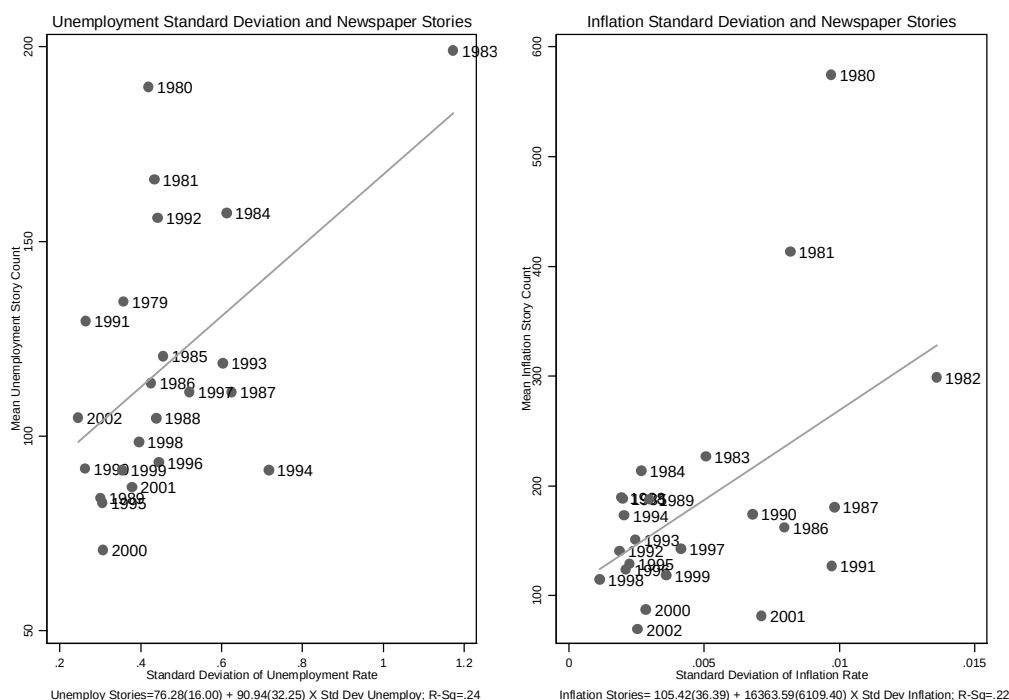


Figure 6.4. Variance in U.S. Macro-economy and Economic News: 1980-2004

Voter Attention to Information about Variance in the Macro-Economy

Is there any reason to believe that the average voter in developed democracies pays any attention to information regarding variance in economic outcomes?¹²⁹ Some have argued that in fact fluctuations in economic outcomes attract considerable attention from the voter. Quinn and Woolley (2001), for example, replicate the Powell and Whitten (1993) aggregate level analysis of incumbent vote shares and find that in, addition to economic growth, their measure of economic volatility has a significant negative effect on incumbent vote shares (although the conclusions they draw regarding its implication for the economic vote are quite different than ours).¹³⁰ Moreover some scholars have argued that the absence of economic volatility as a public good widely favored by the voting public, which might be one reason that democracies have lower levels of economic volatility (Rodrik 1997b). The important point here is that voters – or at least the median voter – must be attentive to variance in economic outcomes.

¹²⁹ Our focus on the developed democracies excludes, for the most part, countries that have experienced dramatic variances in macro-economic outcomes and where there is no question that citizens have been concerned with variance in addition to growth.

¹³⁰ Quinn and Woolley (2001: 636) also cite historical Gallup polls from the U.S. indicating that respondents consider economic instability to be one of their greatest fears for the country.

Overall, however, there has been relatively little effort to investigate empirically whether variations in the economy register with the typical voter. As a result, in the spring of 2005, we conducted a six-nation survey that explored voters' beliefs about the variation in their national economies and its sources.¹³¹ One of the questions in this survey asked respondents the following:

Over the last four years would you say that the economy in [COUNTRY] has experienced very stable growth, somewhat stable growth, somewhat unstable growth, or very unstable growth?

Does this question tap meaningful attitudes about variation in the national economy? One issue here is whether this question is eliciting something other than an overall assessment of economic outcomes which the typical economic evaluation questions are designed to measure. We can get some indication by correlating responses to this question with responses to the standard retrospective national economic evaluation question.¹³² For the combined sample of all six countries the correlation was 0.56 suggesting that in fact these two questions are tapping quite distinct attitudes regarding the economy. Also, if responses to this question show systematic variation as opposed to just random fluctuation we would be inclined to conclude that yes it does tap meaningful attitudes. One approach is to examine whether there is a significant degree of agreement among respondents in their answers. Random answers would show no such clustering; rather randomness suggests answers that are distributed relatively uniformly. Further, we would expect, if voters have well-formed beliefs about variation in the economy, that the proportion of "Don't Know" responses would be similar to their proportion in other surveys that collect other kinds of economic information (e.g., economic retrospections).

Results from each of the six countries clearly suggest that the question is meaningful. Figure 6.5 indicates that in each of the countries we see a clear modal response and relatively small variances around the modal response.¹³³ In Denmark, for example, over half the respondents choose the "somewhat stable growth" response. In no case is the modal response less than 40 percent of the sample. Furthermore, the number of "Don't Know" responses compare favorably with levels in other surveys soliciting more standard types of economic

¹³¹ The survey was pre-tested in the United States in the early spring. Details on the six nation study are available at www.nuffield.ox.ac.uk/economicvoting/sixnationsurvey.

¹³² This wording of this question for the UK survey was as follows: "Now thinking about the economy in the UK as a whole, would you say that over the past year the nation's economy has gotten much better, gotten somewhat better, stayed about the same, gotten somewhat worse or gotten a lot worse?"

¹³³ The response set for this question varies from 1 (Very Stable) to 4 (Very Unstable).

beliefs. The highest level of “Don’t Know” responses was 6.7 percent of the Danish sample and the lowest was 1.1 percent of the France sample. These compare quite favorably, for example, to the range of “Don’t Know” responses to the standard question concerning retrospective evaluations of the national economy that we asked in the same set of surveys: the highest level, again, was 7.5 percent of the Danish sample and the lowest level was 1 percent of the French sample.

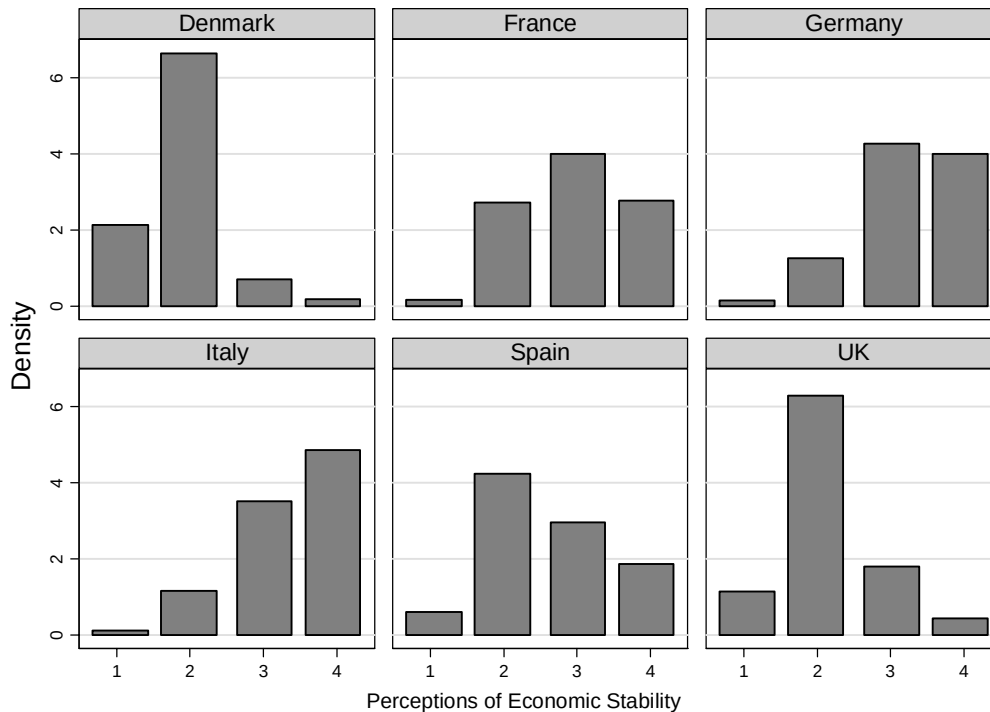


Figure 6.5. Histogram of Volatility of Economic Perceptions, Europe

The competency elements of our theory suggest that perceptions of economic volatility condition the economic vote. Hence part of the theory’s claim to explain cross-national variations in the economic vote is contingent upon significant cross-national variation in average perceptions of economic volatility. The left hand graph in Figure 6.6 shows that average citizen perceptions of the stability of economic outcomes in their country vary quite significantly across European nations. Danish and British respondents clearly perceive their national economies as turning in very stable growth over the previous four years. By contrast the Germans and Italians report very high levels of instability in growth outcomes. In short, these individual-level data lend support to the idea that contextual variation in the economic vote could come, in part, from differences in the competency signal perceived by the voter.

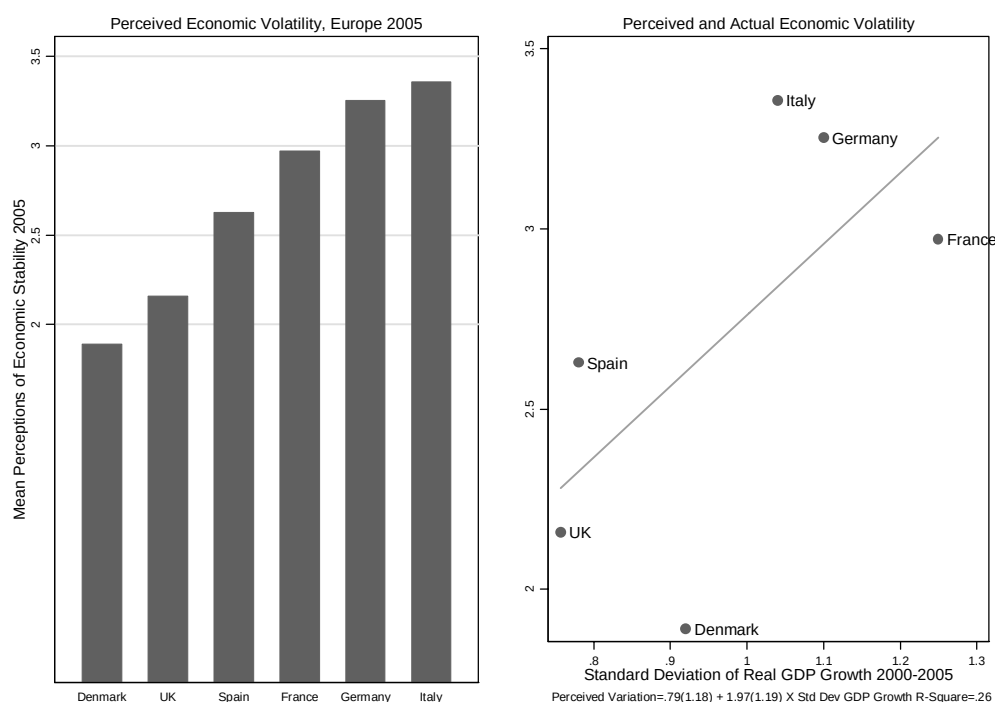


Figure 6.6. Volatility of Real GDP and Economic Perceptions, Europe

Of particular concern to our competency argument is whether cross-national variation in perceptions of economic stability is grounded in real variation in the economies of the different countries. We expect citizens in contexts with highly variable economic outcomes to perceive economic outcomes to be highly unstable. The right hand graph in Figure 6.6 confirms this expectation: Citizens in contexts where the economic outcomes were highly unstable over the 2000-2005 period (measured by the standard deviation of annual growth in real GDP for the period) – Italy, Germany and France, in particular – have perceptions of economic instability that are much higher than citizens in contexts with more stable economic outcomes – the U.K. and Spain. Our evidence, though based only on six countries, is unique in its focus on economic variation and confirms that voters have well-formed beliefs about variation in their national economies and that these beliefs are grounded in economic reality.

The Sources of Variation in Macro-Economic Outcomes

Our competency argument assumes that, besides knowing the overall variance in macro-economic shocks, voters also know the relative contributions of σ_{μ}^2 and σ_{ξ}^2 to that total. While models in the political economy literature often make this assumption (for example, Cukierman

and Meltzer 1986; Rogoff and Sibert 1988), there has been little effort to explore empirically whether citizens have perceptions consistent with this characterization. This section addresses the empirical support for this contention.

If voters have a sense of the total variation in economic shocks (i.e., $\sigma_\mu^2 + \sigma_\xi^2$), and in addition have well-formed beliefs about either one of these variance terms, then they can distinguish between the relative contributions of the two components of this total variation. So, for example, if they have a good sense of how important exogenous shocks are in their national economy (σ_ξ^2) and they have a sense of how much shocks to the economy vary in general ($\sigma_\mu^2 + \sigma_\xi^2$), they should have a sense of how much “room” there is for the competence of the elected government to impact the economy ($\sigma_\mu^2 = (\sigma_\mu^2 + \sigma_\xi^2) - \sigma_\xi^2$). One piece of information that we think helps individuals distinguish exogenous from competency shocks is simply the extent to which the domestic economy depends on global economic outcomes. The greater extent to which citizens believe most domestic economic outcomes are determined by global influences, the less likely they are to conclude that economic outcomes result from the competence of elected officials. Our expectation is that voters have beliefs about the dependency of their national economy on global influences.¹³⁴

To gauge whether citizens have beliefs about the extent the economy is subject to exogenous shocks, we asked respondents in our six countries the following question:

To what extent is growth in the economy in [COUNTRY] dependent on growth in the other European economies? Would you say extremely dependent, very dependent, moderately dependent, a little dependent or not at all dependent?

As in the previous question, there was substantial agreement among respondents in each country regarding the extent of global influences on the domestic economy. In each case, more than 45 percent of respondents chose the modal category and no more than 11 percent indicated they could not answer. The fact that the modal response includes such a relatively large number of respondents and that non response is comparatively low, suggests that voters are getting consistent information from the media about their countries’ position in the global economy.

¹³⁴ Much of the trade policy literature makes a similar assumption. For example, the preferences that producers and holders of capital have regarding trade liberalization policies are typically portrayed as resulting from 1) self-interested assessments of the relative scarcity of their skills or their capital and 2) how greater exposure to global competition would affect the returns to their labor or capital. This presumes that individual citizens have basic information regarding the relationship of the nation’s economy to global trade. And there is a growing body of literature demonstrating that citizens in fact have preferences and beliefs about trade policy (Baker 2005; Mayda and Rodrik 2005) that are consistent with these models.

Further, there is some evidence here that these opinions reflect actual global influences on the domestic economy. Figure 6.7 plots average responses to this question for each country against our measure of trade exposure. Note that three countries with relatively low levels of trade exposure – Italy, France and Spain – have average responses in the middle category (“moderately dependent”). Denmark, which is the one country in our sample with a very high trade exposure, also is the only country with an average response close to the “very dependent” category. Obviously this is a very small number of observations. Nonetheless Figure 6.7 is suggestive of the link between perceptions of dependency and actual trade dependency.

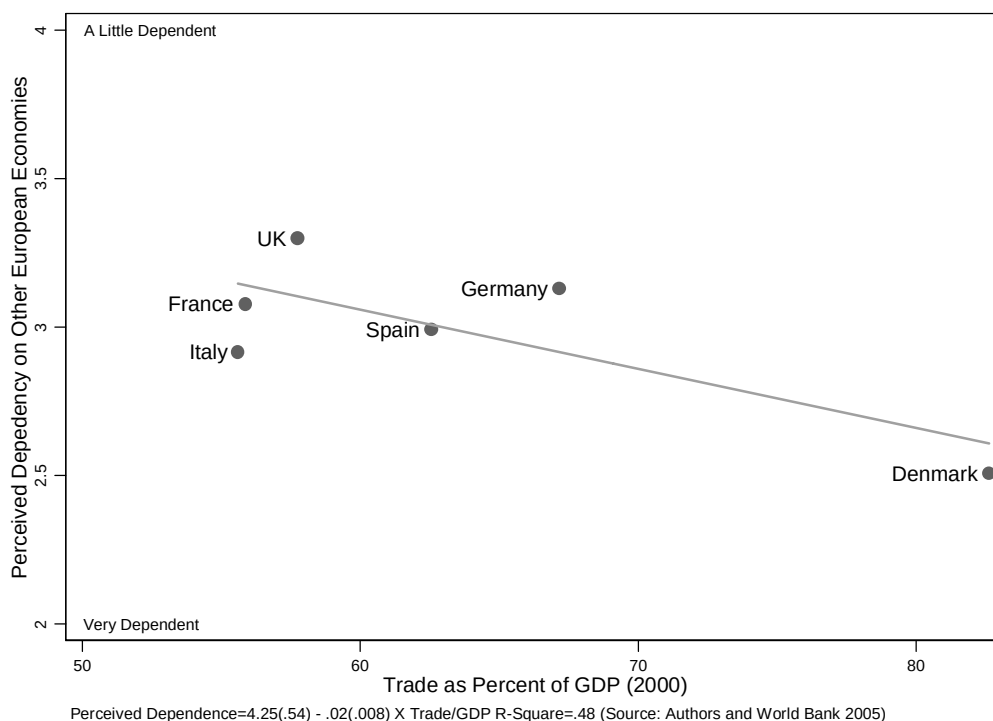


Figure 6.7. Trade and Perceptions of Dependency on European Economies

Having reasonably accurate information about the global dependency of their economy is a resource that citizens can employ in distinguishing competency from exogenous shocks to the macro-economy. Figure 6.7 provides some support for the notion that citizens are equipped for the task. We can further explore whether citizens have the information necessary to distinguish exogenous from competence shocks to the macro-economy with a somewhat larger sample of countries from the PEW Global Attitudes Project. PEW asked the following series of questions:

Do you think change in [INSERT EACH ITEM BELOW] is the result of global connectedness?

- the availability of good-paying jobs
- the working conditions for ordinary workers
- the availability of modern medicines and treatments
- the availability of food in stores
- the gap between rich and poor people

Respondents provided yes or no answers to each question. While not directly related to the macro-economy, as a whole these questions provide an indication of the extent to which different national populations perceive that the global economy affects general economic outcomes. We factor analyzed these items and created a single factor score measuring the extent to which citizens perceived their overall economy to be “globally connected”. All of the items loaded quite highly on a single factor with high values indicating global dependency.¹³⁵ This is reassuring because it provides some confirmation that there is an underlying attitude anchored on one end by a belief that the national economy is heavily influenced by global forces and on the other by a recognition that the national economy is relatively unaffected by global factors.

Again, our argument suggests that populations with heavy exposure to global economic influences should register high levels of “global dependency”. We test this hypothesis in Figure 6.8 which plots each country’s global dependency score (a high mean score indicates the population believes the economy is heavily influenced by global forces) against the country’s trade exposure (total trade – exports and imports – as a percentage of total GDP). The evidence is quite convincing. Countries with high levels of trade exposure tend to be those in which their population registers a high level of awareness of the extent to which the domestic economy is subject to global economic influences.

¹³⁵ Only developed countries from this sample were included in the analysis – countries with GDP per capita in the year 2000 of more than \$5000 1995 USD.

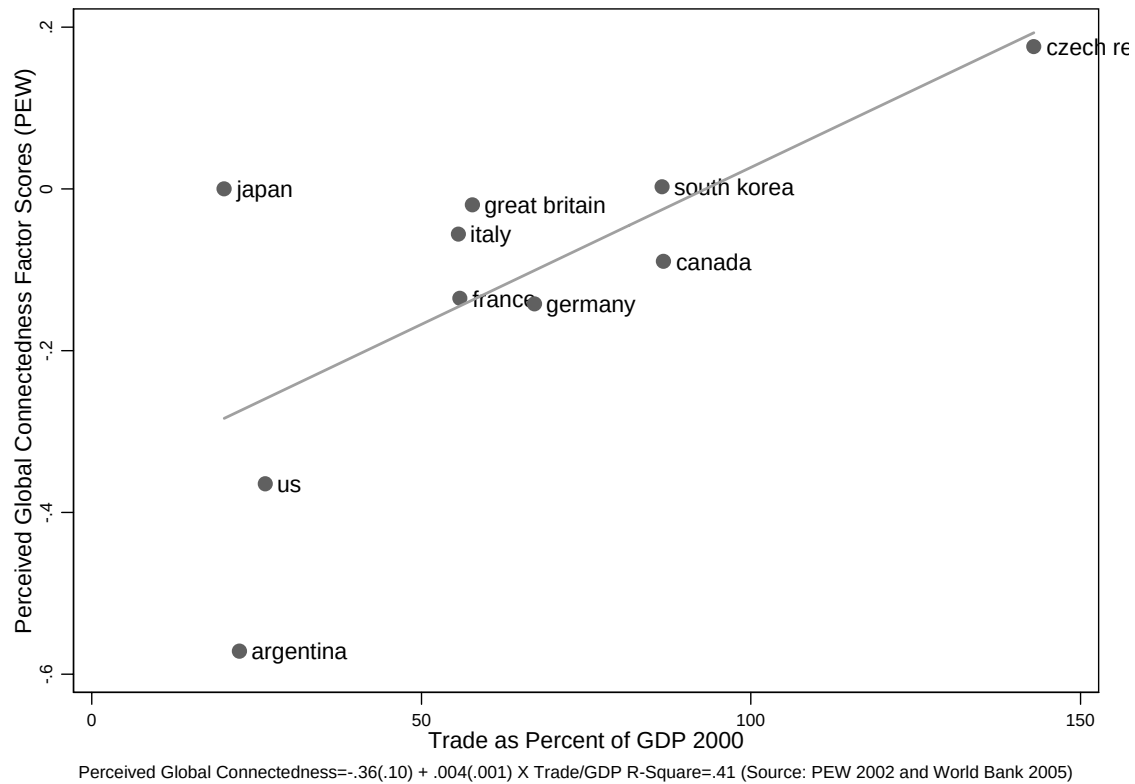


Figure 6.8. Trade and Perceptions of Global Connectedness

One assumption of the competency theory of contextual variation in the economic vote is that voters know the variance in exogenous shocks to the macro-economy (relative to the variance in competency shocks). They can acquire this knowledge by observing the importance of global influences on the domestic economy.¹³⁶ The evidence presented here suggests quite clearly that voters are attentive to information regarding the extent to which the domestic economy is subject to exogenous shocks from the global economy. First, they appear to have meaningful attitudes regarding global influences on the economy. More importantly, cross-national variations in assessments of the magnitude of these exogenous shocks correlate with actual measures of trade dependency suggesting that these attitudes reflect global influences on the voter's national economy.

¹³⁶ We realize, of course, that not all exogenous shocks to the economy arise from non-domestic sources. Indeed, the theory only distinguishes between governmental and non-governmental shocks, where the non-governmental category includes shocks arising from the behavior of any economic actor who voters do not include in the government whose competency they wish to evaluate.

Individual perceptions of economic variation and the economic vote

We can use our six-nation European survey to explore how perceptions regarding fluctuations in the domestic economy impact the economic vote. Individuals whose perceptions of the domestic economy deviate from their perceptions of the global economy should infer that economic fluctuations are due to the competence of domestic politicians. In contrast individuals who perceive their national economies to fluctuate in tandem with the global economy would have little reason to conclude that economic outcomes are the result of domestic decision makers. Here we focus on Europe and treat the overall European economy as the global referent. Our six-nation study asked respondents to assess the stability of both their national economies and the overall European economy.¹³⁷ Table 6.1 presents a cross-tabulation of responses to these two questions. The entries are the numbers of respondents falling in each cell. Respondents falling along the diagonal had identical responses for both their domestic economy and the European economy. Off-diagonal respondents represent those who perceived domestic fluctuations as deviating from those of the overall European economy – either being higher or lower. Approximately 50 percent fall on the diagonal with the other 50 percent falling on the off-diagonal cells.

Table 6.1 Perceptions of Stability of National and European Economies.

National Economy	European Economy				
	Very Stable	Somewhat Stable	Somewhat Unstable	Very Unstable	Total
Very Stable	86	143	116	27	372
Somewhat Stable	105	1114	596	90	1905
Somewhat Unstable	35	543	836	131	1545
Very Unstable	27	286	446	484	1243
Total	253	2086	1994	732	5065

Our theory suggests that those respondents falling in the off-diagonal cells of Table 6.1 (in bold) – i.e., those respondents who perceive their variations in their domestic economic outcomes as deviating from variations in the overall European economy – should conclude that the overall competency signal is stronger than respondents in the diagonal cells. The reasoning here is that fluctuations in the domestic economy that are distinct from those in the overall global economy signal the competency of elected officials. Because the respondents in the off-diagonals

¹³⁷ The Europe economy question was worded as follows: “Over the last four years would you say that the European economies have experienced very stable growth, somewhat stable growth, somewhat unstable growth, or very unstable growth?”

perceive a higher overall competency signal they should, according to our theory, have a higher propensity to cast an economic vote.

We can evaluate this argument empirically by determining whether the respondents in the off-diagonals of Table 6.1 in fact are more likely to cast an economic vote. Respondents in the six-nation survey were asked to report their likely vote choice if an election were held in the coming days.¹³⁸ Responses to this question were used to create a dichotomous incumbent vote variable (respondents indicating a preference for any of the parties in the governing coalition were coded as 1 with the remaining respondents coded as 0). The first column of Table 6.2 reports the probit estimates for this simple economic vote equation (country dummies are included in the regression but their coefficients are not reported). As we would expect, retrospective national economic evaluations are strongly correlated with incumbent vote intention.

Our interest though is assessing whether the off-diagonal respondents – those who perceive domestic economic fluctuations as diverging from those in the overall European economy – are more likely to register an economic vote. Accordingly, we created a dummy variable (*Deviation*) that has a value of 1 for all those respondents falling in the off-diagonal cells. This variable, interacted with the economic evaluation variable, indicates whether the off-diagonal respondents have in fact higher levels of economic voting as we hypothesized. The second column of Table 6.2 reports these results. In fact, as we hypothesized, respondents who perceive distinct differences in the fluctuations of their national economies, compared to the overall European economy, are more likely to cast an economic vote. This is indicated by the positive and statistically significant value on the *Retrospective*Deviation* interaction term.

These results suggest a rather novel account of how global trends shape economic voting behavior. A frequently made argument is that globalization “homogenizes” national economic outcomes and as such could, as our theory even suggests, depress economic voting. But globalization also has two other important effects. First, it raises the awareness voters have of other economies, a conclusion consistent with the results in this section. We believe it also sensitizes voters to the extent to which fluctuations in national economies deviate from global fluctuations. And this of course aids voters in the signal extraction task described in our theory. Voters in this formulation are using information about national and global economic outcomes to condition their economic vote in a fashion that is quite different than traditional accounts. To illustrate we sketch out an example from the recent history of British economic voting.

¹³⁸ The British questioning wording is as follows: “If a general election were held next Sunday which political party would you vote for?”

Table 6.2. Perceived Variations in Economic Variations and the Economic Vote.

	<i>Probit Coefficient (Standard Error)</i>	<i>Probit Coefficient (Standard Error)</i>
<i>Retrospective National Economic Evaluations</i>	.33 (.02)	.29 (.03)
<i>Retrospective * Deviation</i>		.08 (.04)
<i>Deviation (Off Diagonal in Table 7.1=1)</i>		-.33 (.12)
<i>Constant</i>	-1.59 (.07)	-1.39 (.10)
Number of Observations	5,834	5,021
Log Likelihood	-3123	-2700

Note: Country dummies included in both equations.

Competency Signals in the 1979 and 1997 British Elections

Rarely are *variations* in macro-economic shocks invoked by election scholars as an explanation for the importance of the economy in election outcomes. In fact, economic context in general is rarely considered to be important for conditioning the economic vote. A possible exception is the argument that during particularly bad economic times, the economic vote might rise (Nannestad and Paldam 1997b). We examine the 1979 and 1997 British elections with the goal of illustrating how economic context likely provides voters with information about incumbent competence and hence conditions economic voting.

When Margaret Thatcher defeated the Labor Government in 1979 our measure of the economic vote in Britain registered a relatively high value of about 12 percent. Figure 6.9 characterizes the macro-economic context at the time of this election. Inflation, at 14 percent, was clearly a problem, although real GDP growth, at 3.5 percent, was reasonably high and a level of unemployment hovering around 1.3 million people, while historically high, was low compared to subsequent years when the level ultimately reached 3 million. By contrast, when Tony Blair unseated the Tories in the 1997 election, the economic vote had declined considerably to about 5 percent. While real GDP growth in 1997 was similar to the 1979 period, unemployment had reached somewhat higher levels than 1979 although inflation was much lower and more stable. Clearly something about the context of the 1979 election persuaded voters to give priority to their economic evaluations while voters in the 1997 election gave priority to other concerns. We expect this variation results from a much clearer signal regarding the economic competence of the

competing candidates in the 1979, as opposed to the 1997 election. But this has not been the explanation favored by the economic voting literature.¹³⁹

One explanation, frequently invoked in economic voting studies, is that economic context could be entirely unrelated to these variations in the economic vote – in 1997 non-economic issues may have simply out-weighed economic evaluations in the British voters' utility functions.¹⁴⁰ We entertain this as a perfectly plausible null hypothesis. It may, in fact, be the case that other factors, such as Conservative divisions over the European Union or public dissatisfaction with the government's decision to pull Britain out of the ERM, simply overwhelmed economic evaluations (Butler and Kavanagh 1997).

Particularly negative shocks to the macro-economy ($y_t - \bar{y} - \mu_{t,t-1} < 0$) may trigger a heightened concern with the economy thereby increasing the relative importance of the economy to other variables in the vote choice model. The 1979 election occurred in the midst of serious economic decline and extreme popular economic angst, possibly inflating the relative importance of the economy, while the 1997 election occurred during a period of comparative prosperity, possibly dampening its importance for vote choice.¹⁴¹ These significant differences between the macro-economic outcomes in 1979 and 1997 might have exaggerated the importance of economic evaluations in vote decisions in 1979. Figure 6.9 presents the *economic vote of the chief executive* at each of the UK elections along with values for three macro-economic outcomes (CPI, Real GDP Growth and Unemployment). A cursory analysis of the economic outcomes associated with the two elections suggests that only one candidate fits this argument. Real GDP growth is similar during the two elections so we can eliminate this as a possible contextual effect. Unemployment in this case seems to have either no effect or possibly the opposite effect that one might expect – somewhat higher levels of unemployment are associated with lower degrees of economic voting. Inflation seems to offer the most plausible contextual effect – it is high in the 1979 election, when economic voting was also high, and it was much lower in the 1997 election when we see economic voting at its nadir.

¹³⁹ See Lewis-Beck, Jerome and Jerome (2001) for a related account of how relative economic performances (in this case in Germany and France) can shape the economic vote.

¹⁴⁰ Alt (1991: 241) notes that “the contribution of the economy to the governments overall record depends on the extent to which the economy is perceived to be the most important issue facing the country, something intensively investigated by MacKuen and Combs (1981), who also showed the state of the economy influences the perceived importance of economic problems.”

¹⁴¹ For an excellent assessment of economic voting in the 1979 election see Norpoth (1992).

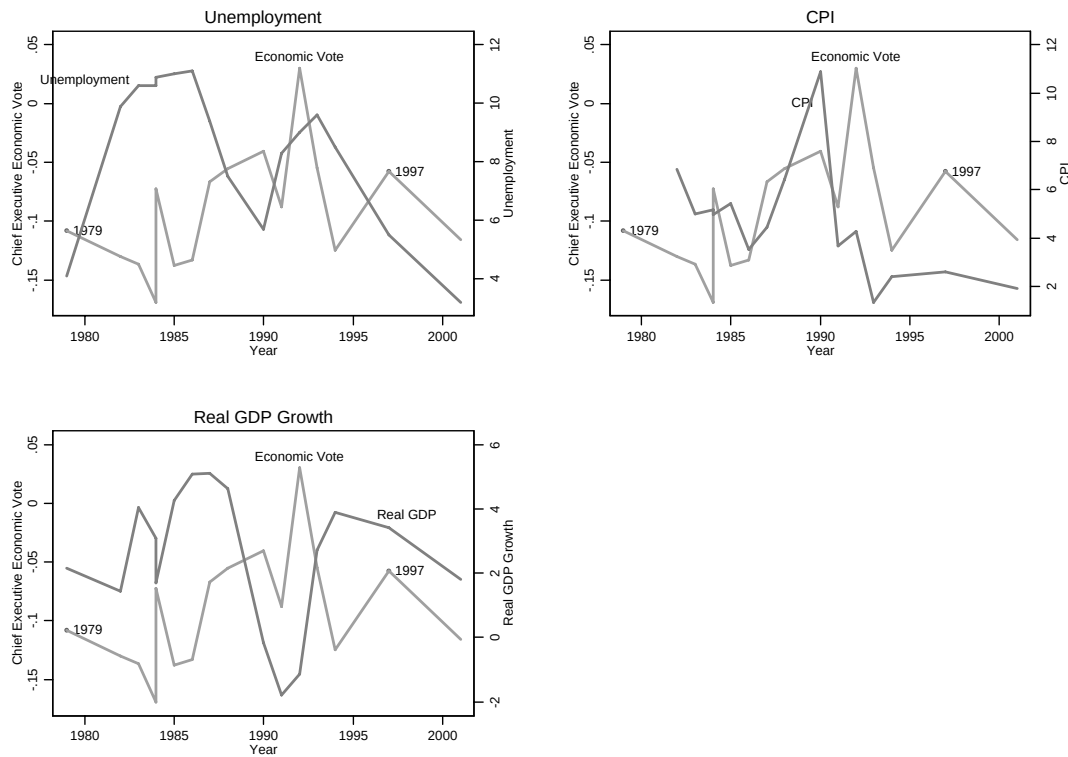


Figure 6.9. UK Macro-economic Outcomes: 1979-2002

This notion that when the economy is performing particularly poorly people are inclined to factor it into their vote decision is certainly persuasive.¹⁴² And to the extent that this is the case, fluctuations in inflation appear to be what triggered British voters' economic vote. This certainly was the conclusion that Margaret Thatcher and the Conservatives came to in 1977 as they prepared for the next election (Norpoth 1992).¹⁴³ Their calculations appeared to pay off: Our estimates suggest that the economic vote played an important role in the 1979 vote decision (based on the 1979 British Election Survey we estimate that a unit deterioration in retrospective economic evaluations reduced the incumbents' vote probability by 11 percent). The implication of this contextual argument is that when the economy is in equilibrium, or possibly even out of equilibrium but in a positive direction, economic evaluations are likely to play a much less important role in the vote decision. In the 1997 election Conservative strategist Michael

¹⁴² Evans (1999a: 151) speculates that this might have contributed to the relative weak impact of economic evaluations on UK vote preference during the 1992-1995 period.

¹⁴³ In the Conservative Party's economic policy statement issued in 1977, *The Right Approach to the Economy*, controlling inflation was clearly identified as the party's prime economic goal.

Heseltine failed to recognize this possibility, predicting, to the end, a Conservative majority of 60 seats, "...regarding it almost as an iron law that a government which delivered steady improvements in real disposable income over 12 months would win the election..." (Butler and Kavanagh 1997: 75). As it turns out Heseltine was wrong – our estimates suggest that the economy played a relatively insignificant role in the 1997 vote decision (the impact for this year is about 5 percent which is slightly under the overall mean size of the economic vote in our overall sample of elections studies). The Conservatives experienced a resounding defeat in the 1997 election in spite of a relatively strong economy. The high levels of inflation seemed to have contributed to the importance of the economic vote in 1979 but the particularly low levels registered in 1997, if anything, seemed to have reduced the weight of the economy in the vote decision.

A more systematic review of the data provides virtually no support for the notion that economic voting rises during periods of particularly bad economic performance. Figure 8.7 plots the level of economic voting in Britain against each of the three macro-economic outcome variables. The evidence in the case of real GDP growth is in fact exactly the opposite direction; as the economy improves, the economic vote rises. Much of this is driven by the paradoxical cases of 1990 through 1992 when economic growth was either at a standstill or declining. Even though the economy was in particularly bad straits our economic vote measures during this period are at their lowest levels for the UK. Unemployment has an effect in the expected direction – rising unemployment associated with higher economic vote – but the slope coefficients is statistically insignificant in a bivariate regression of the two variables. And there is no relationship in the case of inflation. Hence, we would be hard-pressed to conclude that relatively bad economic conditions trigger higher levels of economic voting.

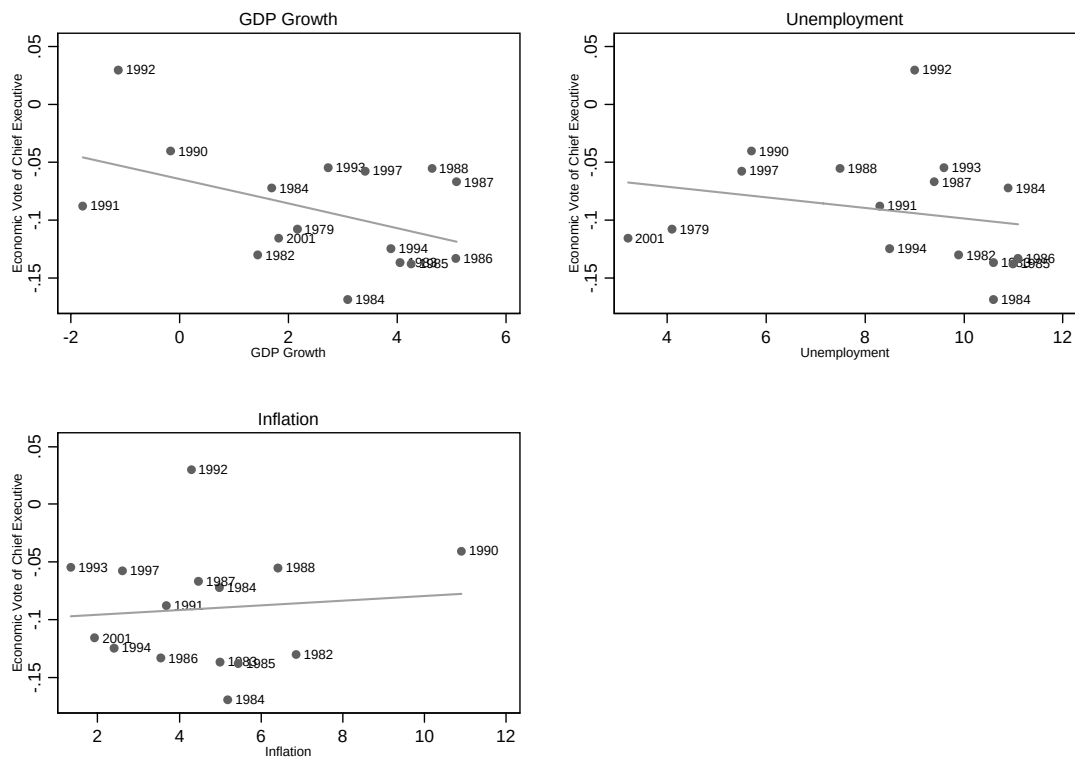


Figure 6.10. UK Economy and the Economic Vote: 1979-2002

In our view a dismal economic performance was not what triggered the importance of the economy in the 1979 vote decision. As Norpoth (1992) points out, by 1979 the British economy had in fact improved considerably and was turning in respectable growth and inflation numbers (particularly compared to the recent past). Voters clearly had negative views of the government's economic performance but what lead them to incorporate these evaluations into their vote decision were crystal clear signals that these fluctuations in economic outcomes were the result of the economic management of the incumbent government. The recent history of economic policy making by both parties had been of a "stop and go" variety that generated highly volatile economic outcomes. This volatility and the attendant commentary by politicians and the press could leave little doubt regarding the government's responsibility for economic outcomes.¹⁴⁴ But just in case this was not sufficient, the contrast between fluctuations in the British economy and variations in the international economy provided glaring evidence that British outcomes were not the result of factors exogenous to the incumbent government.

¹⁴⁴ The incumbent prime minister Callaghan himself, in his 1976 address to the Labour Conference, pointed to these "stop and go" policies as having contributed to the country's economic problems (Callaghan 1987: 426).

In terms of the signals voters received regarding government competence for macro-economic outcomes, the 1997 election was the exact opposite of the 1979 election. The “stop and go” variety of macro-economic management had been abandoned for almost 20 years and fluctuations in the macro-economy had been radically reduced. The British economy was no longer an international pariah as it had been in 1979 – although turning in strong economic performance, the overall economy performed similarly to other developed democracies. Hence, by 1997 the average voter – and maybe just as importantly the media coverage of the economy – was more likely to consider fluctuations in the macro-economy as resulting from exogenous factors – such as developments in the global economy or trends in the global business cycles – rather than government policy. It’s not that the media did not portray the economy as doing well, it’s that there were few credible economic competency signals to which they could respond. The result was not bad media on the economy, as Gavin and Sanders (1997) point out, rather it was that there was very little media coverage of the topic. And in some sense this was the uphill, and ultimately unsuccessful, battle that Prime Minister Major fought throughout much of the 1997 campaign.¹⁴⁵

The relevant contrast between the economic context in 1979 and 1997 is volatility versus stability, not between good and bad economic performance. The competence argument suggests that the decline in economic voting in 1997 results from aspects of the economic context that either decrease variance in the competence signal σ_μ^2 or increase variance in the exogenous economic shocks σ_ξ^2 . We believe that more economic voting occurred in 1979 because economic fluctuations, credibly associated with government policy initiatives, were higher in the period preceding the 1979 election compared to those preceding the 1997 election. This also assumes that exogenous shocks to the macro-economy, unrelated to the incumbent’s policies, are relatively similar in the two periods or possibly lower in the 1970s.

One proxy for the overall competency signal in our theory is a measure of the total fluctuations in domestic macro-economic shocks. We do not pretend that this measure is a precise reading on the competency signal during these elections – since it obviously incorporates both exogenous and competency components – but we present it here as a suggestive proxy. Our expectation then is that this measure of variation in macro-economic outcomes should be high in the 1979 election and low in the 1997 election. Moreover, economic voting should generally be high when there is considerable variation in macro-economic indicators (in the case of our

¹⁴⁵ This is consistent with the results presented earlier that suggest that the media responds to variations in macro-economic outcomes. Hence a relatively long period of positive, but stable, economic performance may in fact undermine the strength of the economic vote.

economic vote measure, the correlation should be negative since large negative values indicate high levels of economic voting). Figure 6.11 plots the economic vote in the UK against variations in the three macro-economic indicators. The magnitude of British economic voting is particularly high in the late 1970s and early 1980s – this is also a period of particularly high variance in all three UK macro-economic indicators. Economic voting then declines quite precipitously in the later 1980s and 1990s – this is also a period in which variations in the macro-economic measures drop. The implication here is that modest fluctuations in the British macro-economy leading up to the 1997 election, particularly since they tended to be much more in line with global variations, lowered the magnitude of the competency signal. This in turn has resulted in less economic voting. By contrast, the late 1970s and the 1980s were clearly periods when variations in British macro-economic outcomes were quite large, and less in step with economic outcomes in the other developed economies. We would argue that this translated into a high competency shock term (σ_{μ}^2) which in turn implied a larger competency signal. As a result British voters in the 1979 election were much more likely to attribute economic outcomes to incumbent competence, which resulted in higher levels of economic voting.

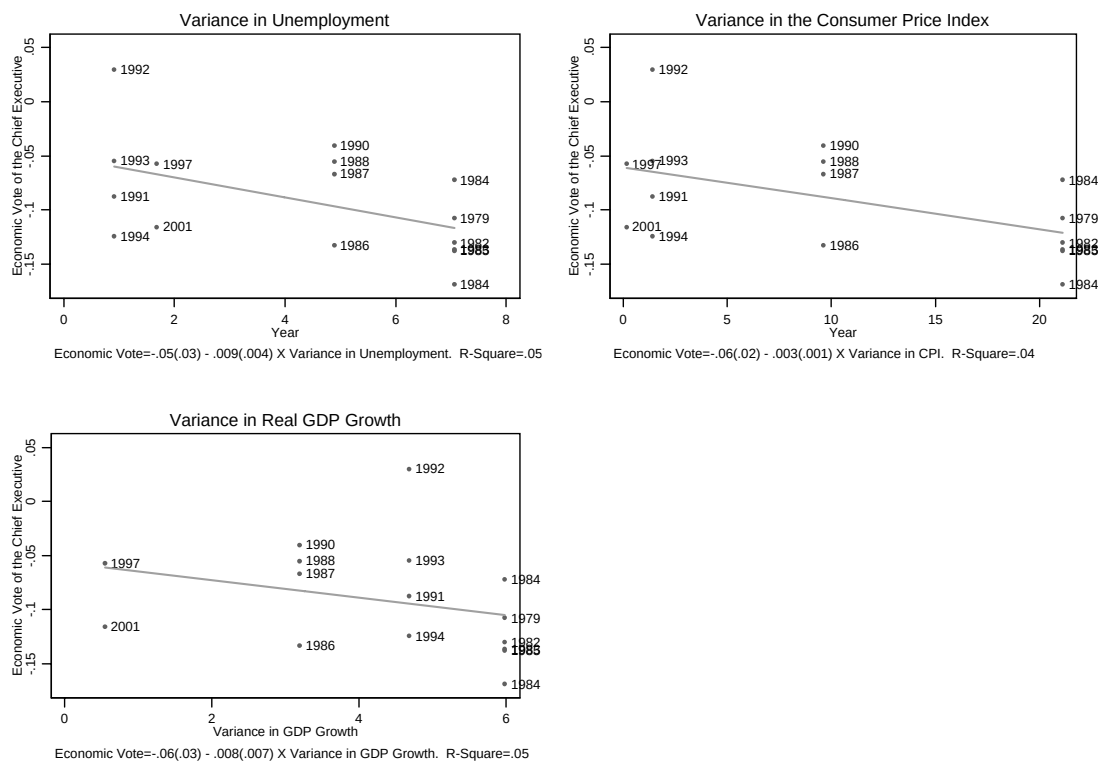


Figure 6.11. UK Variance in Macro-economy and Economic Vote: 1979-2002

The 1979 and 1997 British elections do not, on their own, provide conclusive evidence as to what signals regarding the economic context voters respond to when they weigh the importance of the economy in their vote decision. Nevertheless, they are suggestive and nicely illustrate how economic context affects the economic vote. The economic vote in these two elections diverged considerably as did the overall economic contexts that span the two election periods. We explored the notion that the economy might become more salient in the vote choice model when economic performance is particularly bad. We found little support for this argument. On the other hand, the pattern of fluctuations in British macro-economic outcomes over the past three decades suggests that the signal that seems to matter in the vote choice is variation in shocks to the macro-economy. Consistent with our competency theory, voters seem to use variations in these shocks prior to an election in order to extract information about the incumbent's responsibility for macro-economic outcomes which then conditions their economic vote.

Summary

Voters in our competency model of the economic vote are expected to condition their economic vote on a competency signal. And our theory spells out quite explicitly what information about the economy voters are expected to know as part of this signal extraction exercise. This chapter makes the empirical case that voters in fact are informed about the components of this competency signal, i.e., the typical variation in competence shocks and in exogenous shocks to the economy.

First, the theory assumes that voters are knowledgeable about the total variance in shocks to the macro-economy. The aggregate time series of macro-economic outcomes certainly are convincing evidence of considerable fluctuation in macro-economic outcomes in our sample of mature democracies. Our content analysis of media reports of economic outcomes suggests that the typical voter has access to considerable information regarding these fluctuations in macro-economic outcomes. The analysis of major U.S. print media indicates that the count of economic news stories responds positively to variations in macro-economic outcomes. This suggests to us that the media often frame the reporting of economic information in terms of variation (as opposed to levels) in macro-economic outcomes. Furthermore, the attitudes of respondents in our six-nation survey clearly reflect the manner in which the media frames economic news: Respondents to the surveys have meaningful attitudes about the volatility of macro-economy outcomes. Moreover these attitudes are reasonably well informed in that cross-national perceptions of macro-economic variability correlate quite highly with the magnitudes of actual variations in real GDP.

Second, our theory assumes that voters are able to distinguish variations in competency shocks from variations in exogenous shocks to the macro-economy. Rigorously assessing the voters' capabilities in this regard is challenging. We adopt an indirect strategy here that first assumes that a large part of the exogenous shocks in the economy originate in global fluctuations and that a large part of domestic shocks are attributable to the competence of politicians. Our analysis of dynamic and cross-national variations in aggregate macro-economic time series suggests that the information necessary for this signal extraction task is certainly available to voters. Being cognizant of the extent to which global factors influence the domestic economy contributes to the voter's ability to extract competency signals from shocks to the macro-economy. The results of both our six-nation study and the PEW GAP suggest that voters do have the information to make such distinctions. Specifically, voters have what appear to be meaningful attitudes regarding global influences on their national economies and cross-national variations in these attitudes indicate that they are reasonably informed.

Hence, given the aggregate and survey-based data analyses presented in this chapter, we conclude that voters do have the information to extract a signal about incumbent competence from movements in the domestic economy (as our theory assumes). But did voters respond to the overall competency signal as our theory predicts? Our six-nation survey allowed us to identify those respondents who exhibited individual-level beliefs that the competency theory predicts will lead to higher levels of economic voting – specifically, individuals who perceive that the variation in the national economy differs from variation in the global economy. The empirical results supported the theory: individuals who believed that the variations in shocks to the domestic economy differed from those in the overall European economy had larger economic votes.

However, our theory can tell us more about how the political and economic context conditions the economic vote. Specifically in Chapter 5, we provided an interpretation of the competency model that links the strength of the competency signal to the political and economic institutions that regulate the relative influence of electorally-accountable versus non-electorally accountable decision makers on economic outcomes. We examine the empirical evidence for these links in the next chapter.

Chapter 7

Political Control of the Economy

The previous chapter suggests that voters are attentive to economic fluctuations in a manner consistent with our competency theory. In this chapter we explore whether there are features of the political-economic context that affect these competency signals in a systematic fashion and hence account for the contextual variation in economic voting that we saw in Chapter 3? For example, the evidence presented in the previous chapter suggests that voters understand the extent to which their domestic economy is subject to the influence of global economic factors. Does this help account for the contextual variation in economic voting? While there has been speculation in the economic voting literature about the links between these features of the policy making context and the economic vote (Lewis-Beck 1988; Powell and Whitten 1993), the results have been mixed. Improving our understanding of this link requires a more developed theoretical understanding of how individuals use information regarding the political economic context to inform their vote choice.

Our competency theory provides an explanation for why, and how, fluctuations in the magnitude of the economic vote are related to these variations in political and economic contexts. Recall from the theoretical discussion in Chapter 5 that context matters because it influences the number of electorally versus non-electorally dependent decision makers (or, using our abbreviations, EDDs versus NEDDs) associated with macro-economic policy outcomes, which in turn shapes the size of the incumbent's competency signal. Specifically, the competency signal is a function of the variance in shocks to the economy as well as the variance in the shocks due to the managerial competence of elected officials. The variance of competency shocks will be bigger when the number of EDDs (α) influencing macro-economic outcomes gets bigger. Likewise, the variance of exogenous shocks to the economy (i.e., those not due to EEDs) rises with the number of economically consequential decisions made by NEDDs (β). The argument is summed up by the ratio $\frac{\beta}{\alpha}$. As the ratio of NEDDs (β) to EDDs (α) rises, the overall

competency signal will decline and vice-a-versa. So, for example, in a policy making context where, relative to other contexts, responsibility for making economic policies is more shared with NEDDs (this might be the case in a highly corporatist policy making setting, for example), voters are likely to discount the impact that the incumbent's competency can have on economic growth. Since the importance of the economy in the vote function is determined by the overall competency signal, this ratio of NEDDS to EDDs should be correlated with the economic vote.

This ratio of NEDDS to EDDs effectively defines the policy context that shapes economic outcomes. While not employing our explicit formulation, many students of comparative political economy over the past four decades have implicitly examined how the mix of NEDDS to EDDs has varied both cross-nationally and over time and have explored its implications for a variety of political and economic outcomes. An early contribution to this debate is David Cameron's (1978) classic article exploring the impact of exposure to the global economy on the size of government. Cameron pointed out how economic context – an open economy in particular– can undermine the ability of incumbents to control macro-economic policies. But he argued that the institutional and political responses to the exigencies of an open economy tend to broaden the scope of government which enhances the ability of government to manage the social and economic disruptions associated with global integration. The conventional view is that these two developments balance each other out in terms of their effect on government's ability to affect economic outcomes – an open economy undermines the ability of government to shape macro-economic policies while expanding the role of the state in the economy is a counter weight because it increases the ability of the government to enact compensating measures.

Recently we have seen a number of countries adopting liberal economic policies ostensibly in response to the competitive influences of global economic forces. This has led some to suggest a different twist on Cameron's initial argument: In this view open-economies faced with the exigencies of international competition are forced to liberalize (or privatize) their domestic economies. This in turn reinforces the constraints on governments which reduces the role of elected officials in shaping domestic economic outcomes and consequently the ability of voters to hold decision makers accountable for economic outcomes (Alvarez, Nagler, and Willette 2000; Hellwig 2006). Freeman speculates that "... there is evidence that as privatization and globalization have progressed, democratic citizens have lost faith in their governments' capacities to manage their economies (Freeman 2006)."

These arguments are very much in the spirit of our competency signaling theory. They imply that voters are attentive to global economic influences on the domestic economy; voters somehow understand how the policy making context constrains their economic policy choices; and they use information about the global and domestic economies to condition their individual vote choice. These arguments also identify the two features of political economies – an open versus closed economy and extensive versus limited statist institutions – that are critical features of the macro-economic policy making environment. They help define the balance in economic decision making between electorally accountable decision makers and those who are not. Hence, both features should shape the size of the competency signal. In this chapter we provide an argument relating changes in the global economy and the scope of government

to the magnitude of the competency signal and, given the theory in Chapter 5, to the economic vote. This argument thus provides testable hypotheses about how specific economic and political contexts impact the economic vote.

Globalization, Competence and the Economic Vote

We begin by exploring the impact of globalization on the economic vote. The nature of the international economy has changed remarkably over the past three decades. Global trade as a percent of GDP has increased from 20 percent in the early 1970s to 55 percent in 2003 (IMF 2005).¹⁴⁶ But this varies rather dramatically by country. For example, the IMF trade intensity score for the EU is about .88, for the U.S. 1.05 and for Canada 2.93 (IMF 2005).¹⁴⁷ Amongst the industrialized nations between 1990 and 2003, the magnitude of foreign assets and liabilities has tripled to a remarkable 200 percent (IMF 2005). Not surprisingly, this has created an entire “industry” of scholars devoted to understanding the economic and political implications of these trends.¹⁴⁸ Our interest in rising globalization is two-fold. First, it is clear from our theory that this trend should impact the competency signal in some way – although the precise nature of the impact is not obvious. Second, these changes in the competency signal will in turn have implications for the magnitude of economic voting.

What are the implications for economic voting? Scholars who previously have addressed the question from various theoretical perspectives disagree. Lewis-Beck (1988) speculated that greater openness might sensitize voters to the importance of economic policies and thereby increase the economic vote. Scheve (2004) argued for the same relationship (higher economic voting in an open economy) although he suggested that it results from lower exogenous shocks (our σ_{ξ}^2 term) in open economies and therefore a higher overall competency signal. Hellwig’s (2001) analysis suggested exactly the opposite: Greater openness seems to dampen the positive impact of economic performance on incumbent vote. And this conforms to the earlier speculations of Lewis-Beck and Eulau (1985: 6) and, more recently, Hibbs (2006).

These are essentially disagreements over the relative magnitudes of the competence and exogenous shock terms in the competency signals generated in open and closed economy contexts. In Chapter 5 we have argued that the variation in these magnitudes across contexts

¹⁴⁶ Global trade here is defined as the sum of exports and imports of goods and services.

¹⁴⁷ Trade intensity is defined as the ratio of manufacturing trade to total manufacturing GDP (IMF 2005).

¹⁴⁸ Rodrik (1997a) summarizes many of the arguments that make up this literature. Fischer (2003) provides a review and careful assessment of the debates regarding the impact of globalization.

reflects the differences in the number of EDDs to NEDDs. As economies become increasingly integrated into the global economy, the variance in the competency shocks is likely to shrink because the number of electorally dependent decisions affecting the macro-economy decline (i.e. $\alpha_{op} < \alpha_{cl}$, where *op* refers to an open economy and *cl* refers to a closed economy). At the same time the number of non-electorally dependent decision makers increases ($\beta_{op} > \beta_{cl}$).

Capital mobility, for example, is one of the defining features of an open economy. And the Mundell-Fleming condition is the classic statement on how increased capital mobility reduces the number of EDDs affecting macro-economic policies and increases the impact of NEDDs.¹⁴⁹ This condition essentially states that a country can have at most two of the following three conditions: capital mobility, fixed exchange rates, and monetary policy autonomy (Frieden 1991). Hence, for example, having opted for capital mobility, if authorities reduce interest rates in order to stimulate the local economy, capital will flow out of the country in search of higher rates abroad and before this initiative can have its effect, domestic interest rates will have climbed back to world levels (Frieden 1991: 431). In order to retain autonomy over monetary policy authorities would need to abandon a fixed exchange rate regime. This condition does not eliminate the ability of governments in an open economy to enact monetary and fiscal policies (Oatley 1999; Garrett 1998a; 1998b; 2000) but rather restricts their options given the decision to, or not to, maintain fixed exchange rates.¹⁵⁰ The Mundell-Fleming condition suggests that the price paid by incumbent governments for participating in the global economy – capital mobility, for example – is a more restricted ability to manage the domestic economy.¹⁵¹ In terms of our model, the Mundell-Fleming condition suggests that once a country opens up to the global economy by allowing for capital mobility, it essentially cedes a decision option, either the ability to control its exchange rate or the ability to shape monetary policy – and one of these decisions is in effect relinquished to non-electorally dependent decision makers. As Clark et al (1998) point out, not

¹⁴⁹ See the early works of Mundell (1962; 1963; 1964). Our discussion is based on Frieden (1991).

¹⁵⁰ And while governments are not obligated to commit to fixed exchange rate regimes under a system of capital mobility many have taken this option to ward off financial speculation which in turn has restricted their fiscal policy options (Garrett 2000).

¹⁵¹ It is important to point out that we are not drawing any conclusions here regarding the impact of open economies on partisan differences in macroeconomic policies (in fact the evidence of Oatley 1999 suggesting that partisan differences persist is in our minds quite convincing) nor are we suggesting that open economies necessarily constrain the redistributive and overall taxation policies pursued by governments (Boix 1998 and Garrett 1998 provide convincing challenges to these arguments).

all countries opt for policies or institutions that increase exposure to these global forces, but once they do incumbent governments cede considerable control over managing the macro-economy.¹⁵²

While capital mobility is certain a defining feature of an open economy it is not the most visible manifestation to the average voter. The typical voter is much more likely to be cognizant of the increased trade flows that result from an open economy. The macro-economic indicators that typically attract the attention of the national media and politicians are those related to trade flows and trade balances. Trade flows have an impact similar to that of capital mobility on the ability of domestic decision makers to manage aggregate demand and control inflation (Cameron 1978; Lindbeck 1975; 1976). Aggregate demand in economies with significant dependence on export markets is subject to external demand and to prices that are determined by global markets beyond the influence of domestic policy makers. High levels of imports subject economies to inflationary (or deflationary) shocks that again are beyond the control of domestic governmental officials (Cameron 1978). Because trade typically implies specialization resulting from the forces of comparative advantage, the production structure in open economies is often more concentrated than in closed economies (Rodrik 1998; IMF 2005). This can exaggerate the extent to which domestic economies are subject to external shocks beyond the control of incumbent policy makers. Rodrik (1998), for example, presents empirical data clearly indicting that more open economies have significantly more volatile GDP growth.¹⁵³

Capital mobility and increased trade flows reduce the control that EDDs have on macro-economic policy outcomes and they subject domestic economic outcomes to greater influence by non-electorally dependent, and particularly foreign, decision makers. Accordingly, in an open economy the number of NEDDs affecting the macro-economy rises ($\beta_{cl} < \beta_{op}$) while the number of EDDs should remain constant or possibly be smaller in open economy than in a closed one (*i.e.*, $\alpha_{op} \leq \alpha_{cl}$). Hence, we expect the ratio of NEDDs to EDDs will be smaller in a closed economy,

$$(7.1) \quad \frac{\beta_{op}}{\alpha_{op}} > \frac{\beta_{cl}}{\alpha_{cl}}$$

¹⁵² And more recent efforts by proponents of the “new open economy macro-economics” have provided models of the precise impact of these global interdependencies on domestic output and inflation in open economies (Lane 1997). Good reviews of this literature include Lane (2001) and Sarno (2000).

¹⁵³ See also evidence to this effect for the developing economies in Gavin and Hausmann (1996).

which suggests a lower overall competency signal in economies that are open to global economic influences.¹⁵⁴ Given this, it follows from the theory in Chapter 5 that open economies will have lower levels of economic voting than closed economies.

To evaluate this hypothesis we employ a measure of Trade Openness from the World Bank which is a ratio of total trade to gross domestic product (GDP) (World Bank 2004). We think this is a particularly good measure because it both captures the objective constraints imposed on national economic policy makers and it is also the indicator that typically attracts the most attention of both the media and politicians. Figure 7.1 presents a plot of the *economic vote of the chief executive* against our measure of Trade Openness. First, there clearly is no evidence that the economic vote is higher in open economies as some have claimed (Scheve 2004). In fact, openness of the economy is associated with significantly smaller economic vote. This lends support to our argument that the competency signal in open economies is weaker than in closed economies which in turn leads to lower levels of economic voting. This is an important result because it suggests that recent rising levels of globalization imply that a declining importance for the economy in vote choice.

¹⁵⁴ And as Ebeid and Rodden (2006) nicely demonstrate this effect can differ at the sub-national level as a function of the structural features of provincial or state economies.

designed to maintain international competitiveness (government investment in human capital).¹⁵⁶ Adserà and Boix (2002) demonstrate the empirical relationship between openness and size of the government sector in democracies although they point out that the expansion of the government sector should be seen as a strategy for building a free-trade coalition. Some see this expansion of the state sector as enhancing democratic governance: a large role for the state increases the ability of citizens to hold decision makers accountable for the economy.

Alternatively, some suggest a liberal convergence in economic policies resulting from the competitive pressures of globalization: Because of global constraints, national governments are forced to liberalize the economy in response to similar initiatives by competing nations, resulting in what some characterize as a race to the bottom phenomenon. These liberal policies typically reduce the scope of government in the economy either through the introduction of more liberal regulatory regimes or privatization of state-owned entities. This “privatization” of domestic economies results in a more limited role of the state in shaping macro-economic outcomes. The conventional wisdom in this case is that a reduced role of government in managing the macro-economy results in lower democratic accountability.

These arguments raise three interesting puzzles regarding globalization, the government’s role in the economy and the economic vote: What happens to the scope of government in the face of rising globalization? Theoretically how does the scope of government shape the incumbent’s competency signal? What empirical relationship is suggested between scope of government and the economic vote?

We begin with the theoretical puzzle: the implications of an extensive, as opposed to limited, state sector for the competency signal. A policy context with an extensive state sector increases the number of domestic institutions and other constellations of political actors actively involved in making economic policies – in short, it produces a more “dense” policy making environment. This results in more “decisions” and more “decision-makers” that contribute to the competency signal that the voters perceive. For example, increasing the scope of government in the economy results in more government entities making decisions that affect the economy. These would include government-owned firms, authorities overseeing infrastructure or energy projects, health services providers, etc. At the same time, there are more decision makers shaping these policies: executives of government-owned firms, labor unions, competing firms,

¹⁵⁶ While it is true that more open economies tend to have an extensive state sector, its important to point out that this is an historical legacy and that increasing the role of government is not the only policy response to increased exposure to global competition. Garrett (1998) and Boix (1998), for example, describe the very different policy responses to global competition of the Left and Right in the advanced democracies.

local mayors, contractors, professional associations, NGOs, consumers, etc. Most of this expanded group of decision makers is not politically accountable. Accordingly more of the decisions affecting the economic competency signal perceived by voters are shaped by agents who are not politically accountable. This follows in part because in a context with an extensive state sector the benefits and costs associated with changing economic priorities are more likely to be borne by greater numbers of interested parties (Becker 1983; Buchanan and Tullock 1962).

Short of significant institutional changes designed to reinforce principal-agent oversight, as the scope of government increases, this proliferation of non-electorally accountable decisions is an inevitable result of the expansion of the government sector. This more dense policy making environment has an interesting, and somewhat counterintuitive, implication for the relationship between the breadth of the state sector and the economic vote. If we focus on the ratio of NEDDs

to EDDs, $\left(\frac{\beta}{\alpha}\right)$, as it rises the competency signal declines, as does the economic vote. We expect this ratio to be large in contexts with extensive as opposed to limited state sectors because an extensive state sector implies that a larger number of interested, and non-electorally-dependent, actors affect economically relevant decisions¹⁵⁷

A dense policy making environment implies a proliferation of actors that affect (or veto) macro-economic policy initiatives, many of whom are not electorally dependent or are not subject to discipline by elected officials – thus cannot (in the specific sense of our theory) be considered part of the “government” whose competence the voter is trying to determine. It is not simply the proliferation of veto players (Alesina and Drazen 1991; Henisz 2004; Olson 1982; Tsebelis 2002) that matters here but rather that an extensive state sector attracts the participation of many NEDDs as opposed to EDDs in the policy process. This argument has two important implications for the competency model. First, an extensive state sector will increase the number of the NEDDs. Second, while an extensive state sector may also raise the number of EDDs it will do so at a more moderate rate than the increase in NEDDs. As a result we expect the ratio $\left(\frac{\beta}{\alpha}\right)$ will be higher in extensive state, as compared to limited state, contexts.

Contexts with a more extensive state sector are hypothesized to have a smaller competency signal and less economic voting. The breadth of the state sector of course can be measured in a number of different ways. Accordingly, in order to test this argument we identify three distinct dimensions of state influence over the macro-economy and explore their impact on

¹⁵⁷ It is important to point out here that our argument has no necessary implications for the quality of economic policy outcomes – our only concern here are unanticipated shocks to the macro-economy.

the competency signal and the economic vote. First, we explain how a large state role in the economy dampens the magnitude of the overall competency signal and we provide empirical evidence that it reduces the size of the economic vote. Second, we explore how corporatist arrangements – which include an explicit expansion of the set of actors involved in macro-economic management – reduce the overall competency signal and provide empirical evidence that this reduces economic voting. Finally, we explain how increasing government regulatory oversight of the private sector reduces the overall competency signal and demonstrate empirically that it results in lower economic voting.

Size of the State Sector

The size of the state sector refers to the breadth of the state's role in economic transactions and covers such activities as wealth and income transfers, consumption of goods and services, and ownership of productive entities. Since World War II, developed democracies have experienced both a rise in the size of the state sector and a more activist role of the government in managing the macro-economy. Franzese (2002a) documents the considerable post-World War II growth in the size of government in the economies of the OECD nations. During this same period, Keynesian notions of macro-economic management became economic orthodoxy for the governments of most developed democracies (Hall 1989). The size of the government sector has become synonymous with “state capacity” – the ability of the state to manage the political economy (Krasner 1984). And as we pointed out earlier, many argue that open economies have embraced a state sector in large measure in order to compensate constituencies negatively affected by global economic shocks (Adserà and Boix 2002).

A reasonable inference to draw from this literature is that as the state's capacity to manage the economy increases so should the level of economic voting (Anderson 1995). As Lewis-Beck and Eulau speculated, “In the nations with more government economic involvement, one could anticipate a stronger association between economics and the vote, since their citizens would be more likely to attribute economic responsibility to government (Lewis-Beck and Eulau 1985: 5).” Tufte (1978) provided an early catalyst for this notion with his powerful images of Nixon's manipulation of spending on transfer programmes just prior to the 1972 U.S. presidential elections. A reasonable generalization is that big government, which implies more redistribution programmes, means an expanded opportunity to engage in Nixonian-type manipulations of the macro-economy. The implication here is that big government results in government making more economically consequential decisions and so results in a larger economic vote. Moreover, since the size of the government sector varies significantly cross-nationally (Franzese 2002a) this raises

the possibility that these differences might help explain cross-national variations in the economic vote.

Our argument regarding the impact of an expansive state sector on the overall competency signal, however, suggests exactly the opposite: big government should result in less economic voting.¹⁵⁸ Our case for the notion that big government reduces the competency signal rests on two assumptions proposed in the previous section:

$$(7.2) \quad \begin{aligned} &\beta_s < \beta_l \\ &\frac{\alpha_l}{\alpha_s} < \frac{\beta_l}{\beta_s} \end{aligned}$$

where l represents contexts with large government sectors and s represents contexts with small government sectors.

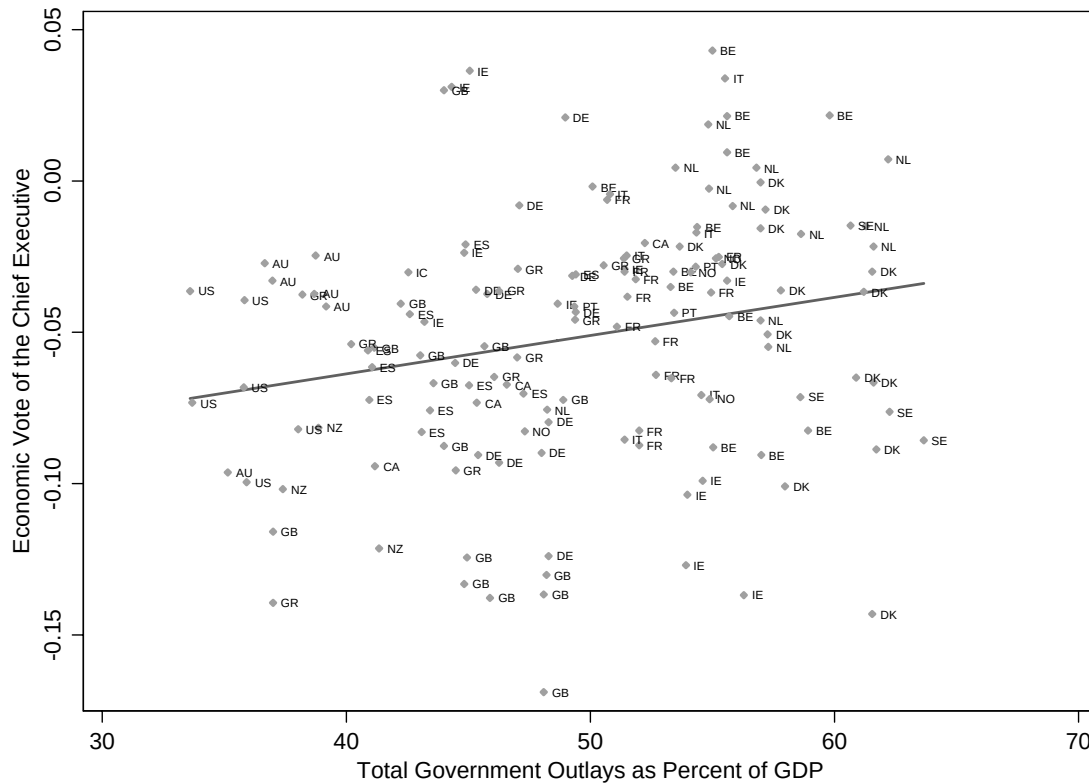
The first assumption simply suggests that there are more NEDDs made in contexts with big government. As government grows and becomes more complex, the number of NEDDs (interest groups, commissions, elected local government officials, for example) affecting macro-economic policies will increase significantly as will the volume of their decisions. We see this as a rather non-controversial – and easily verifiable – assumption. The second assumption simply recognizes that there are more EDDs in big government contexts but imposes the condition that the ratio of EDDs in large versus small government contexts is smaller than the ratio of NEDDs in large versus small government contexts. This results in the ratio of NEDDs to EDDs in contexts with small government being less than it is in contexts with large government,

$$(7.3) \quad \frac{\beta_s}{\alpha_s} < \frac{\beta_l}{\alpha_l}$$

The ratio in Equation (7.3) indicates that the exogenous shock term in large government contexts will be inflated because voters perceive NEDDs as having a disproportionate impact on shocks to the macro-economy. This of course depresses the overall competency signal, which we predict will result in a lower economic vote. Accordingly, economic voting should be negatively correlated with the size of the government sector. Figure 7.2 plots the *economic vote of the chief executive* against the magnitude of the government sector. It suggests that economic voting is

¹⁵⁸ We are not alone in arriving at this prediction. Hibbs (1993) argues that in large welfare states (which, as he points out, are also typically highly exposed to global economic influences) other policies or welfare state spending will dwarf the macro-economic demand management “signals” of the incumbent. As a result, there will be less economic voting in these contexts. Pacek and Radcliff (1995) argue that the size of government moderates economic voting although their explanation is quite different than ours.

higher in contexts with more limited government (remember that a large negative value indicates high economic voting). Note that countries with relatively small government sectors such as the U.S., the U.K., Greece, Canada, and Spain tend to have much higher levels of economic voting. Conversely, the big government states such as Denmark, the Netherlands, Sweden and Belgium have lower economic voting.



Source: General government total outlays as percent of nominal GDP (OECD, 2002)

Economic Vote of Chief Executive = $-0.11 (3.93) 0.001 (2.06) \times (\text{Government Outlays/GDP})$. $R^2=0.04$.
Observations=146.¹⁵⁹

Figure 7.2. Size of Government and Economic Vote of the Chief Executive

Big government implies a lower overall competency signal because the ratio of NEDDs to EDDs is greater than in contexts with a smaller government sector. As the competency theory predicts, this results in lower levels of economic voting in contexts with a large government

¹⁵⁹ Estimates are from an OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. T-statistics are in parentheses.

sector. Let's now take a closer look at the notion that this results from a relatively larger number of non-electorally dependent decision makers affecting macro-economic policy outcomes.

Corporatism and Coordinated Wage and Price Bargaining

Falling within our rubric of statist institutions are formal and informal arrangements requiring the participation of key economic actors in major economic policy making – typically referred to in the literature as corporatism or coordinated wage and price bargaining. A number of post-World War II developed economies adopted corporatist policies designed in large part to moderate the magnitude of shocks to the macro-economy. For the most part these were open economies that were particularly vulnerable to global economic shocks (Katzenstein 1985). These initiatives are described by Hall as a “...process of social or economic policy making in which considerable influence over the formulation or implementation of policy is devolved onto the organized representatives of producer groups, often by means of peak-level bargaining about wage settlements” (Hall 1999: 138).¹⁶⁰ By expanding the role of government to include the negotiation of “social contracts,” which typically focused on wages but could include a range of other policies, the expectation was that government could ensure satisfactory levels of real economic growth, restrain inflationary pressures, and avoid excessive levels of unemployment (Franzese 2002a, chapter 4).¹⁶¹

The institutional actors associated with these efforts to promote coordinated wage and price bargaining (CWB) are a good example of the NEDDs in our competency signaling model. Corporatist, or CWB, institutions are explicitly designed to increase the number of non-electorally dependent decision makers influencing macro-economic policy outcomes. In corporatist contexts, key macro-economic policies – price and wage policies – are subject to relatively non-partisan negotiation amongst a country's leading interests groups. And corporatist institutions significantly constrain how capital is employed within the country, ranging from investment and disinvestment decisions, wage settlements and employee benefits. The implications of these institutional features for our competency signal are quite straightforward. First, they clearly imply that the overall number of economically important decisions in a corporatist setting (*cw*), compared to a more market oriented setting (*m*), is shifted from EDDs to

¹⁶⁰ Students of European political economy, in particular Schmitter and Lehmbruch (1979) and Lehmbruch and Schmitter (1982) have elaborated a set of measures of neocorporatism and have demonstrated that policy outcomes are correlated with variation in levels of neocorporatism.

¹⁶¹ There is a general consensus in the literature that levels of CWB have in fact declined over the past two decades although there is some debate as to how quickly and since when (Franzese 2002b; Golden et al 1995).

NEDDs (even though these decisions might be seen as “governmental” in some sense). This has the effect of increasing the number of NEDDs (β_{cw}) and decreasing the number of EDDs (α_{cw}) in a corporatist context. Accordingly,

$$(7.4) \quad \begin{aligned} \beta_m &< \beta_{cw} \\ \alpha_m &> \alpha_{cw} \end{aligned}$$

This implies that the ratio of NEDDs to EDDs will clearly be higher in a corporatist than a more market oriented setting,

$$(7.5) \quad \frac{\beta_m}{\alpha_m} < \frac{\beta_{cw}}{\alpha_{cw}}$$

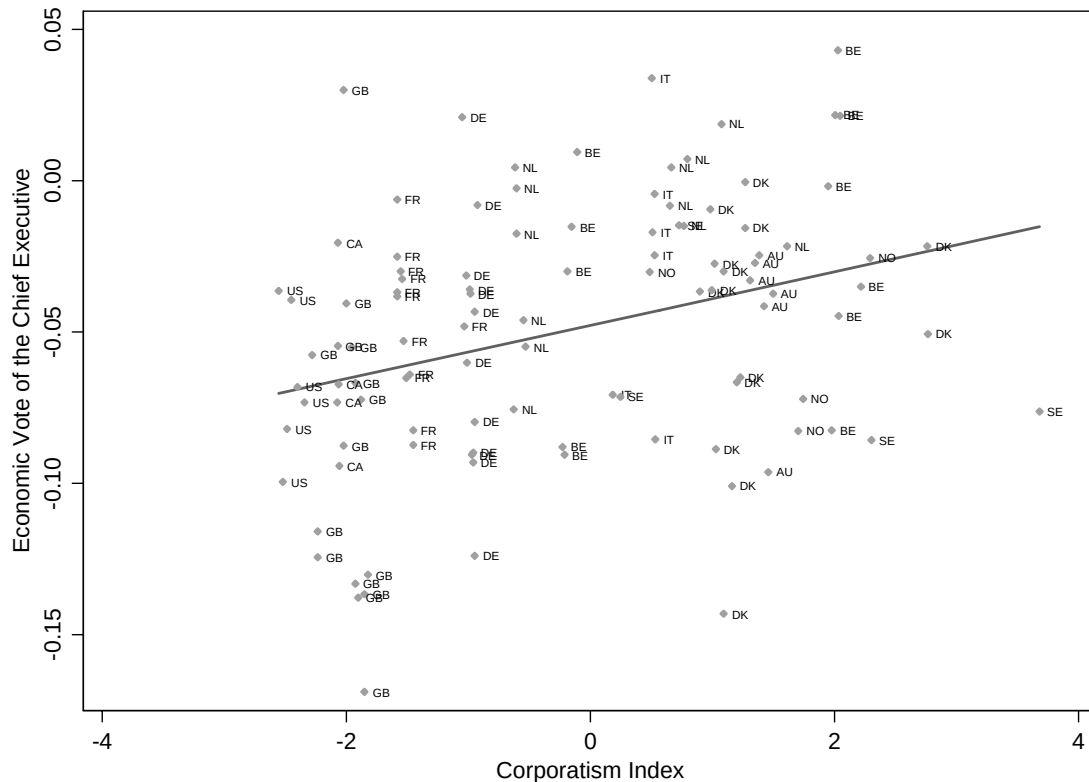
And this in turn suggests that the overall competency signal is higher in contexts with more market-oriented institutions as opposed to those with high degrees of corporatism. Because they have an higher overall competency signal, contexts with a low level of corporatism should have a larger economic vote than contexts with high levels of corporatism.

Some readers might find it odd that we suggest that there is more electorally accountable economic decision makers in unregulated versus corporatist labor markets. Any confusion can be resolved by drawing a careful distinction between governmental and electoral accountability. In a free market for labor, the government makes some decisions and is electorally accountable for them (say investment in education). But in a corporatist system these decisions, and many others, become party to the overall societal bargain that is as much the responsibility of labor unions and business as it is of the politicians.

Corporatism is measured by an index developed by Golden (2000). The index consists of the factors scores from a principal component factor analysis of four measures that Golden (2000) demonstrates empirically as representing an underlying corporatist dimension: government involvement in private-sector wage bargaining (a fifteen-point scale with 1 indicating no government involvement at all and a 15 indicating government imposed wage freezes and prohibition of supplementary bargaining); bargaining level (a four-point scale with 1 indicating plant-level bargaining and 4 indicating a centralized bargaining level); confederal involvement (an eleven-point scale measuring the extent to which private-sector wage negotiations are determined by bargaining amongst the confederation of unions – a low score indicating no

confederation involvement at all); and adjusted union density (union membership with unemployed, self-employed and retired members excluded).

Figure 7.3 presents a plot of the *economic vote of the chief executive* against an index of corporatism constructed by Golden (2000). Clearly high levels of corporatism tend to reduce economic voting. This reinforces our contention that the competency signal is higher in contexts with limited statist institutions, which in turn generates higher levels of economic voting.



Source: Corporatism Index (Miriam Golden, 2000)

Economic Vote of Chief Executive = $-0.05 (8.65) 0.008 (2.64) X (\text{Government Outlays/GDP})$. $R^2=0.04$.
Observations=107.¹⁶²

Figure 7.3. Corporatism and Economic Vote of the Chief Executive

Regulation Density

The third dimension of an extensive state sector is regulation density. Our reasoning here is similar to our argument regarding size of government. Regulating a complex economy poses a daunting

¹⁶² Estimates are from an OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. T-statistics are in parentheses.

agency problem for elected decision makers (Spiller 1990). We assume that there are more NEDDs in contexts with dense regulations. As regulatory oversight of the private sector grows and becomes more complex elected officials delegate increasing amounts of authority to non-elected regulatory agents; hence the number of NEDDs affecting macro-economic policies will increase significantly. Initiating, or increasing, the regulatory oversight of a particular industry, for example, will result in more NEDDs participating in economic policy decisions. This manifests itself in a variety of different forms: industry lobbying efforts, the organization of consultative entities designed to facilitate the government's regulatory oversight tasks or the formation of independent entities that are given responsibility for regulatory activities. All of these different byproducts of increased regulatory oversight have in common the fact that they diffuse policy making responsibility to a wider range of NEDDs.

Again this reasoning can be summarized in our theoretical notation: there are more NEDDs made in contexts with dense regulation. Secondly, while there are more electorally dependent decisions in dense regulatory contexts, the ratio of electorally dependent decisions in dense (dr) versus limited (lr) regulatory contexts is smaller than the ratio of non-electorally dependent decisions in dense versus limited regulatory contexts. This results in the ratio of NEDDs to EDDs in contexts with limited regulation being less than it is in contexts with dense regulation,

$$\beta_{lr} < \beta_{dr}$$

$$(7.6) \quad \frac{\alpha_{dr}}{\alpha_{lr}} < \frac{\beta_{dr}}{\beta_{lr}}$$

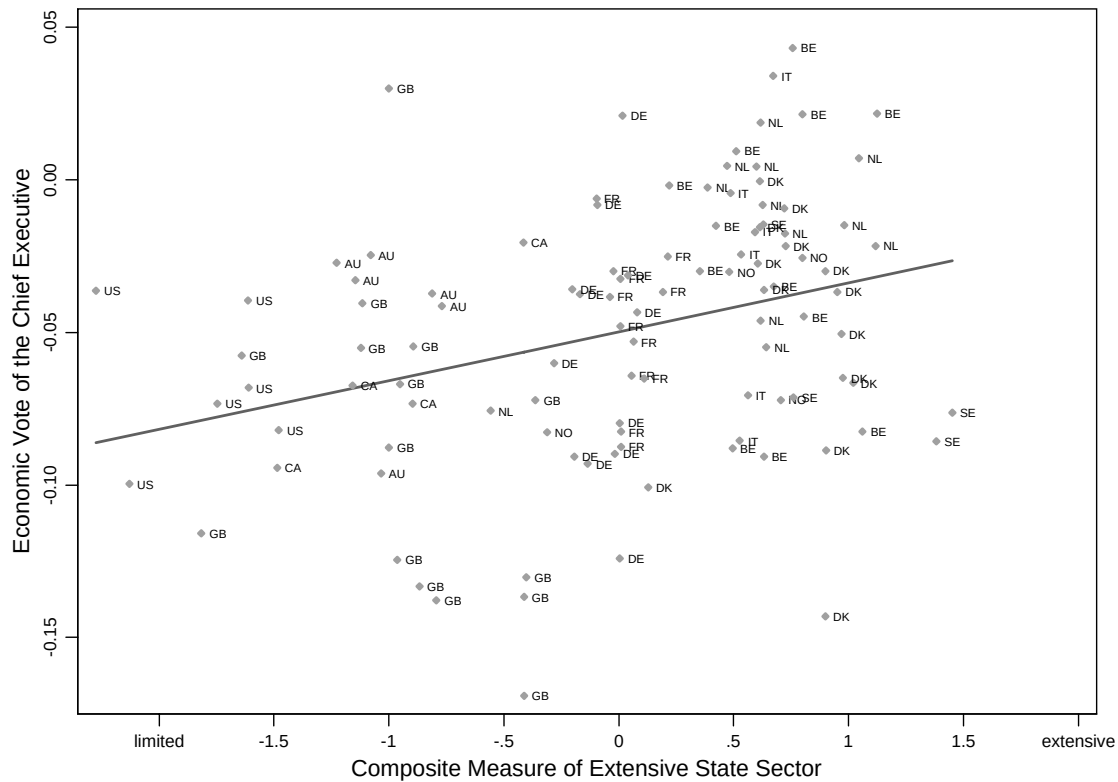
$$\frac{\beta_{lr}}{\alpha_{lr}} < \frac{\beta_{dr}}{\alpha_{dr}}$$

Figure 7.4 plots the economic vote against regulation density. Regulatory density is measured by a composite index of the extent of government regulation of credit markets, labor markets and businesses – the index ranges from 0 to 10 with 0 indicating a maximum of government regulatory oversight (Gwartney and Lawson 2006). Note that a low score on the regulation density variable suggests a highly dense regulatory environment. Accordingly the plot suggests that the economic vote is suppressed in contexts with high levels of regulation: hence countries with a relatively low degree of government regulatory oversight tend to have high levels of economic voting. These results reinforce our contention that the competency signal is higher in contexts where the state sector is more limited, which results in more economic voting.

and hence economic voting, are lower in contexts with an extensive state sector. The hypothesis is worth pursuing further.

We can think of our three measures of the policy context as dimensions of an underlying structural feature of political economies – a limited versus extensive state sector. Factor analysis of these three items indicates that they are tapping this underlying feature of a limited state sector. The factor analysis results in a single dimension with each item having a reasonably high loading. The factor loadings of the three measures are as follows: Regulation Density is -.67; Corporatism is .59; and Government Size is .80. The Alpha reliability statistic for these three items is 0.63, again confirming that it is reasonable to treat these items as measuring a similar underlying concept.

Figure 7.5 examines the relationship between the *economic vote of the chief executive* and the factor scores resulting from our factor analysis of the three statist items. The relationship is quite strong and in the expected direction: contexts with a more extensive state sector have lower levels of economic voting.



Limited government factor score from factor analysis of CWB, Regulation Density and Government Outlays/GDP

Economic Vote of Chief Executive = $-0.05 (9.10) + 0.02 (3.08) \times \text{Composite Measure of Extensive State Sector}$. $R^2=0.10$. Observations=107.¹⁶⁴

Figure 7.5. Extensive State Sector and Economic Vote of the Chief Executive

A plausible alternative argument here is that our measure of scope of the state sector is actually capturing the extent to which the state has adopted social transfer policies designed to cushion the impact of economic shocks and that this safety-net effect is actually moderating the magnitude of the economic vote rather than the impact of the ratio of NEDDs to EDDs. This is an argument made by Pacek and Radcliff (1995). We can gain some insight into whether this is the case by identifying what proportion of government expenditures represent social benefits and social transfers. If much of the relationship between our measure of the scope of government and the economic vote is the result of a safety-net effect then including this as a control variable in the simple regression in Figure 7.5 would significantly attenuate the relationship. In fact, when we

¹⁶⁴ Estimates are from an OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. T-statistics are in parentheses.

include this variable in the equation, the relationship between the limited state sector variable and the economic vote is slightly stronger.

$$(7.7) \quad \begin{aligned} \text{Economic Vote} = & -.11 (.03) + .02 (.0006) * \text{Limited State Sector} \\ & + .16 (.07) * (\text{Social Benefits/Government Expenditures}) \\ R^2 = & .12. \quad N=108 \\ \text{Standard errors are in parentheses.} \end{aligned}$$

Interestingly, though, the percent of government expenditures accounted for by social benefits and social transfers has a positive and significant coefficient confirming the earlier findings of Pacek and Radcliff (1995).

Another test of this alternative argument is to substitute the measure of government expenditures as a percent of GDP employed in Figure 7.2 with a measure that includes only expenditures not associated with social benefits or transfers. This results in a weaker relationship with the economic vote but one that is still positive as hypothesized although slightly below conventional levels of significance – the t-statistic is 1.5.

There is no disputing the fact that the economic vote is lower in contexts with a more extensive state sector. Moreover, this clearly challenges conventional notions that big government, in itself, signals incumbent responsibility for economic outcomes and hence should raise levels of economic voting. Rather, the signal that conditions the economic vote is the relative magnitude of the variance of the competency shock to the variance of exogenous shocks to the macro-economy. As the number of EDDs relative to the number of NEDDs in a national context rises, the overall competency signal gets larger. It turns out that this ratio is larger in contexts with limited statist institutions and hence, some might say paradoxically, the overall competency signal is stronger in contexts with limited statist institutions, as is the economic vote.

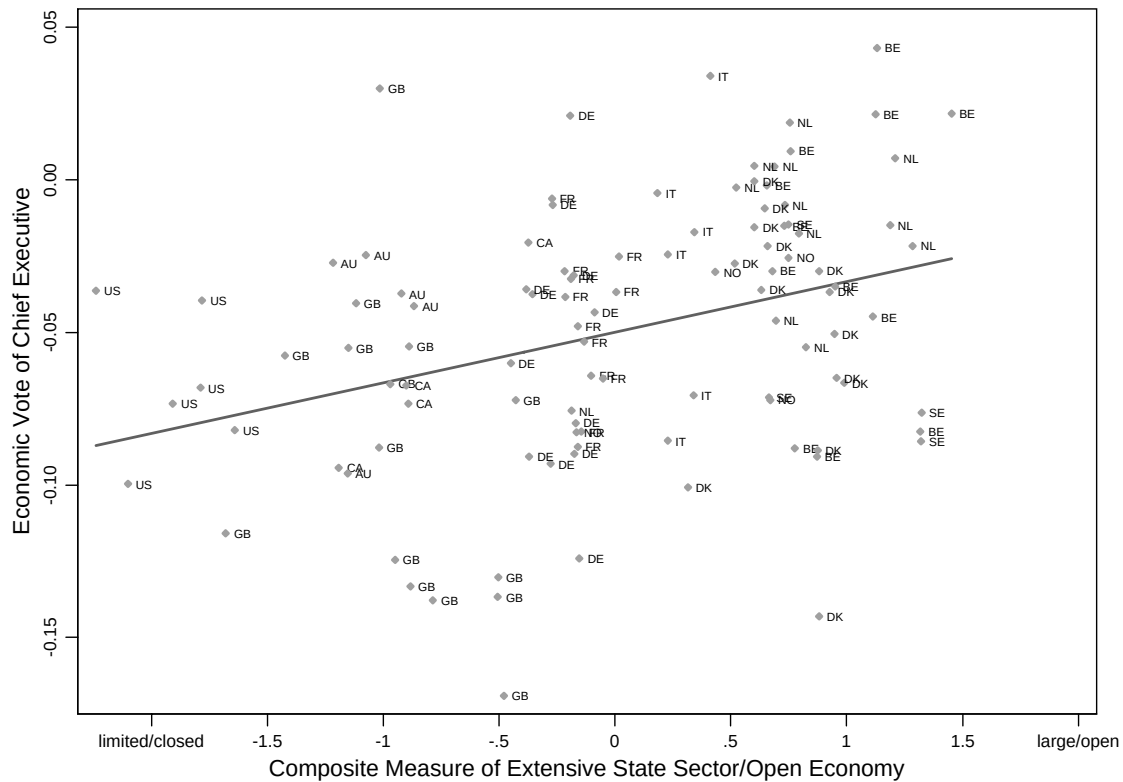
An Open Economy and an Expansive State Sector Depress the Economic Vote

Globalization and an expansive state sector represent two of the most prominent changes experienced by developed democracies in the post-WWI period. There is a debate as to whether increased globalization has resulted in 1) policy convergence around liberal economic policies, designed to enhance competitiveness, or 2) the expansion of the scope of government, seen as a way to cushion the social dislocation resulting from global competition. Given our empirical results this controversy has implications for overall levels of economic voting in developed democracies. The notion that openness to the global economy and the size of the government sector are positively correlated implies that these two variables reinforce each other's impact on

the economic vote. Alternatively, if these two variables are negatively correlated – resulting from the fact that increased globalization engenders a liberal economic policy response – then their contextual effects on the economic vote would cancel out. With globalization likely to continue increasing this has interesting implications for economic voting and, by extension, democratic accountability. The former scenario – a positive correlation between globalization and government size – implies declining levels of economic voting while the later scenario – a negative correlation – suggests stable levels of economic voting in the face of rising globalization.

We now explore these two possible empirical scenarios. The first scenario suggests that democracies may fall along a single dimension characterized by open economies with an expansive state sector at one extreme and economies less open to global trade and with a limited state sector, at the other. A factor analysis of all four variables described above (the open economy item and the three limited government items) confirms that this is the case – we obtain a single factor dimension with each item exhibiting a high loading.¹⁶⁵ Figure 7.6 shows that this composite measure of scope of the state sector and trade openness is significantly correlated with the *economic vote of the Chief Executive*. Hence, countries with exposure to global trade will likely also have an expansive state sector; and both of these characteristics will have negative affects on the economic vote. It is not unreasonable then to distinguish national contexts in terms of this dimension: contexts with an open economy and an expansive state sector will have relatively low levels of economic voting while contexts with a closed economy and a limited state sector should have a high economic vote. If globalization continues to increase, and its relationship with size of government persists, this suggests declining levels of economic voting in the future.

¹⁶⁵ The item factor loadings are as follows: trade openness is .60; government outlays is .83; corporatism is .65; and regulation density is -.62.



Composite measure of extensive state sector/open economy factor analysis of Open Economy, CWB, Regulation Density and Government Outlays/GDP

Economic Vote of Chief Executive = $-0.05 (9.62) + 0.02 (3.42) \times \text{Composite Measure of Extensive State Sector/Open Economy}$. $R^2=0.12$. Observations=106.¹⁶⁶

Figure 7.6. Extensive State Sector/Open Economy and Economic Vote of Chief Executive

Multi-level Model Results

This is an important result because it supports the underlying competency model that we described in Chapter 5; it confirms rather counterintuitive predictions derived from the theory in Chapter 5 regarding the impact of globalization and scope of the government sector on the economic vote; and it suggests future trends in economic voting for the developed democracies. Accordingly, we assess the robustness of the finding by estimating a pooled multi-level model. To implement this we combined individual level data from our 163 electoral surveys (into a pooled or “stacked” data set) to estimate a single logistic model of the individual level vote for or against the Chief Executive party. This vote was modeled as a function of economic perceptions and a much reduced set of control variables that were available in most of our surveys. The

¹⁶⁶ Estimates are from an OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. T-statistics are in parentheses.

requirement that we have a set of common control variables eliminated 14 studies, so that 146 studies remain in the pooled data set. We estimate a sparse individual level model: it includes an *Ideology* variable which is the de-meaned left-right self-placement of the voter (coded so higher numbers indicate a more right leaning voter) and its interaction with a dummy variable (*CE_Ideology*) indicating whether the chief executive's party is a leftist party.¹⁶⁷ We expect then that the coefficient on left-right self-placement will be positive and its interaction with the ideology of the prime minister to be negative. The variable measuring retrospective perceptions of the national economy has been broken into three dummy variables: *worse*, which equals one if voters thought the economy had gotten worse in the last year; *better*, which equals one if they thought the economy had gotten better; and *same*, which equals one if they thought the economy had stayed the same. Two of these, *better* and *worse*, were included in the individual level model. We expect these two dummy variables to have opposite signs with *worse* being negative and *better* being positive. The pooled model is thus limited to one main control variable – the ideology of the voter relative to that of the prime minister. This sparseness in specification is the cost one pays for pooling data from disparate surveys.¹⁶⁸

We also include a variable, *size of chief executive's party*, which is simply the average support for the chief executive's party in the survey. This is constant for each survey and essentially allows the constant-only model (which captures the probability that an ideologically average voter will vote for the chief executive's party) to vary based on the general level of support for the chief executive's party in the population.

More specifically, the basic form of the model illustrated here with the *limited state sector* contextual variable, is:

$$\begin{aligned}
 v_{ik} &\sim \text{Bin}(\pi_{1ik}) \\
 \text{logit}(\pi_{1ik}) &= \beta_{0k} + \beta_{1k} \text{Worse}_{ik} + \beta_{2k} \text{Better}_{ik} + \\
 &\quad + \phi_1 \text{Ideology}_{ik} + \phi_2 (\text{Ideology}_{ik} * \text{CE_Ideology}_k) \\
 \beta_{0k} &= \gamma_{00} + \gamma_{01} \text{Size}_k + \gamma_{01} \text{LS}_k + \omega_{0k} \\
 \beta_{1k} &= \gamma_{10} + \gamma_{11} \text{LS}_k + \omega_{1k} \\
 \beta_{2k} &= \gamma_{20} + \gamma_{21} \text{LS}_k + \omega_{2k}
 \end{aligned}
 \tag{7.8}$$

¹⁶⁷ In order to include as many studies as possible, we had to combine some left-right self-placement questions that gave the responded different response options. This was done by normalizing the various scales and then using the normalized scales.

¹⁶⁸ For a detailed comparison of the two-stage estimation strategy and the conventional multi-level interaction method using these data, see Duch and Stevenson (2005).

where *Bin* just indicates the dependent variable is distributed according to an appropriate binomial distribution with parameter π_{ijk} . *LS* is *limited state sector* and we assume the errors are distributed according to a multivariate normal distribution, with variances $\sigma^2_{\omega 1}$, $\sigma^2_{\omega 2}$, and $\sigma^2_{\omega 3}$ and covariances $\sigma^2_{\omega 2, \omega 2}$, $\sigma^2_{\omega 1, \omega 3}$, and $\sigma^2_{\omega 2, \omega 3}$. Since we expect support for the chief executive to be negatively related to economic perceptions, we expect $\gamma_{10} < 0$ and $\gamma_{20} > 0$ (recall that the baseline category is voters who think the economy has stayed the same). Further, since a more *limited state sector* is expected to increase the size of the economic vote, we expect $\gamma_{11} > 0$ and $\gamma_{21} < 0$ (since this will decrease the difference between the overall impact of better versus worse perceptions).

, which equals one if s, which equals one if theyThe multi-level interaction results reported in Table 7.1 strong studies. The coefficient on the *economy got better* dummy variable is positive and statistically significant; and the *economy got worse* variable has a statistically significant negative coefficient.

Model 2 adds two interaction terms to the core equation: the *limited state sector* variable interacted with each of the *economic perception* variables. The coefficients on these interaction terms are exactly as we would expect: In the case of the *limited state sector* variable interacted with the *economy got better* variable, the coefficient is negative and significant suggesting that the impact of this economic perception variable is lower in contexts with more extensive government involvement in the economy (recall that the *limited state sector index* has a high value for an expansive state sector). The *limited state sector* variable interacted with the *economy got worse* variable has a positive coefficient suggesting that the impact of negative perceptions is moderated in contexts with a large government involvement in the economy. There is more economic voting in contexts with a limited state sector.

Model 3 examines the hypothesized impact of an open economy on economic voting. Again, these results reinforce our earlier conclusions: more open economies have less economic voting. *Trade openness* interacted with the *economy got better* variable has a negative coefficient suggesting that the impact of positive economic perceptions on vote choice is lower in contexts with higher exposure to global economic influences. And the positive coefficient on the interaction with the *economy got worse* variable suggests that the impact of negative economic

¹⁶⁹ The estimates were obtained using the PQL second-order linearization method outlined in Goldstein (1995). Diagnostics on estimated residuals at Level 2 suggest that the assumption of normal variance in the level two coefficients is not violated.

perceptions on the vote is lower in open economy contexts. This confirms that there is more economic voting in contexts less exposed to global influences.

Table 7.1 Multi-level Logistic Regression Model of Incumbent Vote

	Model 1	Model 2	Model 3	Model 4
Left-Right Self Placement (dev. from mean)	0.48 (.01)	0.48 (0.01)	0.49 (0.01)	0.48 (0.01)
Left-Right Self Placement X Leftist PM (Leftist PM is an indicator variable)	-.74 (.01)	-0.87 (0.01)	-0.75 (0.01)	-0.88 (0.01)
Voter Perceives Economy got Better (indicator variable)	.43 (.03)	0.46 (0.03)	0.80 (0.07)	0.45 (0.03)
Voter Perceives Economy got Worse (Indicator variable)	-.55 (.03)	-0.57 (0.04)	-0.86 (0.07)	-0.56 (0.04)
Limited State Sector Index		-0.02 (0.04)		
Voter Perceives Economy got Better X Limited State Sector Index		-0.23 (0.04)		
Voter Perceives Economy got Worse X Limited State Sector Index		0.22 (0.05)		
Trade Openness Ratio			-0.002 (.001)	
Voter Perceives Economy got Better X Trade Openness Ratio			-0.005 (0.001)	
Voter Perceives Economy got Worse X Trade Openness Ratio			0.004 (0.001)	
Composite Measure				-0.03 (0.04)
Voter Perceives Economy got Better X Composite Measure				-0.23 (0.04)
Voter Perceives Economy got Worse X Composite Measure				0.20 (0.04)
Size of chief executive's Party	5.22 (0.22)	6.15 (0.28)	5.24 (0.23)	6.12 (0.28)
Constant	-2.39 (.07)	-2.70 (0.08)	-2.28 (0.10)	-2.70 (0.08)
$\sigma^2_{\omega 1}$	0.09	0.06	0.09	0.06
$\sigma^2_{\omega 2}$	0.08	0.06	0.06	0.06
$\sigma^2_{\omega 3}$	0.10	0.08	0.08	0.07
$\sigma_{\omega 1, \omega 2}$	-0.03	-0.02	-0.03	-0.03
$\sigma_{\omega 1, \omega 3}$	0.04	-0.02	-0.03	-0.02
$\sigma_{\omega 2, \omega 3}$	-0.05	-0.04	-0.03	-0.04
N	201,876	201,876	201,876	201,876

Standard Errors are listed below the logit coefficients. All coefficients are statistically significant at $p < .05$ except the shaded cells.

The results for the *trade openness* and *limited state sector* variables in Models 2 and 3 are consistent with each of the individual bi-variate analyses presented earlier. In the previous section we also provided evidence that countries typically fall along a single continuum characterized by an open economy with an expansive state sector on one extreme and a closed economy with limited government on the other. Accordingly we created a *composite measure* of state sector expansiveness and trade openness. Model 4 in Table 7.1 confirms that this composite

measure has the hypothesized effect on the magnitude of the economic vote clearly suggested by Figure 7.6: The coefficient on this *composite variable* interacted with negative retrospective evaluations is positive and significant while the coefficient on the interaction of this variable and positive evaluations is negative and significant. The *composite variable* interaction term decreases the difference between the overall impact of better versus worse perceptions. This is further strong confirmation that countries with high scores on this *composite measure* – have high exposure to global trade and an expansive state sector – will have less economic voting.

This estimates clearly support the hypothesis, since all the coefficients are statistically significant and in the expected direction. Thus, this pooled, multi-level model tells exactly the same story as the second-stage estimates from the two-stage method: Economic voting is depressed in contexts with a more expansive state sector and an open economy.

Summary

Voters in our competency model condition their economic vote on signals regarding the competency of incumbents. One of the results from Chapter 5 indicates that as the ratio of decisions affecting economic outcomes by non-electorally dependent decision makers (β) to those by electorally dependent decision makers (α) declines, the overall competency signal will rise and vice-a-versa. This theoretical result provides an important explanation for some of the contextual variation in the economic vote that we documented in Chapter 3. The theory does not specify precisely which national contexts have a high versus low ratio of NEEDs to EEDs. This chapter identifies features of the political economic context that correspond to our theoretical notion of EED versus NEED influences over economic outcomes. We also provide empirical evidence that our measures of political economic context explain variations in the economic vote.

While not using our explicit theoretical jargon, Cameron's (1978) early essay on open economies essentially makes one of our principal empirical arguments: Open economies reduce the number of electorally dependent decisions over macro-economic outcomes which implies a weaker competency signal and hence should lead to lower levels of economic voting. The empirical results presented in this chapter clearly suggest this is the case.

As Cameron and others have also pointed out, open economies often enact policies that expand the role of the state in the economy. Some see this broadening of the scope of statist institutions as compensating for the erosion of incumbent control over economic outcomes and for the decline in democratic accountability resulting from globalization (Freeman 2006). To the extent that economic voting can be seen as a measure of democratic accountability, both our theoretical and empirical results are at odds with this perspective. With respect to economic

voting, our theory suggests that broadening the scope of the state in fact further weakens the competency signal of incumbent governments, i.e., reinforces the weakening effect of the open economy on the competency signal. We arrive at this prediction because we assume that adopting more extensive statist institutions increases the ratio of NEED versus EED because it results in a larger number of NEEDs participating in the economic decision making process.¹⁷⁰ Our theory maintains that this increased ratio of NEED to EED lowers the incumbent government's competency signal and should result in lower economic voting which could be an indicator of less democratic accountability. Again, the empirical results are quite supportive: voters in contexts with extensive statist institutions – measured as the size of the government sector, the extent of corporatism and the extent of regulatory density – have less economic voting.

This result has important implications for the debate regarding globalization and democratic governance because it provides a rigorous explanation for how globalization affects vote choice. The net implications of globalization for democratic accountability depend on whether greater exposure to the international economy occurs in a context with an extensive as opposed to more limited state sector. A larger government sector may in fact moderate the negative social consequences of globalization but our competency theory suggests that broadening the scope of government's involvement in the economy reinforces the negative consequences of trade openness on democratic accountability. More specifically, increasing the size of the government sector, embracing corporatist institutions and increasing government regulation reduce the magnitude of the incumbent's competency signal leading voters to discount the importance of the economy in their voting decision. Paradoxically, countries responding to increased exposure to global economic forces by privatizing their economies and adopting liberal economic policies are more likely to counteract the erosion of democratic accountability associated with globalization.

Accordingly the debate as to whether increasing globalization will result in policy convergence around liberal economic policies rather than the expansion of the scope of government has important consequences for trends in economic voting in particular and democratic accountability more broadly. We find that openness to the global economy and the size of the government sector are positively correlated. It is not the case that countries with high levels of trade openness have privatized their economies and adopted liberal economic policies.

Our analyses suggest that democracies fall along a single dimension characterized by

¹⁷⁰ Of course this assumption that contexts with an expansive state sector have a high ratio of NEED to EED is not empirically evaluated in this book. Nevertheless, as we point out, the assumption is consistent with our understanding of how voters learn about the competency signal and it is of course consistent with the empirical data presented in this chapter.

open economies with an expansive state sector at one extreme and at the other extreme democracies with less exposure to global trade and with a limited state sector. These two trends – a more open economy and an expansive state sector – seem to reinforce each other’s negative impact on democratic accountability. This of course is a static snapshot of current global realities for advanced democracies of the world. But if this current correlation between the scope of government and exposure to global economic influences persists, then as globalization increases we can expect declining levels of economic voting and a growing crisis of democratic accountability.

Part III

A Contextual Theory of Rational Retrospective Economic Voting: Strategic Voting

The previous section explored the conditions under which rational voters can extract signals about incumbent competence from movements in the economy. The theory pointed to specific economic contexts that should condition the strength of the competency signal and, consequentially, the importance of the economy in the vote choice. Further, our empirical analysis was largely consistent with those implications. We now propose to explore two aspects of economic voting that were not developed in the model in Chapter 5.

First in Chapter 8, we generalize the model in Chapter 5 – where only one party was responsible for economic decision making – to multi-party contexts in which policymaking authority may be shared among many – or even all – parties (including opposition parties). We show that in political contexts in which policymaking authority is widely shared, the rational voter's ability to extract an informative signal about the competence of any one party declines. Chapter 9 explores empirically how variations in the distribution of policy-making responsibility amongst parties currently in government affect the economic vote.

The model developed in Chapter 5 assumed two parties competing for total control of executive authority. Chapter 8 will generalize this model of rational economic voting to contexts in which there are multi-parties (rather than simply two) competing for election and where executive authority is shared (rather controlled by a single party). This multi-party competition and shared executive authority necessarily introduces incentives for strategic voting. We explore the implications of these incentives by both applying McKelvey and Ordeshook's (1972) model of strategic voting incentives in multi-party contexts to the model in Chapter 5 and by generalizing it to cases in which coalition cabinets can form (or where policy making responsibility is shared amongst all parties). We show in Chapter 10 that the pattern of contention for future executive authority is important for the strength of economic voting. Further, we show that in cases in which the incentives for economic voting created by these two kinds of contexts are opposed, the incentives created by the pattern of competition for future office can trump those associated with the distribution of policy-making responsibility.

Chapter 8

Responsibility, Contention and the Economic Vote

Multi-Parties with a Plurality Race for a Single Party Executive

We begin by considering the case where multiple parties are competing for a single party executive – typically the case in the UK or Canada, for example. Building on the notation introduced in Chapter 5 we now use the letter “ k ” to refer to a generic opposition party (since there can now be more than one) and let $\mathbf{A}$ be the set of all opposition parties. By the same logic that motivated Equation 5.11 in Chapter 5, we can write the voter’s expected utility for any of these parties gaining office as:

$$(8.1) \quad E[u_{k,t+1}] = b\bar{y} \quad \forall k \in \mathbf{A}$$

Since voters are rational expected utility maximizers, in a multiparty context we must allow for the possibility of strategic voting. McKelvey and Ordeshook (1972) have addressed how utility maximization is achieved in exactly this kind of decision-theoretic, multiparty plurality situation. Specifically, McKelvey and Ordeshook show that when voters face a decision problem in which they must choose between multiple parties and care only about the winner of the plurality election, the difference in the expected utility from voting for some party j versus that of abstaining is:¹⁷¹

$$(8.2) \quad E[u | v_j] - E[u | v_0] = \sum_{j' \in \mathbf{J}} P_{jj'}(u_j - u_{j'})$$

Where $\mathbf{J}$ is the set of all parties in the legislature and $E[u | v_j]$ is the expected utility from voting for party j , $E[u | v_0]$ is the expected utility from abstaining, and u_j and $u_{j'}$ are the utilities associated with parties j and j' winning the election. $P_{jj'}$ is a “pivot probability” and is defined as the probability that parties j and j' are tied for first place in the plurality race. The subscript on the summation notation is how we will indicate that the sum is over all parties in the election.¹⁷²

¹⁷¹ They also assume that the probability of a tie between three or more parties is zero.

¹⁷² Any theory in which strategic voting in mass electorates is taken seriously is subject to the criticism that rational voters should never believe their vote is decisive in these kinds of elections. Most scholars who use models of rational voting no longer spend much time defending this assumption, since strategic voting in mass elections is a demonstrated fact (e.g., Cox 1997; Duch and Palmer 2002b; Cain 1978).

McKelvey and Ordeshook (1972) show that utility maximization requires a rational voter to vote for the party for which $E[u | v_j] - E[u | v_0]$ is the largest.¹⁷³ The intuition captured in this equation is straightforward: the act of voting for a party can only change a voter's utility by changing the outcome of the election. The likelihood of a voter's vote for a particular party changing the outcome of the election is captured by the probability that this vote is decisive in the contest between this party and each of the other competing parties. Consequently, the voter's expected utility for voting for a particular party is simply the sum of the ways her vote can be decisive multiplied by the utility of the outcome associated with each kind of decisive vote.¹⁷⁴

To combine this analysis of strategic voting with the previous analysis of economic voting from Chapter 5, we simply define u_j in Equation (8.2) as $E[u_{i,t+1}]$ from Equation (5.10) for the incumbent and as $E[u_{k,t+1}]$ from Equation (8.1) for opposition parties. Thus for the incumbent we can write:

¹⁷³ Palfrey (1989), Myerson and Weber (1993) and Cox (1997) have extended this logic to other voting systems and to a game theoretic rather than decision theoretic setting. With respect to different electoral systems in a decision theoretic setting, the same basic result applies - i.e., the pivot probabilities are the key to the voter's calculus. With respect to the decision theoretic vs. game theoretic treatments, the game theoretic treatments simply make the pivot probabilities endogenous to the game. The main result of this is to show the existence of equilibria that support different behavior than the decision-theoretic models imply. Fey (1997), however, has shown that when one adds to the game theoretic models a realistic account of how voters' beliefs about pivot probabilities might be generated dynamically in an election campaign, the different predictions from the game theoretic and decision theoretic models disappear.

¹⁷⁴ Here is an example of the calculation that leads to this equation in the three-party case: Let $Prob(a=b)$ mean the probability that party a and party b tie for first place; $Prob(a=b-1)$ mean the probability that party a trails party b by one vote in the race for first place; $Prob(a=b+1)$ mean the probability that party a leads party b by one vote in the race for first place; $Prob(a > b+1)$ mean the probability that party a leads party b by more than one vote in the race for first place (with similar notation for other pairings of parties); and $Prob(a < b-1)$ mean the probability that party a trails party b by more than one vote in the race for first place. Assuming ties are broken randomly and the chance of a three way tie is zero, we can write: Expected utility of a vote for party a = $E^a = Prob(a=b)(u_a) + Prob(a=c)(u_a) + Prob(a=b-1)(u_a+u_b)/2 + Prob(a=c-1)(u_a+u_c)/2 + Prob(a=b+1)(u_a) + Prob(a=c+1)(u_a) + Prob(a < b-1)(u_b) + Prob(a < c-1)(u_c) + Prob(a > b+1)(u_a) + Prob(a > c+1)(u_a)$; Expected utility of abstention = $E^0 = Prob(a=b)(u_a+u_b)/2 + Prob(a=c)(u_a+u_c)/2 + Prob(a=b-1)(u_b) + Prob(a=c-1)(u_c) + Prob(a=b+1)(u_a) + Prob(a=c+1)(u_a) + Prob(a < b-1)(u_b) + Prob(a < c-1)(u_c) + Prob(a > b+1)(u_a) + Prob(a > c+1)(u_a)$. If we assume, following McKelvey and Ordeshook, that $Prob(a=b) = Prob(a=b-1)$, then Subtracting E^0 from E^a gives: $Prob(a=b)(u_a-u_b) + Prob(a=c)(u_a-u_c)$, which is the specific form of the equation above that is appropriate to this example.

$$\begin{aligned}
(8.3) \quad E[u|v_i] - E[u|v_0] &= \sum_{k \in \mathbf{A}} P_{ik} (E[u_{it+1}] - E[u_{kt+1}]) \\
&= \sum_{k \in \mathbf{A}} P_{ik} \left(b\bar{y} + b \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) - b\bar{y} \right) \\
&= b \left(\sum_{k \in \mathbf{A}} P_{ik} \right) \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1})
\end{aligned}$$

and for each opposition party we can write:

$$\begin{aligned}
(8.4) \quad E[u|v_k] - E[u|v_0] &= P_{ki} (E[u_{kt+1}] - E[u_{it+1}]) + \sum_{k' \in \mathbf{A}} P_{kk'} (E[u_{kt+1}] - E[u_{k't+1}]) \\
&= P_{ki} \left(b\bar{y} - \left(b\bar{y} + b \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) \right) \right) + \sum_{k' \in \mathbf{A}} \{P_{kk'} (b\bar{y} - b\bar{y})\} \\
&= -P_{ki} b \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1})
\end{aligned}$$

There will be as many of these expressions as there are opposition parties and the voter will vote for the opposition party or incumbent for which this expression is the largest. These equations depend on three substantively interesting terms: the competency signal, retrospective economic performance, and the distribution of the pivot probabilities across pairs of parties, which we will call the *pattern of electoral contention*.

The first thing to notice about the interaction of these terms is that they imply economic voting of the usual kind. That is, if economic voting is important at all to vote choice (i.e., there is some non-zero P_{ik}), voters will vote for the incumbent party when economic performance is sufficiently good and vote against the incumbent party when it is not.¹⁷⁵ To see this in the simplest case, assume that $P_{ik} > 0$ for only one opposition party. In this case the voter will cast her vote for the incumbent party if:

¹⁷⁵ Exactly how good or bad economic performance ($y_{it} - \bar{y}$) must be to make $(y_{it} - \bar{y} - \mu_{it-1})$ positive or negative.

$$(8.5) \quad bP_{ik} \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) - \left(b(-P_{ki}) \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) \right) > 0$$

$$2bP_{ik} \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) > 0$$

where $P_{ik} = P_{ki}$. Since P_{ik} , b , and $\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}$ are all greater than zero by definition, this condition

can only be fulfilled if $(y_{it} - \bar{y} - \mu_{it-1})$ is positive. Given a fixed level for the incumbent's competence in the previous period (i.e., μ_{it-1}), this can only happen if economic performance $(y_{it} - \bar{y})$ is sufficiently positive. If, however, economic performance is sufficiently negative, then this condition will not be fulfilled and the voter will vote for the one opposition party k that has a non-zero probability of being in a tie with the incumbent for an electoral plurality. When P_{ik} is non-zero for more than one opposition party a similar argument applies.

The second thing to notice about equations (8.3) and (8.4) is that if the incumbent party is either sure to win a plurality or sure to lose it, then there will be no economic voting in this model. Specifically, since the retrospective economy plays no role in conditioning the voters' expectations about the performance of the opposition parties (these expectations are just zero), the voters have no way to choose between different opposition parties. Consequently, even if two such parties were tied for selection, this probability is multiplied by a utility difference of zero and so falls out of the model.¹⁷⁶ The obvious empirical implication of this result is that we should expect (for multi-party plurality competition) no economic voting when the incumbent party is either sure to lose power or sure to win it.¹⁷⁷

The third interesting implication of equations (8.3) and (8.4) is that when the economy has been performing poorly, so that the voter's utility for voting for the incumbent is negative, the

¹⁷⁶ We could allow some non-economic utility that voters attain when a particular party obtains office. This would mean the pivot probabilities for pairs of opposition parties would not fall out of the equation. However, nothing about the economy would be in the retained terms and so we would still conclude that there would be no economic voting if the incumbent is not in competition with some party to win a plurality.

¹⁷⁷ It is important to distinguish the chance that an incumbent party will *win* a plurality from the idea that it will be in *contention* for a plurality. In order to be in contention some pivot probability must be non-zero, which means there is a chance the incumbent could tie with some other party. In a similar fashion, Meyerson (1999) draws a distinction between the strength of a party and whether it is a serious contender. Hence a party can be certain to win an election which would imply it is not a serious contender.

voter's choice among the contending opposition parties is determined only by the relative size of the various P_{ik} 's, not by the size of the competency signal or the size of the change in economic performance.¹⁷⁸ Specifically, in this situation, the recipient of the economic votes flowing away from the incumbent will be the opposition party with the largest value for P_{ik} . The empirical hypothesis that comes from this is that in multiparty systems with a competitive plurality race for executive control, a poor economy will not only hurt the incumbent but will help the opposition party that is most competitive.

This observation leads to a fourth implication that at first seems to undermine our earlier argument (in the two-party model) that within the rational retrospective model of economic voting, variations in the quality of the competence signal should impact the economic vote. To see why, look again at the simple example in equation (8.5). Notice that while the inequality depends in one way or another on each of the terms in the product on the left hand side, except the competence signal. Specifically, for the inequality to hold P_{ik} must be non-zero and

$(y_{it} - \bar{y} - \mu_{it-1})$ must be positive. However, since $\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}$ is constrained to be positive by

definition, we eliminate it from the inequality (without changing the direction of the inequality)

simply by dividing both sides by this quantity. To put it another way, $\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2}$ can take on any

of its allowable values and it will not affect whether the inequality in (8.5) is fulfilled or not. It would seem then that in the model described by equations (8.3) and (8.4), the competency signal does not condition the economic vote. This conclusion, however, only obtains because of a particular limitation of the current model that can be easily corrected. Specifically, the inconsequence of the competency signal in equation 8.5 only follows because the expected utility of voting for a party only includes the utility derived from the voters' expectations about economic performance. If there is anything non-economic that voters care about when voting, it will not be modified in the equation by the competency term and so we cannot eliminate the term in the way described above. There are a number of substantively different ways one could add other non-economic considerations to the voter's utility but ultimately each method "corrects" the impression of the previous model that the competence signal is inconsequential. However, the

¹⁷⁸ The absolute size of largest P_{ik} is not important because any non-zero value (that is larger than those of other opposition parties) will attract the entire economic vote. This happens because the vote for or against the incumbent is determined by the sign of $(y_{it} - \bar{y} - \mu_{it-1})$. To see this, return to the simple example in equation 5.5. As long as $P_{ik} > 0$ then we can divide both sides by the first two three terms of the product on the left hand side and cancel them from the condition (unlike the economic performance term, this will never change the direction of the inequality since this term is positive by definition).

different methods have quite different consequences for the notational complexity of the models we will develop later and so we chose the one that is expressed most simply.¹⁷⁹ Specifically, if we assume the voter derives some utility (positive or negative) from the *act* of voting for each party, we can rewrite 8.2 as:

$$(8.6) \quad E[u | v_j] - E[u | v_0] = \gamma_j + \sum_{j' \in J} P_{jj'}(u_j - u_{j'})$$

where u_j and $u_{j'}$ are now understood to include only the parts of the voter's utility that require party j to win the election to be realized, while γ_j summarizes the utility that the voter obtains simply from voting for party j .¹⁸⁰

With this we can rewrite equations 8.3 and 8.4 as:

$$(8.7) \quad E[u | v_i] - E[u | v_0] = \gamma_i + b \left(\sum_{k \in A} P_{ik} \right) \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1})$$

$$(8.8) \quad E[u | v_k] - E[u | v_0] = \gamma_k - P_{ki} b \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\xi^2} \right) (y_{it} - \bar{y} - \mu_{it-1})$$

and the inequality in equation (8.5) as:

¹⁷⁹ One more notationally complex method would be to go back to the voter's utility function in equation 5.4 and add some summary of non-economic utility that a voter gets for party j being in office. However, this term will, like those for economic voting, will be conditioned by the distribution of pivot probabilities and so will add various substantively unimportant but complex terms to all our subsequent models. Further, this method of adding non-economic utility to the model, would require, in later models, that we aggregate it in some way so that rather than applying to a single party it would apply to cabinets composed of several parties. Given none of these complexities add anything to the message about economic voting that cannot be obtained by simply allowing utility for the act of voting for a party, we do not pursue this strategy here.

¹⁸⁰ For readers who object to the inclusion of this kind of utility, we emphasize that this is only the simplest way for us to produce the desired result (i.e., making it clear why the competency signal matters to economic voting). Further, while it may or may not be viewed as good *modeling* practice to include this kind of term in a utility function, it is hardly *substantively* controversial to suggest voters get psychological or emotional benefits for voting for some parties (and would suffer emotional harm if they were to vote for others).

$$(8.9) \quad (\gamma_i - \gamma_k) + 2bP_{ik} \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\varepsilon^2} \right) (y_{it} - \bar{y} - \mu_{it-1}) > 0$$

$$\cancel{(\gamma_i - \gamma_k)} \left(\frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\varepsilon^2} \right) + 2bP_{ik} (y_{it} - \bar{y} - \mu_{it-1}) > 0$$

This revision of the model, then, makes it plain that the competence signals condition the importance of economic voting relative to other impacts on the vote. When it is large (closer to one) it decreases the weight of $(\gamma_i - \gamma_k)$ relative to $(y_{it} - \bar{y} - \mu_{it-1})$ in the voter's expected utility of voting for a party. Thus, our earlier interpretation of the competence signal as a weight on economic voting stands, as long as we allow some non-economic sources of utility. We have included these sources in the simplest way, but the interpretation does not depend on that choice. In the models that follow, we will retain this revision.

Finally, this modification of the model slightly changes the interpretation of the role P_{ik} discussed above. Whereas in that discussion the hypothesis was that if the performance of the economy was sufficiently poor, the economic vote would flow away from incumbents to the opposition party with the largest P_{ik} , even if P_{ik} were very small. All that was required was that $P_{ik} > 0$. If this condition as met, the utility associated with voting for the incumbent would be negative and the utility associated with party k would be positive and so voters would vote for k , even if its chance of tying the incumbent were very small. In the current model, however, $P_{ik} > 0$ is no longer enough for voters to prefer the challenger to the incumbent when $(y_{it} - \bar{y} - \mu_{it-1})$ is negative. Since $(\gamma_i - \gamma_k)$ can now be positive when the economy is performing poorly, a small P_{ik} may make the overall economic term so small (recall that $0 \leq P_{ik} \leq 1$) that a voter who gets some positive utility for voting for the incumbent rather than party k , will nevertheless vote for the incumbent. Thus, the addition of non-economic factors simply makes the size of P_{ik} matter to the voter's choice. This is important empirically, because it tells us whether economic voting in these systems should require close races for pluralities or only some chance that the incumbent could lose.

Specifically, the earlier model implied that economic voting should be important as long as some challenger had *any* chance of competing for an electoral plurality; however, the new one implies that economic voting should be important when some challenger has a real chance of winning, where the degree of competition required depends on the strength of the typical voter's

psychological attachment to the incumbent (or aversion to the challenger). For example, think about how a plurality contest like this might have worked out in post-war Italy. The Christian Democrats dominated the Italian system and many voters had deep ties to the party based on their religious beliefs. In contrast, the closest contender for an electoral plurality was usually the Communists, which remained for many years one of the least reformed in Europe. In this case, it is likely that for the typical voter $(\gamma_i - \gamma_k)$ was quite large, so we would expect economic voting to be important (in a hypothetical plurality race for control of government) only if race was quite close indeed.

Economic voting in its traditional form continues in this generalization of the model – voters will vote against the incumbent when economic performance is sufficiently weak and for the incumbent when it is sufficiently strong. When economic performance is poor the voter will vote for the opposition party that is most likely to contend with the incumbent for an electoral plurality. The less the incumbent party contends for an electoral plurality (i.e., it is a sure loser or a sure winner), the smaller the economic vote.

Multiple Parties with a Generic Selection Process for a Single Party Executive

Most executives are not selected directly in a plurality election as assumed in the previous discussion. Usually, there is some selection process following the election in which the electoral result may be but one factor that determines the composition of the government. In order to examine economic voting in these cases, we must generalize the logic of strategic voting from the previous section. We do that in two steps. In the next section, we discuss the case of the selection of coalition governments, which is the usual situation in which post-election selection of governments is an issue. However, that discussion will be clearer if we first consider the logically possible (but empirically rare) case in which a single-party government is selected in a post-election process that is only partially based on the electoral result.¹⁸¹

McKelvey and Ordeshook conceive of the pivot probability, P_{ik} , as the probability that parties i and k are tied in the race for *election* to the executive. Their analysis, however, can be applied to a more general executive selection processes in which we define P_{ik} as the probability that parties i and k are tied in the race for *selection* to the executive. This change in language is used to indicate that the selection process may only partly depend on the electoral outcome. For example, it could be that a single-party executive is selected in a bargaining process in which having more votes is helpful but not determinant because other factors like ideological centrality

¹⁸¹ Such a system might result when a Head of State has discretion in appointing a government and is not constrained to choose the plurality winner.

also impact the selection. One real world example of this might be systems in which a Head of State has some discretion in choosing the new prime minister. In such cases, the Head of State may pay close attention to the electoral result but sometimes choose a party that is not the plurality winner for other reasons. Votes can still be decisive in this setting – decisiveness, however, does not correspond to breaking ties for electoral pluralities. Instead, a vote is decisive if it breaks a tie between two parties in their overall chances of being selected to be the governing party. For example, selection could depend on two factors, electoral strength and ideological centrality. Suppose that, absent voter i 's vote, one party receives 20% of the vote and another 30%, but the smaller party is closer to the median position than the other to the exact extent necessary to make the selection between the two parties indeterminate. Voter i 's vote now becomes decisive in the selection of the executive, even though the two parties' electoral support differs substantially.

With this generalization of the interpretation of the P_{ik} 's, the analysis of the previous section can be directly applied to any process of executive selection that gives executive power to a single party based in part on their electoral support (where, of course, the chance of selection is increasing in that support). Consequently, we can conclude that rational, forward-looking voters in systems that *select* single-party executives will not cast economic votes unless the incumbent party is in a close race with some other party for selection to the executive.

Here, we consider the unusual case in which multiple parties compete for exclusive control of the executive, but the selection of the executive is not based only on which party wins an electoral plurality. We conclude that when economic performance is poor, the voter will vote for the opposition party that is most likely to contend with the incumbent for *selection* to the executive. And, the less the incumbent party contends for selection to the executive (i.e., it is a sure not to be selected or it is sure to be selected), the more muted the economic voting.

Multiple Parties with a Coalitional Executive

The discussion above generalizes the notion of pivot probabilities to cover cases in which single-party incumbents are selected rather than simply elected, but it does not address the more usual situation in which post-election executive selection procedures result in a multi-party coalition cabinet. We begin by generalizing the analysis of strategic voting in the last section to cases of coalition government and then use this to generalize Alesina and Rosenthal's model of retrospective economic voting as well.

In our generalization, we continue to focus on the role that the vote plays in the *selection* rather than the *election* of an executive; however, we further alter the concept of a pivot

probability to apply to the situation in which a vote may be decisive in bringing a potential *coalition* (rather than a single party) to power. This requires that we consider the role that the vote for a *party* plays in the selection among potential *cabinets*.¹⁸² The details and formalization of the concept of a pivot probability in this setting are worked out in Appendix A. In that discussion, we define a cabinet as a vector, $\mathbf{g} = \{g_1, g_2, g_j, \dots, g_J\}$, where g_j is one if party j is a member of cabinet $\mathbf{g}$ and zero otherwise. Further, we define a generalized pivot probability, $P_{j, \mathbf{g}\mathbf{g}'}$, as the probability that two cabinets, $\mathbf{g}$ and $\mathbf{g}'$, are tied for selection *and* that a vote for party j will break that tie in favor of cabinet $\mathbf{g}$. More precisely, the formal definition in Appendix B models the cabinet selection process as having two stages. In the first, each potential cabinet is assigned a selection value based on some selection function that depends on the electoral support of all parties and other non-electoral factors. Second, the cabinet with the highest selection value is the one that forms. Thus, the idea of being “tied for selection” in this setting means two cabinets have the same selection values and have higher values than any other cabinet. The voter’s belief that two cabinets are tied for selection is just $P_{\mathbf{g}\mathbf{g}'}$. Further, with this generic model of the selection process, we can think of the probability that a vote for party j breaks a tie for selection between $\mathbf{g}$ and $\mathbf{g}'$ in favor of cabinet $\mathbf{g}$ as the probability that the derivative of the selection function for $\mathbf{g}$, with respect to the vote total of party j , is bigger than the corresponding derivative for $\mathbf{g}'$. In the appendix, we write this requirement as $P(D_{\mathbf{g},j} > D_{\mathbf{g}',j})$, so $P_{j, \mathbf{g}\mathbf{g}'}$ is just shorthand for $P_{\mathbf{g}\mathbf{g}'}P(D_{\mathbf{g},j} > D_{\mathbf{g}',j})$.¹⁸³ In later sections we examine more closely the form of the derivatives in this definition and use them to define different political contexts in which we expect more or less economic voting. However, at this stage in the argument, all we need to understand is that the generalization of rational economic voting to the coalition context requires that we consider the voter’s beliefs about the efficacy of her vote in the process of cabinet selection, as well as her beliefs about which cabinets are likely to tie for selection, given that process. Further, our strategy for generalizing the rational retrospective model to coalition contexts is to incorporate these beliefs into a generalized version of the pivot probability commonly found in models of rational voting.

¹⁸² We concur with Myerson’s argument that, “Finally, we saw the importance of taking account of how the winners will bargain after the election to control the government. Rational voters should be concerned about their potential effect on this bargaining process, and this concern may be an essential factor to explain the systematic differences between party structures in presidential and parliamentary systems (Myerson 1999: 39).”

¹⁸³ We assume that these probabilities are independent.

In what follows, we will always indicate the set of incumbent parties as $\mathbf{g}$ and generic alternative cabinets as $\mathbf{g}'$ or $\mathbf{g}''$. The set of all such alternative cabinets is $\mathbf{A}$ (note that this set does not include the incumbent cabinet $\mathbf{g}$). In Appendix B we show that - analogous to Equation (8.6) - we can write:

$$\begin{aligned}
 (8.10) \quad E[u | v_j] - E[u | v_0] &= \sum_{\mathbf{g}' \in \mathbf{A}} P_{j, \mathbf{g} \mathbf{g}'} (E[u_{\mathbf{g}, t+1}] - E[u_{\mathbf{g}', t+1}]) \\
 &+ \sum_{\mathbf{g}' \in \mathbf{A}} P_{j, \mathbf{g}' \mathbf{g}} (E[u_{\mathbf{g}', t+1}] - E[u_{\mathbf{g}, t+1}]) \\
 &+ \sum_{\mathbf{g}' \in \mathbf{A}} \sum_{\mathbf{g}'' \in \mathbf{A}} P_{j, \mathbf{g}' \mathbf{g}''} (E[u_{\mathbf{g}', t+1}] - E[u_{\mathbf{g}'', t+1}]) + \gamma_j
 \end{aligned}$$

This expression is just the voter's expected utility gain or loss for voting for party j rather than abstaining. The voter will select the party for which this expression is the highest. It identifies all the ways that a vote for party j can be decisive in bringing one coalition to power rather than another and sums the utility change associated with casting a vote for party j rather than abstaining in these situations.¹⁸⁴

To arrive at an expression for the utility of voting for each party in a coalitional situation, we must derive specific expressions for the expected utilities on the right hand side of equation (8.10). To do this, we need to generalize Alesina and Rosenthal's model of retrospective economic voting to the case of coalition governments. However, since the voter's expected utility depends, in their model, on economic policy and outcomes, we must first ask what inflation policy a coalition government will choose once in office? Further, because the voter's expected utility for a potential cabinet also depends on its expected competence, we need to ask how the voter can forecast both the competence of the incumbent cabinet as a whole and the competence of other potential cabinet. With answers to these questions, we can then come back to the equations above to explore how voters translate the utility they expect to accrue under a potential cabinet into a vote for a given party.

Policy Making in Coalition Cabinets

In this section our task is to generalize Alesina and Rosenthal's model of retrospective economic voting to provide an expression for the voter's expected utility if the incumbent government were to stay in power, allowing for the possibility that this incumbent government is a coalition among two or more parties. As we pointed out earlier, no government (whether single

¹⁸⁴ In chapter 10 we provide some evidence that individuals have well formed beliefs about the outcomes of the coalition formation process as well as the process itself.

or multiparty) can gain from choosing a level of inflation higher than zero. Consequently, we can conclude that any assumption about how the parties in a coalition cabinet aggregate their policy preferences should result in the same prediction – the coalition government will choose a zero inflation policy. In contrast to inflation, however, economic growth in the model is not determined only by the policy the government chooses, but also the government’s level of administrative competence. If we assume that each party has its own “competence shock” then we will need to specify how the overall competency of the government is determined in order to generalize the model to coalitional cases. Doing so reveals an important dimension of political context that will condition the size of the economic vote in this model.

Assume each party in the cabinet has its own competency shock that is persistent in the same way as in the single-party incumbent model (i.e., as in Equation 5.3). Thus, if we indicate the set of incumbent parties as $\mathbf{g}$, with a typical member i , we can write:

$$(8.11) \quad \varepsilon_{it} = \mu_{it} + \mu_{it-1} \quad \forall i \in \mathbf{g}$$

In contrast, there is only one non-political shock and since there cannot be a different level of economic growth for each party, the different party competences must be aggregated in some way to produce an overall level of growth. Suppose that the overall competence of the government in a given period is a weighted average of the competence of the incumbent parties, with weights corresponding to the amount of *administrative responsibility* that the party holds. Call these weights, for each incumbent party at time t , $\lambda_{i,t}$. We can thus write:

$$(8.12) \quad \varepsilon_{\mathbf{g},t} = \sum_{i \in \mathbf{g}} \lambda_{i,t} \mu_{i,t} + \lambda_{i,t-1} \mu_{i,t-1}$$

This expression is the overall competence of the government, where we assume $\sum_{i \in \mathbf{g}} \lambda_{i,t} = 1$ and that the competence shocks for each incumbent party, $\mu_{i,t}$, are drawn from a normal distribution with mean zero and variance σ_{μ}^2 .¹⁸⁵ The weights can describe a variety of policy-making processes - from a prime ministerial dominant system (λ_{pm} is close to one) to a system of equal power sharing ($\lambda_i = 1/n$ for all n incumbent parties) to a system of proportional responsibility (λ_i is proportional to legislative or cabinet seat shares). We will have more to say about the influence

¹⁸⁵ It is not necessary that the competency shocks for different parties come from the same distribution, but allowing these to be different introduces partisan differences into the model.

of the distribution of administrative responsibility on economic voting below, but presently we simply want to use this device to derive the voter's expected utility if the incumbent government retains power the next period (i.e., $E[u_{g,t+1}]$).

If we assume voters know the distribution of administrative responsibility across incumbents then what will $E[u_{g,t+1}]$ be? Recall that any cabinet will choose a zero inflation policy and that voters will correctly anticipate this policy so that growth under any cabinet, g , is just the natural rate of growth plus a shock. The shock consists of two parts, the government competency shock, which is just the weighted sum in Equation (8.12), and an exogenous random shock that represents the politically uncontrollable aspects of growth. Growth in the current period is thus:

$$\begin{aligned}
 (8.13) \quad y_{g,t} &= \bar{y} + \eta_{g,t} \\
 &= \bar{y} + \varepsilon_{g,t} + \xi_t \\
 &= \bar{y} + \sum_{i \in g} \{ \lambda_{i,t} \mu_{i,t} + \lambda_{i,t-1} \mu_{i,t-1} \} + \xi_t
 \end{aligned}$$

Rearranging this gives:

$$(8.14) \quad \sum_{i \in g} \{ \lambda_{i,t} \mu_{i,t} \} + \xi_t = y_{g,t} - \bar{y} - \sum_{i \in g} \{ \lambda_{i,t-1} \mu_{i,t-1} \}$$

Where everything on the right hand side of this equality is observed and so the sum of the terms on the left is also observed, though not the individual components. Denote the sum on the left hand side as $k_{g,t} = \sum_{i \in g} \{ \lambda_{i,t} \mu_{i,t} \} + \xi_t$. Since $k_{g,t}$ is observed (while the split between its two random

components is not) the voter's expectation about $\mu_{g,t} = \sum_{i \in g} \{ \lambda_{i,t} \mu_{i,t} \}$ given $k_{g,t}$ (i.e., the voter's

expectation about the government's total level of competence in the current period, given the observed level of growth and last period's total government competence). To calculate this conditional expectation, we need to know the distribution of both $k_{g,t}$ and $\mu_{g,t}$. $\mu_{g,t}$ is a weighted sum of normally distributed random variables, each with zero mean and variance σ_μ^2 .

Consequently,

$$(8.15) \quad \mu_{\mathbf{g},t} = \sum_{i \in \mathbf{g}} \lambda_{i,t} \mu_{i,t} \sim N \left(0, \sigma_{\mu}^2 \sum_{i \in \mathbf{g}} \lambda_{i,t}^2 \right) \\ \sim N \left(0, \sigma_{\mu_{\mathbf{g},t}}^2 \right)$$

Thus $k_{\mathbf{g},t}$ is the un-weighted sum of two normally distributed random variables, both with zero means and variances $\sigma_{\mu_{\mathbf{g},t}}^2$ and σ_{ξ}^2 , respectively. The distribution of $k_{\mathbf{g},t}$ is thus:

$$(8.16) \quad k_{\mathbf{g},t} = (\mu_{\mathbf{g},t} + \xi_t) \sim N \left(0, \sigma_{\mu_{\mathbf{g},t}}^2 + \sigma_{\xi}^2 \right)$$

Given that both, $k_{\mathbf{g},t}$ and $\mu_{\mathbf{g},t}$ are distributed normally, their joint distribution is bivariate normal and the optimal forecast of $\mu_{\mathbf{g},t}$ given $k_{\mathbf{g},t}$ is just this conditional expectation, which is computed from the appropriate conditional distribution of the bivariate normal. Using standard results on the bivariate normal distribution:

$$(8.17) \quad E \left[\mu_{\mathbf{g},t} \mid k_{\mathbf{g},t} \right] = E[\mu_{\mathbf{g},t}] + \frac{\sigma_{\mu_{\mathbf{g},t}, k_{\mathbf{g},t}}}{\sigma_{k_{\mathbf{g},t}}^2} \left(y_{\mathbf{g},t} - \bar{y} - \sum_{i \in \mathbf{g}} \{ \lambda_{i,t-1} \mu_{i,t-1} \} \right) - E[k_{\mathbf{g},t}] \\ = \frac{\sigma_{\mu_{\mathbf{g},t}}^2}{\sigma_{\mu_{\mathbf{g},t}}^2 + \sigma_{\xi}^2} (y_{\mathbf{g},t} - \bar{y} - \mu_{\mathbf{g},t-1})$$

Where the second line uses the fact that the covariance between $k_{\mathbf{g},t}$ and $\mu_{\mathbf{g},t}$ is equal to $\sigma_{\mu_{\mathbf{g},t}}^2$.¹⁸⁶

Finally, substituting for $\sigma_{\mu_{\mathbf{g},t}}^2$ from Equation (8.15) and noting that (in an analogous argument to that in footnote 118) $E \left[\mu_{\mathbf{g},t} \mid k_{\mathbf{g},t} \right] = E \left[\mu_{\mathbf{g},t} \mid y_{\mathbf{g},t} \right]$, we have an intuitive generalization of Equation (5.9) to the case of coalition cabinets:

¹⁸⁶ $\text{Cov}(k_{\mathbf{g},t}, \mu_{\mathbf{g},t}) = E[(k_{\mathbf{g},t} - E[k_{\mathbf{g},t}])(\mu_{\mathbf{g},t} - E[\mu_{\mathbf{g},t}])] = E[k_{\mathbf{g},t} \mu_{\mathbf{g},t}] =$
 $E \left[\left(\sum_{i \in \mathbf{g}} \{ \lambda_{i,t} \mu_{i,t} \} + \xi_t \right) \left(\sum_{i \in \mathbf{g}} \{ \lambda_{i,t} \mu_{i,t} \} \right) \right] = E \left[\sum_{i \in \mathbf{g}} \{ \lambda_{i,t} \mu_{i,t} \} \sum_{i \in \mathbf{g}} \{ \lambda_{i,t} \mu_{i,t} \} \right] = \sum_{i \in \mathbf{g}} \lambda_{i,t}^2 E[\mu_{i,t}^2] + \sum_{i \in \mathbf{g}} \sum_{i' \neq i \in \mathbf{g}} \lambda_{i,t} \lambda_{i',t} E[\mu_{i,t} \mu_{i',t}]$
 $= \sigma_{\mu}^2 \sum_{i \in \mathbf{g}} \lambda_{i,t}^2 = \sigma_{\mu_{\mathbf{g},t}}^2$

$$(8.18) \quad E[\mu_{g,t} | y_{g,t}] = \left(\frac{\sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2} \right) (y_{g,t} - \bar{y} - \mu_{g,t-1})$$

Thus, allowing parties to have different levels of administrative responsibility makes the voter's overall evaluation of competence of the incumbent government (or the quality of the signal of that competence) a function of the distribution of administrative responsibility in the cabinet. Notice that like our earlier result, the *responsibility augmented competency signal* varies between zero and one – approaching one as the variance in the non-political shock to the economy approaches zero. Further, the augmented competency signal reduces to the competency signal we derived for the case of unified government only if $\lambda_{i,t} = 1$ for some party (in which case all other parties must have $\lambda_{i,t} = 0$). Thus, the models are the same when one party has all administrative responsibility.

We can now write the voter's expected utility if the incumbent government stays in power as:

$$(8.19) \quad \begin{aligned} E[u_{g,t+1}] &= \frac{1}{2} E[\pi_{g,t+1}^2] + bE[y_{g,t+1}] \\ &= b(\bar{y} + E[\eta_{g,t+1}]) \\ &= b(\bar{y} + E[\xi_{t+1}] + E[\varepsilon_{g,t+1}]) \\ &= b(\bar{y} + E[\varepsilon_{g,t+1}]) \\ &= b(\bar{y} + E[\mu_{g,t+1} + \mu_{g,t}]) \\ &= b(\bar{y} + E[\mu_{g,t}]) \\ &= b(\bar{y} + E[\mu_{g,t} | y_{g,t}]) \\ &= b\bar{y} + b \left(\frac{\sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2} \right) (y_{g,t} - \bar{y} - \mu_{g,t-1}) \end{aligned}$$

Looking back at Equation (8.10), we now need only one more component to be able to evaluate the conditions under which a rational voter in a coalitional system will use the retrospective economy in her vote choice. Specifically, we need an expression for the voter's expected utility

in the next period if an alternative government comes to power. This opens up some new theoretical possibilities that we address in the next section.

Evaluating Alternatives to the Previous Government

In comparing the expected utility of a future under the current government with the expected utility of the future under an alternative government, voters will compare $E[u_{g,t+1}]$ with $E[u_{g',t+1}]$ for all $g' \in \mathbf{A}$. In the last section, we saw what the first expectation should be, given our generalization of Alesina and Rosenthal's model of retrospective economic voting. In this section we provide an expression for the second expectation – the expected utility of an alternative government coming to power in the next period. There are different ways we can think about this term, so we start with a simple assumption that closely matches the treatment of opposition competence in Alesina and Rosenthal's single-party incumbent model. Specifically, we first assume that voters have no information about the competence of non-incumbent cabinets and so assign these cabinets as a whole the unconditional expectation of the known distribution of competence, which is zero by assumption. Thus, since all governments will pursue zero inflation policies, this simple assumption about how voters evaluate the competence of alternative governments produces a correspondingly simple expression for the voter's expected utility if any alternative government comes to power:

$$\begin{aligned}
 E[u_{g',t+1}] &= \frac{1}{2} E[\pi_{g',t+1}^2] + b E[y_{g',t+1}] \\
 &= b(\bar{y} + E[\eta_{g',t+1}]) \\
 &= b(\bar{y} + E[\xi_{t+1}] + E[\varepsilon_{g',t+1}]) \\
 &= b(\bar{y} + E[\varepsilon_{g',t+1}]) \\
 &= b(\bar{y} + E[\mu_{g',t+1} + \mu_{g',t}]) \\
 &= b(\bar{y} + E[\mu_{g',t}]) \\
 &= b(\bar{y} + E[\mu_{g',t} | y_{g,t}]) \\
 &= b\bar{y}
 \end{aligned}
 \tag{8.20}$$

Using the above expression along with the expression for $E[u_{g,t+1}]$ in Equation (8.19), we can express the generalized version of McKelvey and Ordeshook's result as:

$$\begin{aligned}
& E[u | v_j] - E[u | v_0] = \gamma_j + \sum_{g' \in \mathbf{A}} P_{j,gg'} (E[u_{g,t+1}] - E[u_{g',t+1}]) \\
& + \sum_{g' \in \mathbf{A}} P_{j,g'g} (E[u_{g',t+1}] - E[u_{g,t+1}]) + \sum_{g' \in \mathbf{A}} \sum_{g'' \in \mathbf{A}} P_{j,g'g''} (E[u_{g',t+1}] - E[u_{g'',t+1}]) \\
(8.21) \quad & = \gamma_j + \sum_{g' \in \mathbf{A}} P_{j,gg'} (E[u_{g,t+1}] - b\bar{y}) + \sum_{g' \in \mathbf{A}} P_{j,g'g} (b\bar{y} - E[u_{g,t+1}]) \\
& + \sum_{g' \in \mathbf{A}} \sum_{g'' \in \mathbf{A}} P_{j,g'g''} (b\bar{y} - b\bar{y}) \\
& = \gamma_j + b \left(\frac{\sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2} \right) (y_{g,t} - \bar{y} - \mu_{g,t-1}) \sum_{g' \in \mathbf{A}} (P_{j,gg'} - P_{j,g'g})
\end{aligned}$$

There are J of these expressions and the rational voter votes for the party j for which this expression is the largest.

Like equations (8.7) and (8.8) from our discussion of non-coalitional contexts, equation (8.21) has three pieces that interact to produce economic voting and to condition its importance across contexts: the *responsibility augmented competency signal*, retrospective economic performance, and the *pattern of cabinet contention*. In the rest of this section we explore these pieces beginning with the *responsibility augmented competency signal*. This signal is:

$$(8.22) \quad \frac{\sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{i \in g} \lambda_{i,t}^2}$$

As we discussed at length in the previous section, the *competency signal* did not vary over parties and so could not affect the voter's choice between them. Incorporating information about the distribution of responsibility within the incumbent administration does nothing to change that conclusion since this distribution (and therefore the sums in equation 8.22) is constant across parties.¹⁸⁷ As before, however, the *responsibility augmented competency signal* does help regulate the relative weight of the retrospective economy compared to other factors in the voter's utility, so can impact the overall importance of economic voting in different contexts.

In the last section we also discussed the interpretation of the variance parameters in the competency signal and that interpretation remains true in equation 8.22, holding the distribution

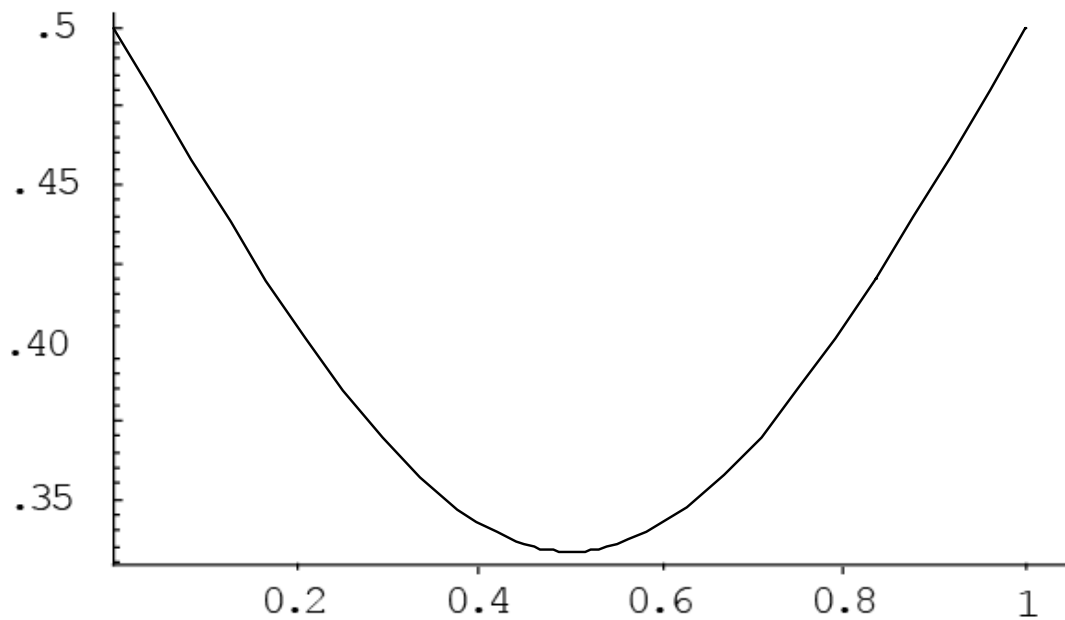
¹⁸⁷ Further generalizations of the model, however, eliminate this implication.

of responsibility constant. Thus, in the current discussion we are principally concerned with how the overall weight in equation 8.22 changes when we hold the variance parameters constant and vary the other part of the signal - the distribution of responsibility across parties. This variation is substantively interesting because it represents different power sharing arrangements within cabinets. Hence it can help us answer questions about how rational retrospective economic voting may be different in substantively interesting contexts like those in which a prime minister is dominant or those in which cabinet decision making is consensual.¹⁸⁸ This question (and others like it) can be answered in the context of this model by examining how the *responsibility augmented competency signal* changes when the distribution of the $\lambda_{i,t}$'s change.

Consider first the simple case in which only two parties have any administrative role (e.g., a two-party cabinet). The share of administrative responsibility for the first party is λ_1 and for the second is $\lambda_2 = 1 - \lambda_1$. Figure 8.1 graphs equation (8.22) as a function of λ_1 , where we set both the variances in the competency signal to one.

¹⁸⁸ Since our discussion is limited at this point to cabinets, we do not address the consequences of more general power sharing arrangements that extend policymaking responsibility to opposition parties. This generalization, however, is easy to incorporate just by taking the sum in the text over all parties instead of just government parties (indeed, we prove the result for the more general case in Appendix B). Since the sum is constant in any case, it does not impact the other parts of the model. We do not do so here, however, because it introduces an awkward disconnect between the rest of the model (which is defined in terms of cabinets) and the responsibility augmented competency signal, which would be defined in terms of the distribution of responsibility generally. Thus, we hold off on this particular generalization until we provide the, more complicated, generalization of the whole model to focus on competition between alternative distributions of responsibility rather than alternative cabinets.

Figure 8.1



This graph suggests that, at least in the case of a two-party cabinet, the competency signal will be weakest (closer to zero) when power is shared equally between the two parties. More generally, in Appendix B we show that if we minimize equation (8.22) with respect to the vector of λ_i 's in the government, the minimum always occurs when all the λ_i 's are equal.¹⁸⁹ Thus we can conclude that, in general, more power sharing in cabinets decreases the strength of the competency signal.

This result mirrors Powell and Whitten's (1993) clarity of responsibility hypothesis, but gives a very different logic for the result. Their argument was based on the traditional accountability model of retrospective voting: when voters are faced with a situation in which administrative competence is shared, they find it difficult to attribute responsibility to any party. The intuition for the clarity of responsibility argument is poorly informed voters which reduces the voter's ability to discern responsibility for economic outcomes. This in turn reduces the correlation between economic evaluations and vote choice. In contrast, the rational retrospective model of economic voting assumes that voters know the distribution of administrative responsibility perfectly and so it cannot be their uncertainty about this distribution that weakens

¹⁸⁹ This result is proved for a more general result in which the sum in the ratio is over all parties rather than just cabinet parties. In that case minimizing with respect to the $J-1$ λ_j 's (recall that the λ_j 's sum to 1) reveals that the minimum occurs when $\lambda_j = 1/J$ for all J parties. We can restrict attention to just the government parties by holding $\lambda_j = 0$ if party $j \notin \mathbf{g}$ to get the result reported here. None of these results depend on the variances in the equation.

the economic vote. Instead, power sharing decreases the importance of the economy on vote choice only because it leads voters to attribute more weight to exogenous factors shaping the observed movements in the economy.

The *pattern of cabinet contention* in the model is captured in the last term in the product in equation 8.21, which includes generalized pivot probabilities for all the pairs of potential cabinets that include the incumbent cabinet. All other pairs of pivot probabilities (between two non-incumbent cabinets) fall out of the expression. As in the case of single party cabinets, this occurs because of our assumption that the retrospective performance of the economy is only informative about the competence of the incumbent cabinet, not alternatives. Thus, the voter assigns the same competence to all non-incumbent alternative cabinets and there is no difference in the voter's expected utility from living under one or the other of these cabinets. The empirical implication of this piece of the model is correspondingly similar to our earlier one for the single party case: economic voting will disappear if the incumbent cabinet is not in competition to return to power.¹⁹⁰

The generalized pivot probabilities that remain in equation (8.21) represent the aspects of the *pattern of cabinet contention* that are relevant to this model and so we will examine them somewhat more carefully. Recall from the discussion of equation 8.10 that $P_{j,gg'} = P_{gg'}P(D_{g,j} > D_{g',j})$ and $P_{j,g'g} = P_{gg'}P(D_{g,j} < D_{g',j})$, where $P_{gg'}$ is the probability that cabinet g and g' are tied for selection and the D 's are derivatives of the selection functions for g and g' with respect to party j 's vote total.¹⁹¹ In appendix A, we show that $D_{g,j}$ is equal to $w_g g_j$, where w_g is the derivative of the selection function for cabinet g with respect to the electoral support for cabinet g and g_j is equal to one if party j is a member of cabinet g and zero otherwise. We assume that voters know which parties are in the incumbent cabinet and which parties are in any alternative cabinet they are considering (i.e., they know g_j and g'_j). However, they are uncertain how changes in electoral support for a given potential cabinet impact its selection (i.e., they do not know w_g or $w_{g'}$ for certain). Their beliefs about these quantities are governed by the probability distributions $f(w_g)$ and $f(w_{g'})$, respectively. We assume that voters believe that

¹⁹⁰ Our assumption ignores the fact that some alternative cabinets may contain cabinet members whose competence in the current cabinet may provide information about the competence of alternatives. We address this limitation in a generalization below.

¹⁹¹ We make *assumption #1* from the appendix in this discussion.

electoral support cannot hurt a party's selection value, so that $f(w_g) = 0$ if $w_g < 0$ and $f(w_{g'}) = 0$ if $w_{g'} < 0$. With this we can rewrite equation 8.14 as:

$$\begin{aligned}
 E[u | v_j] - E[u | v_0] &= \gamma_j + Z \sum_{g' \in \Lambda} P_{gg'} \left(P(w_g g_j - w_{g'} g'_j > 0) - P(w_g g_j - w_{g'} g'_j < 0) \right) \\
 (8.23) \quad &= \gamma_j + Z \sum_{g' \in \Lambda} P_{gg'} \left(2P(w_g g_j - w_{g'} g'_j > 0) - 1 \right)
 \end{aligned}$$

Z is just the part of the utility difference that does not change over parties. To get the last line we use the fact that $P(X > 0) = 1 - P(X \leq 0)$.

To facilitate our interpretation of the impact of the *pattern of cabinet contention* on economic voting, we will ignore the γ 's in equation 8.23 (assume they are all 0) and assume in all of our examples that $Z > 0$. Further, we will initially assume that there are only two cabinets in contention to form the government, the incumbent g and an alternative g' . Given this, the first thing to notice about equation (8.23) is that the direction in which the distribution of cabinet contention impacts utility is determined by $P(w_g g_j - w_{g'} g'_j > 0)$. If this probability is less than 0.5, the whole equation will be negative, if it is equal to 0.5 it will be zero, and if it is greater than .5 it will be positive. Most of our effort in understanding this equation, then, will focus on this probability. We begin by defining the random variable $Y = w_g g_j$ and $Y' = w_{g'} g'_j$. Thus, we can write $P(Y = y) = f(y) = f(w_g)$ if $g_j = 1$ and $P(Y = y) = 0$ if $g_j = 0$.¹⁹² Similar expression apply for $P(Y' = y)$. Finally, define the random variable $K = Y - Y'$. We want to derive the distribution $P(K = k)$ and use it to find expression for $P(K > 0)$. Since the distributions of Y and Y' have discrete and continuous parts, we must write the distribution of K for each case. This distribution is as follows:

¹⁹² Recall that we have assumed voters assign zero probability to the event that increased electoral support for a cabinet decreases its selection values. Thus, $f(w_g) = 0$ for $w_g < 0$ and likewise with Y .

$$(8.24) \quad P(K = k) = \begin{cases} 0 & \forall k \neq 0 & \text{if } g_j = 0 \& g'_j = 0 \\ 1 & k = 0 \\ P(Y = k) & \forall k & \text{if } g_j = 1 \& g'_j = 0 \\ P(Y' = -k) & \forall k & \text{if } g_j = 0 \& g'_j = 1 \\ M_k & \forall k & \text{if } g_j = 1 \& g'_j = 1 \end{cases}$$

where,

$$M_k = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) f(y') \delta((y - y') - k) dy dy'$$

$\delta(x)$ is the delta function and is equal to 0 for $x \neq 0$. This last expression is a function of k and just sums up the probability of all the events in which $Y - Y'$ can be equal to k . Now that we have an expression for the density of K we need $P(K > 0)$. For each of our cases this is:

$$(8.25) \quad P(K > 0) = \begin{cases} 0 & \text{if } g_j = 0 \& g'_j = 0 \\ \int_0^{\infty} P(Y = k) dk = 1 & \text{if } g_j = 1 \& g'_j = 0 \\ \int_0^{\infty} P(Y = -k) dk = 0 & \text{if } g_j = 0 \& g'_j = 1 \\ \int_0^{\infty} M_k dk & \text{if } g_j = 1 \& g'_j = 1 \end{cases}$$

With this we can write:¹⁹³

¹⁹³ The argument for the first line above is somewhat more complicated than the others since it must account for the use of the equality, but this is just a technical issue and has no substantive import.

$$(8.26) \quad P_{gg'}(2P(w_g g_j - w_{g'} g'_j > 0) - 1) = \begin{cases} 0 & \text{if } g_j = 0 \& g'_j = 0 \\ P_{gg'} & \text{if } g_j = 1 \& g'_j = 0 \\ -P_{gg'} & \text{if } g_j = 0 \& g'_j = 1 \\ P_{gg'} \left(2 \left(\int_0^\infty M_k dk \right) - 1 \right) & \text{if } g_j = 1 \& g'_j = 1 \end{cases}$$

The integral in the last line will be between zero and one, so the value of the whole term will vary between $-P_{gg'}$ and $P_{gg'}$. An example will help convey the intuitive way in which economic voting occurs in this model. Suppose that there are four parties (a , b , c , and d) and the incumbent cabinet, $\mathbf{g}$, includes a and b . Only one viable alternative government exists, a coalition between c and d (call this $\mathbf{g}'$). Using Equation (8.26), we can calculate the utility difference in casting a vote for each party and abstaining as follows:

$$\begin{aligned} E[u | v_a] - E[u | v_0] &= \gamma_a + P_{gg'} Z \\ E[u | v_b] - E[u | v_0] &= \gamma_b + P_{gg'} Z \\ E[u | v_c] - E[u | v_0] &= \gamma_c - P_{gg'} Z \\ E[u | v_d] - E[u | v_0] &= \gamma_d - P_{gg'} Z \end{aligned}$$

where Z is just the part of Equation 8.21 that is constant over parties.

Since we have assumed that $\mathbf{g}$ and $\mathbf{g}'$ are in competition to form the government, $P_{gg'} > 0$ in each of the above expressions, so they differ only in their signs. Thus, this example produces economic voting of a familiar kind – when the economic performance has been sufficiently strong (Z is positive) voters will vote for one of the parties in the incumbent cabinet but when it has been weak (Z negative), they will vote for some opposition party. More generally, if no members of the incumbent cabinet are in an alternative cabinet that is in contention with the incumbent cabinet for selection, this model implies that the economic part of the voter's expected utility of voting for each party will have different signs for incumbent and opposition parties. This is an important implication since there are many real world cases in which the membership of the incumbent cabinet and the likely alternatives are distinct. For example, contending cabinets in Irish politics have usually not included the same parties – Fianna Fail ruling alone, Fianna Gael

ruling alone, or Fianna Gael and Labor in coalition. Likewise, the Scandinavian party systems often pit socialist coalitions against bourgeoisie coalitions with no overlapping memberships.

Notice, however, that (ignoring the γ 's) all we can say about the economic voter's choice is that he will choose an incumbent party or an opposition party – we cannot say which incumbent or opposition party he will choose because the expected utility associated with the economy is the same for all opposition and all incumbent parties, respectively.¹⁹⁴ This is clearly a limitation of the current model since it does not capture important cases in which we think that there is very likely a difference in the way that votes for different parties in a potential cabinet impact the chance of the cabinet's selection. One example we have already mentioned is the *formateur* system in which it is the vote of the largest party (in contention for a plurality) that is crucial to the cabinet's formation. In later sections, we partially address this limitation by allowing the role that each party plays in a potential cabinet to differentially impact its electoral support.

More complicated cases than the one in the example can occur when more than one cabinet is in contention for selection. For example, suppose we have the same situation as the example above, except that besides the coalition between c and d , a single party cabinet, $\mathbf{g}''$, consisting only of c is also an alternative. The relevant utility differences are just:

$$\begin{aligned} E[u | v_a] - E[u | v_0] &= \gamma_a + (P_{\mathbf{g}\mathbf{g}'} + P_{\mathbf{g}\mathbf{g}''})Z \\ E[u | v_b] - E[u | v_0] &= \gamma_b + (P_{\mathbf{g}\mathbf{g}'} + P_{\mathbf{g}\mathbf{g}''})Z \\ E[u | v_c] - E[u | v_0] &= \gamma_c - (P_{\mathbf{g}'\mathbf{g}} + P_{\mathbf{g}''\mathbf{g}})Z \\ E[u | v_d] - E[u | v_0] &= \gamma_d - P_{\mathbf{g}'\mathbf{g}}Z \end{aligned}$$

Thus, the voter still has the same utility difference for all incumbent parties, but now if Z is negative (a poor economy) the expected utility of voting for party c will be greater than voting for party d , so (ignoring the γ 's) the voter will vote for party c . The reason for this is simply that the voter wants to cast the vote that contributes most to replacing the incumbent. Voters are indifferent as to which of the alternative cabinets comes to power (since they are all judged equally competent) and so party c 's membership in more alternatives gives it an advantage. The empirical hypothesis that follows from this is that in the absence of overlapping memberships

¹⁹⁴ The reason for this is that the voter's utility stemming from economic performance is defined over potential cabinets and the vote for any party in a cabinet has the same effect on the chances of selection as a vote for any other party (that is, electoral support of the cabinet responds to a change in a vote for each party in the cabinet in the same way - the derivative is 1 for all parties in the cabinet).

between the incumbent and alternative cabinets, the greater the number of competitive alternative cabinets to which an opposition party belongs, the more important the economic vote will be (relative to other factors) in the voter's utility for that party. Notice, however, that the voter is still indifferent over which party in the incumbent cabinet he chooses if the economy has been good.

More generally, if the membership of the competitive alternative cabinets is distinct from the incumbent cabinet, the utility of any opposition party is $-Z \sum_{g' \in A} P_{g'g}$ with all noncompetitive alternatives adding nothing to this sum. Thus, if the economy has performed sufficiently poorly then the voter will vote for the opposition party that is most competitive overall to be part of an alternative cabinet. Thus economic voting should be more important to these parties. In contrast, the performance of the retrospective economy will not lead the voter to favor one particular incumbent party or another.

When we turn to the case of a party that is a member of both the incumbent cabinet and a contending alternative cabinet, the voter's utility of casting a vote for the party is more complicated. Consider an example in which there are three parties labeled a , b and c and the incumbent cabinet consists of parties a and b . The only contending alternative cabinet is composed of b and c . In this case the utility difference for voting for each party versus abstaining is:

$$\begin{aligned} E[u | v_a] - E[u | v_0] &= \gamma_a + P_{gg'} Z \\ E[u | v_b] - E[u | v_0] &= \gamma_b + \left(P_{gg'} - P_{gg'} + P_{gg'} \left(2 \left(\int_0^\infty M_k dk \right) - 1 \right) \right) Z \\ &= \gamma_b + P_{gg'} (2P(K > 0) - 1) Z \\ E[u | v_c] - E[u | v_0] &= \gamma_c - P_{gg'} Z \end{aligned}$$

Notice that $0 \leq (2P(K > 0) - 1) \leq 1$ so that the voter's utility for the party, b , that is in both the incumbent coalition and the competitive alternative is always between $P_{gg'}$ and $-P_{gg'}$. This means that however the economy performs, there will always be a party that the voter prefers at least as much as party b (party a when the economy is good and party c when it is not). This result is intuitive, since the voter who wants to either keep or remove the incumbent cabinet can do so more efficiently when the whole contribution of the vote is felt in the electoral support of either the incumbent or the alternative but not both.

More generally, the implication of this model is that the economy will be less important to the vote of parties who are members of both the incumbent cabinet and competitive alternative cabinets than it will be for parties whose membership in the incumbent cabinet is distinct from competitive alternatives. This conclusion suggests, for example, that we should expect economic voting to be relatively muted for a party like the Danish Radical Liberals, who have a history of joining coalitions with the Social Democrats on the left as well as participating in rightwing bourgeoisie coalitions.

Finally, if we consider the most general case in which we allow for multiple alternative cabinets and overlapping memberships between the incumbent cabinet and these alternatives, we can break the portion of the voter's utility associated with the economy into four parts that help make the overall message plain. Specifically, we can write this portion of the utility as:

$$(8.27) \quad Z \left(g_j \sum_{\mathbf{g}' \in A} P_{\mathbf{g}\mathbf{g}'} - \sum_{\mathbf{g}' \in A} g'_j P_{\mathbf{g}\mathbf{g}'} + g_j \sum_{\mathbf{g}' \in A} g'_j P_{\mathbf{g}\mathbf{g}'} (2P(K > 0) - 1) \right)$$

where the g_j 's are defined as usual (i.e., $g_j = 1$ if party j is a member of cabinet $\mathbf{g}$). Thus, when Z is positive, for every pair of cabinets consisting of the incumbent cabinet and an alternative cabinet, the voter's utility for party j goes up if j is in the incumbent cabinet but not in the alternative. The voter's utility goes up by some amount or less (and may even go down) if j is a member of both the incumbent and the alternative cabinet. This makes it clear that when Z is positive the voters will vote for the incumbent party with the least "contamination" from membership in viable alternative cabinets. Similarly, if the economy has performed poorly, for every pair of cabinets, the voter's utility for party j goes up if the party is in the alternative cabinet but not the incumbent cabinet; but only goes up by the same amount or less (and may even go down) if the party is a member of both the incumbent and the alternative cabinet. However, since in this case one party may be a member of multiple cabinets that are contesting with the incumbent, it could be that the voter prefers a party that is incumbent but included in a number of alternatives, than one that is not incumbent but is in only one alternative. In the most general case, then we cannot conclude (as we did in the case of only one alternative) that when the economy is poor, the voter will prefer to vote for an opposition party over any incumbent party.¹⁹⁵

¹⁹⁵ For this to occur $P(K > 0)$ would need to be $< .5$ for some $\mathbf{g}$ and $\mathbf{g}'$. This means that it must be more likely that $w\mathbf{g}' > w\mathbf{g}$ than the opposite. For example, suppose that the incumbent coalition was a socialist party, center party coalition, but the center announced that it would not rejoin the government but would join one

In this section we extended the model of rational retrospective economic voting to situations in which there are multiple parties competing for positions in potentially multi-party cabinets. We did so under the assumption that voters get no information about the competence of alternative governments from the retrospective economy (even if these alternatives contain incumbent parties). If the membership of the incumbent cabinet and contending alternatives is distinct, we predict conventional economic voting. If some parties are members of both the incumbent cabinet and contending alternatives, economic voting in the form above is not guaranteed. However, the conditions necessary to cause voters to want to vote for incumbents parties when the economy has been poor are stringent and so unlikely. The interpretation of the competency signal from earlier sections generalizes in this model to the *responsibility augmented competency signal*. This change adds to our earlier conclusion regarding the competency signal the additional implication that a more equal division of policy-making responsibility in the incumbent cabinet should depress the economic vote. Our earlier conclusions about the distribution of contention generalizes to this model of coalitional executives; specifically, economic voting will be muted if the incumbent government is not in contention to return to office (either because it is sure to be selected or sure not to be). Any opposition party's share of the economic vote will tend to increase in the number of competitive alternatives of which it is a member as well as in the extent to which these coalitions are likely to tie with the incumbent for selection. The share of the economic vote of parties that are members of both the incumbent coalition and viable alternatives should decrease as the party's memberships in the incumbent and competitive alternative cabinets are less distinct.

Multiple Parties with a Coalitional Executive and Opposition Policy Making Responsibility

In the previous sections we maintained two assumptions that should be relaxed if we are to apply the model of rational retrospective voting to as wide a set of contexts as we would like. The first of these assumptions restricted policymaking responsibility to parties in cabinets. Clearly, however, other parties influence policymaking and in many contexts (e.g., minority government and independent legislatures) this influence is substantial (Strom 1990; Powell 2000; Hallerberg 2000). Thus, the rational voter's beliefs about the distribution of responsibility and her consequent voting behavior should reflect this wider notion of responsibility. Second, in the

of several possible right wing coalitions. In this case, the voter may well believe that $w_g' > w_g$ for each of the possible alternatives coalitions. Suppose they are all competitive for selection, then in the face of a poor economy the voter's utility for the center party will be $Z * (x_1 P_{gg'} + x_1 P_{gg''} + \dots)$ where the x 's are all negative and Z is negative which results in a positive number. This may well be larger than the voter's utility for opposition parties that are in one or more of the coalitions. This scenario though seems unlikely.

previous sections we maintained Alesina and Rosenthal’s assumption that the retrospective economy provides no information about the competence of non-incumbent parties. Further, in generalizing the model to multiparty cabinets we adopted the additional assumption that the economy provides no information about the overall competence of non-incumbent cabinets, even when these alternative cabinets include members of the current cabinet. Such an assumption should hardly be maintained in a model in which voters use all the information available to them in forming their beliefs. Below, we relax these two assumptions in turn and then discuss whether doing so maintains or changes the various conclusions of previous sections.

We begin by redefining a *distribution of responsibility* to be a vector of shares of policymaking responsibility (with elements λ_j) such that all the parties have shares greater than or equal to zero and the shares sum to one across parties. One is free to define opposition and cabinet parties as those for which $\lambda_j = 0$ or $\lambda_j > 0$, respectively. However, by allowing λ_j to be greater than zero for parties that are not in the cabinet, one can capture more interesting institutional variation than we have so far allowed (i.e., minority governments). Thus, in what follows we will drop our “g” notation for indicating a cabinet and instead use λ to refer to a distribution of responsibility across all J parties in the legislature (i.e., $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_J\}$). As above, a λ will be used to refer to the status quo or “incumbent” distribution of responsibility (or for a generic distribution of responsibility when there is no chance of confusion), while λ' and the like refer to non-incumbent, alternative distributions of responsibility.

This generalization can be readily incorporated into the theoretical framework developed in previous sections. Specifically, we can write an expression for the expected utility of living in the next period under the current distribution of responsibility that follows exactly the logic that we developed for the expected utility of continuing the current cabinet. First, we have, mirroring equation (8.19):

$$\begin{aligned}
 (8.28) \quad E[u_{\lambda,t+1}] &= \frac{1}{2} E[\pi_{\lambda,t+1}^2] + bE[y_{\lambda,t+1}] = b(\bar{y} + E[\eta_{\lambda,t+1}]) \\
 &= b(\bar{y} + E[\xi_{t+1}] + E[\varepsilon_{\lambda,t+1}]) \\
 &= b(\bar{y} + E[\varepsilon_{\lambda,t+1}]) = b(\bar{y} + E[\mu_{\lambda,t+1} + \mu_{\lambda,t}]) \\
 &= b(\bar{y} + E[\mu_{\lambda,t}]) = b(\bar{y} + E[\mu_{\lambda,t} | y_{\lambda,t}])
 \end{aligned}$$

It can be shown in an argument analogous to the one for cabinets developed earlier that the conditional expectation in the last line is:

$$(8.29) \quad E[\mu_{\lambda,t} | y_{\lambda,t}] = \left(\frac{\sigma_\mu^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2} \right) (y_{\lambda,t} - \bar{y} - \mu_{\lambda,t-1})$$

This is essentially the same result as in equation (8.18), except that now the summations are over all parties instead of just cabinet parties. Plugging this into equation (8.28) above, we get a similarly familiar expression for the utility of the status quo distribution of responsibility for another period:

$$(8.30) \quad E[u_{\lambda,t+1}] = b\bar{y} + b \left(\frac{\sigma_\mu^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2} \right) (y_{\lambda,t} - \bar{y} - \mu_{\lambda,t-1})$$

Finally, retaining (until the next subsection) the assumption that voters assign zero competence to all alternatives to the status quo distribution of responsibility, the appropriate expression for the utility difference between voting for party j and abstaining (that mirrors equation (8.21) and is proved along with it in Appendix A) is:

$$(8.31) \quad \begin{aligned} E[u | v_j] - E[u | v_0] &= \gamma_j + \sum_{\lambda' \in \Lambda'} P_{j,\lambda\lambda'} (E[u_{\lambda,t+1}] - E[u_{\lambda',t+1}]) \\ &\quad + \sum_{\lambda' \in \Lambda'} P_{j,\lambda'\lambda} (E[u_{\lambda',t+1}] - E[u_{\lambda,t+1}]) + \sum_{\lambda' \in \Lambda'} \sum_{\lambda'' \in \Lambda'} P_{j,\lambda'\lambda''} (E[u_{\lambda',t+1}] - E[u_{\lambda'',t+1}]) \\ &= \gamma_j + \sum_{\lambda' \in \Lambda'} P_{j,\lambda\lambda'} (E[u_{\lambda,t+1}] - 0) + \sum_{\lambda' \in \Lambda'} P_{j,\lambda'\lambda} (0 - E[u_{\lambda,t+1}]) \\ &\quad + \sum_{\lambda' \in \Lambda'} \sum_{\lambda'' \in \Lambda'} P_{j,\lambda'\lambda''} (0 - 0) \\ &= \gamma_j + b \left(\frac{\sigma_\mu^2 \sum_{i \in \mathbf{J}} \lambda_{i,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{i \in \mathbf{J}} \lambda_{i,t}^2} \right) (y_t - \bar{y} - \mu_{\lambda,t-1}) \sum_{\lambda' \in \Lambda'} (P_{j,\lambda\lambda'} - P_{j,\lambda'\lambda}) \end{aligned}$$

where Λ' is the set of all non-status quo distributions of responsibility. There will be J of these equations and the voter will vote for the party j for which this expression is the largest. The pivot probabilities, $P_{j,\lambda\lambda'}$ and $P_{j,\lambda'\lambda}$, are interpreted as the chance that λ and λ' are in a tie for selection and a vote for party j will break the tie in favor of the status-quo government λ . The formal definition of $P_{j,\lambda\lambda'}$ is in Appendix A. This derivation parallels the one for cabinets, but

defines the *electoral support* for any distribution of responsibility as the sum of the support for all parties, weighted by their share of responsibility in that distribution. Thus, we have $P_{j,\lambda\lambda'}$ equal to $P_{\lambda\lambda'}P(D_{\lambda,j} > D_{\lambda',j})$, where $P_{\lambda\lambda'}$ is the probability that λ and λ' are tied for selection and the D 's are derivatives of the selection functions for λ and λ' with respect to party j 's vote total.¹⁹⁶ Given our definition of *electoral support* for λ , each $D_{\lambda,j}$ is equal to $w_\lambda \lambda_j$, where w_λ is the derivative of the selection function for cabinet λ with respect to the electoral support for λ , and λ_j is party j 's share of the distribution of responsibility λ .¹⁹⁷

This change in focus from cabinets to distributions of responsibility does not change anything substantively interesting in our earlier conclusions about the role of the *responsibility augmented competency signal* in rational retrospective voting.¹⁹⁸ Thus, it is still the case that the *responsibility augmented competency signal* is minimized when parties have equal shares of responsibility. In the current model, however, this equality is across all parties and not just those in the cabinet. Given this, we focus the following discussion of equation (8.31) on the role that the *distribution of contention* plays in conditioning economic voting.¹⁹⁹ Given the definitions above and in Appendix A, we can rewrite equation (8.31) as:²⁰⁰

$$\begin{aligned}
 E[u | v_j] - E[u | v_0] &= \gamma_j + Z \sum_{\lambda' \in \Lambda'} (P_{j,\lambda\lambda'} - P_{j,\lambda'\lambda}) \\
 &= \gamma_j + Z \sum_{\lambda' \in \Lambda'} (P_{\lambda\lambda'}P(D_{\lambda,j} > D_{\lambda',j}) - P_{\lambda'\lambda}P(D_{\lambda',j} > D_{\lambda,j})) \\
 &= \gamma_j + Z \sum_{\lambda' \in \Lambda'} P_{\lambda\lambda'} (P(w_\lambda \lambda_j - w_{\lambda'} \lambda'_j > 0) - P(w_{\lambda'} \lambda'_j - w_\lambda \lambda_j > 0)) \\
 &= \gamma_j + Z \sum_{\lambda' \in \Lambda'} P_{\lambda\lambda'} (2P(w_\lambda \lambda_j - w_{\lambda'} \lambda'_j > 0) - 1)
 \end{aligned}
 \tag{8.32}$$

¹⁹⁶ Recall that the post-election process that distributes responsibility among parties simply implements the distribution of responsibility with the highest selection value.

¹⁹⁷ Again we make assumption #1 from Appendix A, which just says that w_λ is non-negative for all distributions of responsibility.

¹⁹⁸ Other than the fact, of course, that we now consider the full distribution across all parties rather than cabinet parties.

¹⁹⁹ This distribution is no longer properly called *the pattern of cabinet contention* because it now means the pattern of contention for selection among different distributions of responsibility, not cabinets.

²⁰⁰ Clearly the incumbent distribution of responsibility is included in Z so will change when we consider different incumbents, but this does not change the interpretation of how that term impacts economic voting given a distribution of responsibility.

where Z represents those parts of the equation that do not differ across parties and we use the fact that $P(x > 0) = 1 - P(x < 0)$.

The first point to make about equation (8.32) is that it maintains our earlier conclusion that contention is a critical ingredient in rational retrospective economic voting. The specific form of contention required in this model is contention for selection between the status-quo distribution of responsibility and one or more alternatives. That is, if the incumbent distribution of responsibility is sure to be maintained or sure to be changed, then there should be no economic voting in this model.²⁰¹

In order to come to any substantive conclusions about the voter's utility difference for voting for different parties rather than abstaining, we need to evaluate the term inside the sum in equation (8.32) – i.e., $P_{\lambda\lambda'} \left(2P(w_{\lambda}\lambda_j - w_{\lambda'}\lambda'_j > 0) - 1 \right)$. Further, notice that as the probability $P(w_{\lambda}\lambda_j - w_{\lambda'}\lambda'_j > 0)$ approaches one, this whole term approaches $P_{\lambda\lambda'}$, but as the probability approaches zero this term approaches $-P_{\lambda\lambda'}$. Consequently, interpreting when the voter's utility difference for voting for party j rather than abstaining will be positive or negative (given Z) reduces to evaluating $P(w_{\lambda}\lambda_j - w_{\lambda'}\lambda'_j > 0)$.

We assume that voters know how much responsibility party j holds in the status quo distribution of authority as well as in any alternative distribution they are considering. Given this we can write $P(w_{\lambda}\lambda_j - w_{\lambda'}\lambda'_j > 0)$ more conveniently as $P\left(\frac{w_{\lambda'}}{w_{\lambda}} < \frac{\lambda_j}{\lambda'_j}\right)$, where $\frac{w_{\lambda'}}{w_{\lambda}}$ is a random variable and $\frac{\lambda_j}{\lambda'_j}$ is a known constant.²⁰² This probability is in the form of a cumulative probability function and, since any such function must integrate to one and be non-decreasing, it must be the case that as the ratio on the right hand side increases, this probability approaches one. Thus, we

²⁰¹ If we imagine that no distribution of responsibility will ever be exactly maintained, this would suggest that we would never see economic voting. This conclusion, however, depends on our assumption that the only distribution of responsibility to which the voter assigns non-zero competence is the incumbent one. In the next subsection we relax this assumption and so let the voter's information about the competence of different distributions of responsibility change continuously with the similarity of the alternative to the status quo. This is sufficient to soften our conclusion to one in which economic voting should be greater when distributions of responsibility “close” to the status quo are competitive (with alternatives that are not close to the status quo distribution).

²⁰² We assume $\lambda'_j > 0$. It is not difficult to account for the case in which $\lambda'_j = 0$ but doing so does not change the substantive message or add any additional insight, so we do not present the additional qualifications that would be required to include this case in the current discussion.

can conclude that, given two parties j and i , this probability will be larger for party j if λ_j / λ'_j is larger than λ_i / λ'_i .²⁰³ If we now go back to equation 8.32 and focus on the case in which there is only one alternative distribution of responsibility, we can conclude that as λ_j / λ'_j gets larger (i.e., party j 's share of responsibility in λ increases relative to its share in λ') the change in utility associated with voting for party j rather than abstaining will move away from $-P_{\lambda\lambda}Z$ and toward $P_{\lambda\lambda}Z$.

This is a natural generalization of our earlier result for cabinets. In that case, if a party was a member of the incumbent cabinet but not the alternative, the voter's utility for voting from that party was $P_{\lambda\lambda}Z$. In contrast, if the party were a member of the alternative and not the incumbent cabinet the utility was $-P_{\lambda\lambda}Z$. In the current case, "incumbency" is not a discrete category so we get an appropriately continuous result in which having relatively more responsibility in the incumbent distribution of responsibility than in the alternative pushes the voter's utility toward $P_{\lambda\lambda}Z$ and away from $-P_{\lambda\lambda}Z$.

Notice that this change overturns our earlier result that the voter's utility of voting for different members of the incumbent cabinet was the same, as long as these parties were not members of a viable alternative cabinet. In this generalization, this result does not hold because the distribution of responsibility distinguishes between shares of responsibility among different cabinet members and so voters distinguish between the contributions that votes for different cabinet members will have on the selection of the incumbent distribution of responsibility.²⁰⁴ More specifically, they believe these contributions are proportional to the parties' shares of responsibility. In this way, the generalized model produces an implication that is close to Anderson's (1995) suggestion that increased cabinet responsibility should enhance a party's economic vote. This implication, however, comes from a completely different logic than

²⁰³ Notice that this does not require that we assume that the voter's beliefs about the ratio of derivatives on the left hand side of the inequality are the same across the parties being compared, since this is true by definition. Specifically, w_λ is the derivative of the selection function for λ with respect to the total electoral support for λ and so will be the same across all parties. It would be strange to assume voters' beliefs about this quantity would vary across parties, given the quantity itself does not vary.

²⁰⁴ If we assume that opposition members have no responsibility, then we can properly call this the *distribution of responsibility within the cabinet* and say that the voter believes the impact of her vote on the reselection of the cabinet (meaning the exact replication of the shares of responsibility of the incumbent parties in the cabinet) to be proportional to the parties share of responsibility.

Anderson suggests.²⁰⁵ Indeed, it does not even stem from the voter's attempt to extract a signal about the competence of different parties from the previous economy (as did our earlier conclusion about the impact of power sharing on the economic vote of the *whole* cabinet or distribution of responsibility). Instead, this implication stems directly from our assumption that voters believe the impact of a vote for party j on the selection of a distribution of responsibility λ is proportional to party j 's share of responsibility in λ and that they want to vote rationally to maximize that impact.²⁰⁶

To stimulate better intuition for the implications of this generalization, we can examine an example that parallels the first one we provided following equation (8.26). In that equation we had four parties, two of which were in the incumbent cabinet and two of which were in the only viable alternative cabinet. To parallel this, but make it relevant to the analysis of more widely shared responsibility, suppose that responsibility, λ , assigns 60% the responsibility to party a , 20% to party b , and 10% to parties c and d respectively (i.e., $\lambda = \{.6, .2, .1, .1\}$). There is only one viable alternative distribution of authority: $\lambda' = \{.1, .1, .2, .6\}$. Given equation (8.32), the difference in the voter's expected utility of voting for each party rather than abstaining is:²⁰⁷

$$\begin{aligned} E[u | v_a] - E[u | v_0] &= \gamma_a + \left(P_{\lambda\lambda'} \left(2P \left(\frac{w_{\lambda'}}{w_{\lambda}} < 6 \right) - 1 \right) \right) Z \\ E[u | v_b] - E[u | v_0] &= \gamma_b + \left(P_{\lambda\lambda'} \left(2P \left(\frac{w_{\lambda'}}{w_{\lambda}} < 2 \right) - 1 \right) \right) Z \\ E[u | v_c] - E[u | v_0] &= \gamma_c + \left(P_{\lambda\lambda'} \left(2P \left(\frac{w_{\lambda'}}{w_{\lambda}} < \frac{1}{2} \right) - 1 \right) \right) Z \\ E[u | v_d] - E[u | v_0] &= \gamma_d + \left(P_{\lambda\lambda'} \left(2P \left(\frac{w_{\lambda'}}{w_{\lambda}} < \frac{1}{6} \right) - 1 \right) \right) Z \end{aligned}$$

If $f \left(\frac{w_{\lambda'}}{w_{\lambda}} \right)$, the probability distribution governing voters' beliefs about the ratio $\frac{w_{\lambda'}}{w_{\lambda}}$ is the same over parties, then (again ignoring γ) the voter will vote for party a when Z is positive and party d when Z is negative.

More generally, if there is only one alternative to the status quo distribution of responsibility and the economy has performed well, the voter will vote for the party with the

²⁰⁵ In fact, Anderson (1995) reviews three different reasons one might expect economic voting to differ among coalition partners, none of which have much in common with the explanation proposed here.

²⁰⁶ This implication differs from Anderson (1995) to some degree since it applies to all parties and not just cabinet parties. Empirically, then, we expect the strength of economic voting for a party to be proportional to the party's overall policy influence, not just its share of influence in cabinet.

²⁰⁷ Recall that in Equation 8.32 $P(w_{\lambda} \lambda_j - w_{\lambda'} \lambda'_j > 0)$ can be expressed as $P \left(\frac{w_{\lambda'}}{w_{\lambda}} < \frac{\lambda_j}{\lambda'_j} \right)$,

highest ratio of shares of responsibility in the two competing distributions of responsibility. Likewise, when the economy has performed poorly, the voter will vote for the party with the lowest ratio. As in the case of cabinets, the message is clear: a party will benefit more from rational economic voting when a vote for the party unambiguously helps or hurts the selection value of the incumbent distribution of responsibility. That is, the party that offers the voter a *distinctive* choice.

Allowing more than one competitive alternative distribution of responsibility complicates the calculation of utilities in the model. However, it does not change the message that a party's share of the economic vote will be greater when the party's role in the incumbent distribution of responsibility is more distinct from its role in competitive alternatives

In this section we extended the model of rational retrospective economic voting to situations in which there are multiple parties competing for shares of policymaking responsibility. We continue to do so under the assumption that voters get no information about the competence of alternative distributions of responsibility from the retrospective economy (even if these alternative distributions assign responsibility to some parties that held responsibility under the status quo distribution). This model generalizes our previous finding that if the membership of the incumbent cabinet and contending alternatives is distinct, we get economic voting in a traditional form. In the current case, however, "incumbency" is not a discrete category so we get an appropriately continuous result in which having relatively more responsibility in the incumbent distribution of responsibility than in the alternative pushes the voter's utility toward $P_{\lambda\lambda}Z$ and away from $-P_{\lambda\lambda}Z$. Thus, for a negative Z (a poor economy), the voter will turn away from parties that hold more responsibility in the incumbent distribution of power than in contending alternatives.

Multiple Parties with a Coalitional Executive, Opposition Policymaking Responsibility and Rational Beliefs about the Competence of Alternative Governments

The assumption that voters have no information about the competence of alternative cabinets or distributions of responsibility seems unjustified in a model in which voters use information rationally. Specifically, some alternatives to the status-quo will necessarily include parties that were in the previous cabinet or for which λ_j was positive in the previous period. Consequently, the voter may have information about the competence of these parties that should bear on the evaluation of possible governing alternatives.

In order to allow for this possibility in the model, we begin by writing an expression for the expected utility of living under some alternative distribution of responsibility in the next period that does not assume expected competence is zero:

$$\begin{aligned}
 (8.33) \quad E[u_{\lambda',t+1}] &= \frac{1}{2} E[\pi_{\lambda',t+1}^2] + bE[y_{\lambda',t+1}] = b(\bar{y} + E[\eta_{\lambda',t+1}]) \\
 &= b(\bar{y} + E[\xi_{t+1}] + E[\varepsilon_{\lambda',t+1}]) \\
 &= b(\bar{y} + E[\varepsilon_{\lambda',t+1}]) = b(\bar{y} + E[\mu_{\lambda',t+1} + \mu_{\lambda',t}]) \\
 &= b(\bar{y} + E[\mu_{\lambda',t}]) = b(\bar{y} + E[\mu_{\lambda',t} | y_{\lambda,t}])
 \end{aligned}$$

Notice that we write λ' everywhere except in the conditional on the last line. This is important. This conditional expectation is the expected value of the current competence of the alternative distribution of administrative responsibility (λ') given the observed economy under the status-quo distribution (λ). In order to calculate this expectation we will have to unpack the voter's overall assessment of the current competence of the government and get an expression for expectations about the competence of each individual party. With this in hand, we can construct the voter's expectations for the competence of various alternative governments. In doing this, we should expect the current economy to provide no information about the competence of parties that have no administrative responsibility under the status quo distribution of responsibility. Thus, the voter's belief about the competency of such parties should just be the mean of the competency distribution (i.e., zero). However, if a party does have a share of the incumbent distribution of responsibility, the voter should use information about the economy in producing expectations about that party's competence. If we write the voter's expectation about the competence of party i , given the economy under incumbent distribution of responsibility λ as $E[\mu_{i,t} | y_{\lambda,t}]$ then we expect that in general $E[\mu_{i,t} | y_{\lambda,t}] \neq 0$ for parties in which $\lambda_i \neq 0$. What is $E[\mu_{i,t} | y_{\lambda,t}]$? In Appendix C we solve the signal extraction problem that leads to the following intuitive result:

$$(8.34) \quad E[\mu_{i,t} | y_{\lambda,t}] = \left(\frac{\sigma_\mu^2 \lambda_{i,t}}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{j \in J} \lambda_{j,t}^2} \right) (y_{\lambda,t} - \bar{y} - \mu_{k,t-1})$$

Thus, the voter's expectation about the competence of party i is just the competency signal associated with the incumbent distribution of responsibility λ , weighted by party i 's role in that

distribution. This expression is in line with our intuition that if λ_i is zero, then the conditional expectation of party i 's competence will also be zero. Further, the expression implies that:

$$(8.35) \quad E[\mu_{k,t} | y_{k,t}] = \sum_{i \in \mathbf{J}} \lambda_i E[\mu_{i,t} | y_{k,t}]$$

With these two expressions we are now in a position to propose how the voter can estimate the current competence of any alternative government. Specifically, we suggest that:

$$(8.36) \quad E[\mu_{k',t} | y_{k,t}] = \sum_{i \in \mathbf{J}} \left\{ \lambda'_{i,t} E[\mu_{i,t} | y_{k,t}] \right\} \\ = \sum_{i \in \mathbf{J}} \left(\frac{\sigma_\mu^2 \lambda_{i,t} \lambda'_{i,t}}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2} \right) (y_{k,t} - \bar{y} - \mu_{k,t-1})$$

Thus, the voter assessment of an alternative government's current competence is formed by weighting the current competence of each party by the role the party will play in the alternative government and summing over all parties. If the alternative distribution of executive responsibility is the same as the status quo distribution (i.e., $\lambda = \lambda'$) this expression reduces to the expression in Equation (8.29) as it should. In contrast, if no party with a share of responsibility in the status-quo distribution of responsibility also has a share responsibility in the new distribution, this expectation is zero.

With this, we can write the utility the voter expects if some alternative distribution of executive authority should obtain as:

$$(8.37) \quad E[u_{k',t+1}] = b(\bar{y} + E[\mu_{k',t} | y_{k,t}]) \\ = b\bar{y} + b \left(\frac{\sigma_\mu^2 \sum_{i \in \mathbf{J}} \lambda_{i,t} \lambda'_{i,t}}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2} \right) (y_{k,t} - \bar{y} - \mu_{k,t-1})$$

To finish the derivation and get an expression equivalent to the last line of equation 8.32 but allowing non-zero competence for alternatives, we just substitute the last line of equation 8.37 into the first line of equation 8.31:

$$\begin{aligned}
E[u | v_j] - E[u | v_0] = & Z \sum_{\lambda' \in \Lambda'} P_{j, \lambda \lambda'} \left(\sum_{i \in J} \lambda_{i,t} (\lambda_{i,t} - \lambda'_{i,t}) \right) + \\
(8.38) \quad & Z \sum_{\lambda' \in \Lambda'} P_{j, \lambda' \lambda} \left(\sum_{i \in J} \lambda_{i,t} (\lambda'_{i,t} - \lambda_{i,t}) \right) + \\
& Z \sum_{\lambda' \in \Lambda'} \sum_{\lambda'' \in \Lambda'} P_{j, \lambda' \lambda''} \left(\sum_{i \in J} \lambda_{i,t} (\lambda'_{i,t} - \lambda''_{i,t}) \right)
\end{aligned}$$

where,

$$Z = b \left(\frac{\sigma_\mu^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_{j \in J} \lambda_{j,t}^2} \right) (y_{k,t} - \bar{y} - \mu_{k,t-1})$$

Again, there will be J of these expressions and voters will vote for the party for which this expression is largest. Z differs from our earlier definitions, but is still constant over parties. Unlike our previous generalizations, in which the our conclusions about the responsibility augmented competency signal stayed the same from model to model, so that we focused on the implications for the distribution of contention, this generalization changes both our previous conclusions about contention and our conclusions about the competency signal. To make our discussion of these changes more transparent, we will begin with a discussion of the case in which each pair of distributions of responsibility that are in contention include the status-quo distribution. This case focuses attention on the responsibility augmented competency signal because where there are no pairs of competitive alternative distributions, our conclusions regarding the distribution of contention are exactly the same as those in the earlier section. Specifically, under this assumption we can write equation (8.38) as:

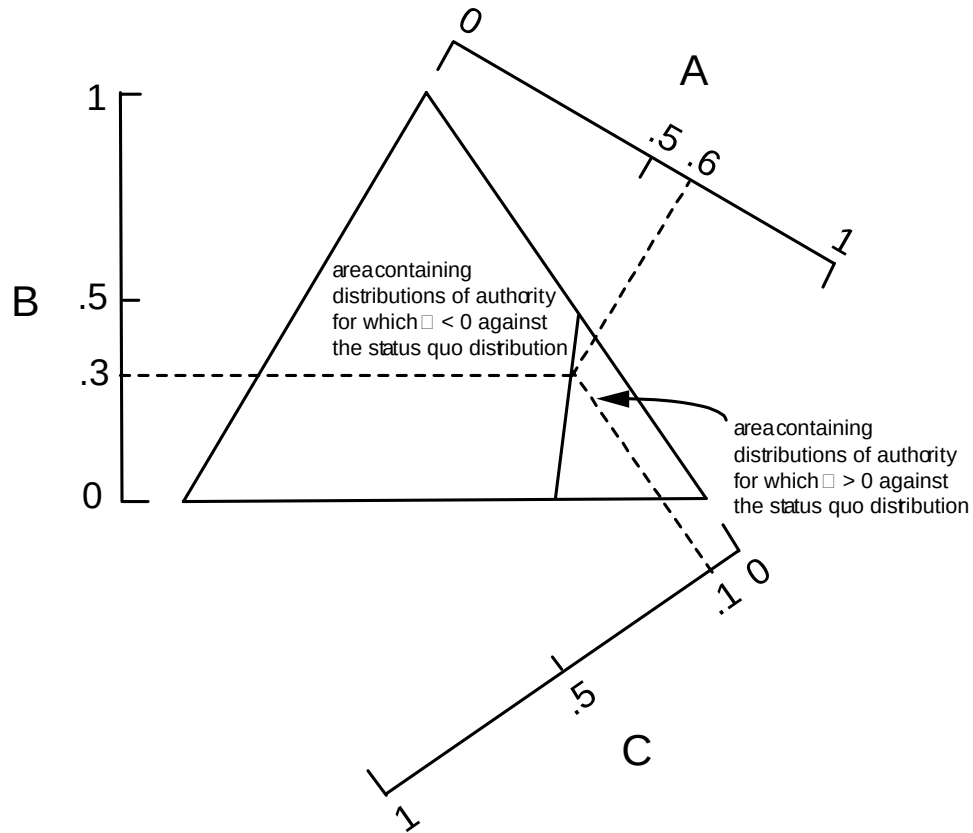
$$\begin{aligned}
E[u | v_j] - E[u | v_0] = \\
(8.39) \quad & Z \sum_{\lambda' \in \Lambda'} P_{\lambda \lambda'} \left(2P(w_\lambda \lambda_j - w_{\lambda'} \lambda'_j > 0) - 1 \right) \left(\sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda'_i) \right)
\end{aligned}$$

The first two terms in the product impact economic voting in exactly the same way as in the last section. However, with the addition of non-zero competence for alternative distributions of responsibility, we have altered the competence signal that voters use to weight economic voting. Specifically, while the signal is still constant over all parties, it is now not always positive. In equation (8.39) the sign of the last term depends on which alternative distributions of authority are in contention. If the competitive alternatives are very different from the status quo

distribution (giving parties with large shares of responsibility under the status quo small shares under the alternative and visa versa) then $\sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda_i')$ will be positive. Indeed, it reaches its maximum, given a fixed λ , for distributions in which $\lambda_i \lambda_i' = 0$ for all parties, meaning that no parties hold responsibility in both the status quo and alternative distributions. In contrast, it is equal to zero when there is no difference between the alternative and status quo distributions of responsibility.

The easiest way to understand what this means for economic voting is to examine the example provided in Figure 8.2. In this example, there are three parties and the status quo distribution of responsibility assigns 60% of the responsibility to party A, 30% to party B, and 10% to party C. This ternary diagram maps this point in the lower right side portion of the triangle and shows all the other possible distributions of responsibility (the area of the triangle). $\sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda_i')$ is positive for any distribution of responsibility, λ' , in the large area to the left of the solid line bisecting the triangle. In contrast, $\sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda_i')$ is negative for the distributions of responsibility to the right of the solid bisecting line.

Figure 8.2



Note: $\omega = \sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda_i')$ in Figure 8.2 and labeled axes are shares of party responsibility.

If there are only two distributions of responsibility in contention (the status quo and some alternative), this means that if the other term in equation (8.39) is positive for some party and, for example, the economy is improving, the utility associated with voting for this party will be positive as the competitive alternative is to the right of the bisecting line and negative if it is to the left. The “preferred” alternatives to the right are those that, in general, concentrate even more responsibility on the parties that had the most responsibility in the status-quo distribution. Focusing only on the area to the right of the bisecting line, almost all of these preferred

alternatives increase the responsibility of party A, which was most responsible for the good economic performance. It is interesting, however, that, in this example, there is a relatively small area in which preferred alternatives increase the responsibility of the second most responsible party, B. Instead, the improvement in A's position is more often at the expense of B, if only because B has more responsibility to lose than does C. Substantively this is intriguing. It suggests that when we allow voters to use information about the current competence of incumbents to infer the competence of alternative distributions of responsibility, then economic success may lead to a concentration of the vote on the party with the most responsibility as a way to produce a new cabinet that "rewards" this party at the expense of the others. If we were to make the reasonable assumption that, in this example, party A held the prime ministry and party B was a cabinet partner, then the implication is that the economy would cause votes to flow to the prime minister and away from both the opposition party C and the smaller cabinet partner B. The implication, more generally, is that economic voting in its traditional form (i.e., electoral rewards for incumbents and punishments for opposition) should apply most strongly to the leading parties in the cabinet and least strongly to partners, who may (in this model) have the opposite result. In the empirical chapters that follow, we will be interested in whether the economic vote (in its traditional form) is more consistent for parties with a greater share of responsibility.

Of course, it is also clear from the diagram that $\sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda_i')$ can be positive for some alternatives that give less responsibility to the leading party. This small area is the triangle to the right of the bisecting line that is above the dashed line leading to A's axis. In this area, the alternatives award greater responsibility to B at the expense of A. However, it is also the case that in this area the ratio λ_A / λ_A' (that is, party A's share in the status quo over its share in the alternative) is greater than one, while B's share is less than one. But, recall that in the last section we saw that $P_{\lambda\lambda'} \left(2P(w_\lambda \lambda_j - w_{\lambda'} \lambda_j') > 0 \right) - 1$ would be bigger for party j than for party i if $\lambda_j / \lambda_j' > \lambda_i / \lambda_i'$. Thus, in this area parts of equation 8.32 other than $\sum_{i \in J} (\lambda_{i,t}^2 - \lambda_i \lambda_i')$, which is positive, will likely be bigger (and more likely to be positive) for A than for B, so that the overall utility of voting for party A in this region may well be bigger than B even in this region. Finally, we can consider the third term in Equation 8.38, which allows the possibility of economic voting when the status quo distribution of responsibility is not in contention with any cabinet, but rather two alternatives are in contention with each other. The interpretation of this term is

essentially the same as the one above. Specifically, if the status-quo distribution of responsibility is not in contention then we can write:

$$(8.40) \quad E[u | v_j] - E[u | v_0] = \sum_{\lambda' \in \Lambda'} \sum_{\lambda'' \in \Lambda'} P_{\lambda', \lambda''} \left(2P(w_{\lambda'} \lambda_j - w_{\lambda''} \lambda_j' > 0) - 1 \right) \left(\sum_{i \in J} (\lambda_i \lambda_i' - \lambda_i \lambda_i'') \right)$$

Notice that it still contains the elements of the status quo distribution of authority since it is only through their responsibility for current economic outcomes that voters have any information at all about the competence of the parties in these contending alternatives. Clearly, if no party in either alternative held responsibility in the status quo distribution, these terms will be zero and there will be no economic voting. Further, if λ_j' and λ_j'' are very similar to one another for most parties this term will be small and economic voting will be muted. So again, we find that distinct alternatives enhance economic voting.

Also, notice that if those parties in the two alternatives that had some experience in the previous distribution of authority, have equal shares of power in the new distribution, this sum will be zero. For example, suppose that the incumbent distribution of responsibility was a “grand coalition” in which two large parties shared power equally and all other parties had no responsibility at all. Further, suppose that this coalition breaks up and cannot reform, and that the only contending alternatives are two one-party governments in which one or the other former partners rules alone.²⁰⁸ Under these circumstances, the sum in the equation 8.40 is $(.5*1 - .5*0) + (.5*0 - .5*1) = 0$. The intuition of this result is clear: because the only information the voter has about the competence of the parties comes from economic outcomes under the status quo distribution of responsibility and voters apportion responsibility for those outcomes to the two large parties equally, they must conclude that the two alternatives they must decide between have the same level of competence. Because of this, they cannot distinguish between them and so there will be no economic voting in this case (since in our example these are the only contending distributions).

²⁰⁸ The same result applies to any coalitions of parties that include only one of the two large parties but give some responsibility to other parties that did not serve in the previous distribution of responsibility. This follows from the fact that these other parties can contribute nothing to the sum in Equation (5.2) since they had not responsibility in the status quo distribution.

The most prominent cases of “grand coalitions” in the western democracies are those in Austria and Germany and these look a lot like the example that we gave. The theory would thus suggest that economic voting during the reign of such coalitions should be muted. In this section we extended the model of rational retrospective economic voting to situations in which there are multiple parties competing for shares of policymaking responsibility but the voter can use information about the participation of parties in the incumbent distribution of responsibility to infer the competence of new alternative distributions that involve those parties.

The model leads to two main conclusions that finish our generalization of the rational retrospective model of economic voting. Economic voting in a general form is apparent in this model. First, if all the contending distributions of authority include the status quo distribution, the expected utility of voting for a party, when the economy has been good, is greater for parties that have a relatively greater share of the incumbent distribution of responsibility and is larger overall when the differences between the contending alternatives and the status quo distribution are large. When the incumbent distribution of responsibility is not competitive the previous point still holds but economic voting overall is larger as there are greater differences between non-status quo distributions of authority that are in contention. Again, then economic voting depends both on sufficient representation of the incumbent distribution of responsibility in the new distribution and on distinctiveness between alternatives.

Summary

Most economic voting takes place in contexts with multiple parties. This chapter extends the competency results from Chapter 5 to accommodate economic voting in multiparty contexts. Our assumption that voters are fully informed about the political and economic context implies that voters will condition their economic vote on the distribution of policymaking responsibility and on the pattern of cabinet contention.

Multiple parties also introduce the possibility of shared responsibility in governing. And voters in our model are assumed to be fully informed about how power is shared by parties in governing coalitions but also more broadly in the legislative process. The results presented in this chapter make it clear that the distribution of administrative responsibility should condition the extent of economic voting as well as its distribution across parties. Specifically, as responsibility is more equally shared, economic voting in an election should generally decrease. Further, as a given party’s share of responsibility decreases, the economy should be down weighted in the voter’s utility for that party. In general (where there are more than two parties) this should lead to

less of an economic vote for that party. In Chapter 9, we explore how economic voting varies with these beliefs about the distribution of administrative responsibility.

Multiple parties introduce the likelihood of strategic voting. Our generalization of the rational retrospective economic voting model to the case of multiparty competition allows our rational voters to condition their votes on the likely results of the election – specifically the distribution of policymaking responsibility that will likely result from the election outcome. We showed that the economy would only be important to the utility comparisons between parties that were “in competition” to receive a share of responsibility after the election. In simple plurality systems this meant competing to win the most votes, but in other systems this meant being a part of competitive alternative cabinets or, more generally, competitive distributions of responsibility. Further, if the incumbent (party, cabinet, or distribution of responsibility) was not in contention, either because it was sure to win or sure to lose, economic voting would be muted. Finally, in the models of contexts that allowed coalition cabinets and shared responsibility, the distinctiveness of the alternatives mattered to both the overall magnitude of the economic vote and its distribution across parties. In general, those parties whose role in the incumbent cabinet (or in the incumbent distribution of responsibility) was distinct enough from its role in contending alternative cabinets received a greater share of the economic vote than those whose roles were not distinct. In Chapter 10, we explore the implications of these arguments by evaluating the extent to which the economic vote is important to parties that are (or are not) in contention for different roles in the cabinet.

Chapter 9

The Distribution of Responsibility and Economic Voting

Two theoretical propositions come out of the discussion of administrative responsibility, each of which produces a straightforward hypothesis. The first concerns the distribution of the economic vote across parties: Parties with a greater share of the status-quo distribution of administrative responsibility will receive a greater share of the economic vote than parties with a smaller share. The second concerns the overall size of the economic vote across all parties in an election: As the status-quo distribution of administrative responsibility over parties becomes more equal, the overall economic vote declines.

The main task in testing these empirical hypotheses is the measurement of voters' beliefs about the share of administrative responsibility that each party holds. A variety of indicators of the status quo distribution of policymaking responsibility have been discussed in the literature: the current distribution of cabinet membership, the current distribution of cabinet portfolios, the coalition status of the government, the majority status of the government, the influence of the opposition on the government, the extent of collective cabinet responsibility, the distribution of legislative seats, the distribution of ministries specifically dealing with economic matters, and the role of the president. Further, the values of these variables are so widely reported and so well known that we can assume voters' beliefs about them closely mirror the empirical reality, at least on average. Thus, in this chapter, we will examine how these different indicators of the distribution of responsibility are related to both the overall economic vote and its distribution across parties.

Of course, we are not the first to examine the empirical relationship between administrative responsibility and economic voting. As we emphasized in Chapter 1, Powell and Whitten (1993) used aggregate electoral data to explore whether various measures of power sharing, such as coalition and minority status, predicted depresses economic voting. Likewise, using a different sample of aggregate data, Powell (2005) explored how power sharing in presidential regimes influenced the magnitude of the economic vote. Stevenson (1996) examined the impact of majority status, coalition status, and role in the cabinet on the economic vote of individual parties using aggregate data. Palmer and Whitten (2000) used a composite measure of shared responsibility in their aggregate analysis of individual level data. Anderson (1995) examined how a party's share of cabinet portfolios conditions the relationship between the macro-economy and aggregate party popularity in five countries. At the individual level, Andersen (2000) examined the effect of the incumbent party's share of cabinet portfolios in an

interactive model using survey data for 13 European surveys. Hellwig (2001) estimated the impact of the effective number of parliamentary parties on individual level economic voting using nine surveys from the Comparative Study of Electoral Systems project. Lewis-Beck offered an explanation for cross-country differences in individual level economic voting in his five-country study that was based on variation in the number of parties in the cabinet. The results of each of these previous studies provide support for one or both of the hypotheses given above. Taken together with these previous analyses, the results in this chapter leave little room to doubt that the distribution of administrative responsibility is an important factor conditioning both the magnitude of the economic vote across elections as well its distribution across parties. The rest of this chapter is divided into two main sections corresponding to the two empirical hypotheses described above.

Administrative Responsibility and the Distribution of the Economic Vote across Parties

Our first hypothesis suggests that parties with a greater share of the status-quo distribution of administrative responsibility will receive a greater share of the economic vote than parties with a smaller share. There are many ways that a party can gain such administrative responsibility. Accordingly, in testing this hypothesis we adopt a number of different approaches to measuring each party's administrative responsibility.²⁰⁹ First, there are institutional factors that affect the distribution of administrative responsibility between the executive and the legislature. In presidential systems, administrative responsibility will be shaped by whether or not the presidential party commands a legislative majority and whether it competes for executive power with other executive parties. Because of the fusion of legislative and executive powers in parliamentary systems and their institutional separation in presidential systems, the level of administrative responsibility of the typical prime minister is likely to be greater than that of the typical president (in a presidential system), even when the latter controls a majority in the legislature. Secondly, in parliamentary systems, the distribution of administrative responsibility should reflect characteristics of the incumbent cabinet: The administrative responsibility of a Prime Ministerial party that controls all the cabinet seats and/or commands a legislative majority will typically be greater than the administrative responsibility of Prime Ministerial parties that do not. Thirdly, a typical party's share of administrative responsibility should be closely connected, in parliamentary systems, to its position within or outside of the cabinet. Whether the party is the

²⁰⁹ More precisely the theory is about the voter's beliefs about the distribution of administrative responsibility across parties. In using measures of the actual distribution of administrative responsibility in place of these beliefs, we assume that these beliefs reflect the real distribution.

prime ministerial party, a coalition partner, or an opposition party will affect its distribution of administrative responsibility and hence its economic vote. One useful gauge of a parliamentary party's level of administrative responsibility is its share of cabinet portfolios, which according to our theory should be strongly correlated with its share of the economic vote. In the sections that follow, we present empirical tests of these propositions using the size or the share of the economic vote for individual parties as the dependent variable in our analyses.

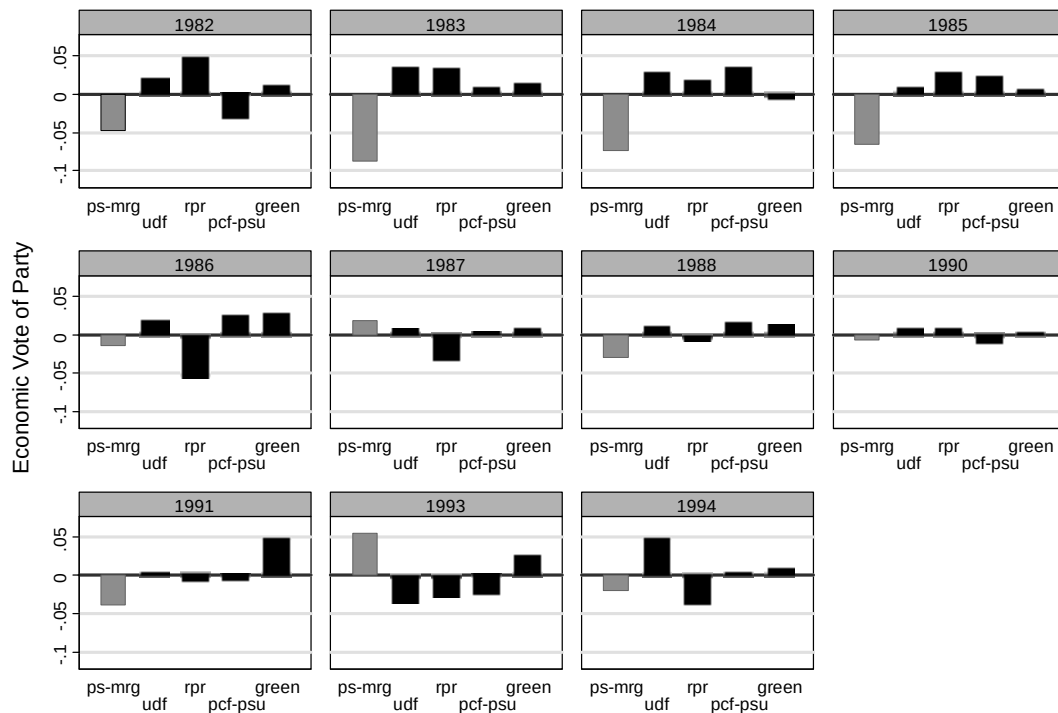
Presidential Parties and the Distribution of the Economic Vote

There are only two countries in our sample that have directly-elected presidents who have significant administrative responsibility: the United States and France. We begin, then, with an analysis of economic voting for presidential parties in these cases. For the United States, the available data include both surveys that ask voters about their presidential vote and those that ask them about their legislative vote. We can explore economic voting for the party of the president in either case. Likewise, surveys exist that ask French voters which parties they support in the legislature as well as for president. However, because the surveys that include the presidential vote question have not also included an appropriate economic perceptions question, our analysis of French economic voting for the party of the president only includes cases in which voters were asked about their legislative vote.

Economic Voting for Presidential vs. Non-Presidential Parties

If we assume that voters think administrative responsibility is disproportionately in the hands of presidents, our first hypotheses translates directly into the expectation that presidential parties will have a greater share of the economic vote than non-presidential parties. Since in two-party systems the economic vote of one party is just the opposite of the economic vote for the other, in the two-party U.S. system we simply expect that changes in party support due to worsening economic perceptions will be negative for presidential parties but positive (and the same size) for non-presidential parties. This has certainly been the prevailing wisdom in the U.S. case, where estimates of economic voting have typically found that the President's party is hurt by poor economic performance (Erikson 1989; 1990; Kramer 1971; Norpoth 2001). Similarly, our own estimates of the *economic vote of the chief executive* in the United States (reported in Figure 3.2 in Chapter 3) were consistently negative. Figure 9.1 reports our estimates of the magnitude of the economic vote for French parties in legislative elections. This chart allows us to compare the size (or with some work the share) of the economic vote of French presidential parties to the size of the economic vote for all other parties. The estimated changes in support due

to worsening economic perceptions for presidential parties are listed first and have gray colored bars.



Gray bars in this graph are for Presidential parties.

Figure 9.1 The Economic Vote of French Presidential Parties versus Non-Presidential Parties

This figure makes it clear that, like the U.S. case, the size of the economic vote for French presidential parties is typically larger than the economic vote of other parties.²¹⁰ The economic vote of the president's party is negative in nine of eleven cases and is the largest negative economic vote in six of those cases. Of course, there are exceptions to this trend. Most prominently, in 1986, 1987, 1993, and 1994, the Rassemblement pour la République (RPR), as well as other parties in 1993, experienced more negative economic voting than did President Mitterrand's Socialist Party. Interestingly, these are exactly the periods in which Mitterrand was forced to cohabitate with a prime minister from the RPR. We will come back to this fact below, when we ask how the economic vote of prime ministerial parties compares to that of presidential

²¹⁰ We have also examined the *share* of the economic vote more directly by inspecting percentages of the economic vote rather than its size. However, we save the presentation of this analysis for a later section in which we present the French results along with those of other systems.

parties. However, for now, the main message from Figure 9.1 is that the economic vote of the French presidential party is typically greater than the economic vote of other parties in the same elections, with the possible exception of prime-ministerial parties during periods of cohabitation.

Economic Voting for Presidential Parties under Divided vs. Unified Government

The sharing of administrative responsibilities in presidential systems is usually referred to as “divided government.” This occurs when different parties control the executive and legislative branches of government (Elgie 1999; Shugart and Carey 1992). There is evidence that divided government weakens the economic vote for presidential parties – much of it based on aggregate-level analyses. For example, Lewis-Beck and Nadeau (2000) produce such evidence for French presidential voting. Likewise, Powell (2005) finds that divided government reduces the magnitude of the economic vote in his sample of 23 presidential democracies. Employing a sample of elections from both mature and new democracies, Hellwig and Samuels (2007) confirm that the economic vote for the President becomes insignificant, while the PM party’s economic vote becomes much larger, in periods of cohabitation (or divided government). There is convincing evidence from the U.S. that divided government either at the national or state level results in a smaller economic vote for the executive (president or governor). Leyden and Borrelli (1995) find that state unemployment impacts vote choice more strongly in states with unified as opposed to divided government. Lowry, Alt and Ferree (1988) demonstrate that the impact of a state’s fiscal condition on vote choice (for both gubernatorial and state legislative candidates) is more pronounced after periods of unified party control of the state government. On the other hand, Norpoth’s analysis of individual-level data from the 1992 and 1996 U.S. national elections suggests that divided government does not “divert” the economic vote from the presidential party to the majority party in Congress (Norpoth 2001). Our estimates of the size of the economic vote in different electoral contexts in both the U.S. and France allow us to assess the impact of divided government on economic vote magnitudes.

Divided government has been the norm in the US during the period under study here (only the short period of unified Democratic government in the early 1990’s is an exception) and has occasionally occurred in France.²¹¹ In France these periods also produce shared control of the executive in a situation referred to as “cohabitation.” Unlike in the US, divided government in France also means that an opposition prime minister and cabinet will be installed and this locus of executive authority will compete with the president for control. Indeed, with no legislative

²¹¹ The current era of divided government in the U.S. began in 1956 when the Republicans won the presidency but failed to carry Congress (Fiorina 1992). See also Brady (1993) for a discussion of divided government in the U.S.

majority and the government ministries controlled by the opposition, French presidents during periods of cohabitation may be especially powerless.²¹² This contrasts sharply with the experience of divided government in the United States, in which the president's party retains control of the executive apparatus even when faced with a majority opposition in the legislature. Our expectation is that voters will attribute less administrative responsibility to presidents under periods of divided government than during periods of unified government and that this difference should be more severe for French presidents who must also "cohabit" with the opposition within the executive.

In order to test these expectations it was necessary not only to compare the average economic vote of presidential parties under unified and divided government, but also to control for whether the election study asked voters about their presidential or legislative vote. The results of these tests are in Table 9.1. The table reports the average economic vote for presidential parties in each indicated context and the accompanying tests are from an appropriately specified dummy variable regression (see the footnote to the table).

²¹² During the time period covered in our analysis there were two periods of cohabitation: 1986-1988 when Mitterrand (Socialist) was President and Chirac (RPR) was his prime minister; and 1993-1995 when Mitterrand was president and Edouard Balladur (RPR) was prime minister.

Table 9.1. Average Economic Vote for Presidential Parties in Different Contexts

	Divided Government or Cohabitation (C1)	Unified Government (C2)	p-value for rejecting the hypothesis that C1 = C2
US Presidential Election	-0.06 (0.02) n = 5	-0.10 -- -- n = 1	0.29
French Legislative Election	0.01 (0.03) n = 4	-0.05 (0.03) n = 8	0.004
US Legislative Election	0.012 (0.038) n = 9	-0.016 (0.03) n = 2	0.19

Tests are F-tests for equality of the coefficients in a dummy variable regression in which each cell represents a separate indicator variable in the regression (with appropriate handling of the constant and robust standard errors). In cases in which difference of means tests (with unequal variances) could be calculated (i.e., with more than one observation in a cell) the results from these tests are essentially identical to the F-tests.²¹³

While we have limited data on presidential parties, the data that we do have all support the proposition that economic voting for presidential parties will be muted under periods of divided government. Consistent with our hypotheses, in each type of election considered, the average economic vote for presidential parties under divided government is smaller (less negative) than under unified control. Further, the biggest difference in the average economic vote between unified and divided control is, as we expected, in the French case (where divided government also implies executive cohabitation). Of course, the small number of cases forces us to be cautious in our interpretation. While the difference between periods of divided and unified government for the French case is both large and statistically different from zero, the two results for the US case are not as large and are not quite statistically significant at conventional levels. Taken together, however, the strong French results and the directionally consistent US ones support the view that divided government suppresses the economic vote.

²¹³ The differences between the averages in the first and third rows of the table are statistically significant (as are those between the first and second). This supports the often-noted idea that economic voting is more important in presidential elections than in legislative elections (Kiewiet 1983 and Erikson 1989; 1990) for the US and Lewis-Beck and Nadeau (2000; 2004) for France). This was clear in the description of U.S. economic voting provided in Chapter 3. However, since we lack data on economic voting in French presidential elections the evidence provided here does not contribute much strength to what is already the common wisdom. It is worth noting, however, that if voters think administrative responsibility is disproportionately in the hands of presidents and thus question the utility of casting economic votes in legislative races, this result supports our argument.

Economic Voting for Presidential versus Prime Ministerial Parties

The principal institutional difference between presidential and parliamentary systems is the formal sharing of administrative responsibility between the executive and the legislature in the former case and the fusion of these responsibilities in the latter case. We expect that the combining of legislative and executive functions in the parliamentary case will result in higher levels of economic voting for prime ministerial parties compared to presidential parties, even in the case when the president's party commands a majority in the legislature. First, a number of scholars who compare the institutional powers of prime ministers to those of presidents have noted that single-party prime ministers controlling a legislative majority of seats in the legislature tend to be more powerful than presidents (e.g., Linz 1990; 1994; Sartori 1994). In these situations, prime ministers can not only dictate the administration of public policy but also, because of the strong party discipline in most parliamentary systems, dominate the legislative agenda to an extent that is unobtainable by most presidents (Ogg 1936).²¹⁴

Second, even when single party prime ministers do not control a majority in the legislature, they retain considerable agenda setting powers in most parliamentary systems—most importantly the more or less exclusive right to initiate legislative proposals. In contrast, the presidents in our sample are largely dependent on legislative parties to pursue their legislative agenda and so without a parliamentary majority the president's agenda is unlikely to move easily through the legislature.

The evidence comparing the economic vote of single party prime ministerial parties to that of presidential parties is presented in Table 9.2 and favors the hypotheses that the economic vote of prime ministerial parties is larger than that of presidential parties. Specifically, when we compare the economic vote of presidential parties and prime ministerial parties in elections in which voters were asked about their legislative vote choice, all the evidence (from either the US or France and for both unified and divided government) indicates that the economic vote of single party prime ministerial parties is larger than that of presidential parties.²¹⁵

There are two reasons that we think this mixed evidence favors the hypothesis that the economic vote of prime ministerial parties is larger than that of presidential parties. First, the

²¹⁴ A clear exception here is the presidential regime in Russia where the rules regarding presidential vetoes and decrees give him almost unchallengeable power (Parrish 1998)

²¹⁵ In the US case, we chose only to examine evidence from a comparison of election studies of the same type (i.e., all legislative elections in both the presidential and pm systems). Thus, the US case is the economic vote for the President's party when voters are asked about their choices in the legislative election. This focus allows us to avoid conflating differences in office (presidential or prime ministerial parties) with differences in the type of vote choice question.

evidence in favor of the hypothesis is based on many more cases than the evidence against it (23 vs. 6 cases, respectively).

Table 9.2. Average Economic Vote for Presidential and Prime Ministerial Parties

	<i>Economic Vote for Single-Party Prime Ministers</i>	<i>Economic Vote for Presidents</i>	
	<i>All are Legislative Election Studies</i>	<i>French Legislative Election Studies</i>	<i>US Legislative Election Studies</i>
<i>Divided Government/ Cohabitation</i>	-.052 (0.03) 19	.01 (0.03) 4	0.012 (0.038) 9
<i>p-value for difference in means test between column 1 and each other column</i>	-- --	p < 0.015	p < 0.001
<i>Unified Government</i>	-.068 (0.04) 49	-.053 (0.03) 8	-.016 (0.03) 2
<i>p-value for difference in means test between column 1 and each other column</i>	-- --	p < 0.11	p < 0.10

Numbers in cells are mean *economic vote of the chief executive*, the standard deviation of this variable (in parenthesis), and the number of observations for the case, respectively.

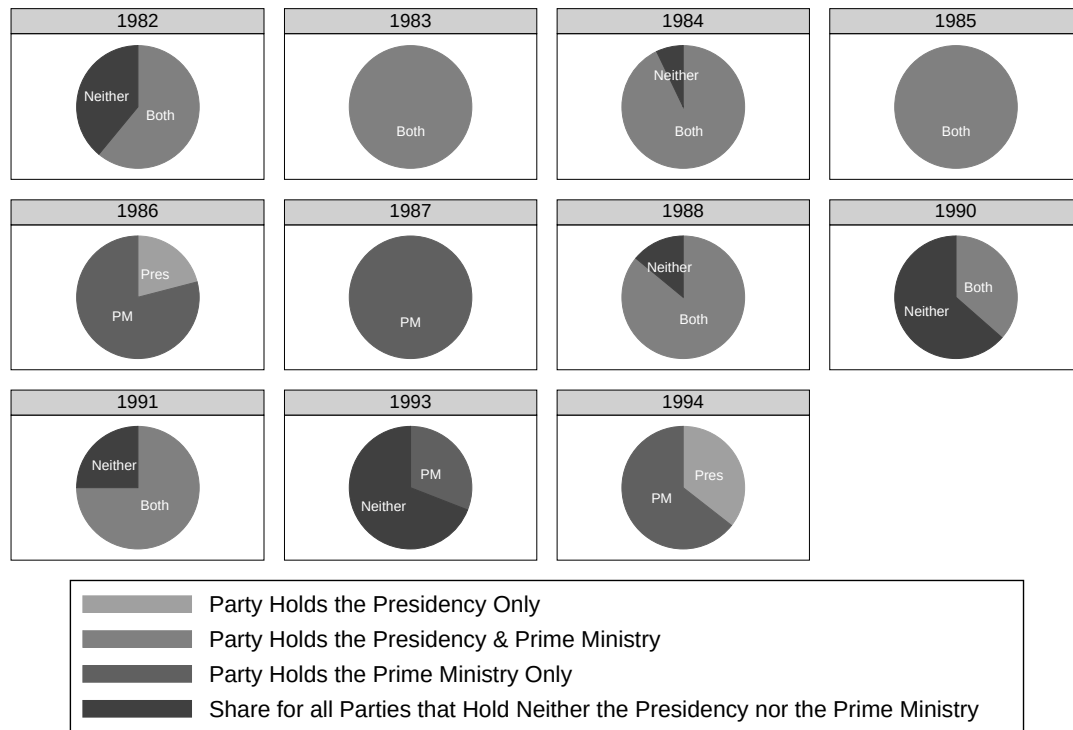
Only cases of presidential parties or parties of single-party majority prime ministers are included. For France, which has a PM and president, only the economic vote of the presidential party is included.

“Unified Government” for parliamentary systems means the party of the prime minister controls a majority in the legislature. “Divided Government” for parliamentary systems means that the party of the prime minister does not control a majority in the legislature.

The p-value is from a difference in means test allowing for unequal variances. Specifically, it is a test that value in the first column is smaller (more negative) than the corresponding row value in each of the other columns

Another way to examine the question of whether presidential or prime ministerial parties have a larger economic vote is by looking more closely at the one case in which these kinds of parties can exist in the same system at the same time. If our hypothesis is true, then we would expect the party holding the prime ministry to capture a larger share of the economic vote in a given election than the party of the president, when these two offices are controlled by different parties. The relevant data is provided in Figure 9.2. This figure shows the total share of the negative economic vote that was captured by prime ministerial, presidential and non-executive parties for each of the election studies we have for France. The “negative economic vote” is just

the sum of the economic vote for all parties that experienced a decline in their support when the economy worsened (all the parties with negative projecting bars in Figure 9.1). If a party type does not appear in one of the pie graphs, no party of that type experienced a negative economic vote (and following the discussion in Chapter 2, we can think of the economic vote in these cases as zero).



The area of each slice represents the share of the “negative economic vote” that accrued to the party (or parties) indicated by the label on the slice.

Figure 9.2 The Share of the Economic Vote of French Presidential and Prime Ministerial Parties

There are four cases in our data in which there was an incumbent French president that was from a different party than the incumbent prime minister (at the time of the survey). These were in 1986, 1987, 1993, and 1994. Notice first that in each of these four cases (and, indeed, in all eleven cases) the prime ministerial party experienced a negative economic vote as expected. In contrast, during two of the cases of cohabitation (1987 and 1993), the presidential party did not experience a negative economic vote at all. Further, in the other two cases (in which the prime ministerial and the presidential parties were the only two to have a negative economic vote) the prime ministerial party’s share of the economic vote was almost three times larger than that for

the presidential party. Thus, in the head to head competition for economic votes that the French cases highlights, we find evidence consistent with the idea that French voters see the prime minister as more responsible for economic policy and concentrate their economic votes accordingly.²¹⁶

In sum, if we make the plausible assumption that single-party prime ministers have more administrative responsibility than presidents, our evidence about the relative level of economic voting for prime ministerial parties and presidential parties tends to support our first hypothesis – that a greater share of administrative responsibility should lead to a greater share of the economic vote. Only when the type of voter preference survey is conflated with the type of executive office do we find evidence that presidential parties experience the same (or larger) level of economic voting as prime ministerial parties.²¹⁷

The Economic Vote of Prime Ministerial Parties Leading Different Types of Cabinets

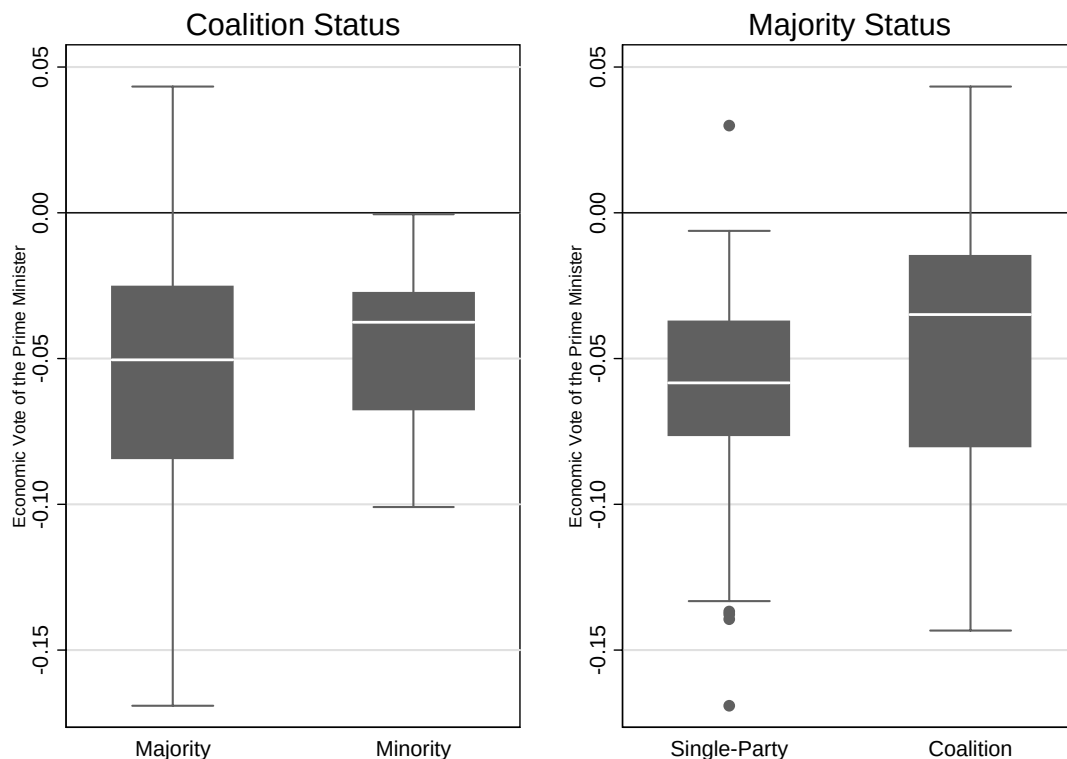
We have seen that the magnitude of the economic vote responds, as expected, to differences in the distribution of responsibility in different kinds of presidential systems and between presidential parties and single-party prime ministers. Following the same logic, we can look within parliamentary systems, and examine whether the economic vote for prime ministerial parties varies with the degree to which administrative responsibility is concentrated on them or distributed to other parties as well. Our assumption is that single-party prime ministers hold a greater share of administrative responsibility than prime ministers who head coalition cabinets. Further, prime ministers that do not control a majority in the legislature hold less administrative responsibility than those who do. If these assumptions are correct (and voters' beliefs actually reflect the distribution of administrative responsibility), our theory implies that the economic vote for prime ministers should be highest for a single party prime ministers that command a legislative majority and lowest for coalition prime ministers that do not command a legislative

²¹⁶ In a related finding, Lewis-Beck and Nadeau (2000) find that in two French presidential elections under cohabitation (1988 and 1995), the economic vote on the second ballot is much higher when an incumbent prime minister (whose party controlled the cabinet) was running – this was the case in 1988 but not in 1995. This suggests that under cohabitation, voters recognize the PM party as responsible for economic management and are more likely to include the economy in their vote choice decision if the PM candidate is actually one of the presidential candidates. In general, Lewis-Beck and Nadeau (2000) conclude that economic voting is moderated under cohabitation. Likewise, in a poll of 400 French voters aimed at assessing the distribution of responsibility for economic outcomes to the president or prime minister, Lewis-Beck (1997b) finds the economy to have a greater influence on approval for the prime minister than for the president.

²¹⁷ A cautionary note here, though, is the finding in Hellwig and Samuels (2007) suggesting that the economic vote under presidential regimes is similar, if not higher, than in parliamentary ones.

majority. The order of the intermediate cases (i.e., single-party minority prime ministers and majority coalition prime ministers) will depend on which effect (coalition government or majority status of the government) impacts the distribution of administrative responsibility more.

We begin by examining how the size of the economic vote for prime ministers varies by majority and coalition status separately. The data are provided in Figure 9.3 and, as we expected, they show that the economic vote for prime ministerial parties in coalition cabinets is muted relative to the economic vote for prime ministerial parties that govern alone (the right hand panel). Likewise, majority cabinets tend to have a bigger economic vote than minority cabinets, though this difference is not large (the left hand panel).



Shaded areas of boxplots include the 25-75 percentiles and the “whiskers” extend to 1.5 the interquartile range. The centerline is the median.

Figure 9.3 Economic Voting for Prime Minister Parties by Majority and Coalition Status

In order to better understand how the coalition and the majority status of the cabinet affects the prime minister’s retrospective economic vote, we regressed the economic vote of prime ministers on dummy variables for the coalition and majority status of the cabinet. Further, we examined if the relationship changed when we allowed these variables to interact. This

produced the coefficients and standard errors in Table 9.3 and the predicted values and p-values for significant differences in Table 9.4.

Table 9.3
Economic Voting for Prime Ministerial Parties: The Impact of Coalition and Majority Status

<i>Dependent Variable: Negative of the Economic Vote</i>	OLS Coefficient (t-ratio)	OLS Coefficient (t-ratio)
<i>Coalition Government</i> (1 = "yes", 0 = "no")	-0.023 (-2.38)	-0.026 (-2.31)
<i>Minority Government</i> (1 = "yes", 0 = "no")	-0.014 (-2.11)	-0.021 (-2.32)
<i>Coalition Government * Minority Government</i>	-- -- -- --	0.017 (1.19)
<i>Constant</i>	0.067 (9.93)	0.069 (7.85)
Number of Observations	146	146
Adjusted R-Squared	0.08	0.09

OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys.

These results make at least one conclusion unequivocally clear: the support of single party majority cabinets depends more heavily on the economy than any other cabinet type. Not only are the coefficients on coalition and minority status negative and highly significant in both equations, but the corresponding predicted values for the economic vote of single party majority prime ministerial parties are also statistically different from (and larger than) the economic votes of all the other kinds of prime ministerial parties (Table 9.4).

Beyond this, the estimates can also tell us something about differences in the magnitude of the economic vote between other types of parties, though these contrasts are not nearly as clear as the difference between single party majority prime ministers and all the other kinds of parties. Specifically, while the estimates of the magnitude of the economic vote for the four different cabinet types are in the same order in both model specifications, the statistical significance of these differences depends on the specification. If one is willing to make the restriction that the insignificant interaction term in Table 9.3 is really zero then we find that leading a coalition cabinet dampens the economic vote of the prime ministerial party more than leading a minority cabinet. This is reflected both in the differences in the size of the coefficient estimates for minority and coalition status in Table 9.3, as well as the larger differences in the predicted values between the rows of Table 9.4 (going from majority to minority) versus differences between the columns (going from single party to coalition).

Table 9.4. Predicted Size of the Economic Vote: Different Types of Parliamentary Cabinets

<i>P</i> = 0.043, <i>p</i> = 0.028		
	Single Party, Majority	Single Party, Minority
	.067	.053
	.069	.048
	50	19
<i>p</i> = 0.024 <i>p</i> = 0.028	Coalition, Majority	Coalition, Minority
	.044	.030
	.042	.038
	66	11
	<i>p</i> = 0.04, <i>p</i> = 0.73	
	<i>P</i> = 0.02 <i>p</i> = 0.31	

The first and second numbers in each cell are the predicted sizes of the economic vote for the type of party based on the coefficients in columns 1 and 2, respectively, of Table 9.3. The last number is the number of cases in the category. Italicized numbers are *p*-values for rejecting the hypothesis that the predicted values in the adjacent columns or rows are equal. The *p*-values for the difference in the northwest and southeast cells are 0.005 and 0.008 respectively and for the northeast and southwest cells are 0.41 and 0.57, respectively.

The fact that in both specifications prime ministerial parties, leading single party minority cabinets, have the second largest economic vote is perhaps surprising given the much discussed difficulty that minority cabinets have in policy making, relative to majority cabinets. However, an examination of the specific cases of single party minority government on which these estimates are based may suggest an explanation. Specifically, these cases are all large parties that hold a dominant position on at least one side of the political spectrum in each country. Some are “almost majority” governments, while others (like the Scandinavian Socialists) tend to rule when they face a badly divided opposition. Laver and Shepsle (1990) provide one explanation for the existence of such governments, which suggests that parties who hold a very strong bargaining position with respect to post election policy making will sometimes be able to rule alone even when they only hold a minority of seats. Thus, our sample of single party minority governments may be “self selected” so that they are exactly those parties that are well-positioned vis-à-vis other parties to dominate policymaking. Some evidence for this comes from the fact that all of our single party minority cabinets except one hold a plurality of seats in the legislature as well as the median position on at least one of the two policy dimensions described in Mueller and Strom (2000). More convincing evidence, however, comes from our use of Laver and Benoit’s *Winset* program to calculate whether these cabinets tend to be the kind of “strong” cabinets that Laver and Shepsle’s theory identifies as capable of ruling alone despite holding a minority of seats.

When we make this calculation for each case, we find that twelve of the nineteen single party minority cabinets in our sample are formally “strong” parties by Laver and Shepsle’s criteria.²¹⁸

Thus, when we look closely at these single party minority cabinets, it is less surprising that they should mimic their majority counterparts. What is perhaps more surprising is that voters seem to be aware of this strength and so treat these parties almost as if they were similar to majority governments.

The Distribution of the Economic Vote among Parties Playing Different Roles in (and out of) Government

In the analyses so far, we have only compared the economic vote of different kinds of prime ministerial parties. However, our hypothesis that a party’s share of administrative responsibility should impact its share of the economic vote applies to parties playing various roles in government and opposition. In this section, we examine whether these differences in roles are indeed associated with differences in the importance of economic voting among the parties.

The Distribution of the Economic Vote Among Prime Minister, Partner and Opposition Parties

Our theory implies that economic voting should be more important for prime ministerial parties than it is for their partners and more important for these partners than it is for opposition parties. In order to assess these hypotheses empirically we need to explore differences in economic voting among different kinds of parties that compete with each other in the *same* election.²¹⁹ When we look at the economic vote for parties within the same election, however, we face a practical problem in examining the data: the economic vote must sum to zero across parties in an election. Clearly, if there are only two parties in the system the size of the economic vote for the prime ministerial party and the opposition party must necessarily be equal (in absolute value) and so there is little point to examining the share of the economic vote that each party received (they will always have equal shares).²²⁰ In contrast, in the multiparty case there is room for the

²¹⁸ Specifically, in a two dimensional space defined by economic policy on one hand and social policy on the other, no party preferred any possible alternative cabinet to the one in which the ruling party controlled all the portfolios (we used positioning data from the manifestos project but in some cases substituted it for Laver and Benoit (2005) or Laver and Hunt’s (1990) data when the case was close to the period to which that data applied).

²¹⁹ As in the other parts of this book, whenever we refer to “cabinet partners” or a “cabinet partner” we mean non-prime ministerial parties who hold seats in the cabinet. When we need to refer generically to any party in the cabinet (including the PM), we use the term cabinet party or government party.

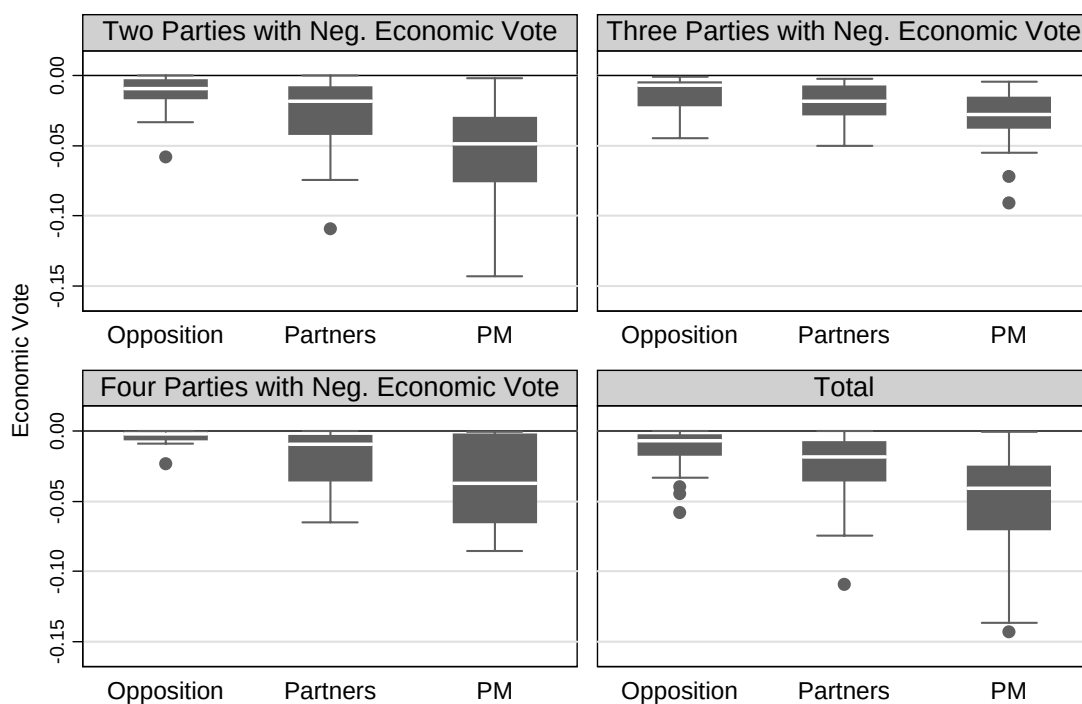
²²⁰ These shares will either both be equal to 0 (if change in the support for the prime minister due to a worsening economy is in the “wrong” direction or is really zero) or both equal to 1 (when changes in support for the prime minister are non-zero and in the expected direction).

total economic vote to be distributed among different parties in various patterns. Our hypothesis is that the lion's share of the negative economic vote should go to the prime ministerial party, followed by other cabinet parties, with the least economic voting going to opposition parties.²²¹ When examining distributional hypotheses of this sort, we want to rely on techniques that focus our attention on the *shares* of the overall economic vote that different parties receive and that are sensitive to the compositional nature of economic voting (i.e., that shares sum to one).

Thus to examine these data, we first divide the parties in each election into two groups: those that experienced a negative economic vote and those that experienced a positive economic vote.²²² As shown earlier, the prime ministerial party will usually be included in the group of parties that has a negative economic vote, as will the cabinet partners though not as reliably. In the data examined below (which excludes two-party cases and multiparty cases in which only one party had a negative economic vote) the economic vote was negative for 80% of the prime ministers and 65% of the cabinet partners. Of course opposition parties also experience negative economic voting in some of our cases, but this is much less common (41% of opposition parties). Below we compare the extent of negative economic voting for prime ministers, cabinet partners, and opposition parties in two different ways. First, we simply examine the average size of the negative economic vote for parties of each type that are in comparable electoral situations. Specifically, we can only make this kind of comparison if we control for the number of parties in a given election that have a negative economic vote (e.g., if four parties are splitting -10 percent of the vote then they will tend to have smaller shares than if two parties split this same change). Figure 9.4 provides the relevant data for elections in which the negative economic vote was split among two, three, and four, parties respectively.

²²¹ Recall that as we developed the theory in chapter 8, we successively relaxed the definition of “incumbency” until it was possible for the theory to predict negative economic voting for opposition parties that had some role in policy making. Nevertheless, if this responsibility is less than that of cabinet partners the hypothesis here follows.

²²² As discussed above a “negative economic vote” is just a negative change in support due to a worsening in economic perceptions. By definition the negative economic vote in a given electoral situation will equal the positive economic vote and so we examine only the negative half here. Further, as we have shown, the economic vote for prime ministers is overwhelmingly negative so this is the most relevant comparison.



one case with five parties that had negative economic votes is not graphed

Shaded areas of boxplots include the 25-75 percentiles and the “whiskers” extend to 1.5 the interquartile range. The center line is the median.

Figure 9.4. Share of the Negative Economic Vote: Prime Ministers vs. Cabinet Partners and Opposition Parties.

The results convincingly confirm our expectation that prime ministerial parties will have a larger economic vote than other parties and that cabinet partners have a larger negative economic vote than opposition parties.

We can also examine the data in a different way that separates information on the overall size of the economic vote from the distributional information about the share of the economic vote held by each party. To do this we calculated the sum of the negative economic vote in each election study and then calculated the share of that total that accrued to each party (assigning zero to parties whose votes were positive). Figure 9.5 shows the average of these shares for parties playing different roles in (or out of) the cabinet. Since only cases with the same number of parties receiving a negative economic vote can be so compared, we provide a separate graph for each set of cases.

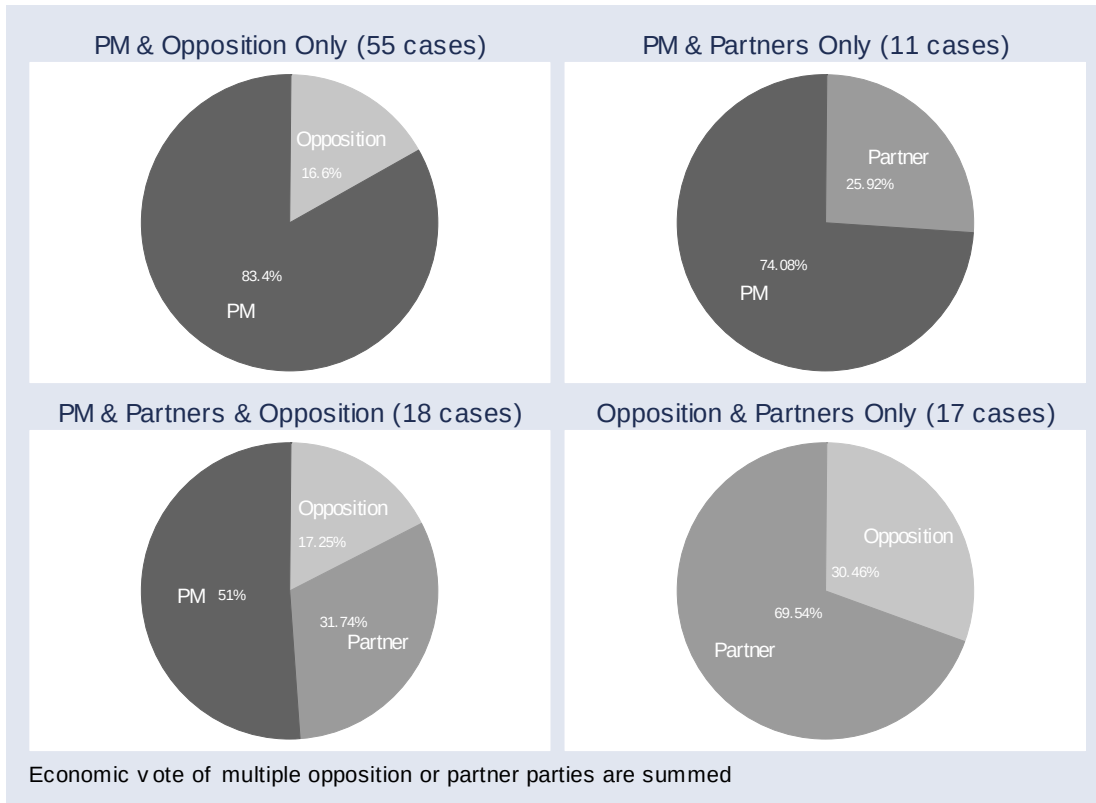


Figure 9.5. Average Distribution of the Negative Economic Vote: Prime Ministerial, Partner and Opposition Parties

This graph reveals that whatever the overall size of the negative economic vote in an election, most of it goes to the prime ministerial party - no matter whether the opposition, other cabinet partners, or both also get a negative economic vote. Further, this evidence also shows that cabinet partners get a bigger share of the economic vote than opposition parties. Assuming that voters attribute more administrative responsibility to cabinet partners than to opposition parties, this clearly supports the hypothesis that the distribution of the economic vote is a function of the distribution of administrative responsibility across parties.

We can also examine a statistical model of the relationship in Figure 9.5. To do so, we need to account for the compositional nature of the vote choice in the statistical model, which we can do by applying the techniques of compositional data analysis to the problem (King and Katz 1999). Specifically, for each of the four situations in Figure 9.5, we create a dependent variable that corresponds to the percent of the negative economic vote that each party received (each dependent variable is a vector with elements that are the percent of the vote that each party type received in a given electoral situation). Each of these dependent variable vectors will necessarily sum to one across party types and, consequently, they will not even approximate the assumptions necessary to use regression reliably. Hence, we transform the elements of these vectors by

forming ratios of the shares for different party types (just choosing an arbitrary type as the denominator of the ratio) and then taking logs of these ratios. We then regress these log-ratios on a set of constants.²²³ This exercise provides a reliable test of whether the apparent differences in the proportions Figure 9.5 are statistically different from zero. The result of these tests confirm, with high levels of confidence, that the differences are indeed different from zero (even the apparently small difference between the shares for opposition and partner parties in the lower left graph can be statistically distinguished from zero, with a p-value of .056).

Parties with Control of the Economic Affairs and Finance Ministries

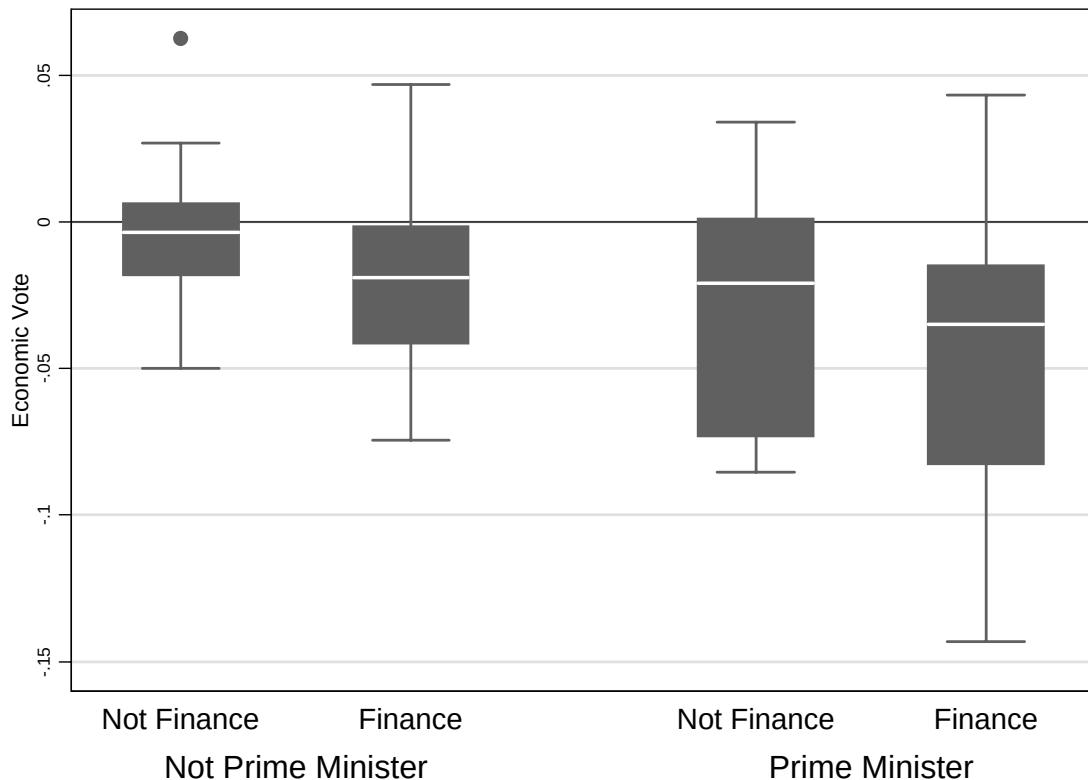
Our theory suggests that the importance of retrospective economic voting to a party support depends on the ability of voters to discern that party's competence based on movements in the economy. This only follows, however, because in our theory competence in economic management is ultimately consequential for *economic* outcomes. Thus, it is really not the distribution of administrative responsibility *in general* (which of course concerns a variety of policy outcomes unrelated to the economy) that should condition retrospective economic voting, but rather the distribution of administrative responsibility for economic management specifically. The recognition that our use of the term "administrative responsibility" is really just a short-hand for "administrative responsibility for economically consequential decisions" allows us to propose that the voter should attribute more administrative responsibility to parties that hold key positions in economic management than to those holding other positions. Concretely, we expect that parties controlling ministries such as finance, economic affairs, and the treasury will have a larger economic vote than other parties, even if these others have parties have significant administrative responsibility in other areas.

In this section we explore whether parties that hold these positions have a systematically different economic vote than similar parties that do not. Clearly, this will not be an interesting question for single-party governments in which the same party holds all the cabinet ministries, including the prime ministry. Likewise, it would not be an interesting question if prime ministerial parties in coalition cabinets also consistently controlled the finance portfolios. In fact, the prime minister's party frequently does not control the finance ministry: if we look at all coalition cabinets from 1960 to 2002 (for the countries in our sample), the prime minister's party controlled the finance ministry 59% of the time.²²⁴ When, in addition to the finance ministry,

²²³ For the case in which there are three party types there are two log-ratios, so we need to estimate a system of two equations. We use seemingly unrelated regression to do so.

²²⁴ This is calculated over months thereby allowing for changes in the composition of cabinets.

there was also an economic affairs ministry it was controlled by the PM's party 34% of the time.²²⁵ Figure 11.6 shows (for coalitional governments) the extent of economic voting for finance ministers compared to other cabinet partners, controlling for whether the party also held the prime ministry. The economic vote of parties that hold the finance ministry is greater (more negative) than other parties, which clearly supports our hypothesis and suggests that voters make rather subtle distinctions in the distribution of administrative responsibility.²²⁶



Shaded areas of boxplots include the 25-75 percentiles and the “whiskers” extend to 1.5 the interquartile range. The centerline is the median.

Figure 9.6. Economic Vote for Parties Holding the Finance Ministry, Coalition Governments

²²⁵ Cabinets almost always include a finance ministry, but have a separate economic affairs ministry only about half the time.

²²⁶ The regression results for Figure 9.6 are as follows: *Economic Vote for Parties Holding Finance Ministry in a Coalition Government* = 0.005 (1.96) + 0.023 (1.74) X Dummy Variable Indicating Party Holds Prime Ministry + 0.015 (2.84) X Dummy Variable Indicating Party Holds Finance Ministry - 0.0004 (-0.03) Prime Ministry*Finance Ministry. $R^2=.19$. Observations=156.

The predicted values and significance tests provided in Table 9.5 show that there is always a larger economic vote for parties holding the finance ministry compared to those who do not, regardless of whether the party also holds the prime ministry. However, this difference is only statistically significant for cabinet partners. These results suggest an interesting modification to our earlier conclusions about the impact of holding the prime ministry on a party's economic vote. Specifically, while in the last section we saw that prime ministers have a substantially larger economic vote than their cabinet partners (a conclusion widely shared by students of comparative economic voting), the analysis in this section suggests that this difference may be due in part to the fact that most prime ministerial parties also control the finance ministry. In contrast, when different parties hold these two positions the electoral fate of the finance minister is almost as dependent on the economy (an economic vote of .02) as is the electoral fortunes of the prime minister (an economic vote of .028).²²⁷

Table 9.5
Predicted Size of the Economic Vote:
Parties Holding the Finance Ministry, the Prime Ministry, or Both

		$p = 0.08$			
$p = 0.004$	Not PM, Not Finance Ministry .005 (.0001, .011)			PM, Not Finance Ministry .028 (.002, .053)	$p = 0.30$
	Not PM, Finance Ministry .020 (.012, .03)			PM, Finance Ministry .043 (.024, .061)	
		$p = 0.02$			

Numbers in the cells are predicted amount of economic voting for the type of party based on the coefficients from the regression model from footnote 220. Italicized numbers are p-values for rejecting the hypothesis that the predicted values in the adjacent columns or rows are equal. The p-value for the difference in the northwest and southeast cells is 0.0001 and for the northeast and southwest cells is 0.51

Given the strength of our results for the importance of the position of finance minister in the distribution of the economic vote, one may wonder whether there is any evidence that voters actually appreciate the different roles that different parties play in economic policymaking.

²²⁷ An interesting implication for future research is that if prime ministers recognize this, they should use it when negotiating the distribution of cabinet portfolios. They should offer the finance ministry to other parties when they expect the economy to decline and keep it for themselves when the economy is likely to be strong.

While the best evidence would ask voters directly about these beliefs, some indirect evidence can be gleaned from a new data set on newspaper reporting of economic news that we are currently developing for the larger research project of which this study is a part. That dataset codes newspaper stories about the economy for a variety of variables, including references and attributions of responsibility to different political actors. With these data we can examine how often newspaper stories in different countries mention the prime minister and/or the finance minister in connection with economic news. This provides direct evidence about the availability of the kinds of information that inform voters of the relative influence of parties holding the prime ministry and finance ministry on policymaking.

Thus far in the project, we have collected data from twenty-eight newspapers (four papers each in Canada, Belgium, Norway, US, UK, France, and Ireland) from 1980-2002. We distinguished two types of newspaper stories: 1) all stories concerning the economy and 2) other kinds of stories that mentioned the incumbent government. Our measure of the extent to which voters receive information about the finance minister's connection to the economy is the relative frequency that the finance minister is mentioned in economic stories compared to all stories (including non-economic ones).²²⁸ Table 9.6 summarizes the results and shows that newspapers are four times more likely to mention the finance minister in a story about the economy than they are to mention the prime minister. Further, this is in contrast to other kinds of stories in which the incumbent government is mentioned, where the PM is slightly more likely than the finance minister to be mentioned.

Table 9.6. Mentions of Prime Minister and Finance Minister in Newspaper stories about the Economy

	Prime Minister Mentioned	Finance Minister Mentioned	Neither Mentioned	Newspaper Stories
All stories in which a member of cabinet is mentioned	2,012 (49%)	1,836 (45%)	608 (6%)	4,068 (100%)
All Economic Stories	50 (3%)	113 (16%)	1685 (91%)	1848 (100%)

The 4,456 mentions in the first row come from 4068 stories (in some stories both are mentioned).

This evidence suggests that the finance minister tends to be highlighted in media discussions of the government (about half the economic stories that mention anyone in the government mention the finance minister) and is the dominant government figure associated with

²²⁸ Since this analysis is only meant to be illustrative, we forego a detailed discussion of the methodology for this extensive project. In short, however, we sampled newspaper stories from the front page of the four leading newspapers in each country. We coded all the stories that appeared either the day of or the business day after the official release of either monthly or quarterly economic statistics on unemployment, inflation, or GDP/GNP.

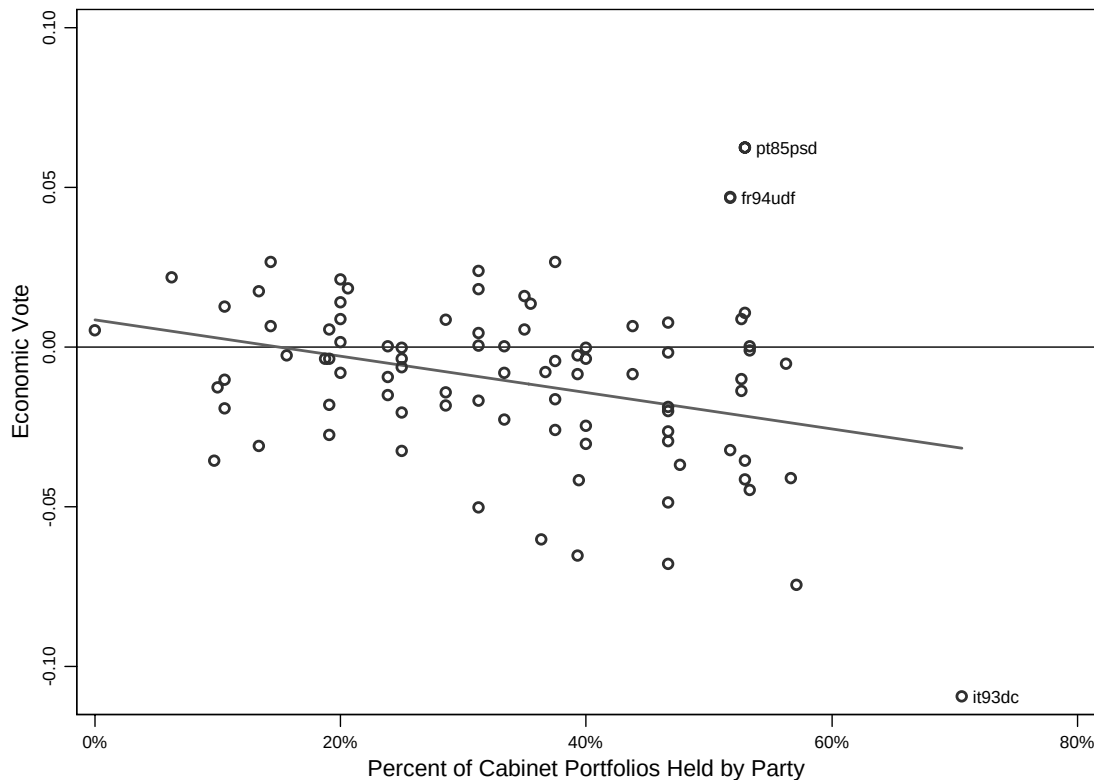
reporting about the economy. In our view, this gives some credence to the notion that the finance ministry is not just an important institutional player in the making of economic policy, but that the media provides the kind of information that would allow voters to understand this as well.

Overall this section provides convincing evidence that is consistent with the idea that voters attribute more responsibility for economic policy making to this party than to others and that it has a correspondingly larger economic vote.

The Distribution of Cabinet Seats

Voters appear to establish their assessment of party competence – which in turn conditions their economic vote – based on reasonably sophisticated assessments of the administrative roles of parties within the governing coalition. Each party's share of cabinet seats strikes us as an indicator of a party's administrative responsibility that is even more accessible to voters. We have already established that prime ministerial parties register a greater economic vote than other parties. Hence what is of particular interest here is whether the economic vote of cabinet partners is positively related to their share of cabinet seats. Figure 9.7 plots the relationship between the economic vote for cabinet partners (excluding prime minister parties) and their share of cabinet seats. The relationship is in the predicted direction with a statistically significant slope coefficient and so is consistent with our hypothesis.²²⁹

²²⁹ As readers might expect, if we include the PM parties in the plot and regression the relationship is much stronger.



Economic Vote for Cabinet Partners = 0.007 (1.25) - 0.057 (2.42) X % of Cabinet Seats Held. $R^2=.10$.
Observations=84.²³⁰

Figure 9.7. Economic Vote and Share of Cabinet Seats, Cabinet Partner Parties Only

In this section we have examined how four different dimensions of administrative responsibility impact the distribution of the economic vote across parties. For presidential systems, we found that presidential parties that command a legislative majority have a greater economic vote than presidential parties that do not. Similarly, we found that the economic vote of prime ministerial parties is largest when the prime minister leads a majority single-party cabinet, smaller when the prime minister leads a majority coalition or minority single-party cabinet, and smallest when the prime ministerial party leads a minority coalition cabinet. Likewise, we showed that a party's share of the economic vote is tied closely to its role in government: Prime ministerial parties experience the most economic voting, coalition partners

²³⁰ OLS regression is with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. Numbers in parentheses are t-ratios.

less, and opposition parties the least.²³¹ Further, control of the finance ministry increases a party's economic vote. Finally, we found that each party's share of cabinet portfolios, a summary gauge of a party's level of administrative responsibility, impacts its economic vote.

The Distribution of Administrative Responsibility and Overall Economic Voting

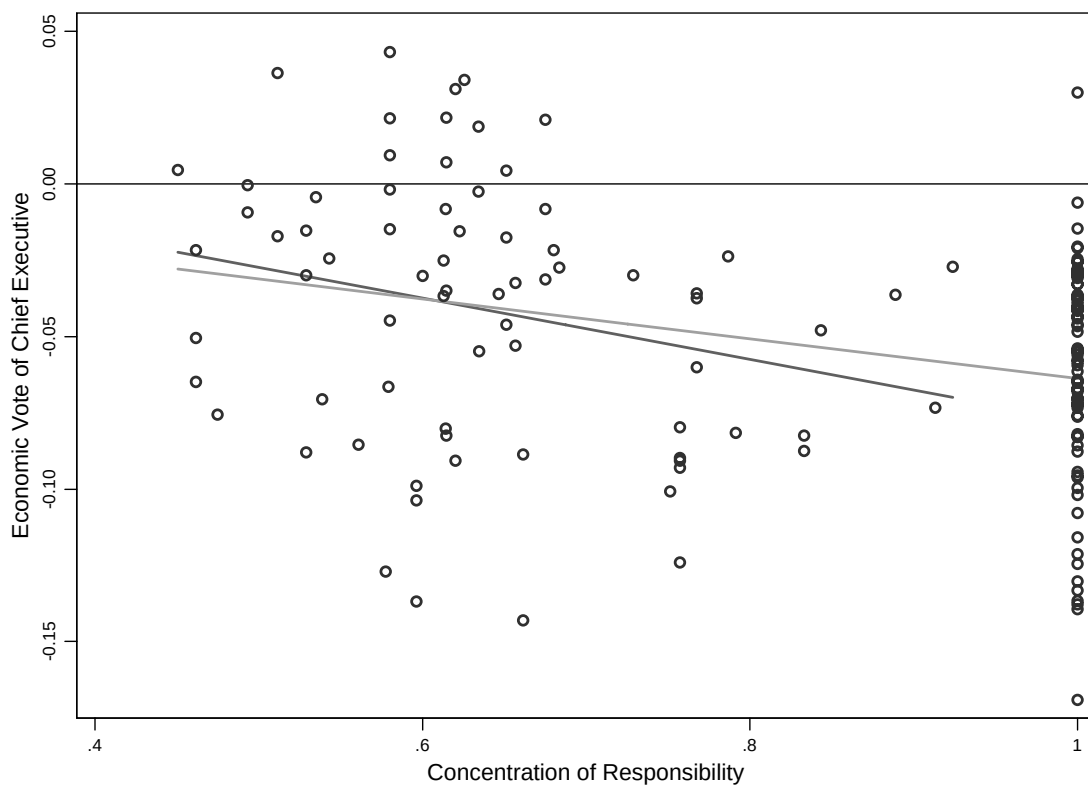
The previous section was concerned with how the distribution of the economic vote across parties depends on the distribution of administrative responsibility. In this section we change our focus: instead of focusing on parties, we ask how aggregate features of the distribution of administrative responsibility (i.e., whether it is concentrated or diffuse) impact the overall level of economic voting in an election. Specifically, the second general hypothesis that we proposed at the beginning of this chapter was that as the status-quo distribution of administrative responsibility over parties becomes more equal, the smaller the overall economic vote.

To test this hypothesis we construct a measure of the concentration of administrative responsibility, where we initially use each party's share of cabinet ministries to indicate its share of administrative responsibility and later construct a more general measure that accounts for opposition influence on policymaking. Specifically, given n parties in an election, we can define an associated hypothetical distribution of responsibility in which all parties get equal shares of responsibility. Formally, this is just a vector – call it δ – in which each party's share of responsibility is $1/n$. Our theory suggests that this vector of equal shares is the distribution of responsibility that will lead to the *least* amount of economic voting in an election. We can then define a corresponding vector consisting of the real shares of administrative responsibility for each party in a given case. If we call this vector λ , then the vector distance between λ and δ is an intuitive measure of the degree to which responsibility is concentrated in the system.²³² If administrative responsibility is equally shared across parties, then the elements of λ will be equal to the elements of δ and the vector distance will be zero no matter how many parties are in the election. Likewise the distance between the two vectors will be greatest when administrative responsibility is concentrated on one party. One potential drawback of the measure is that while its lower bound is always zero, its upper bound – which, again, occurs when one party has all the

²³¹ Of course, here we mean only those opposition parties that experience a “negative economic vote” when the perceptions worsen. Other opposition parties may (and usually do) have large “positive economic votes”.

²³² The vector distance is just the square root of the summed squared differences in the elements of the two vectors.

administrative responsibility - will differ depending on the number of parties. Specifically, the more parties in the system, the bigger this upper bound will be. If one thinks that voters see power as more concentrated as the number of parties excluded from power is bigger, then this differing upper bound is appropriate. However, if (for example) voters perceive power as equally concentrated when one party holds all the power in a two-party system and when one party holds all the power in a five-party system, then we need to normalize this distance by dividing by the upper bound. Below, we report results using the second strategy, though the choice is not empirically consequential. We call this measure *concentration of responsibility* and it can take values between zero (equality) and one (complete concentration of responsibility in one party).



The flatter regression line corresponds to the following regression: Economic Vote of Chief Executive = - 0.002 (0.08) - 0.066 (2.81) * Concentration of Responsibility. $R^2 = .10$. Observations = 152.²³³ The steeper line excludes the cases in which concentration of responsibility equals one.

Figure 9.8. Concentration of the Distribution of Responsibility and Economic Voting for the Chief Executive

²³³ Estimates are from OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. Numbers in parentheses are t-ratios. For the U.S. case, the presidential party is coded as holding all administrative responsibility and only presidential election studies are used.

The results of our analysis of the impact the concentration of responsibility on the economic vote of the prime minister is provided graphically, along with the corresponding bivariate regression results, in Figure 9.8. The results of the analysis support the hypothesized negative relationship – more concentrated responsibility results in higher levels of economic voting. The flatter of the two regression lines includes all the cases and the steeper one excludes the cases on the right hand side of the figure in which power is completely concentrated in a single party.

This evidence thus supports our second hypothesis. However, it is based on a measure of the concentration of responsibility that is built from the distribution of cabinet portfolios exclusively. Consequently, it does not capture dispersion in the distribution of administrative responsibility that is due to an institutionally strong opposition or the majority/minority status of the government.²³⁴ With respect to the first of these factors, we would expect administrative responsibility to be more widely shared when legislative institutions give the opposition significant power over policy making and an institutional ability to review and alter the decisions of the executive. With respect to the second factor, we have already discussed the idea that oppositions may be empowered relative to the executive when the cabinet does not control a majority of seats in the legislature. This general expectation, as per our previous discussion, may be different for single-party minority governments versus coalition minority governments, since the former may obtain sole control of the government exactly because they occupy an unassailable bargaining position in the legislature.

With these ideas in mind, we seek to develop a measure akin to the concentration of authority measure used above, but one that weights opposition parties more heavily in situations of single-party minority or coalition minority government and institutionally powerful oppositions. We do so in two steps. First, we construct an index of opposition influence that combines information on the institutional strength of the opposition in each of our countries with the majority status of the government. Next, we normalize this measure to vary between zero and one and use it to build a linear combination of the cabinet seat share and legislative seat share of each party. The justification for the second step is simply that the distribution of policymaking power in a system in which the executive was powerless relative to the legislature would be well approximated by the distribution of legislative seats. In contrast, in a system in which the cabinet had all the power, legislative seats would be meaningless and only the distribution of cabinet seats would be relevant. Thus, for intermediate cases in which opposition influence is between

²³⁴ In contrast, the coalition status of the government is captured in the distribution of cabinet portfolios, since single-party cabinets will necessarily have all portfolios held by a single party.

these extremes, we combine these two distributions, mixed by an appropriate weight for the extent of opposition influence.

We develop an appropriate weight for mixing the cabinet seat and legislative seat distributions by focusing on two institutional distinctions. First, we distinguish three kinds of cabinets and give them decreasing scores for susceptibility to opposition influence: coalition minority governments (the most susceptible to opposition influence), single-party minority governments (the next most susceptible to opposition influence) and all majority governments (the least susceptible to opposition influence). Secondly, we measure the institutional strength of the opposition using information from Powell (2000). Powell (2000) uses four institutional features of legislatures to assign countries to High, Medium, and Low categories of institutional strength of the opposition. The four features are (1) whether the legislature has a large number of standing committees that mirror the government ministries, (2) whether committee chairs are controlled by the government or shared by the opposition, (3) whether the government controls the legislative agenda, and (4) whether the government limits the ability of the committees to amend proposed legislation. The resulting classification of countries is provided in Table 9.7.²³⁵ It is identical to Powell’s classifications except for Italy, since our coding of Powell’s four variables for our cases gave Italy a score that put it in the “High” category as opposed to “Medium” (where Powell has classified it).²³⁶

Table 9.7. Classification of Institutional Strength of the Opposition

High	Medium	Low
Belgium, Denmark, Germany, Italy, Netherlands, Norway, Spain, Sweden	Canada, United States	Australia, France, Greece, Ireland, New Zealand, United Kingdom

From Powell (2000) with one exception.

With these data and the data on the majority status of the government, we created a weight for each case in the data that took a value of one when the strength of the opposition was High and the cabinet was a minority coalition and zero when the strength of the opposition was Low and the cabinet was a majority (either coalition or single party). Intermediate cases fell between these extremes.²³⁷ The distribution of this weight for the countries included in the

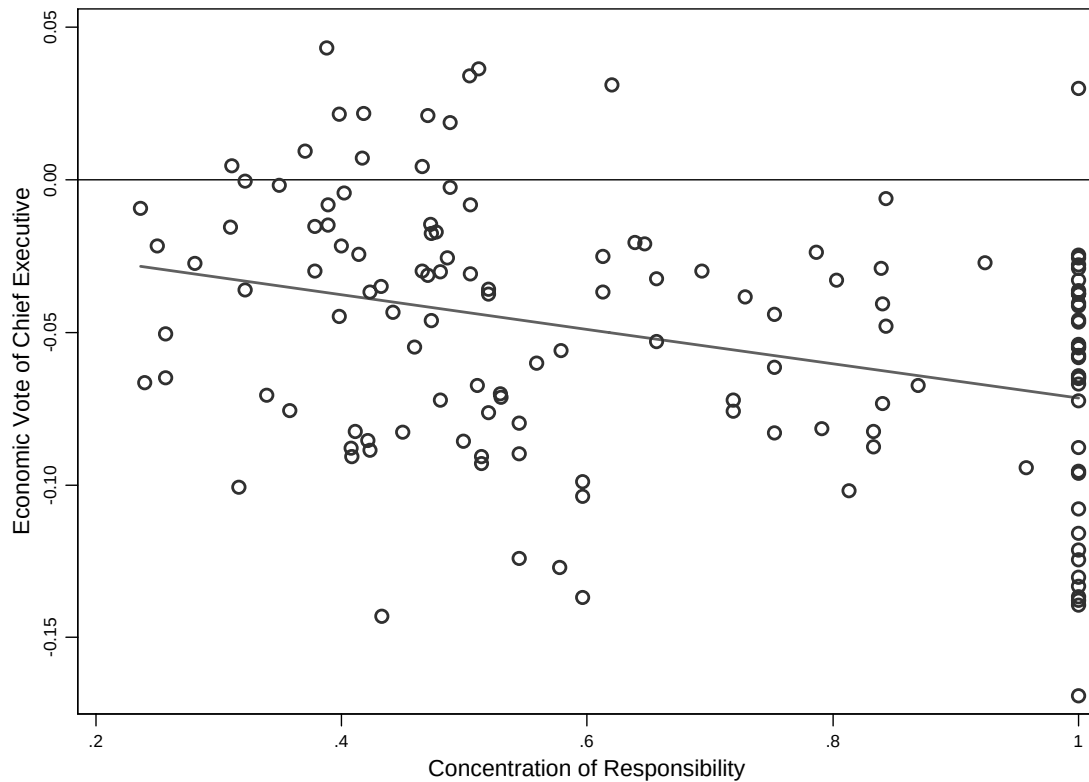
²³⁵ These institutional features vary little over our period and so we classify each country for the whole period.

²³⁶ Giving the four factors equal weight gives Italy the same score as most the other countries in the “High” category and a greater score than the others in the “Medium”.

²³⁷ Institutional strength of opposition was coded 0 for Low, 1 for Medium, and 2 for High. Majority Status of the government was coded 0 for majority, 1 for single-party minority, and 2 for coalition minority. The

analyses below is in the appendix to this chapter. To create our new measure of the concentration of administrative responsibility, we used this weight, as described above, to form a linear combination of the vector of cabinet seat shares and legislative seat shares for the parties. Specifically, if c_i is party i 's cabinet seat share, s_i is its legislative seat share, and ϕ is the opposition influence weight for the case, then the party's share of administrative responsibility is $(1 - \phi)c_i + \phi s_i$. This measure will sum to one across all the parties and can be used in the procedure described in the last section to create a measure of the extent to which the real distribution of responsibility differs from an equal distribution. This new measure of concentration of responsibility is graphed against the economic vote of the prime minister in Figure 9.9.

opposition influence weight was calculated as follows: (institutional strength of opposition score + majority status score)/4.



Economic Vote of Chief Executive = $-0.015 (1.18) - 0.057 (3.40) \times \text{Concentration of Responsibility}$ with Opposition Influence. $R^2=0.10$. Observations=137.²³⁸

Figure 9.9. Concentration of the Distribution of Administrative Responsibility and Economic Voting for the Chief Executive, Opposition Influence Included

This graph looks similar to Figure 9.8, as it should since the variable on the x-axis is constructed, in part, from the one used in that figure. The result of accounting for opposition influence in our measure of the concentration of responsibility is reflected in the differences between the figures. Specifically, the current graph includes cases with considerably more dispersed responsibility than the previous one. For example, all of the Norwegian cases in our sample had single-party minority governments and a strong legislative committee system, but score a one on our previous measure of concentration of responsibility because one party holds all the cabinet portfolios. However, the current measure scores the four Norwegian cases around 0.5.

Despite these (and other) differences in the data, the impact of using this index on the relationship is not large and the implied substantive effect of concentration of responsibility on

²³⁸ Estimates are from an OLS regression with standard errors robust to heteroscedasticity and to non-independence between observations of the same party in different surveys. We could not calculate the measure for the Portuguese cases and one case from Iceland. For the U.S. case, the presidential party is coded as holding all administrative responsibility and only presidential election studies are used.

the economic vote is not significantly different in the two cases. However, the new measure is directionally consistent with our expectations – that is, the impact of an empowered opposition on the distribution of administrative responsibility is to make it more equal and that this depresses the magnitude of economic voting in the system. Overall, however, this impact is quite small compared to the impact of the distribution of responsibility that stems from the role that parties play in cabinet.

The Concentration of Responsibility in an Interactive Model

As we discussed in Chapter 4, the two-stage analysis reported above can be replicated employing a one-step approach in which we combine our individual level datasets and estimate a random coefficients, multi-level model with a more limited, uniform, specification of the individual level vote choice models. By including interactions between economic perceptions and *concentration of responsibility* we can capture how differences in this contextual variable impact the individual decision to vote for the chief executive's party. We estimate the same core model described for the random coefficient model in Chapter 7.²³⁹ The basic form of the model, with the size variable, is:

$$(9.1) \quad \begin{aligned} v_{ik} &\sim \text{Bin}(\pi_{ik}) \\ \text{logit}(\pi_{ik}) &= \beta_{0k} + \beta_{1k} \text{Worse}_{ik} + \beta_{2k} \text{Better}_{ik} + \\ &\quad + \phi_1 \text{Ideology}_{ik} + \phi_2 (\text{Ideology}_{ik} * \text{CR_Ideology}_k) \end{aligned}$$

$$(9.2) \quad \beta_{0k} = \gamma_{00} + \gamma_{01} \text{Size}_k + \gamma_{01} \text{CR}_k + \omega_{0k}$$

$$(9.3) \quad \beta_{1k} = \gamma_{10} + \gamma_{11} \text{CR}_k + \omega_{1k}$$

$$(9.4) \quad \beta_{2k} = \gamma_{20} + \gamma_{21} \text{CR}_k + \omega_{2k}$$

where *Bin* just indicates the dependent variable is distributed according to an appropriate binomial distribution with parameter π_{jk} . *CR* is *concentration of responsibility* and we assume the errors are distributed according to a multivariate normal distribution, with variances $\sigma^2_{\omega 1}$, $\sigma^2_{\omega 2}$, and $\sigma^2_{\omega 3}$ and covariances $\sigma^2_{\omega 2, \omega 2}$, $\sigma^2_{\omega 1, \omega 3}$, and $\sigma^2_{\omega 2, \omega 3}$. Since we expect support for the chief executive to be negatively related to economic perceptions, we expect $\gamma_{10} < 0$ and $\gamma_{20} > 0$ (recall that the baseline category is voters who think the economy has stayed the same). Further, since greater *concentration of responsibility* is expected to increase the size of the economic vote,

²³⁹ We measure *concentration of responsibility* using the first of the two measures that were described in the last section because we wanted to maximize the number of cases in this analysis.

we expect $\gamma_{11} < 0$ and $\gamma_{21} > 0$ (since this will increase the difference between the overall impact of better versus worse perceptions).

Table 9.8. Multi-Level Model of Effective Number of Concentration of Executive Responsibility and Economic Voting

Left-Right Self Placement (dev. from mean)	0.49 (0.01)
Left-Right Self Placement X Leftist PM (Leftist PM is an indicator variable)	-0.75 (0.01)
Voter Perceives Economy got Better (indicator variable)	0.46 (0.03)
Voter Perceives Economy got Worse (Indicator variable)	-0.58 (0.03)
Concentration of Executive Responsibility (deviation from mean)	0.84 (0.15)
Voter Perceives Economy got Better * Concentration of Executive Responsibility	0.58 (0.15)
Voter Perceives Economy got Worse * Concentration of Executive Responsibility	-0.97 (0.15)
Size of chief executive's Party	4.61 (0.25)
Constant	-2.21 (0.07)
$\sigma^2_{\omega 1}$	0.07
$\sigma^2_{\omega 2}$	0.08
$\sigma^2_{\omega 3}$	0.07
$\sigma_{\omega 1, \omega 2}$	-0.04
$\sigma_{\omega 1, \omega 3}$	-0.02
$\sigma_{\omega 2, \omega 3}$	-0.03
N	201,876

Standard Errors listed below logit coefficients. All coefficients are statistically significant at $p < .1$ except the shaded cells. In the case of the US, the dependent variable is vote for the party of the president. Congressional election studies are excluded.

Table 9.8 provides the estimates for the equations in Equations (9.1) – (9.4).²⁴⁰ These estimates clearly support the hypothesis, since all the coefficients are statistically significant and in the expected direction. In particular, *Concentration of Administrative Responsibility* interacted with positive economic evaluations is positive and significant and its interaction with negative evaluations is negative and significant. Thus, this pooled, multi-level model tells exactly the same story as the second-stage estimates from the two-stage method – shared power decreases the *economic vote of the chief executive*.

²⁴⁰ The estimates were obtained using the PQL second-order linearization method outlined in Goldstein (1995). Diagnostics on estimated residuals at Level 2 suggest that the assumption of normal variance in the level two coefficients is not violated.

Summary

This purpose of this chapter was to explore the influence of the distribution of administrative responsibility on both the extent of economic voting and its distribution across parties. Our contextual theory of rational retrospective economic voting suggests that voters know the distribution of administrative responsibility and that, along with other information in the responsibility augmented competency signal, this information alters the weight that voters assign to movements in the previous economy in their vote calculus. These theoretical expectations were expressed in two general empirical hypotheses relating the distribution of the economic vote across parties to each party's share of the economic vote as well as to the overall size of the economic vote.

The empirical results in this chapter are uniformly consistent with these hypotheses. Specifically, we find that in presidential systems economic voting is higher under unified as opposed to divided government; single party prime ministers tend to experience more economic voting than presidents, prime ministers leading single party majority cabinets have a larger share of the economic vote than prime ministers leading coalition or minority cabinets; in coalition cabinets, prime ministerial parties receive the lion's share of the overall economic vote and their cabinet partners most of the rest; parties that control the finance ministry have a larger economic vote than parties who do not, and the size of this effects is almost as large as the impact of holding the prime ministry; and that a governing party's economic vote is closely tied to its share of cabinet seats. Finally, consistent with our second hypotheses about the distribution of responsibility and the overall level of the economic vote, we find that there will be more economic voting in an election in which the distribution of administrative responsibility is more concentrated around a small number of parties.

All these empirical results support what has become an impressive body of empirical evidence (at both the aggregate and individual level) linking administrative responsibility to economic voting (e.g., Anderson 2000; Powell and Whitten 1993). Indeed, our result fills one of the most important gaps in that literature – a comprehensive examination of the individual level evidence. Given the strength of our results and its consistency with previous work, we think it is safe to conclude that the distribution of responsibility is simply one of the most important factors conditioning the economic vote in western democracies.

Our results are also consistent with the theoretical model described in Chapter 8 and so may lend some support to the view of economic voting described there. However, we recognize

that the empirical work in this chapter does nothing to distinguish our explanation from other theoretical possibilities. Most of these possibilities build informally on the clarity of responsibility argument, which assumes that the main factor that prevents economic voting in situations of diffuse responsibility is the voter's inability to determine who is responsible for policy making. Our theoretical story differs in that we assume voters are full informed about the distribution of responsibility across parties and they use this information in a rational fashion to condition the magnitude of their economic vote. Thus the economic vote is depressed when responsibility is diffuse, not because of the voter's ignorance about the roles that parties play in government but because they must use the same amount of information (observed movements in the economy) to make more complicated inferences about party competence. Of course, the existence of these different theoretical possibilities does nothing to detract from the strength and importance of the empirical result itself, but simply argue for future work that seeks different kinds of empirical leverage that would allow us to distinguish between them.

Appendix 9.1

Influence of the Opposition weight Used to Construct Concentration of Responsibility Measure that was used in Table 9.12 and Figure 9.10

	Opposition Influence Weight				
	<i>0</i>	<i>0.25</i>	<i>0.5</i>	<i>0.75</i>	<i>1</i>
<i>Australia</i>	6	0	0	0	0
<i>Belgium</i>	0	0	12	0	0
<i>Canada</i>	0	4	0	0	0
<i>Denmark</i>	0	0	3	1	10
<i>France</i>	9	2	1	0	0
<i>Germany</i>	0	0	12	0	0
<i>Greece</i>	11	0	0	0	0
<i>Ireland</i>	8	3	0	0	0
<i>Italy</i>	0	0	6	0	0
<i>Netherlands</i>	0	0	12	0	0
<i>New Zealand</i>	2	1	0	0	0
<i>Norway</i>	0	0	0	4	0
<i>Spain</i>	0	0	6	4	0
<i>Sweden</i>	0	0	0	4	0
<i>United Kingdom</i>	16	0	0	0	0
<i>Total</i>	52	10	52	13	10

Chapter 10

The Pattern of Contention and the Economic Vote

The previous chapter established that voters condition their economic vote on the current distribution of policymaking responsibility amongst parties competing for election. In this chapter we examine whether the data also support our theoretical proposition that voters condition their economic vote on the extent to which parties are “in contention” for policymaking responsibility in the future. As we did in Part 2, we begin this examination by asking whether the assumptions we make about voter beliefs are plausible. Our theory assumes that voters know the extent to which different parties and coalitions of parties contend for significant shares of policymaking responsibility. To support this assumption we present survey evidence from three countries that shows that voters are surprisingly well informed about the pattern of cabinet contention, even in very complex coalitional systems. With this assurance that, in the few cases for which we have direct evidence, our information assumptions are plausible, we next propose two different methods for measuring beliefs about cabinet contention in the larger number of cases in our sample (for which direct survey data about beliefs is not available). Finally, in the rest of the chapter we use these measures to test three general hypotheses (all implied by the theory in chapter 8) about how the pattern of contention for policymaking responsibility conditions economic voting.

Voter Beliefs about the Pattern of Contention

Our model of economic voting assumes that voters have knowledge about which parties or coalitions of parties are likely to form the government or otherwise participate in policy making. Only with such knowledge can they assess whether their vote can make a difference in shaping the future distribution of administrative responsibility. We do not, however, have direct measures of the voters’ assessment of these patterns of contention for most of the studies in our sample.²⁴¹ Consequently, in this chapter we pursue an alternative strategy: we measure voters’ beliefs about that pattern of contention by identifying and measuring the sources of these beliefs. If we can do so successfully, we can test our theoretical hypotheses by examining whether

²⁴¹ There are though some survey based studies demonstrating that voters perceive these patterns of contention and condition their voting behavior in a fashion that is consistent with strategic voting theories (Duch and Palmer 2002b; Johnston et al 1992:197-211).

variation in the sources of beliefs leads to the expected variation in retrospective economic voting.

However, before we begin the task of building indirect measures of voter beliefs about the pattern of contention and using them to explore our theoretical hypotheses, we first provide evidence that voters are actually capable of forming sensible expectations about the outcome of the potentially complex political process that results in the formation of a government and the distribution of policymaking authority. This evidence comes from a set of surveys that were conducted in the week before the Danish elections of 2001, the Norwegian elections of 2001, and the Dutch elections of 2002 - three elections in which the coalition possibilities were unusually varied and unsettled in the weeks before the elections. The surveys, which asked respondents which party was most likely to obtain the prime ministry following the election and which parties were most likely to join a given prime minister in the cabinet. Our brief summary of the survey results suggest that even in these very complex cabinet formation situations, voters had little trouble understanding and forecasting the likely outcome of the coalition formation process.²⁴²

The 2001 Norwegian election was by all accounts particularly unpredictable. Eight major parties stood for election (and were included in the survey) and in the months leading up to the election, the incumbent Labour government steadily lost support until it was running even with the Conservatives as the election approached.²⁴³ For the first time, it was possible that Labour would not win the most votes. This fact and the failure of any group of parties to commit to rule together after the election made the situation particularly uncertain. Indeed, as late as September 8, *AftenPosten* reported that, “For the time being, the poll data fails to show any natural constellation of political parties that would be able to form a new government.” Carl I Hagen, the leader of the Progress Party summarized the situation succinctly, “It’s a real mess.” (*AftenPosten*, September 10, 2001)

In Denmark, the election held on November 20, 2001 was also unusual.²⁴⁴ Leading up to the election, the Social Democratic incumbent government was losing ground to the Liberal Party and for the first time since 1920, it looked like the Liberals would replace the Social Democrats as the largest party in Denmark. At the same time, far right parties like the Danish Peoples Party

²⁴² Complete details of the survey are available on www.nuffield.ox.ac.uk/economicvoting, and in Stevenson (2005).

²⁴³ The parties were the Labour Party, the Conservatives, the Christian People’s Party (KrF), the Center Party, the Liberals, the Socialist Left, and the Red Alliance.

²⁴⁴ The main parties in the election were the Social Democrats, the Liberals, the Conservatives, the Socialist People’s Party, the Radical Liberals, the Center Democrats, the Christian People’s Party, the Danish People’s Party, and the Progress Party.

(DPP) were doing very well in the polls and made the issue of reforming Denmark's liberal immigration laws (and the ability of these immigrants to access a generous welfare system) the main issue of the campaign. The electoral weakness of the Social Democrats and the popularity of relatively extreme parties opened up the post election coalition possibilities because it made the formation of a non-extremist rightist majority or a leftist majority more difficult.

Finally, the Dutch election of May 2002 was perhaps even more uncertain (and certainly more dramatic) than the Danish and Norwegian elections. Support of the incumbent "purple" coalition of the PvdA, D'66, and the Liberals (VVD) had been declining and it seemed unlikely the incumbent government would re-form, though certainly some of the partners might be in the next government. Indeed, the incumbent government itself had been a novel coalition that would have been unheard of in the many years of dominance by the Christian center parties. However, what really introduced uncertainty into the coalition possibilities was the rise of two right wing populist parties, Livable Netherlands and List Pim Fortuyn.²⁴⁵

The first evidence about voter beliefs in these elections that we will examine comes from a question asking voters which party would most likely provide the prime minister following the election. The distribution of responses in each of the three cases is provided in Table 10.1. It is clear that in each case responses were concentrated on only two or three parties, with essentially no responses identifying other parties (Pym Fortuyn providing the only possible exception).

We can get an even better sense of how well voters do at identifying the likely prime ministers by comparing the aggregate distribution of responses to predictions from a statistical model of prime ministerial selection that mirrors those that have been used successfully in the literature to predict identity of the prime minister in large samples of elections (Warwick 1996).²⁴⁶

²⁴⁵ Of course, the assassination of Pim Fortuyn just days before the election also complicated the situation, but the survey, which was in the field at the time, was suspended as a result of this event and so all the respondents (over half the target of 1000) were interviewed previous to this.

²⁴⁶ The variables usually used include functions of each party's size, change in size, ideological position, and history of incumbency. The specific model used to produce the predictions in Table 10.1 is from Stevenson (2003)

Table 10.1 Voters Beliefs about Prime-Ministerial Contention

	<i>Number of respondents who chose party</i>	<i>% of respondents who chose party</i>	<i>Probability of becoming PM for various plausible values of predictors¹</i>
Norway 2001			
Labour	319	33 %	8 – 82 %
KrF	306	32 %	6 – 45 %
Conservative	139	14 %	2 – 55 %
Total for top 3	764	79 %	96 – 99 %
Progress	7	.7 %	< 2 %
Center	2	.2 %	< 2 %
Socialist Left	1	.1 %	< 2 %
Liberal	1	.1 %	< 2 %
Red Alliance	0	0 %	< 2 %
Another Party	0	0 %	--
Don't Know	191	20%	--
Total	775	100 %	100 %
Denmark 2001			
Social Democrats	174	17 %	36 %
Liberals	760	73 %	56 %
Conservatives	2	.2 %	1%
Total for top 3	936	89 %	93 %
Socialist PP	0	0 %	< 1 %
Radical Liberals	0	0 %	< 1 %
Center Democrats	0	0 %	< 2 %
Christian PP	0	0 %	< 1%
Unity List	0	0 %	< 1%
Danish PP	0	0 %	< 3 %
Progress	0	0 %	< 1%
Another Party	0	0 %	--
Don't Know	110	11%	--
	936	100 %	100- 100 %
Netherlands 2002			
PvdA	115	20 %	12 %
Liberals	88	15 %	6 %
CDA	190	32 %	52 %
Total for top 3	393	67 %	70 %
D66	3	.5 %	1 %
Green Left	8	1.5 %	4 %
United Christians	0	0 %	2 %
SGP	0	0 %	2 %
Pim Fortuyn	29	5 %	16 %
Live. Netherlands	0	0 %	2 %
VSP	0	0 %	3 %
Another Party	0	0 %	--
Don't Know	156	26 %	--
	589	100 %	100 %

The predictors were largest party, seat shares, changes in seat shares, median party, and previous PM (Stevenson 2003). For Norway, predictors were set at plausible values given the pre-electoral polls (for vote shares and who would be the largest party). Also, the identity of the previous PM was varied between the Labour Party and the KrF to reflect the fact that both had been incumbent prime ministers since the last election. Predictors were set at plausible values given the pre-electoral polls (for vote shares and who would be the largest party). For other cases, the variation in predictions from different plausible values of the predictors was not large; so only one value is given.

Clearly, voter's beliefs about which party will become prime minister are quite close to the predictions that political scientists make. In Norway and Denmark, both the statistical models and the surveys concentrate almost all the probability on the same parties. While in the Netherlands, both our models and the voters are more uncertain. It seems then that, at least with respect to beliefs about which parties are in contention for the post of prime minister, voters are very well informed. Indeed, the distribution of beliefs seems to pick out the same parties as our statistical models and evidence about as much overall uncertainty as the statistical models.

Respondents in the study were also asked to indicate which parties they thought would most likely join a given prime minister in the cabinet. Table 10.2 only summarizes these results for Denmark since the results for Norway and the Netherlands suggest the same substantive message.

Table 10.2 Beliefs about Cabinet Partnership, Given Beliefs about Which Party will become PM, Denmark, 2001

Party	If Social Democrats Obtain the Prime Ministry		If Liberals Obtain the Prime Ministry	
	% of respondents indicating party was a likely partner	Predicted Probability of Becoming a Cabinet Partner ¹	% of respondents indicating party was a likely partner	Predicted Probability of Becoming a Cabinet Partner ¹
Social Democrat	--	--	.021	0.025
Liberal	.063	0.022	--	--
Conservatives	.040	0.098	.762	0.319
SPP	.454	0.451	.009	0.011
Rad. Liberals	.563	0.642	.084	0.024
Center Dem	.149	0.153	.289	0.218
Christian PP	.023	0.123	.270	0.277
Unity List	.098	0.107	0	0.004
Danish PP	.028	0.012	.233	0.347
Progress	0	0.010	.005	0.310

Based on statistical models of cabinet partner selection in Stevenson (2003). The Predictors were the party's seat share, changes in seat shares, median party, and being in the previous cabinet with the designated PM, and ideological distance from the designated PM.

This table reveals a strong correspondence between the respondents' beliefs and the predictions of the statistical model. There are no large disagreements for Social Democratic cabinets and for the Liberals the discrepancies are only for the Progress Party and the Conservatives. In the case of the Conservatives, voters were fairly certain the Liberals would ask the Conservatives to join them in a coalition government if they obtained the prime ministry and this was widely discussed as a viable partnership (which, in fact, occurred). However, because of the fairly substantial ideological distance between the parties (at least in the data on which the

predictions rely), the statistical models find this outcome less likely. With respect to a Liberal/Progress Party partnership, voters disagreed with the predictions of the empirical model believing strongly that Progress would not join a Liberal cabinet. This outcome results from the empirical model's inability to distinguish between the Danish People's Party and the Progress Party (while voters seem quite able to make the distinction). Both parties are on the extreme right, but the Danish People's Party had recently supplanted Progress as the principal alternative in this region of the political space. Voters seem to have been aware of this changing of the guard while the empirical model missed it.

It seems then that even these differences underscore the main message from this example: Voters have a good sense of which parties are likely to join a given prime minister in the cabinet. Indeed, if they put these beliefs together with their beliefs about who is in contention to become prime minister, they should have little trouble distinguishing likely cabinet combinations from unlikely ones. Overall, the evidence suggests that we are on firm ground proceeding under the assumption that voters, even in complex coalitional systems, do have well formed beliefs about the likely outcomes of coalition formation.²⁴⁷

Our task, then, is to measure these beliefs. Since we do not have survey measures of the voters' beliefs about government formation for the large number of cases in our sample, we instead use a more indirect approach. Specifically, we try to identify the sources of voter beliefs about the pattern of cabinet contention and then use measures of these sources as proxies for the beliefs themselves. We develop two sets of measures. The first measurement strategy adopts an "historical approach": It assumes that voters' expectations about the current pattern of governmental contention among the parties will reflect historical patterns of cabinet participation. We develop a number of specific "historical" measures; each is built from information on the history of office holding between January 1960 and the date of the survey corresponding to the data point. Thus, the average voter's "memory" in our surveys extends from 20 to 40 years into the past and always includes most of her adult political life (the average respondent is about 40 years old at the time the surveys).

The second general measurement strategy we use assumes that voters form their beliefs about the future distribution of executive authority rationally using some "model" of the process of government formation and all the available information relevant to that process. Thus, we can build plausible measures of the rational voter's beliefs about the likely outcomes of government

²⁴⁷ Kedar (2005a and 2005b) incorporates a similar view of the voter's knowledge of coalition formation outcomes and presents data suggesting that their voting decisions compensate for expected policy outcomes that result from the composition of these coalitions.

formation from estimates of well-specified statistical models that predict which parties will govern based on observable variables available to the voter at the time of the election (we call this the “rational approach”). The specifics of the statistical models will vary depending on the systems under consideration (i.e., the voter’s “model” of government formation – and hence the best statistical model - will likely be different in systems in which the executive is usually composed of a coalition versus a single party) and are explained in the relevant sections below.²⁴⁸

Patterns of Contention amongst Opposition Parties

The first hypothesis that we will examine concerns how the pattern of contention impacts the distribution of economic voting among opposition parties. Specifically, for multiparty oppositions, all the versions of our contextual theory of rational economic voting that made a distinction between opposition and incumbent parties implied that *economic voting will be more important to the support of opposition parties that are members of more, and more competitive, alternative governments than it will be for other opposition parties that are not* (see, for example, equation 8.27).²⁴⁹

To examine this hypothesis empirically, we begin with the historical approach to measuring voter beliefs about the governmental contention of opposition parties. Formally, this hypothesis says that for opposition parties, as $g'_j \sum_{g' \in A} P_{gg'}$ gets bigger, opposition party j 's economic vote will get bigger; where g'_j is an indicator variable equal to one if party j is a member of alternative cabinet g' and 0 otherwise. In other words as j is a member of more alternatives with larger values of $P_{gg'}$, then it will have a larger economic vote. To build a measure of this quantity for some party j , we need an estimate of $P_{gg'}$ for each possible alternative

²⁴⁸ In comparing these measures with the historical measures there are two issues to consider. The first is measurement error. On one hand, while our estimates of quantities like the predicted probability that a party gets into government will certainly deviate from the true values of these probabilities, our record of the percentage of time in office for each party will not deviate at all from the true value of this measure. Thus, if the true probabilities of government participation and the percentage of time in office were equally valid measures of the voter’s beliefs about the distribution of contention, then the historical approach would have less measurement error. Of course, we have no way of knowing which of our measurement concepts (the true predicted probabilities or the true historical record) is a better proxy for voter beliefs and so can not unequivocally prefer the historical measure.

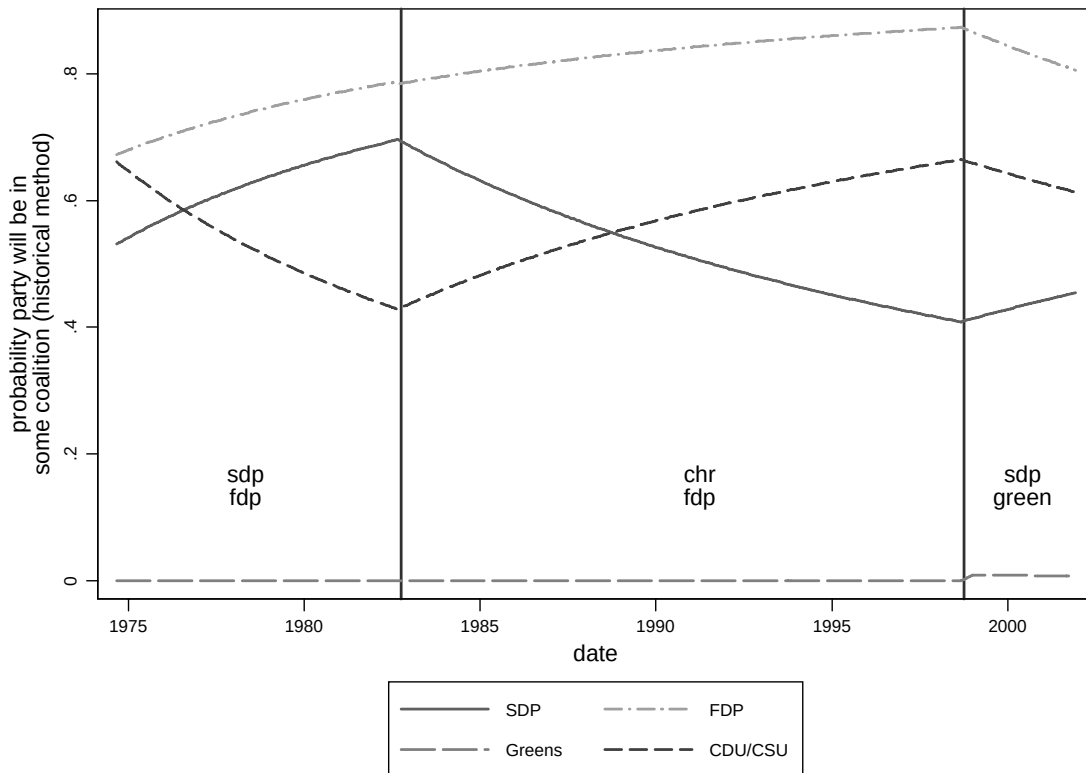
²⁴⁹ We will use the term “government” generically to mean a single-party executive, a coalition cabinet, or a distribution of responsibility. When a hypothesis does not refer to all these types of governments we will use the more specific designations. We will also call the extent to which a party is a member of more, and more competitive, alternative governments its level of governmental contention, or cabinet contention as appropriate, or sometimes (when there is no confusion) just its level of contention.

cabinet and then we can simply sum these estimates over all the alternative cabinets of which party j is a member. In the historical method, our strategy is to estimate $P_{gg'}$ as the historical frequency with which the cabinet alternative g' was actually in office (going back to January 1960).²⁵⁰ Clearly, this will be zero for any alternative that has never been in office and will move toward one as the coalition has spent more time in office. We calculate the measure monthly from 1960 and use the data point for each party for the month in which the election survey was conducted. For example, if the election survey under consideration was conducted in January of 1980 and a specific alternative coalition had been in power for 18 months in the 1970's then we assign its chances of forming as 18/240 (where the denominator is the number of months between January 1960 and January 1980).

Figure 10.1 gives an example of the measure for Germany and illustrates how well it captures our intuition about which parties in Germany are (and are not) members of the competitive alternative cabinets. Notice, for example, that the score for the FDP is high throughout the period. This is because the FDP is a member of both of the coalition alternatives that have historically ruled in Germany (the CDU/CSU – FDP coalition and the SDP-FDP coalition).²⁵¹ Likewise, for both the SDP and the CDU/CSU the measure suggests about a 50% chance of entering government. The measure also moves over time in an intuitive way – as a party spends more time as the incumbent the measure increases and as it is out of government (thus, by definition, not a member of one of the competitive alternatives) its score decreases.

²⁵⁰ Since $P_{gg'}$ is the probability that cabinet g' ties with the incumbent cabinet g , this measurement strategy assumes that a cabinet's historical frequency of ruling is not just positively related to its chances of forming in the current period, but also to its chances of tying with the current incumbent. Thus, the measure assumes the current incumbent is itself competitive.

²⁵¹ The only reason this number is not one, is because of the short-lived SDP-CDU/CSU coalition in the 1960's and the CDU/CSU-DP coalition that was in office when our measurement period started in January 1960.



Each time point in the line is calculated by first calculating the historical frequency between the time point and January 1960 that each possible alternative cabinet has served and then summing these over all alternative coalitions of which the party was a member. The vertical lines are periods with different incumbents and the incumbent cabinet is listed in the bottom half of the graph.

Figure 10.1. Historical Measure of Cabinet Contention by Party for Germany.

Notice that the historical frequency rule for each possible alternative cabinet sums to one across all cabinets in any given month (but not over parties – so the lines in the graph will not sum to one). Further, when we sum these frequencies across the coalitions of which a given party is a member (at one particular time point), the sum of this number will, by definition, be equal to the percentage of months since 1960 that the party has been in the cabinet. Thus, it is possible to calculate our measure without actually writing out all the possible coalitions and their historical frequencies – instead, we can simply calculate how often each party has been in a cabinet.²⁵² This also has the advantage that when we calculate the measure in this way it is easier to modify it to include discounting, to deal with other special cases in the data, and to account not just for a

²⁵² This works for opposition parties but not for incumbent parties, since for these parties we need to subtract the re-formation probability of the incumbent cabinet from the sum (it is not included in the sum for opposition parties by definition).

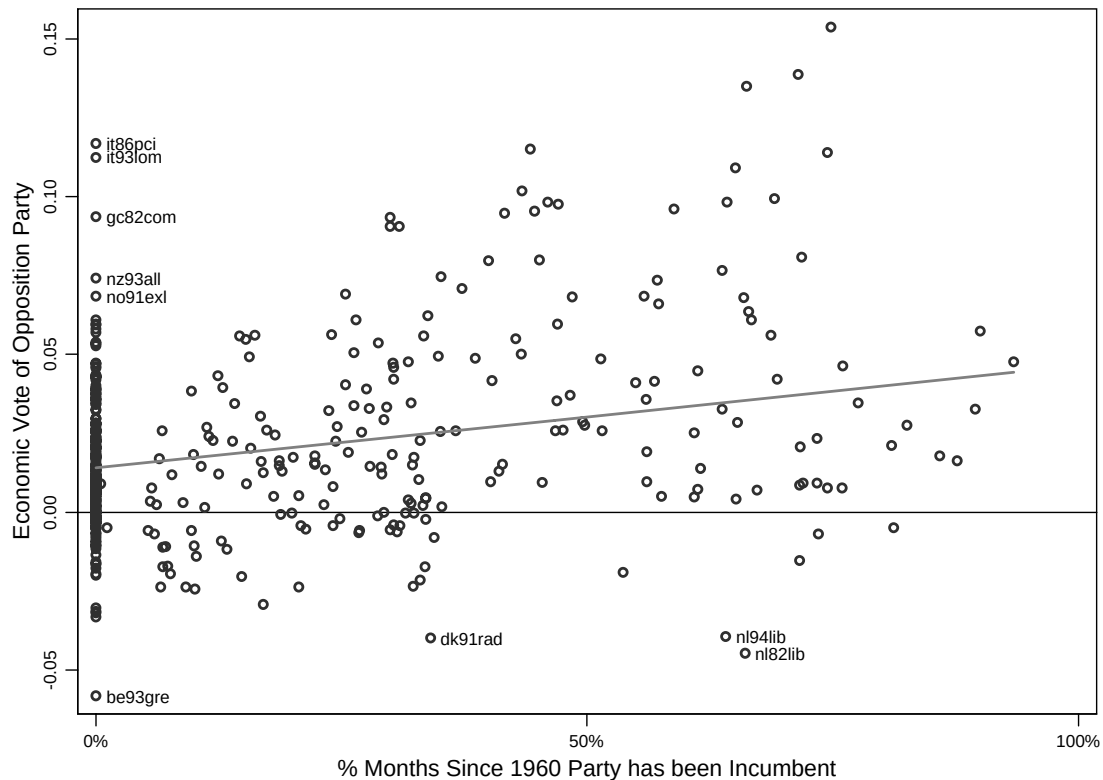
party's record of cabinet participation but also how it participated (i.e., whether it was a PM or a partner).²⁵³

Figure 10.2 graphs, for multiparty systems (and surveys for which there are more than two opposition parties), the economic vote for each opposition party against its previous record of office holding.²⁵⁴ The hypothesis (and measurement assumptions) suggests that voters will not consider opposition parties with short histories of cabinet participation likely members of competitive alternatives and so will have relatively muted economic votes.²⁵⁵ This expectation is confirmed in Figure 10.2 (a large positive value here indicates that the opposition party is rewarded by perceptions of poor economic performance). Indeed, considering that the impact on economic voting of any one context interacts with other contexts in a multiple conjunctural relationship, we should expect to see, if our hypothesis is true, exactly the kind of wedge shape to the data that is apparent in Figure 10.2. That is, when parties have little experience as incumbents, the impact of the economy on their support will be low, but when they have a lot of experience, the impact of the economy on their electoral support *may* be high, but may not be depending on other factors.

²⁵³ We have calculated both the measure described above and the simpler measures that we use in the text and they correlate at .99 for our sample. The only reason this correlation is not perfect is that some of the “parties” in our data set are actually party groups that were combined in the estimation of the economic vote. Such combinations are relatively rare, never cross the opposition/incumbency divide, and make sense ideologically. However, they do mean that the different parties in the group will have different histories of government service and so to use them together we have to average them at any given month. Consequently, there are slight deviations between the measures in some cases (the two measures are identical in 96% of the cases).

²⁵⁴ All our measures of the history of office holding are calculated from monthly data between January 1960 and the date of the survey corresponding to the data point. Thus, the average voter's “memory” in our surveys extends from 20 to 40 years into the past and always includes most of her adult political life (the average voter is about 40 years old at the time the surveys).

²⁵⁵ We use the term “members of alternative governments” to generically include alternatives with only one party.



Only cases in which there were at least two opposition parties are included. The variable on the x-axis is equivalent to a measure (described in the text) that uses the historical method to calculate the probability that each opposition party will enter the cabinet as a member of some alternative.

Figure 10.2. History of Incumbency and the Economic Vote of Opposition Parties in Multi-party Systems

The main exceptions to the wedge-shaped pattern in this graph are the three cases in the upper left corner of the graph that had sizeable economic votes but no previous experience in government (the Italian Lombard League in 1993, the Italian PCI in 1986, and the Greek Communists in 1982).²⁵⁶ The exception of the Italian Lombard league likely reflects the inability of the historical record to capture voter beliefs at the time of this survey since it occurred just as the historical Italian party system was breaking down and a whole set of new alternatives to the

²⁵⁶ The other two notable but less exceptional cases are the New Zealand Alliance in 1993 and the extreme left in Norway 1991, which combines the Left Socialists and the Communists. In the first case, it is likely that the result for the Alliance (which was the alternative to the large Labour party in the opposition) reflects the impact of the voter's overwhelming concern with the 1993 referendum on electoral reform, which the Alliance supported and both large parties did not (Labour in opposition and the National Party in government). Although we controlled for voter's preferences on this referendum in the individual level analysis of this case, it could be that the division between the Alliance on one side and Labour and the National Party on the other reinforced in the minds of voters the role of the Alliance as the chief alternative to the current administration rather than Labour.

previous political parties were emerging. As Verzichelli and Cotta (2000) put it, between 1992 and 1994 “new or previously excluded parties captured a significant part of the political space that the center parties had controlled... Suddenly, all the parties gained a reasonable expectation to win access to the government”.²⁵⁷ Thus, at least with respect the voters’ beliefs about governmental contention, the Lombard League may have been seen as more of a contender than history would otherwise have suggested.

For the other two exceptions, we have no obvious explanations. In the case of the Italian PCI outlier in 1986, the five party government (the DC, PSI, PRI, PSDI, and PLI) at the time included all the parties that had ever served in government after 1946 which may have left economic voters that wanted to vote against the government with little alternative other than to vote for the Communists. However, this plausible explanation for the *distribution* of the economic vote among the Italian parties can not explain why the *size* of the economic vote in this case was so large. Our theory says that rational economic voters faced with no competitive alternative to the incumbent (or alternative governments made up of only incumbents) would simply not use the economy in their vote choice. Indeed in the other three surveys in which the same situation occurred (1987, 1988, and 1990) the overall level of economic voting is quite low as the theory predicts.

The Greek 1982 case presents even more of a mystery since it is only the most extreme case in a more general pattern of unexpected results for Greece. Specifically in almost all of the surveys we have for Greece, our estimates reveal that when the Socialist PASOK is in power, the opposition party that is the primary recipient of the economic vote is not its principal rival, New Democracy, but rather the Communists.²⁵⁸ In contrast, when New Democracy is in power, the economic vote shifts support between New Democracy and PASOK as the theory predicts. We have no explanation for this result; however, as we noted earlier, the Greek case was the case in which our estimates of economic voting were the most uncertain and in which the individual models seemed most sensitive to specifications issues. The problem may lie in the estimation of the economic vote itself.

Despite these three exceptions, the evidence in Figure 10.2 supports the usefulness of the pattern of governmental contention in explaining the distribution of the economic vote across opposition parties. Before providing the statistical model corresponding to Figure 10.2 (and that

²⁵⁷ In addition, the Lombard League had been around for some time at this point while the later re-stabilization of the new political system around two broad coalitions - that would leave Forza Italia and not the Lombard League as the leader of the right - had not yet emerged.

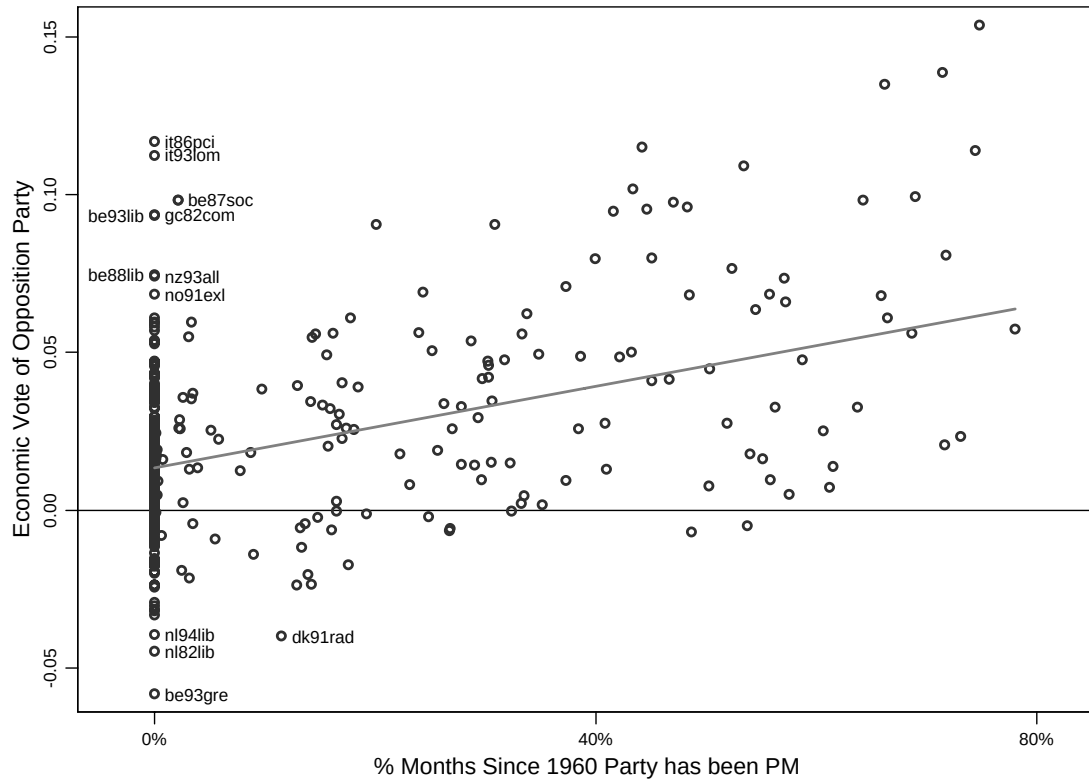
²⁵⁸ In the individual level estimates we combine the Communist Party of the Interior with the Communist Party of the Exterior and add the EDA in 1982 and 1983 and the NAP in 1990 and 1993.

accounts for the various dependencies in the data), we explore some alternative ways to conceptualize the relationship between a party's history of cabinet participation and economic voting and then examine all the corresponding statistical models in the same table.

In some countries voters have come to expect a single party to control the cabinet and in others voters may understand that the primary competition that shapes the membership of a coalition cabinet is the race for the prime ministry rather than for cabinet partnership (i.e., the *formateur* systems mentioned in Chapter 8). Indeed, no matter what kinds of cabinets usually form, the question of which party will provide the prime minister is a focus of speculation and media attention in every parliamentary system. This suggests that voters may pay more attention to, and have better developed beliefs about, the pattern of contention for the prime ministry rather than for cabinet membership in general (a contention supported to some extent by the survey data presented at the beginning of this chapter). Thus, we might expect the relationship between the governmental contention of opposition parties and economic voting to be even stronger when we measure beliefs about governmental contention using each party's history of holding the prime ministry rather than its history of incumbency in general.²⁵⁹ Figure 10.3 provides this graph and confirms our expectation that an opposition party's history of service as the prime minister is strongly related to its economic vote.

In comparing Figure 10.3 to 10.2, three additional exceptions to the expected wedge shaped pattern emerge - all from Belgium. However, as we will see below, when we examine the other hypotheses implied by our theory, we will find that these exceptions are really quite consistent with the more general logic of the theory that accounts for the various separate hypotheses in this chapter. Specifically, in systems (like Belgium) in which there is a perennial prime ministerial party that tends to form coalitions with different sets of smaller coalition partners (e.g., the Belgian Christians), the theory suggests that the economic vote should be concentrated on these contenders for cabinet partnership rather than on the dominant prime minister. Accordingly, in these contexts there will be a number of contenders for cabinet partnership that should have relatively large economic votes even though they have had very little, if any, history of holding the prime ministry.

²⁵⁹ Like the previous measure, there are two equivalent ways of calculating this. First, one could estimate the probability with which each coalition alternative forms (counting coalitions with the same membership and different prime ministers as different alternatives) and then sum all these probabilities for cases in which a given party is prime minister. When the probability of the coalition forming is equal to its historical frequency, however, the resulting measure for each party will be equivalent to the historical frequency with which the party was prime minister.



Only cases in which there were at least two opposition parties are included. The variable on the x-axis is equivalent to a measure that uses the historical method to calculate the probability that each opposition party will enter the cabinet as the prime minister in some alternative.

Figure 10.3. History of Holding the Prime Ministry and the Economic Vote of Opposition Parties in Multi-party Systems

There are number of additional ways that we can examine the relationships in Figures 10.2 and 10.3. Specifically, we may first want to alter our measures of cabinet participation to account for the fact that most scholars who invoke adaptive or “historical” belief formation usually assume that people who form their beliefs in this way discount older information relative to more recent information. We accommodate this in our models by substituting the history of government service in the graphs above with a measure that discounts this history so that service in the past is weighted less in voter beliefs than more recent service.²⁶⁰

A second modification to the above analyses is to focus on shares of the economic vote for opposition parties rather than absolute levels of the economic vote for these parties. This

²⁶⁰ To do this, we use an exponential weighting function, δ^m , where m is the number of months between the current month (for which a score is being calculated) and the month in the past that is under consideration (ultimately, we sum over all months since 1960). We choose $\delta=.99$, which means that service five years in the past is discounted by about one half and ten years in the past by about a third. The specific details of this calculation are available from the authors.

removes any influence of the magnitude of the economic vote from the measure and focuses instead on the distribution of the economic vote (whatever its overall size) across parties.²⁶¹ This distributional focus is useful given that our theory implies that differences in the distribution of contention across parties should impact both the absolute size and the share of a party's economic vote.

Table 10.3 provides estimates of regression models for different combinations of these alternative specifications (in addition to estimates for the relationships graphed in Figures 10.2 and 10.3). All the models follow the procedures for specifying empirical models of contextual effects outlined in Chapter 4. Again, however, we should remember that these estimates are only approximate since they assume the impact of the independent variables are additive, when we know that the appropriate functional relationship should account for multiple conjectural causation. In our graphs this was apparent in the triangular patterns in the data, which also showed clearly that the linear approximation tends to understate support for multiple conjectural hypotheses because the additive model incorrectly counts many of the cases that occur at the wide end of the triangular data pattern as evidence against the hypothesis.

Table 10.3. History of Governmental Contention and the Economic Vote of Opposition Parties

<i>Dep. Var.</i> <i>Ind. Vars.</i>	<i>Econ Vote</i>	<i>Econ Vote Share</i>	<i>Econ Vote Share*</i>	<i>Econ Vote</i>	<i>Econ Vote</i>	<i>Econ Vote Share</i>	<i>Econ Vote Share*</i>	<i>Econ Vote</i>
% months party has been incumbent	.03 (2.53)	.40 (2.89)	1.69 (2.90)	-- --	-- --	-- --	-- --	-- --
% months party has been incumbent, discounted	-- --	-- --	-- --	.04 (2.95)	-- --	-- --	-- --	-- --
% months party has been PM	-- --	-- --	-- --	-- --	.06 (4.20)	.67 (4.19)	2.88 (3.92)	-- --
% months party has been PM, discounted	-- --	-- --	-- --	-- --	-- --	-- --	-- --	.04 (2.95)
Constant	.01 (6.65)	.28 (9.78)	-.95 (-6.93)	.01 (5.97)	.01 (7.33)	.28 (11.67)	-.94 (-8.02)	.01 (5.97)
N	401	398	398	401	401	398	398	401
R ²	0.08	0.09	--	0.07	0.17	0.15	--	0.07

Numbers in cells are coefficients and t-ratios. All models except those indicated by the * were estimated using OLS regression with standard errors robust to heteroscedasticity and non-independence between observations of the same party in different surveys. Those marked with an * use GLM to account for the fact that the share of the economic vote is a proportion (as recommended in Papke and Wooldridge, 1996).

²⁶¹ To calculate shares of the opposition economic vote we have to first set opposition economic votes that are in the "wrong direction" to zero. We then simply sum over opposition parties and divide each (modified) share by that sum.

The estimates in Table 10.3 are all statistically significant, in the expected direction, and tell the same substantive story about the relationship between the pattern of governmental contention of opposition parties and the distribution of the economic vote among them. Thus, we can safely conclude that when voters' beliefs about the pattern of governmental contention are shaped only by the parties' records of office holding, the importance of the economy will be weakest for the opposition parties least likely to contend for office.

The history of previous service in government is only one kind of information that voters can use to form their beliefs about which opposition parties are, or are not, in governmental contention. Our alternative measurement approach assumes that voters form their beliefs about the likelihood of different government alternatives (and the resulting chances that a given party will enter government) rationally. This measure mimics the rational voter's likely "model" of government formation. It is derived by estimating a statistical model that predicts which governments will form as a function of the observable characteristics of parties that are generally recognized as affecting government formation.²⁶² We do this differently depending on whether the system under consideration is one in which coalition cabinets usually form (the "coalitional" cases) or one in which single party cabinets are the norm (the "non-coalitional" cases).²⁶³ In non-coalition countries, the "model" that rational voters likely use to forecast the outcome of the government formation process (and the role of the vote in that process) is a simple one: the plurality party will form the government and rule alone. Assuming that voters' beliefs about which parties contend for a plurality are an increasing function of the parties' shares of electoral support, we should expect each opposition party's electoral support at the time of the survey to be positively related to its economic vote.

Our best available measure of electoral support is simply the survey marginals for the vote choice questions that were used in the surveys from which we estimated the economic vote.²⁶⁴ In Figure 10.4, we graph this measure against the economic vote for opposition parties in

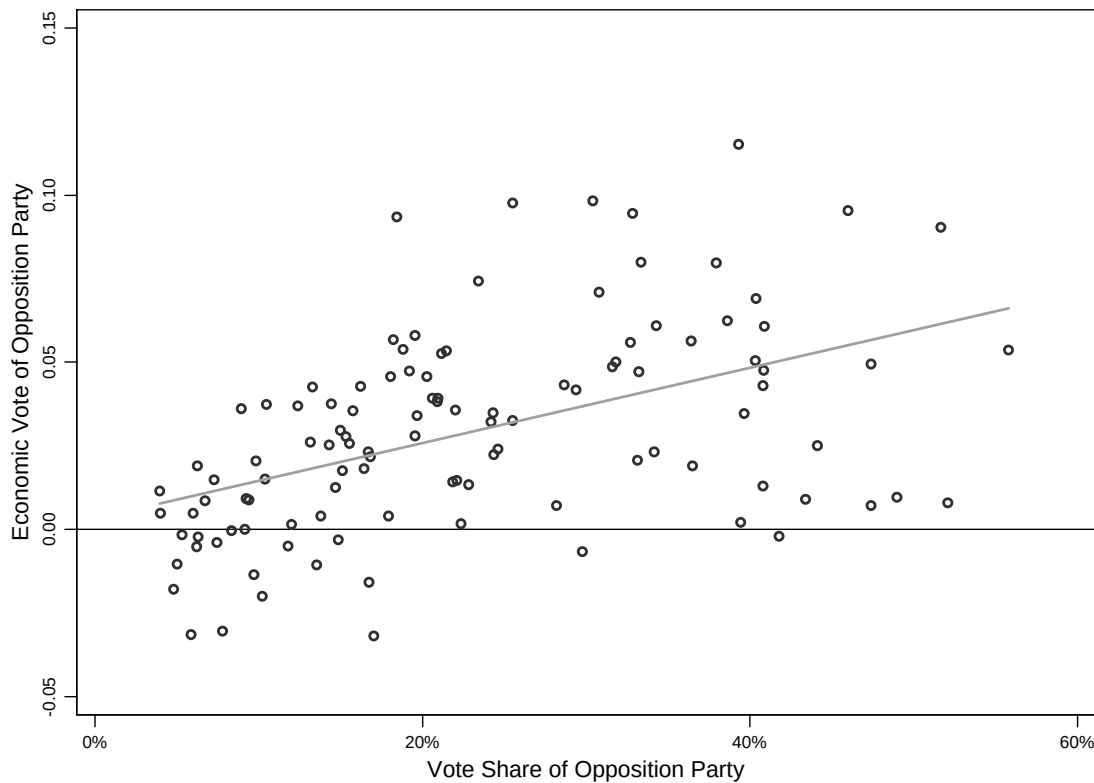
²⁶² As with the historical measure, we are trying to measure $g_j' \sum_{g' \in A} P_{gg'}$, and so require an estimate of $P_{gg'}$.

In the historical approach we estimate this quantity with the historical frequency with which cabinet g' formed. In the current method we estimate this quantity from a statistical model of cabinet formation that incorporates most of the information about party ideologies and the electoral/bargaining situation that might be available to voters to inform their choice.

²⁶³ The non-coalitional countries in our sample are the United States, United Kingdom, Spain, Greece, Canada, Australia, and New Zealand before 1995.

²⁶⁴ For cases in which a survey asked about reported vote instead of vote intention, the survey marginals are likely to be biased in favor of the incumbent. However, in this case we get the same result if we substitute actual electoral support instead.

non-coalitional systems in which there were multiple opposition parties (and at least two of these were included in our individual analysis of the case).



Only cases in which there were at least two opposition parties are included. The variable on the x-axis is the percentage of respondents indicating support of a given party (among those indicating they supported some party) at the time the survey was conducted.

Figure 10.4. Party Vote Share and the Economic Vote of Opposition Parties in Non-Coalitional Systems in Multi-party Systems

This graph (and the corresponding statistical models in Table 10.4 below) clearly supports the expectation that in multiparty non-coalitional systems, the economic vote of opposition parties will be higher the more likely the party is to compete with the incumbent for a plurality. The relationship is statistically significant and positive despite the fact that again the graph shows the tell-tale signs of multiple conjectural causation that were discussed above.²⁶⁵

When we turn from non-coalitional systems to those in which coalition governments often form, the rational voters cannot use the simple plurality winner model to forecast the outcome of cabinet formation. Instead they must consider what characteristics of parties and

²⁶⁵ In this graph the two main exceptions from non-coalitional systems that appeared in Figure 10.2 – the Greek Communists in 1982 and the New Zealand Alliance in 1993 – are no longer exceptions.

coalitions other than their size might impact cabinet formation. We try to measure the voters' rational beliefs about which coalitions will form in coalitional systems by building an empirical model of coalition formation that is based on much of the observable information that voters might use to form their beliefs. We then use this model to predict which coalitions will form in our cases and construct measures of voters' beliefs from these predictions. We use the models and data from Martin and Stevenson (2001) to estimate a slightly modified version of their empirical model of coalition formation in coalitional systems.²⁶⁶

After obtaining the coefficients from this model, we used them along with appropriate data from each of our 163 cases to produce "out of sample" forecasts of the probability that each possible coalition that could have formed (had a cabinet formation taken place at the time of the survey) would have in fact formed.^{267,268} In cases in which our cases overlap with Martin and Stevenson's, our predicted probabilities of formation correspond quite closely to theirs.²⁶⁹ In this

²⁶⁶ Our version of their model forecasts the probability that any potential coalition forms as a function of the minimal winning status of the potential coalition, its majority status, the number of parties in the potential coalition, whether it contained the largest party, the extent of ideological division in the potential coalition, the extent of ideological division in the potential majority opposition (for minority potential coalitions only), the extent of anti-system presence in the potential coalition, whether it contained the median party on a left-right dimension, whether the system had an investiture vote (interacted with majority status of the potential coalition), whether the potential coalition contained the incumbent prime minister, and whether the potential coalition was the incumbent coalition.

²⁶⁷ All the variables except one were constructed as described in Martin and Stevenson 2001. We had to estimate the seat shares that parties would likely obtain if an election were held at the time of the survey. We did this by forecasting seats from the survey marginals for vote choice in the survey. To produce this forecast, we first estimated the functional relationship between votes and seats in each country using Mackie and Rose's electoral data and applied the best fitting function to our survey marginals to generate our predictions. For cases in which a break in the party system occurred (e.g., Italy and New Zealand), we used the actual seat shares from the most proximate election to the survey. Since we are only concerned with coalitional systems, which tend to be highly proportional, the forecast was usually close to linear. For ideology we used the same manifestos scores as in Martin and Stevenson but again used those from the most proximate election. One difference with their model is that in our simplification we dichotomized their continuous anti-system variable and so in creating our predictions also created a dichotomous variable for whether a party was an anti-system party (and whether a coalition contained such a party).

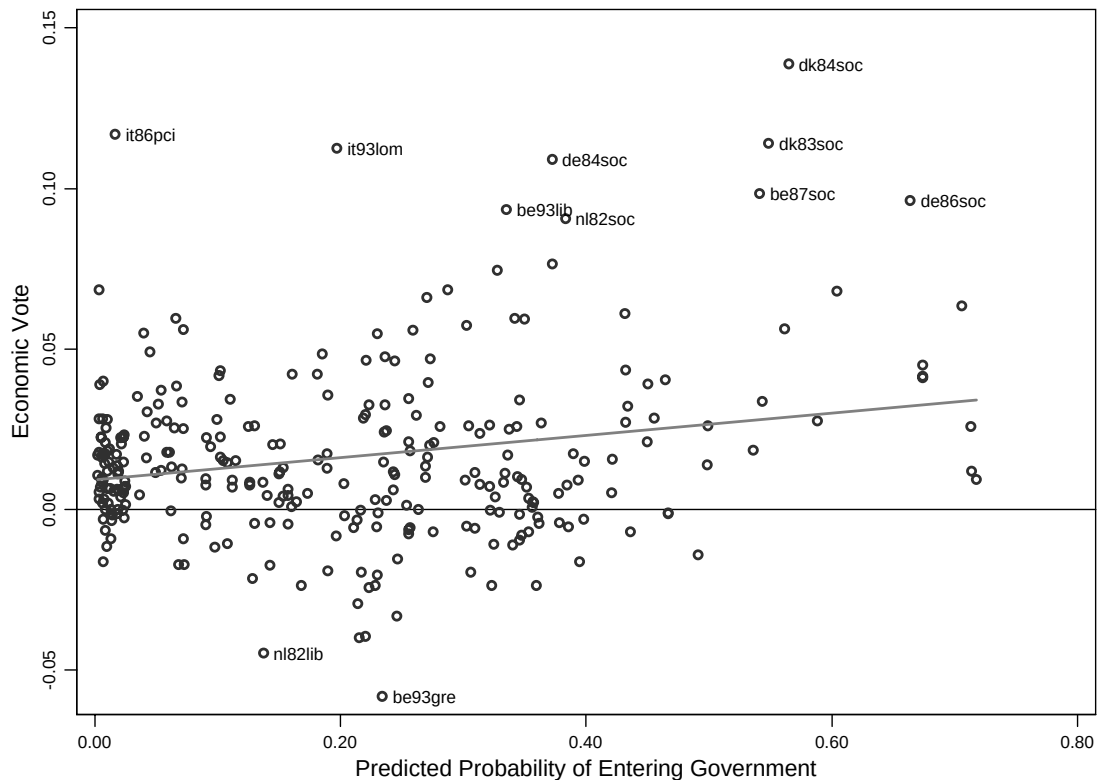
²⁶⁸ An important issue in using the survey marginals to measure the support of parties at the time of the survey (and then to calculate seat shares) is raised by the difference between surveys that ask the voter for their vote intention and those that ask the voter to report their vote in an election that has just happened (although often up to three months previously). It is well known that the survey marginals for such reported vote are generally biased in favor of the parties that "won" the election and (quite likely) for the parties that formed the government. Thus, if we use survey marginals to help produce the predicted probability that different cabinet forms we are likely to get results that over predict the coalition that did form (as well as other coalitions involving the incumbents). Thus, the results in this chapter that rely on predicted probabilities of cabinet formation were run both including and excluding the few surveys that asked vote recall question. The differences in results however were minor.

²⁶⁹ They reported a success rate of about 50% in within sample prediction of the coalition that forms. While this is the best success rate of any empirical model in the literature, it still alerts us to the fact that

section, we use these estimates of the probability that each potential coalition might form to construct a measure of the voter's belief that each opposition party will get into the new cabinet. To do this we simply sum up the formation probabilities of all the coalitions for which a given opposition party was a member. The result is an estimate of the probability that the party will join some cabinet.

Figure 10.5 graphs this measure against the economic vote for opposition parties. The graph shows the expected triangular pattern of data and the linear approximation also makes the positive relationship between the variables plain. However, the main exceptions to the relationship that appeared when we assumed voters formed their beliefs about formation adaptively persist when we assume they form them rationally. That is, the Italian PCI in 1986 and the Lombard League in 1993 both have a relatively high economic vote despite a low estimated probability of entering the cabinet. On the whole, however, the evidence in Figure 10.5 supports the hypothesis that opposition parties that do not contend for cabinet membership will have low levels of economic voting while those with higher levels of contention may have higher levels of economic voting.

this measure is not even a perfect measure of the probability each cabinet will form, let alone a perfect measure of voter beliefs about these probabilities. The predicted probabilities (in addition to the program for producing them) are available from the book website www.nuffield.ox.ac.uk/economicvoting_.



Only cases in which there were at least two opposition parties are included. We calculated the variable on the x-axis by first estimating the probability that each alternative cabinet forms (using our implementation of Martin and Stevenson's 2001 empirical model of cabinet formation); and, next summing these probabilities over cabinets in which a given opposition party is a member.

Figure 10.5. The Predicted Probability of Entering Government and the Economic Vote of Opposition Parties in Coalitional Systems

Table 10.4 presents the statistical models corresponding to Figures 10.4 and 10.5 and the two versions of those models that use shares as the dependent variables. As with the historical measures, when we use measures that try to capture rational belief formation about the pattern of governmental contention, the impact of all of our various measures are statistically significant and in the expected direction. Thus, we are confident that, as our theory predicts, the pattern of governmental contention is an important contextual influence on the distribution of the economic vote among opposition parties.

Table 10.4. Predicted Governmental Contention and the Economic Vote of Opposition Parties

<i>Dep. Var.</i> <i>Ind. Vars.</i>	<i>Non-coalitional Systems</i>			<i>Coalitional Systems</i>		
	<i>Econ Vote</i>	<i>Share of Econ Vote</i>	<i>Share of Econ Vote*</i>	<i>Econ Vote</i>	<i>Share of Econ Vote</i>	<i>Share of Econ Vote*</i>
Share of Vote	.11 (4.16)	1.32 (5.17)	5.82 (4.38)	-- --	-- --	-- --
Predicted probability of entering government	-- --	-- --	-- --	.03 (2.14)	.61 (2.79)	2.74 (2.94)
Constant	.003 (0.71)	.12 (1.89)	-1.67 (-4.97)	.009 (3.55)	.20 (5.58)	-1.34 (-7.28)
N	111	111	111	290	287	287
R ²	0.23	0.30	--	0.05	0.11	--

Numbers in cells are coefficients and t-ratios. All models except those indicated by the * were estimated using OLS regression with standard errors robust to heteroscedasticity and non-independence between observations of the same party in different surveys. Those marked with an * use GLM to account for the fact that the share of the economic vote is a proportion (as recommended in Papke and Wooldridge, 1996)

The Pattern of Contention amongst Incumbent Parties

The hypotheses explored in this section and the next both concern the relative importance of economic voting among incumbent parties. Specifically, our theory suggests that there are two aspects of the pattern of contention in coalitional systems that are important to incumbent parties. First, *economic voting should be smaller for parties that are members of incumbent cabinets that are not in contention with other alternatives - either because the cabinet is unlikely or it is certain to re-form*. In either case, pivot probabilities between the incumbent cabinet and alternatives will be near zero, so we should see little economic voting. In addition to this implication, however, the theory also implies that *economic voting should be smaller for incumbent parties that are members of competitive alternative cabinets than it is for incumbents who are not*. Thus, even if the incumbent cabinet is in contention to re-form, the economic vote of cabinet members that are also members of competitive alternatives should be muted. For example, some frequent incumbent parties, like the Belgian and Dutch Christian parties or the German FDP, owe their longevity as cabinet members not only to the frequent re-formation of the same cabinet but also to the fact that they are members of all the competitive alternatives. Thus, whatever cabinet re-forms, they will be part of it. Our theory tells us that the economic vote of such parties should be depressed relative to their cabinet partners.

Of particular theoretical interest are those cases where these perennial cabinet parties also take a large share of cabinets seats or provide the prime minister (e.g., in the Italian, Dutch, and Belgian cases; but not the German one). For these cases, the implication of our theory for the

pattern of contention works *against* the widely-studied clarity of responsibility argument, which predicts that parties with a greater share of cabinet responsibility will experience a relatively greater economic vote than their partners.²⁷⁰ Thus, this implication will help us to empirically determine whether the pattern of contention adds something to the political context conditioning economic voting that is distinct from the impact of the distribution of responsibility. Specifically, in the clarity of responsibility argument it is exactly those perennial parties that are always in cabinet, usually dominantly so, that should leave the voter most clear about whom is responsible for policy. In contrast, in our theory, even though these parties have high responsibility (and so the responsibility augmented competency signal may be large), that fact will not matter to rational voters because the party is sure to be in whatever government that forms.

In what follows, we will examine evidence about the relationship between voter beliefs about the competitiveness of incumbent parties and the size of their economic votes. We begin, in this section, by examining the second hypothesis noted above: i.e., incumbent parties that are members of competitive alternative cabinets will have muted economic votes relative to their cabinet partners who are members of fewer, or less competitive, alternatives. Before turning to the data, however, there are two points we should clarify about how we have chosen to present the results.

First, as we have already pointed out, our theory implies (1) a quadratic relationship between the vote for any particular incumbent party and the overall level of contention of the incumbent cabinet, and (2) given this overall level of competitiveness, a linear relationship between the extent to which the party is a member of competitive alternatives and its economic vote.²⁷¹ This implies that we should examine the later relationship while controlling for the former which is what we do in the next section. However, in this section our graphs and statistical models of the linear relationship between incumbent party participation in competitive alternatives and the economic vote ignore this control. This is a useful separation pedagogically (because it allows us to delay our rather extended discussion of how we measure the control variable) and does little harm, since the linear relationship that we explore in this section turns out

²⁷⁰ It also works against the effects of the concentration of responsibility that are predicted by our theory and were tested in the last chapter. However, the implications are not *contradictory* - since the theory simply says both influences are in operation simultaneously. This is not the case with the clarity argument, which has no mechanism that would explain a depressed economic vote for perennial prime ministers, to which voters should easily attribute policy making responsibility.

²⁷¹ Of course, given the multiple conjunctural nature of all our hypotheses we should more properly say that the theory implies a set of relationships that should be *approximated* by a quadratic and linear functions, respectively.

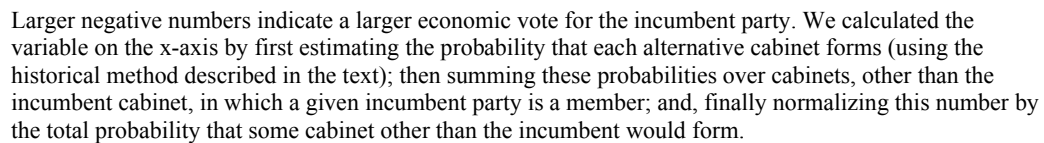
to be very strongly apparent in the data and (as we show in the next section) robust to the control for the overall competitiveness of the cabinet.

Second, in the various analyses that follow, we examine both the size of the economic vote for incumbents and shares of the economic vote among incumbents. This is useful because while the hypothesis applies most clearly to *shares* of the economic vote among the incumbent parties, it also applies to the *size* of the economic vote, *ceteris paribus*. If we consider two parties, one that is in many alternative cabinets and the other is not, and they are in essentially the same position with respect to all other variables, the former should have a smaller economic vote than the latter - even though the overall amount of economic voting in the two systems should be the same. Given this, we present statistical results for both shares and size of the economic vote, though our graphs concentrate on the more visually helpful size measures.

As in the previous section, we begin our empirical analysis with the measurement assumption that the voter uses the historical frequency with which a cabinet formed as an estimate of the probability that it will form in the current period. Thus, the sum of these estimates over all alternative cabinets of which the incumbent party is a member is our estimate of the extent to which the party is a member of competitive alternatives (note that the probability the incumbent cabinet will re-form is not included in this sum). Further, since the hypothesis is about the extent to which a given cabinet member is a member of competitive alternatives *relative* to its cabinet partners, we normalize the measure by the overall probability that some alternative to the cabinet will form. Thus, our final measure is the probability that the incumbent party will enter the new cabinet in some alternative to the incumbent cabinet, given that the incumbent does not re-form.²⁷² We expect the economic vote to decline as the value of this variable is bigger.

Figure 10.6 graphs this measure against the economic vote of incumbent parties and Table 10.5 provides the corresponding estimates from appropriate regression models. Clearly, parties that are members of almost all the competitive alternative cabinets (i.e., near 100% on our measure) have depressed economic votes relative to other parties and we once again can see the expected triangular shape to the spread of data (i.e., the small end of the triangle on the right side of the graph expanding downward toward the left).

²⁷² For example, suppose a party is a member of three cabinet competitive cabinets A, B, and C, where A is the incumbent cabinet and the probabilities of formation are .3, .2, and .1, respectively. In addition, there is one alternative cabinet, D, of which the party is not a member. If D has a formation probability of .4, the party's score on our measure would be $(.2+.1)/(.2+.1+.4) = .3/.7 = .43$. The probability some alternative to the incumbent will form is .6, while the probability that the chosen alternative includes the party is .3, so the probability that the party gets into cabinet, given some non-incumbent alternative forms, is .43.



The two largest exceptions to this pattern are the Italian Christian Democrats in 1993, which had a large economic vote despite the fact that they were a member of all the coalitions that had historically formed in Italy, and (to some extent) the Danish Socialists in 1994. As we pointed in our earlier discussion of the success of the Lombard League in 1993, it is not really surprising that history is not a clear guide to the beliefs of Italian voters in this period. Just as these voters may have perceived the untried Lombard League as a viable alternative governing party, by 1993 they very likely saw the DC's lock on the cabinet as slipping. Thus, while just few years before (i.e., 1987, 1988, and 1991) the Christian Democrats have the expected depressed economic vote of a perennial cabinet member, by 1993 their dominance was already in question.

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in cabinet than they are for the sample of all incumbents. First, our theory implies (and the data analysis in Chapter 9 confirms) that the economic votes of parties holding less responsibility in the cabinet are depressed relative to others; the principal distinction is between prime ministerial parties and their partners. Because of this, the economic votes of partner parties will be closer to zero than that of prime ministers and, as a result, other relationships will necessarily be harder to detect over the background noise.

Second recall that in its most general version (see the discussion following equation 10.37 in Chapter 8), our theory implied that voters wishing to use their vote most efficiently to change (or retain) the status-quo distribution of responsibility should move their votes between those parties with the largest shares of responsibility in the current distribution of responsibility to (or from) those parties with the smallest shares (as long as these parties also have a large share of responsibility in some competitive alternative). This means that voters have an incentive to concentrate economic voting on the coalition members with the most responsibility (e.g., the prime-ministerial party) - not only because the responsibility augmented competency signal is larger for these parties, but also because the pattern of contention makes this strategy the most efficient way to achieve their goals.

Given these theoretical arguments, as well as the empirical results in the last section, we expect that the predicted linear relationship between incumbent party membership in competitive alternative coalitions and economic voting to be more apparent in our sample of prime ministerial parties than in the whole sample of incumbents. This is exactly what we find when we examine the two groups of incumbents separately. Indeed, this should not only be true for the hypotheses we are examining in this section, but also for the other hypotheses we will explore in the next sections; and the data do, in fact, bear this out. Given this, in the rest of this chapter, we focus our discussion on the sample of cases that the theory tells us should most clearly reflect the expected relationships.

To be clear, these hypotheses *do* apply to all cabinet members - it is just that the theory tells us that it should be harder to see the relationships in a sample of cabinet partners than in a sample of prime ministers. And, as we said, we do generally find weaker results (in terms of statistical significance) for cabinet partners compared to prime ministers. It is important to emphasize, however, that the evidence in our sample of cabinet partners never actually contradicts our theoretical expectations. Indeed, the directional evidence is always the same for the partners and prime ministers, it's just that in the former case, we often can not statistically

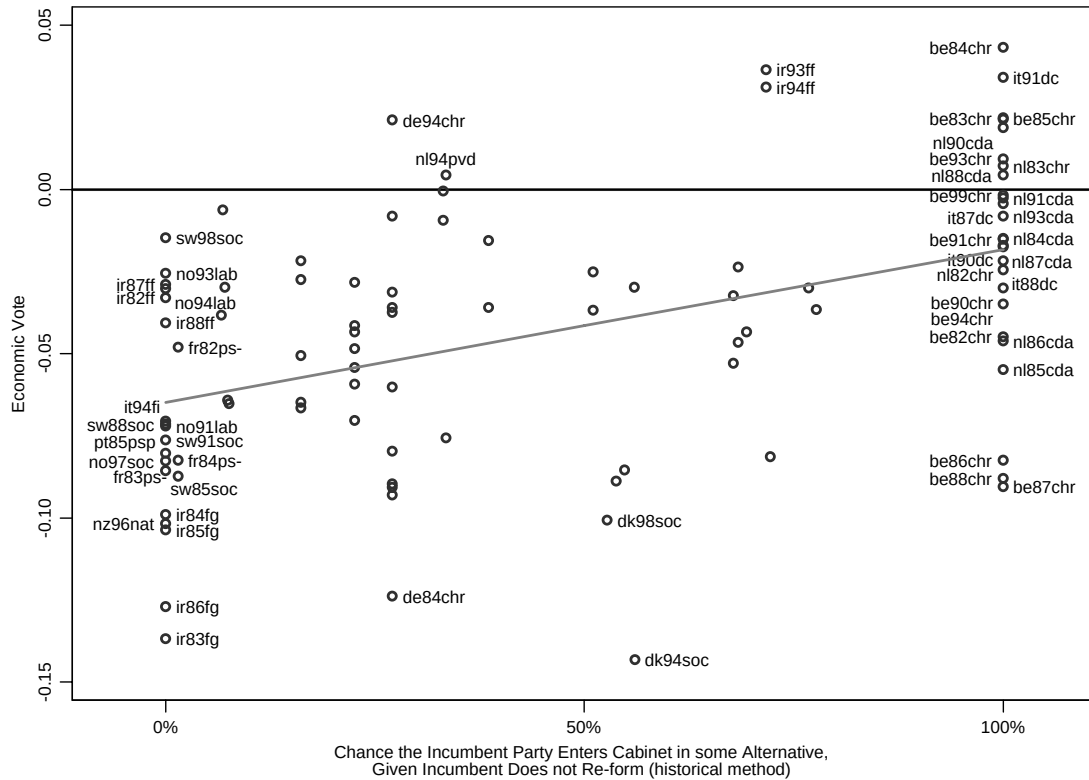
distinguish effects from zero.²⁷³ For example, cabinet partners that are members of competitive alternative cabinets do tend to have smaller economic votes than those who are not, but this difference is less apparent for partners than prime ministers. Further, looking at Figure 10.6, there is really only one case in our data in which a frequent partner party is also a member of most of the competitive alternative cabinets (the German FDP).²⁷⁴ And, as the theory predicts, the average economic vote of this party is rather sharply smaller than other, less perennial partners. Specifically, the mean economic vote for the German FDP is -.00017 compared to an average economic vote of -.0136 for the other partners in Figure 10.6 – a difference that is significant at $p < .005$.²⁷⁵

Thus, while in general the results for cabinet partners mirror the results for prime ministers, the expected relationships are less statistically detectable than for prime ministerial parties (with the exception of the contrast between the FDP and other partners). The extent of this difference is illustrated in Figure 10.7, which graphs the same relationship as in Figure 10.6, but only for prime-ministerial parties. Clearly the expected relationship is much clearer in the second graph, and the empirical models bear this out as well.

²⁷³ A problem made worse by the, now familiar, caveat that we are testing the statistical significance of coefficients from models that are only approximations to the causal structure implied by the theory and that generally work against finding statistical significance.

²⁷⁴ Looking at the right side of Figure 10.6, all the parties except the FDP are either prime ministers (e.g., the Belgian and Dutch Christian Parties for example) or are one time partners that normally hold the prime-ministry (e.g., the Italian Christian democrats in 1993). Thus, the FDP is the only usual partner party that compares with the perennial prime ministers in terms of the number and competitiveness of the alternative coalitions to which it belongs.

²⁷⁵ Notice that just as the perennial prime ministers in Belgium and the Netherlands contribute many of the “wrong signed” economic votes for prime ministers, the FDP provides a disproportionate number of these “wrong signed” economic votes of cabinet partners. Further, this is despite the fact that the party held the finance ministry during this period (recall that we saw in the last chapter that holding the finance ministry enhanced a party’s economic vote, all else being equal).



Larger negative numbers indicate a larger economic vote for the incumbent party. We calculated the variable on the x-axis by first estimating the probability that each alternative cabinet forms (using the historical method described in the text); then summing these probabilities over cabinets, other than the incumbent cabinet, in which a given incumbent party is a member; and, finally normalizing this number by the total probability that some cabinet other than the incumbent would form.

Figure 10.7. Extent to Which Incumbent Prime Ministers are members of Competitive Alternative Coalitions and the Economic Vote in Coalitional Systems

Examining Figure 10.7, there are three different groups of cases to consider. The first are those at the far right of the graph. These cases are ones in which the prime ministerial party is part of every alternative coalition with any significant probability of forming (which, given the historical method used to calculate this probability, means the party is a member of all alternative coalitions that have been in office for any significant length of time since 1960). These are the Dutch and Belgian Christian parties and the Italian Christian Democrats before 1994. On the other side of the graph are cases of incumbent prime ministers that are a part of few of the historically frequent coalition alternatives. Here it is important to remember that the measure does not include the current incumbent cabinet and so it is *not* a measure of the chance that the party will return to government in *any* coalition. For example, in the mid 1980's Fine Gael was in government with the Irish Labour Party, and since it had not coalesced with any other party (or ruled alone) since 1957, it would be unlikely to return to government other than in a coalition

with Labor (at least from the point of view of a voter forming his beliefs about likely cabinets based on history). Thus, our theory suggest that the economic voter who wanted to, for example, retain the Fine Gael-Labour government, could use his vote most efficiently by voting for Fine Gael, since none of this vote would contribute to some other coalition forming. Finally, parties in the middle of the graph are members of some contending alternatives and not others. For example, in 1998 the Danish Social Democrats were in government with the Radical Liberals; however, they had also been a member of five other governments between 1960 and the 1998. Further, several bourgeoisie coalitions had formed during the period which excluded the party all together. Thus, while a vote for the Social Democrats would certainly contribute to keeping the SD-RL coalition in power, it would also contribute to the chances that some other historically plausible cabinet (like the recent SD-Liberal-Center Democratic coalition) would come to power (which is not what an economic voter trying to retain the incumbent coalition would like to see happen).

Table 10.5 provides the statistical model to accompany Figure 10.7 as well as a number of alternative specifications and measures. In each of these specifications, we provide one model that controls for the percentage of cabinet seats held by the prime minister. As mentioned above, a number of results in the theory suggest that a greater share of cabinet responsibility should lead to greater economic voting. While focusing only on prime ministers probably captures most of this impact, there is some variance in the share of the cabinet seats that different prime ministers hold and so it is worth controlling for any additional impact. Looking at the results, the dampening effect of membership in competitive alternatives remains strong when controlling for the prime minister's share of cabinet seats. Further, estimates for the effect of this control variable are either consistent with the results reported in the previous chapter (e.g., in the share equations) or insignificant (in the size equations).²⁷⁶ The relative size of the effects suggests that strategic considerations about cabinet contention are at least as important (if not more so) to the economic votes of prime ministers as is their share of cabinet responsibility.²⁷⁷

²⁷⁶ We should not make too much of this one null result since in all the other equations reported in subsequent tables, the variable is significant and in the expected direction.

²⁷⁷ We really do not want to emphasize this sort of "which variable is more important?" question since it is really only appropriate for additive models (if that); and, of course, we only use the additive model as an approximation to the more complex multiple conjunctural causal structure that is implied by our theory. Nevertheless, we can conclude from these results that there is evidence that it can not just be something like clarity of responsibility driving differences in economic voting, since this evidence clearly indicates an important role for strategic considerations about the pattern of cabinet contention that have nothing to do with the current distribution of responsibility.

Table 10.5. Historical measures of Contention and the Economic Vote of PM Parties²⁷⁸

<i>Dep. Var.</i> <i>Ind. Vars.</i>	<i>Econ Vote</i>		<i>Econ Vote Share</i>		<i>Econ Vote Share *</i>	
Probability of entering government in some alternative other than incumbent coalition, given incumbent does not re-form (historical method)	.05 (5.36)	.05 (3.42)	-.46 (-5.05)	-.25 (-2.95)	-2.51 (-3.30)	-1.51 (-2.31)
Percent of Cabinet Ministries held by Party	-- --	.002 (0.08)	-- --	.60 (2.95)	-- --	4.64 (3.63)
Constant	-.06 (-9.92)	-.06 (-2.69)	.94 (14.18)	.44 (2.18)	2.39 (3.41)	-1.02 (-0.97)
N	98	98	96	96	96	96
R ²	.21	.21	.25	.34	--	--

Numbers in cells are coefficients and t-ratios. All models except those indicated by the * were estimated using OLS regression with standard errors robust to heteroscedasticity and non-independence between observations of the same party in different surveys. Those marked with an * use GLM to account for the fact that the share of the economic vote is a proportion (as recommended in Papke and Wooldridge, 1996)

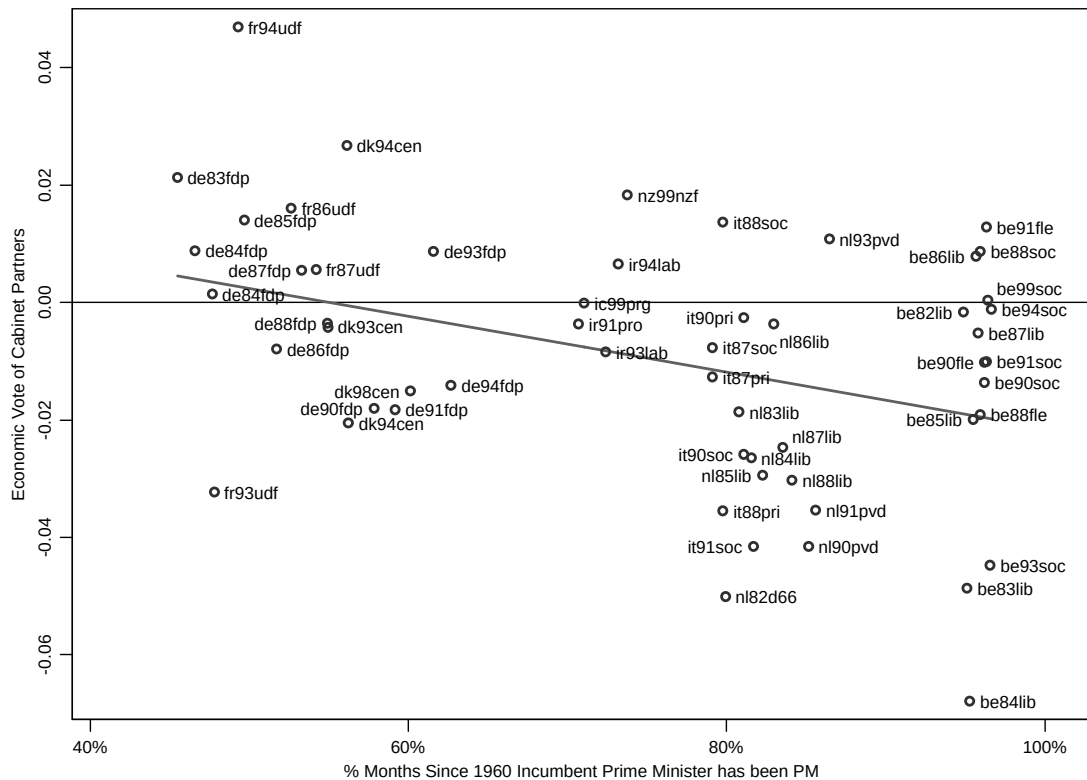
Another particularly revealing implication of this hypothesis turns the story around. Specifically, if the economic vote for perennial prime ministers is depressed relative to their partners and cabinet partners have small economic votes in general, then when we find cabinet partners with large economic votes we would expect them to be from systems with perennial prime ministers. That is, in systems with perennial prime ministers that also tend to switch partners regularly, we should expect the locus of competition over the economy to center on partner parties – since this is the only way the rational voters can use their vote to impact the composition of the cabinet. Thus, we would expect parties with relatively little cabinet responsibility in these systems to have larger economic votes than otherwise expected. Figure 10.8 provides the data necessary to examining this expectation.

The graph plots the economic vote of *partner* parties as a function of their *prime minister's* history of office holding. Thus, partner parties on the right hand side of the graph come from systems with perennial prime ministers and they do tend to have larger economic votes than cases in which the prime minister's party is more competitive.²⁷⁹ This graph (and the corresponding regressions) support the idea that parties like the Dutch and Belgian liberal and socialist parties, which have regularly competed with one another to be the partner of the

²⁷⁸ Models that control for party size give almost identical results for the history variables.

²⁷⁹ Since our hypotheses is about the comparison of partners of perennial PMs to partners of competitive PM's we exclude partners of PMs that had no significant history of office holding (our theory is unclear what we should expect the economic votes of these parties to be relative to other kinds of partners).

dominant Christian party, have larger economic votes than one should expect for their status as partners – a fact strikingly consistent with the logic of strategic economic voting (and not with any sort of clarity of responsibility argument).

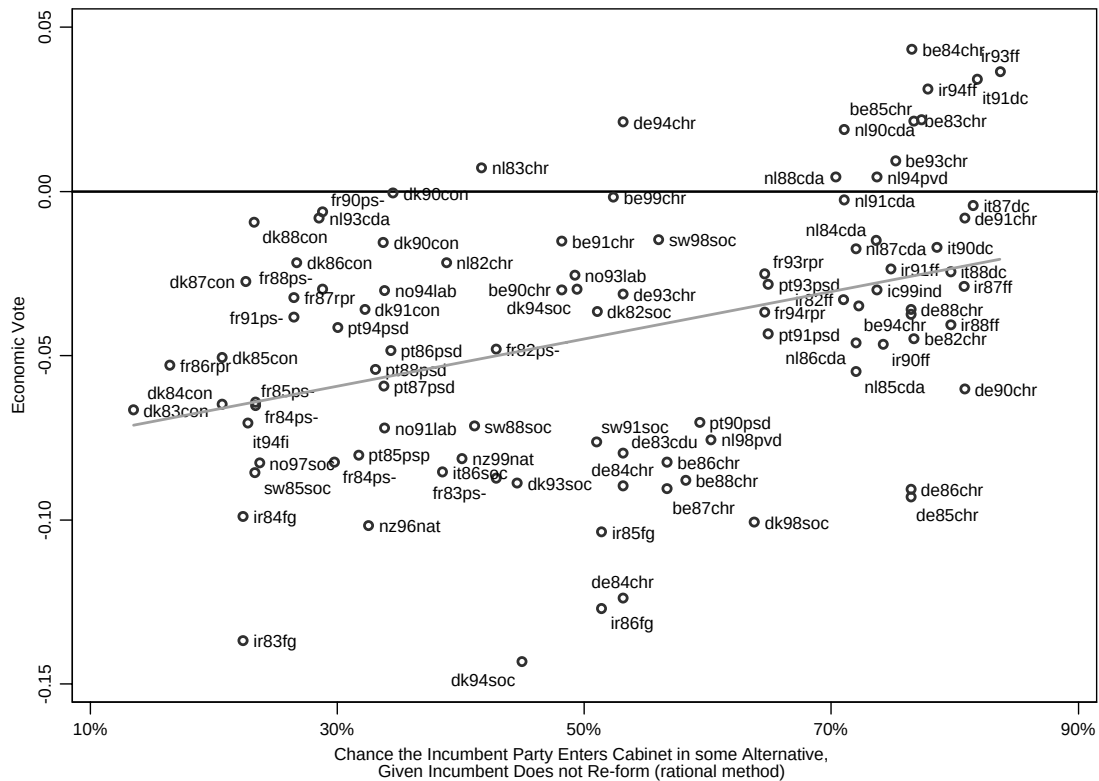


Larger negative numbers indicate a larger economic vote for the incumbent party. The variable on the x-axis applies to the incumbent prime ministerial party that is in office with the graphed partner party. The regression (with appropriate adjustments - see note on Table 10.3) corresponding to this graph is estimates the slope of the line to be $-.05$ with $t\text{-ratio} = -4.32$. Other specifications using shares (specified exactly as in Table 10.3) give even stronger results.

Figure 10.8. The Economic Vote of Cabinet Partners and the Historical Competitiveness of their Prime Ministers.

As we did for opposition parties, we can also explore this hypothesis when voters form their beliefs rationally. Likewise, as in that analysis, we build our measure of the voter's rational beliefs by first using Martin and Stevenson's empirical model to estimate the probability that each potential cabinet forms (as a function of its observable characteristics). Next, for each incumbent party, we sum these estimates over all the cabinets in which the party participates. Finally, we normalize this probability by the total probability that a cabinet other than the incumbent will form. As before, the result is a measure of the probability that any particular incumbent party will continue in some alternative cabinet, given the incumbent cabinet does not form (only this time the probabilities are from our empirical model instead of historical frequencies).

Figure 10.9 graphs the relationship between this measure of the chance of entering the new government and the economic vote of prime ministerial parties in coalitional systems. Clearly, relationship is in the expected direction and, as Table 10.6 shows, is also statistically significant.



Larger negative numbers indicate a larger economic vote for the incumbent party. We calculated the variable on the x-axis by first estimating the probability that each alternative cabinet forms (using the rational method described in the text); then summing these probabilities over all non-incumbent cabinets in which a given incumbent party is a member; and, finally, normalizing this number by the total probability that some cabinet other than the incumbent would form.

Figure 10.9. Extent to Which Incumbent Prime Ministers are members of Competitive Alternative Coalitions and the Economic Vote in Coalitional Systems (Rational Method)

As with our historical measure, the cases dominating the right side of the graph are the perennial prime ministers in Belgium, Italy, and the Netherlands. It is encouraging that the rational measure, which is built quite differently from the historical measure and, indeed, uses no information about historical office holding, agrees with the historical measure that these parties

are very likely to be in whatever new cabinet forms.²⁸⁰ Even more encouraging is the fact that the message of the graph, as well the estimates in Table 10.6, mirrors that of our earlier analysis: the more that the incumbent prime minister is a member of competitive alternative cabinets, the smaller the size (and share) of it economic vote.

Table 10.6. Predicted Cabinet Contention and the Economic Vote of Prime Ministers

<i>Dep. Var.</i> <i>Ind. Vars.</i>	<i>Econ Vote</i>		<i>Econ Vote Share</i>		<i>Econ Vote Share*</i>	
Probability of entering government in some alternative other than incumbent coalition, given incumbent does not re-form (rational method)	.07 (2.83)	.07 (2.99)	-.50 (-1.54)	-.42 (-2.57)	-2.72 (-1.33)	-3.27 (-3.09)
Percent of Cabinet Ministries held by Party	-- --	-.04 (-2.00)	-- --	.82 (4.98)	-- --	6.33 (5.66)
Constant	-.08 (-5.22)	-.05 (-2.45)	.93 (5.56)	.38 (1.99)	2.49 (1.85)	1.12 (-0.96)
N	97	97	95	95	95	95
R ²	.14	.18	.08	.35	--	--

Numbers in cells are coefficients and t-ratios. All models except those indicated by the * were estimated using OLS regression with standard errors robust to heteroscedasticity and non-independence between observations of the same party in different surveys. Those marked with an * use GLM to account for the fact that the share of the economic vote is a proportion (as recommended in Papke and Wooldridge, 1996).

As in the earlier analyses, Table 10.6 provides a number of different statistical models of the general relationship graphed in Figure 10.9. Again, these analyses provide support for our hypothesis that, in coalitional systems, economic voting should be smaller for incumbent parties that are almost certain to be a member of any new cabinet that forms. Further, the relationship persist when we control for the different shares of cabinet responsibility that different prime ministers hold, reassuring us that both effects are important.

The Re-selection of the Incumbent Government and the Overall Economic Vote

One of the implications of our theory was that overall economic voting should be muted when the incumbent government is not in contention to be reselected to govern. Contention has two sides – an incumbent could fail to be in contention either because it is not likely to be reselected or because it is sure to be reselected. We expect, then, that the relationship between voter beliefs about the reselection chances of the incumbent and economic voting will be

²⁸⁰ The rational and historical measures are correlated at .42 for coalitional prime ministers. The un-normalized measures are correlated at .53 for these same cases.

quadratic – reaching a minimum when the voter is either sure the incumbent government will be reselected or sure it will not be.

There are two caveats to this hypothesis that are important to account for in our empirical analysis. First, we should recall that for any given incumbent party, the competitiveness of the government as a whole is only one influence on its economic vote. In the last section, we also saw that the extent to which the party is a member of alternative coalitions is also important. Thus, in the analysis in this section, which concentrates on the relationship between the competitiveness of the incumbent cabinet as a whole and the economic vote of the prime minister (our preferred measure of overall economic voting), we need to control for the extent to which the prime-ministerial party is a member of competitive alternative cabinets (by including measures from the last section as control variables in our regressions).

The second caveat that we need to consider is that while several versions of the theory presented in chapter 8 implied that there should be no economic voting if the incumbent cabinet was not in contention to re-form, this exclusive focus on the incumbent cabinet was relaxed when we (1) generalized the theory to allow for opposition parties to have some policy making responsibility (so that we had to think in terms of a “status-quo distribution of responsibility” rather than an “incumbent cabinet”), and (2) allowed voters in our theory to infer non-zero levels of competence for distributions of responsibility other than the status-quo distribution. Thus in our most general model, we replaced the implication that the incumbent (or status-quo) distribution of responsibility must be in contention for economic voting to occur with a continuous version of that hypothesis. That is, economic voting will be depressed if the competitive distributions of responsibility are either very dissimilar to the status quo distribution or very similar to it.²⁸¹ Substantively, this generalization is important because it allows us to think about, for example, cases in which the exact incumbent cabinet is not competitive but similar cabinets are (e.g., a small party that was an incumbent is sure not to return but all the other parties are likely to). From the economic voter’s point of view, it may well make sense to cast an economic vote that helps this “almost incumbent” cabinet either remain in office or be removed. Further, since the theory says that these similar cabinets need to be competitive (i.e., it can neither

²⁸¹ Recall that the logic for the first part of this hypothesis (i.e., the economic vote should be depressed if the alternatives similar to the status-quo distribution are not at all competitive) comes from the fact that if no parties in an alternative distribution of responsibility held any responsibility in the status-quo distribution, then the voter has no information about the competence of any party that matters to the new distribution. Likewise, the logic for the second part of the hypothesis (i.e., if all the alternative distributions of responsibility that are in contention are very similar to the status quo then the economic vote will be depressed) comes from the fact that if an alternative just reproduces the status quo distribution of responsibility, the voter has no real choice and so has no incentive to cast an economic vote.

be sure that a cabinet similar to the incumbent will form or that none will), we expect a quadratic relationship between the competitiveness of cabinets similar to the incumbent and the overall economic vote.

In what follows, we develop a measure of the overall competitiveness of cabinets similar to (or exactly the same as) the incumbent cabinet. One can think of this measure as the “familiarity” of the incumbent cabinet in that it attempts to account not just for how often the exact previous cabinet has been in power in the past, but also how often similar cabinets have been in office. As with our other historically based measures, this measure begins with the assumption that voters believe those governments that have tended to be selected in the past will also be selected in the future. Clearly, such voters must give all incumbent governments some chance of reselection, since by virtue of being incumbent the government has been selected previously. However, there are clearly differences across different incumbent governments in the extent to which they have a history of selection and reselection. Our measurement assumption is that voters consider it more likely that familiar cabinets will reform than is the case with unfamiliar cabinets. Thus, the expectation from our theory is that economic voting will be depressed when the cabinet is either very familiar or very unfamiliar.

Our measure of familiarity not only accounts for the history of the specific incumbent cabinet but also how different it is from other familiar combinations (perhaps the only difference is the addition of a small and inconsequential party to an otherwise familiar governing combination). For example, we want to be able to distinguish the familiarity of a government like the Dutch incumbent cabinet in 1981 (the CDA, PvdA, and D66), in which the smallish D66 joined the familiar Christian-Socialist coalition, from that of the PvdA, D66, Liberal coalition of 1994 which was, most observers would agree, much less familiar to voters. Indeed, the latter coalition represented a combination of major political forces (the liberals and socialists) that had never previously coalesced. Our measure of the incumbent government’s “familiarity” consists of scoring each government according to how often its members have previously governed and then taking the average of these experiences.²⁸² We then weight the average by the share of legislative seats each party brings to the government. This down-weights the impact of adding small parties to coalitions in which the larger parties are familiar incumbents. The measure thus corrects the problem of counting all unique governing formulas as equally unfamiliar (a score of 0 would indicate the most unfamiliar coalition – one in which none of the members had ever

²⁸² We could calculate how often each government had served rather than its members. However, by building up our measure from the experience of the members rather than the government as a whole, we can more easily discount the “uniqueness” of coalitions that simply add small parties to familiar alternatives.

served in cabinet). However, the measure is still lacking in at least one important respect. The problem is illustrated by the second example we gave above. Adopting this measurement strategy would assign the Dutch incumbent coalition in 1994 (between the PvdA, Liberals, and D66) a familiarity score of .58, which, compared to similar scores for other alternatives, seems rather high. This relatively high score results because the Liberals and the Socialists had served, by that time, quite often in government - although always as junior members of a Christian led coalition. This change in status is not well reflected in our measure, so the government comes out as more familiar than, we think, most voters were likely to find it.

We adjust our measure to address this problem by calculating not just the experience each incumbent party has in government, but also the experience they have in the specific *role* they are playing in the government, where we distinguish only between being the prime minister and being a non-prime ministerial cabinet partner. With this, instead of counting the PvdA in 1996 as having served in government 34% of the months since 1960, it is scored as having served 0% of the months since 1960 as PM. This changes the overall familiarity score to .08 (we continue to weight by the percent of seats the party contributes to the cabinet). Finally, we discount service in governments so that more recent stints in government are weighted more in voter's beliefs about reformation than government participation that occurred farther in the past.²⁸³

We find that this measure captures most of the variation in familiarity observers would expect. Figure 10.10 maps the variation in familiarity of the incumbent government for all the countries in our sample in which multiple parties usually compete. The big changes in the graphs correspond nicely to what we would expect and across the graphs the differences in familiarity scores also reflect, in our view, a reasonable understanding of the differences across countries in how familiar various incumbent cabinets were likely to have been to voters.

²⁸³ We use an exponential weighting function, δ^m , where m is the number of months between the current month (for which a score is being calculated) and the month in the past that is under consideration (ultimately, we sum over all months since 1960). We choose $\delta=.99$, which means that service five years in the past is discounted by about one half and ten years in the past by about a third. The specific details of this calculation are available at www.nuffield.ox.ac.uk/economicvoting.

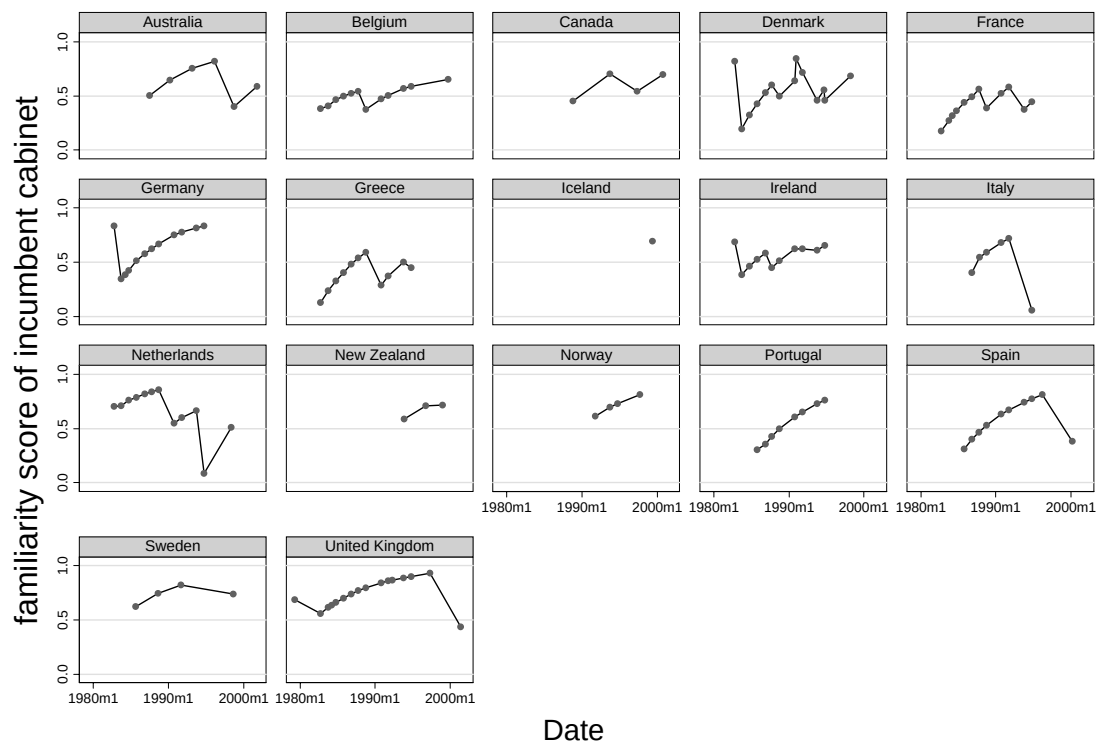


Figure 10.10: A Familiarity Index for Incumbent Cabinets

The least familiar cases are what we would expect: the Dutch case discussed above, the first PASOK government in Greece, and the first Italian cabinet to emerge from the new party system. Likewise the most familiar cases are found after long-periods of rule by the same cabinet – i.e., the German Christian-Liberal coalition, the decade of rule by the Spanish PSOE, and the British run of Conservative governments. The big changes in the graphs also identify the important transitions in the pattern of office holding, during which voters who form their beliefs adaptively would have had considerable uncertainty about the future. For example, in Denmark, the large drop in the familiarity of the incumbent cabinet in the early 1980's reflects the emergence of the rightist bourgeoisie coalitions as a viable alternative to the Social Democrats.

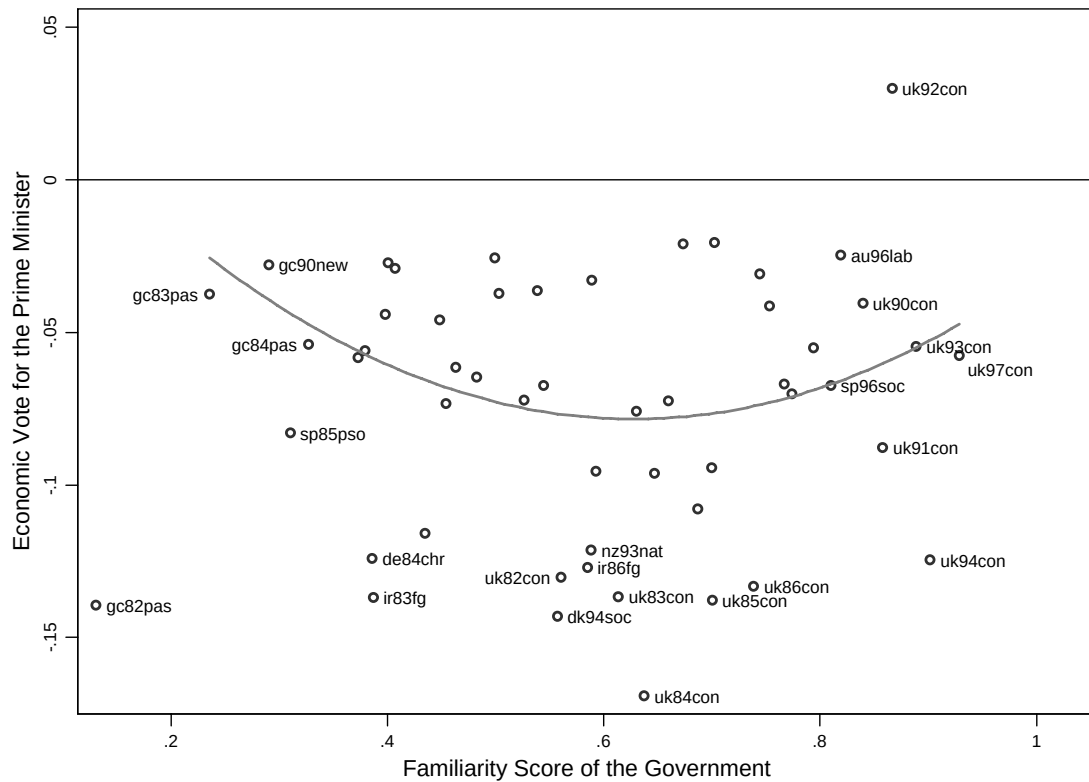
With this measure in hand, we can now graph the familiarity score against our measures of economic voting in each of our cases.²⁸⁴ We use the economic vote of the prime minister as our

²⁸⁴ Of course, these graphs are illustrative since we have emphasized that it is important in examining the competitiveness of the incumbent cabinet to control for the extent to which the prime ministerial party is a member of contending alternative cabinets. We include this control in the empirical models corresponding to the graphs, but since these models tell, more or less, the same story as the graphs, it is worthwhile (we think) to present the pictures.

measure of the economic vote.²⁸⁵ Since the interpretation of the familiarity score may be qualitatively different in coalitional and non-coalitional systems, we graph these two cases separately. Specifically, in coalitional systems, changes in the familiarity of the incumbent are often induced by a new coalitional formula coming to power, but this is not the case in non-coalitional systems in which the same pairs of parties compete election after election for office. Thus in the later case, our measure of familiarity reflects the extent to which in recent years the control of government has, or has not, been volatile.

Figure 10.11 presents the graph of economic vote against familiarity for non-coalition cases. On the left of the graph are incumbents that follow a long period in which another party, now in opposition, was in power; the middle of the graph are cases in which power is switching more regularly between parties; and the right side of the graph are cases in which the incumbent has dominated the government for some time. Clearly, measuring familiarity in this way, we get a strong quadratic relationship between familiarity and the economic vote as we expect (with the now familiar exception of the Greek 1982 case). Further, the multiple conjectural nature of our hypotheses implies, in this case, that economic voting will be low at high and low values of our familiarity score and *may* be large in the middle. Thus, the fact that the bowl shape in the figure is “filled in” is completely consistent with the theory, though it tends to depress the significance of the quadratic approximation.

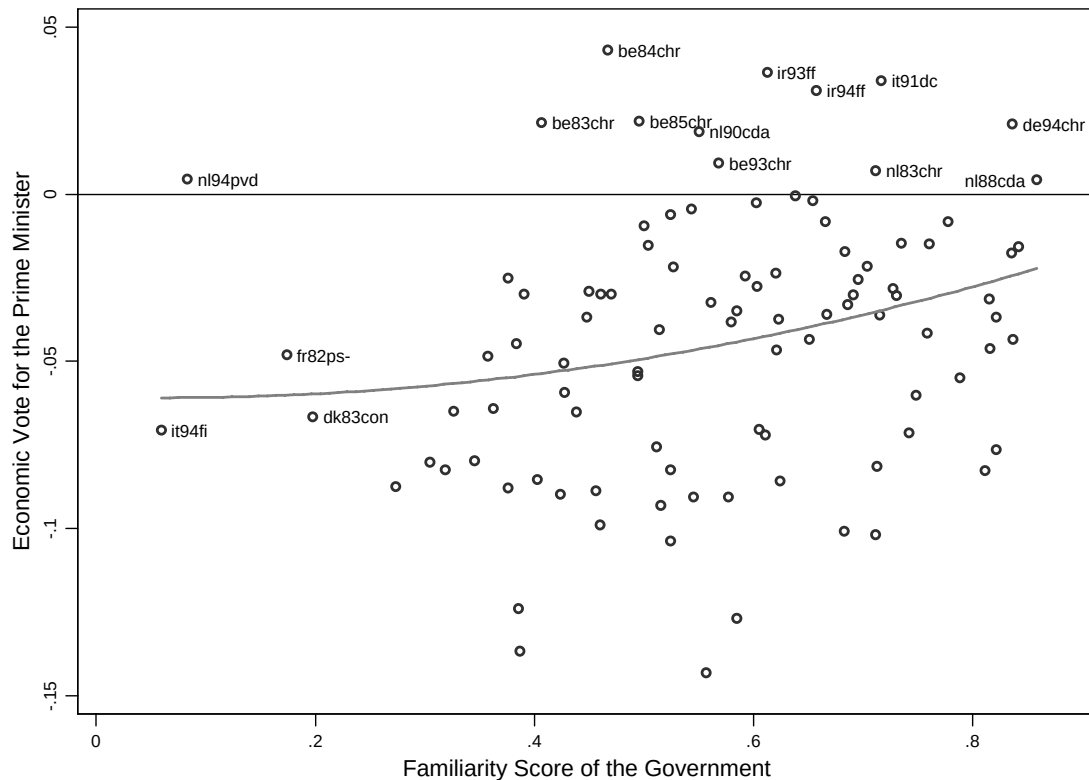
²⁸⁵ Other measures of the overall size of economic voting in the election (such as the mean economic vote of the government parties - with wrong signed economic vote set to zero) produce similar results.



Larger negative numbers indicate a larger economic vote for the incumbent party. Fitted Line excludes the Greek 1982 case.

Figure 10.11. Familiarity of the Incumbent Government and the Economic Vote in Non-Coalitional Systems

In coalitional cases, we can interpret our familiarity score in roughly the same way as we did in the case of non-coalitional cases, but now some of the differences in scores reflect changes in coalition partners rather than how long the incumbent has been in office. On the far left of Figure 10.12 are some of the more unique governments in the data - the Dutch PvdA-Lib-D66 coalition discussed above, the bourgeoisie coalition that emerged in the early 80's in Denmark, the first Italian cabinet lead by Forza Italia, and the return of Socialist coalitions to power in France in the early 1980s. With the partial exception of the Forza Italia government, these (admittedly few) cases of extremely unfamiliar cabinets fit the predicted quadratic relationship. Further, on the other side of the graph, the systems with the most familiar governments in our sample (the German and Dutch Christian led coalitions in the early 1990s) have similarly depressed economic votes.



Larger negative numbers indicate a larger economic vote for the incumbent party.

Figure 10.12. Familiarity of the Incumbent Government and the Economic Vote in Coalitional Systems

This evidence suggests that at low and high levels of familiarity, economic voting is low and that at moderately high levels of familiarity economic voting *may* be high, but this is not guaranteed. However, because of the relative scarcity of cases with familiarity in the range of .1 to .3, the plotted quadratic approximation does not “bend” much on the left side of the graph (though, with the exception of the 1994 Italian case, the outline of the “filled-in bowl” shape predicted from the theory is nevertheless apparent). Table 10.7 reports the estimates corresponding to the graphs along with specifications controlling for the extent to which the prime minister is a member of contending alternative cabinets. In the case of coalitional systems, we present not only the quadratic specification, but also a linear model fit to all the cases with familiarity scores greater than .2 (since we would not expect a positive linear relationship to fit the few cases with very low familiarity).

These regression results provide strong support for the hypothesis in non-coalitional systems. The coefficients of the quadratic approximation are statistically different from zero in both equations, despite the “filled-in bowl” shape apparent in Figure 10.11 (and implied by the

theory). In the case of coalitional systems, the evidence supports part of the hypothesis (i.e., very familiar cabinets in coalitional systems have depressed economic votes compared to those less familiar); but, since there are so few cases of unfamiliar cabinets, we can not be very confident about our evidence for the left side of the predicted quadratic relationship.

One other thing to notice about these results is the strong positive relationship between the extent to which the prime minister is a member of contending alternative cabinets and the economic vote. This result assures us that the results in the last section that focused on this variable are robust to the inclusion to this measure of the competitiveness of the incumbent cabinet.

Table 10.7
Familiarity of the Incumbent Government and Economic Voting

	<i>Non-Coalitional Systems</i>		<i>Coalitional Systems</i>		
Familiarity Score	-.38 (-1.97)	-.41 (-1.98)	-.007 (-.06)	.08 (2.59)	.06 (2.25)
Familiarity Score squared	.30 (1.92)	.33 (2.01)	.061 (0.49)	-- --	-- --
Probability of entering government in some alternative other than incumbent coalition, given incumbent does not re-form (historical method)	-- --	.05 (1.60)	-- --	-- --	.04 (5.20)
Constant	.04 (0.76)	.05 (0.87)	-.06 -2.09	-.09 (-4.27)	-.10 (-5.27)
N	53	53	98	94	94
R ²	0.04	.06	0.06	.09	.26

Dependent Variable is the *economic vote of the prime minister*. Numbers in cells are coefficients and t-ratios. All models were estimated using OLS regression with standard errors robust to heteroscedasticity and non-independence between observations of the same party in different surveys. Similar results were obtained from models using the share of the economic vote (for coalitional systems).

A Summary Measure of the Pattern of Contention

Similar to the earlier empirical chapters on the distribution of responsibility and the quality of the competence signal, we end this chapter by developing a summary measure of overall cabinet contention that we can use (in interaction with economic perceptions) in a one-stage, individual level model of economic voting. Any such measure needs to (1) capture those aspects of the competition for office that we have already shown are most relevant to economic voting, and (2) apply to both coalitional and non-coalitional systems. One measure that we think meets these criteria is the effective number of contenders for the office of chief executive. While the measure does not capture all of the various aspects of competition that we have argued should matter to economic voting, it has two advantages as a summary measure. First, its focus on the competition for chief executive is consistent with the other analyses in this chapter which show that differences in the distribution of contention for the chief executive tend to have a greater impact on economic voting than differences in the distribution of contention for other roles in government. Thus, the type of contention that is most likely to be relevant to economic voting is contention among parties for the job of chief executive. Second, our theory clearly implies that economic voting in an election will be greater when incumbent parties (and those with greater responsibility) are competitive to return to power in some government (i.e., there is uncertainty as to whether they will or will not enter a new governing coalition). Since chief executive parties normally hold a large share of administrative responsibility, the extent of economic voting in the whole election should be closely tied to the extent to which the chief executive's party is competitive (as opposed to some other party).

The effective number of contenders for chief executive is calculated from our data on the historical record of service in that role. Specifically, for each month during our sample period, we calculate the percentage of months since 1960 that each party has served as chief executive and normalize this number so that it sums to one across all the parties competing in a country in a given month. This number then is the share of the historical record of chief executive office holding that is held by each party in each country calculated for each month in our sample period. Thus, to get the effective number of parties competing for the office of chief executive, we simply apply the standard measure of the effective number of parties to these shares (i.e., we take the reciprocal of the sum of the squared shares). Figure 10.13 plots this measure for each country in our sample from 1980 to 2002. The measure captures the important changes in the pattern of contention that we would expect to see. For example, the most dramatic example of change is in

Denmark, where the sudden viability of rightist bourgeoisie coalitions led to the emergence (over the 1980's and 1990's) of the Conservative Party as a real prime ministerial alternative to the (previously) dominant Social Democrats. Similar, but less dramatic, cases of increasing prime ministerial competition in this period occur in Sweden (essentially the same story as in Denmark), and France (the emergence of the Socialists Party as a viable head of government for the first time since World War II).²⁸⁶ In contrast, our few cases of Norwegian data show a decline in competition that reflects the Norwegian Labour Party's dominance of the prime ministry between 1986 and 1997.

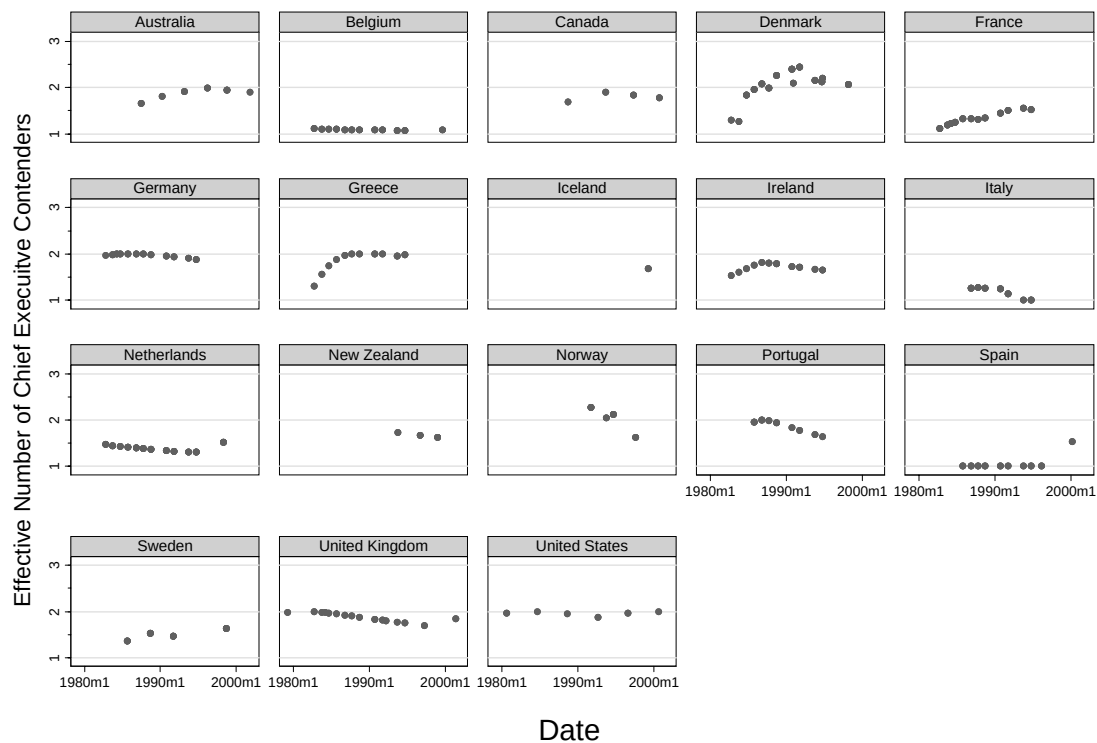


Figure 10.13. Effective Number of Chief Executive Contenders

Figure 10.14 graphs our measure of the effective number of chief executive contenders against the economic vote of the incumbent chief executive party. Unsurprisingly, given our earlier results, the estimated relationship is consistent with our expectations. As competition for

²⁸⁶ The apparent large increase in competition in Greece results from the fact that their history of democratic competition did not begin until the mid 70's, so that, though there was change in the prime ministry relatively early this does not register in our measure until the mid 80's when New Democracy and PASOK experienced their first alternation in power.

the top job in government increase (which in these countries usually means moving from one toward two contenders for the job), we see a larger economic vote.

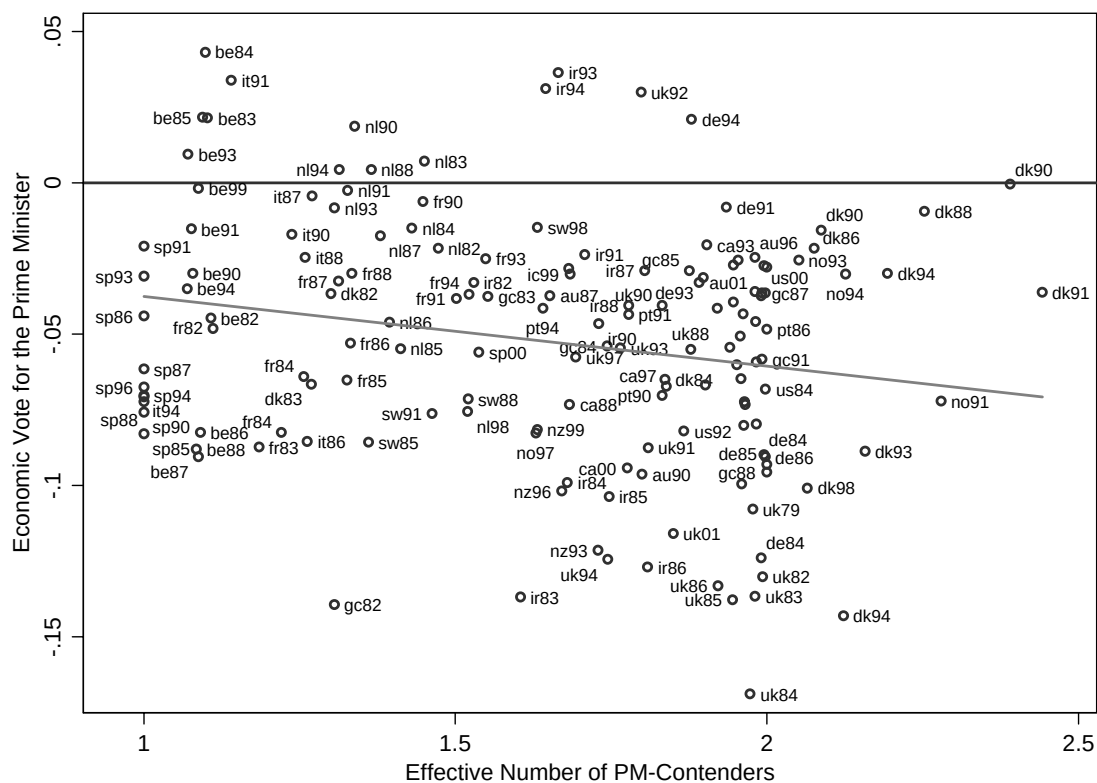


Figure 10.14. Effective Number of Chief Executive Contenders and the Economic Vote of the Chief Executive

The results of the estimation are provided in Table 10.8. These estimates clearly support the hypothesis, since all the coefficients are statistically significant and in the expected direction. In particular, the *PM Contenders* interaction with positive economic evaluations is positive and significant and its interaction with negative evaluations is negative and significant. Hence an increasing effective number of *PM Contenders* results in a greater difference in incumbent vote probabilities between those with a positive versus negative evaluation of economic performance. These pooled, multi-level results tells exactly the same story as the second-stage estimates from the two-stage method – increasing the number of parties that are contenders for the chief executive position increases the *economic vote of the chief executive*.

Table 10.8. Multi-Level Model of Effective Number of PM Contenders and Economic Voting

Left-Right Self Placement (dev. from mean)	0.49 (0.01)
Left-Right Self Placement X Leftist PM (Leftist PM is an indicator variable)	-0.75 (0.01)
Voter Perceives Economy got Better (indicator variable)	0.45 (0.03)
Voter Perceives Economy got Worse (Indicator variable)	-0.57 (0.03)
Effective Number of Prime Minister Contenders	-.06 (.08)
Voter Perceives Economy got Better X Effective Number of Prime Minister Contenders	.34 (.08)
Voter Perceives Economy got Worse X Effective Number of Prime Minister Contenders	-.21 (.09)
Size of chief executive's Party	5.30 (0.24)
Constant	-2.42 (0.07)
$\sigma^2_{\omega 1}$	0.09
$\sigma^2_{\omega 2}$	0.07
$\sigma^2_{\omega 3}$	0.10
$\sigma_{\omega 1, \omega 2}$	-0.02
$\sigma_{\omega 1, \omega 3}$	-0.04
$\sigma_{\omega 2, \omega 3}$	-0.04
N	201,876

Standard Errors listed below logit coefficients. All coefficients are statistically significant at $p < .1$ except the shaded cells. In the case of the US, the dependent variable is vote for the party of the president. Congressional election studies are excluded.

Summary

Our theory of rational economic voting stipulates that voters must believe that parties are “in contention” for significant governing responsibility in order for the economy to affect their vote for that party. This chapter makes a strong empirical case for this theoretical claim. First we provide convincing evidence that voters have well-informed beliefs about the pattern of contention that correspond quite closely to both the predictions of empirical models of cabinet formation as well as the historical record of cabinet participation. Using these two measures of voter beliefs about contention we were able to test three generally hypotheses from our theory.

First, we showed that economic voting has its largest impact on those opposition parties that voters believe are most likely to replace the incumbents. This was the case when we measured voter beliefs about contention using the predictions of empirical models of cabinet formation and employing the historical record of cabinet participation.

The second hypothesis derived from our theory is that economic voting should be weaker for incumbent parties that are members of most of the viable alternative cabinets (i.e. they are perennial cabinet members). The evidence is quite convincing for prime ministerial parties: Perennial prime ministers have a smaller economic vote than prime ministerial parties who are in competition for the position. This result is especially revealing since while consistent with our theory, it is opposite what one would expect from the more standard “clarity of responsibility” argument found in the literature (since perennial parties should be easy targets for voters seeking to attribute responsibility).

Finally, overall economic voting should be muted when the incumbent cabinet (or alternatives similar to that cabinet) is not in contention to be reselected to govern. While the results were not as compelling as those for the other two hypotheses they certainly were directionally correct suggesting that when the incumbent government as a whole (or a close alternative) is either sure to be, or not to be, re-selected, economic voting is depressed.

Overall, this chapter makes it clear that an important piece of the explanation for contextual variation in economic voting must be the strategic environment in which parties compete for administrative responsibility. When voters are faced with a situation in which parties are continually returned to government, election after election, economic voting will be depressed. Similarly, even if there is significant competition amongst opposition parties, if incumbent parties are not also competitive then the economic vote will be depressed.

Part IV
Conclusion and Summary

Chapter 11

Conclusion

This book was initially motivated by a simple concern with understanding contextual variation in the economic vote. Accordingly, we set out first to describe the extent of contextual variation in the economic vote across countries, over time, and across parties. In order to explain this contextual variation we develop a theoretical explanation rooted in the long-tradition of rational choice explanations of economic voting. These explanations provide a set of plausible (and testable) empirical hypotheses about the kinds of political and economic contexts that are likely to condition the economic vote. Finally, we leverage the evidence from our map of economic voting, along with measures of variation in political and economic context, to test these hypotheses. In this concluding chapter, we first summarize our results and evaluate the extent to which we have fulfilled these goals and ask what work remains to be done. Further, we discuss how the lessons we have learned about economic voting in this project might speak to researchers interested in comparative political behavior more generally.

Summary and Evaluation

Variation in the Economic Vote

Part I provides the most comprehensive and comparable description of contextual variation in individual economic voting yet available to political scientists. Our estimates of the economic vote are consistent with the limited comparative evidence already available in the literature. We also produce quite reasonable estimates of economic voting for those parties, countries, and time periods in which previous comparative work has not been available. Moreover, our empirical analyses make it clear that that our estimates of economic voting do not depend, in any significant way, on the particular measurement decisions we have made, or on our particular conceptualization of economic voting.²⁸⁷

Substantively, our map of economic voting confirms, clarifies, and extends the common wisdom about economic voting in western democracies. Economic voting is pervasive – indeed, there is almost no variation in its broad *nature* across different contexts – that is, it is both overwhelmingly incumbency-oriented (i.e., a poor economy hurts incumbents and helps opposition parties) and more important to the party of the chief executive than to other incumbents. At the same time, however, there is significant variation in the *magnitude* of the

²⁸⁷ That is, it does not depend on our measurement and conceptual choices from the set of plausible alternatives.

economic vote across political and economic contexts. Thus, while worsening economic perceptions almost always lowers support for a chief executive party, the extent of this effect may be larger in some contexts and smaller in others.

Contextual Theory of the Economic Vote

A second important objective of this book is to explain this contextual variation. Accordingly, we develop a theory that identifies three broad sources of contextual influences on the magnitude of the economic vote: the extent of political (or electoral) control of the economy; the concentration and distribution of policymaking responsibility over parties; and the pattern of contention among the parties for future policymaking responsibility. Each of these effects is derived from extensions of our basic competency model of the economic vote that identify features of context that inform voters about the size of the competency signal and about strategic voting considerations.

Our theory starts with the observation that in order to understand how context might shape vote choice one must necessarily begin with an individual-level model of vote choice and must ensure that the assumptions of that model are general enough to apply across a wide variety of political and economic contexts.²⁸⁸ Our choice was to build our individual level model of economic voting on the rich theoretical tradition that has assumed economic voting results from the actions of instrumentally rational voters – a literature that has clearly motivated much of the previous theoretical work on the economic vote. Within the family of rational models of the economic voter, however, we rely on a selection rationale for rational economic voting rather than on the more usual sanctioning perspective. This choice was motivated by our conclusion that the selection approach was a more promising starting point for studying contextual effect than was the sanction approach. In part, we are persuaded by theoretical work, such as that by Fearon (1999), which demonstrates that selection incentives overwhelm sanctioning incentives when models of electoral selection and electoral sanction are combined formally. Specifically, when voters perceive any relevant differences between candidates in an election, it is rational for them to vote over those differences rather than to try to use their vote to discipline incumbents.²⁸⁹ As we discuss below, however, our belief

²⁸⁸ This is not a trivial exercise (in our own model this required generalizing the Alesina and Rosenthal model from two-party, winner-take-all systems to multiparty, coalitional systems) and it is one that does not receive as much attention as it should in the typical comparative study of voting behavior.

²⁸⁹ Recent work has argued convincingly that electoral systems must logically produce distinctions between candidates and that these distinctions are as important a part of electoral choice as any desire to ensure accountability. For example, Manin (1997, 141-2) emphasizes the inevitability of differences between politicians even when there are strong incentives for them to adopt common ideological positions.

need not hinder others from exploring context starting from the sanction perspective. Our work here, however, should instruct any such effort by demonstrating that theoretical progress in this area is best achieved when one both clearly articulates the individual model being used and explicitly generalizes the logic of that model to apply to a wide variety of electoral contexts. Thus far, the comparative literature on economic voting that has taken a sanctioning perspective, has failed in one or both of these areas.

Our theoretical development takes as a starting point Alesina and Rosenthal's (1995) model of rational retrospective voting. Like them we assume that voters are motivated by the desire to select the most competent economic managers and voters use information about economic outcomes to assess the future economic competencies of competing candidates. The only economic outcomes that inform the vote choice in our model are those that result in unexpected shocks to the economy (either positive or negative). Voters observe shocks to the macro-economy but cannot observe the mix of exogenous and competence components that make up these shocks. Voters do, however, know the variances of the distributions of these different kinds of shocks and so are able to solve a well-defined signal extraction problem that produces a competence signal.

This much of the story is, of course, familiar to those readers conversant with theoretical developments in the literature on political business cycles. However, we suggest a re-interpretation of the critical variance terms that make up the competence signal that drives the model. Specifically, instead of thinking about a single exogenous or competence shock to the economy, we think of these shocks as being the sum of many smaller shocks – each associated with some economically consequential decision. As a result, we replace the usual variances of the distribution of competence and exogenous shocks with expressions for the sum of many much smaller shocks. Assuming these smaller shocks have equal variances, the critical factor that determines the size of the overall variance terms governing competence and exogenous shocks to the economy are the number of small shocks that contribute to each sum. Further, since only the decisions of electorally accountable decisionmakers can contribute to “competence” shocks, the overall size of the variance terms in the competence signal (and thus its value) is determined by the ratio of electorally accountable economic decisions to non-electorally economic decisions. Thus, the task of identifying how political and economic institution may impact economic voting (for this part of the model) reduces to connecting these institutions to voters' beliefs about the relative importance of electorally and non-electorally accountable decisions on economic performance.

We then generalize the logic of this model to apply to contexts in which multiple parties compete, coalition cabinets may form, and opposition parties may share policy making responsibility with incumbents. This generalization leads to the identification of the other two general contexts mentioned above – the concentration and distribution of policymaking

responsibility over parties; and the pattern of contention among the parties for future policymaking responsibility. The first comes from generalizing Alesina and Rosenthal's model of rational retrospective economic voting to allow for multi-party policy making. This leads to a solution to the voter's signal extraction problem that is different from that described above. This solution, which we call the responsibility augmented competency signal, is maximized when responsibility is maximally concentrated in one party and minimized when responsibility is shared equally among them all. In addition, when we further generalize the model to allow for multiparty competition over the distribution of future policymaking authority, we show that not only will economic voting generally be more important when policy making responsibility is more concentrated, but also that the economic vote (whatever its overall size) will be more important to the support of specific parties when they have a larger share of policy making responsibility.

As we emphasized above, the predictions of this part of the model are the same as those produced by the clarity of responsibility argument.²⁹⁰ Since our theory not only makes these predictions but also provides predictions about areas of context for which the clarity argument is silent, there is an opportunity for us to evaluate the added empirical content that our theory provides over the clarity approach. And, as described below, there is at least one hypothesis that stems from our theory that directly contradicts the implications of the clarity approach so provides the opportunity for a kind of critical test between them.

Given that voters in our model are instrumentally rational, our generalization of the model to multi-party contexts requires that we consider the impact of strategic voting incentives on economic voting. These considerations lead us to identify the pattern of contention for policy making responsibility as the final general context that should impact economic voting. More specifically, we show that economic voting should have its largest impact on those opposition parties that voters believe are most likely to replace the incumbents – because they are members of the most competitive alternative cabinets. Further, overall economic voting should be muted when the incumbent cabinet (or alternatives similar to that cabinet) is not in contention to be reselected to govern. As a result, when voters are faced with a situation in which parties are continually returned to government, election after election, economic voting will be depressed. Or in the rare event that an incumbent cabinet is certain to lose an election, economic voting will also be muted. Finally, given

²⁹⁰ Once again it is worth emphasizing that the mechanism that produces these implications in our model is very different from any version of the clarity argument we have seen. While in those arguments the hurdle to economic voting created by shared responsibility stems from the voter's ignorance of who is responsible for policy making. In our model, however, voters know this information perfectly. Instead, our result comes from the fact that shared responsibility degrades the signal that economic performance can provide about the competence of incumbents.

some level of incumbent contention, the vote of any particular incumbent party's share of the economic vote will be muted relative to its partners if it is a member of many and more competitive alternative cabinets.

This last result is especially revealing since, while consistent with our theory, in the case of "perennial governing parties" it contradicts the clarity of responsibility argument. Perennial governing parties are always in government because they are members of all the viable governing coalitions (e.g., the Belgian, Dutch, and Italian Christian parties) and as such, our theory says they should get a smaller share of the economic vote than their partners. This, of course, is exactly opposite to what one would expect from the more standard clarity of responsibility argument found in the literature (since perennial parties – particularly large ones – should be easy targets for voters seeking to attribute responsibility).

Hypotheses

The third goal of our project is to use our explanation of economic voting to provide a set of testable empirical hypotheses about the kinds of political and economic contexts that are likely to condition the economic vote. Of course, this process begins with our elaboration of the direct implications of the theoretical model that we have just reviewed. However, this is only the first step in developing our empirical hypotheses about how specific kinds of political and economic institutions should impact economic voting. Specifically, the direct implications of the theory always have the following general form: "economic perceptions will be more important to the support of voters who believe X than they will for voters who believe Y". For example, one implication of our theory is that economic perceptions will be more important to the support of voters who believe the distribution of policy making responsibility is concentrated than those who do not. Likewise, our theory implies that economic perceptions will be more important to the support of voters who believe the ratio of electorally accountable to non-electorally accountable decisionmakers is large than those who do believe this ratio is smaller.

To move from these direct implications about the beliefs of individuals on the economic vote to empirical implications about the impact of political and economic institutions on the economic vote for parties, we must supplement the implications of the theory with assumptions about how specific political and economic institutions shape the average voter's beliefs about such things as the distribution of policy-making responsibility among parties or the ratio of electorally and non-electorally accountable decisionmakers. So for example, we assume that voters believe that the distribution of policymaking responsibility is more concentrated in single party cabinets than in coalition cabinets and so test the hypotheses that the economic vote of single party cabinets will be larger than that of coalition cabinets. Likewise, we assume that

voters believe that the ratio of electorally accountable to non-electorally accountable decisionmakers is smaller in open economies than it is in more closed economies and so hypothesize that economic openness will depress the economic vote. Other such auxiliary assumptions are found in each of the main empirical chapters in Parts II and III and, when combined with the direct implications of the theory, result in the various empirical hypotheses that we test.

Evidence

Each of our empirical hypotheses relates a particular economic or political variable to the magnitude and/or share of the economic vote for a party or parties; and, the final goal of our book is to use our data on the magnitude of economic voting to test them. First, in Chapter 7 we explore a set of hypotheses about how political and economic institutions impact economic voting by changing the ratio of electorally accountable and non-electorally accountable economic decisionmakers. Specifically, we assume that voters will perceive a higher ratio of non-electorally accountable to electorally accountable decision makers in open economies than in more closed ones and so economic voting will be less important in open economies. This comports with much of the political economy literature suggesting that domestic governments have declining economic policy making latitude in open economies. A similar reasoning leads us to a more counterintuitive prediction: Voters will perceive a higher ratio of politically non-accountable to politically accountable decision makers in contexts with an expansive state sector. This in turn will result in a lower competency signal and lower economic voting. These two results are consistent with our theoretical argument and they predict lower levels of economic voting in contexts with open economies and with an expansive state sector – both of which turn out to be confirmed by the empirical analysis.

This result has important implications for trends in economic voting because these are characteristics of the political and economic context that 1) are changing quite dramatically over time (national economies are becoming increasingly integrated into the world economy) and 2) are increasingly distinctive cross-nationally (nations differ considerably in terms of the scope of their state sectors). We suggest how the interaction amongst these two trends might affect trends in economic voting and democratic accountability more generally. The net implications of globalization for the economic vote depend on whether greater exposure to the international economy occurs in a context with an extensive as opposed to more limited state sector. A larger government sector may in fact moderate the negative social consequences of globalization but our competency theory suggests that broadening the scope of government's involvement in the economy reinforces the negative impact that trade openness exerts on the economic vote. More specifically, increasing the size of the government

sector, embracing corporatist institutions and increasing government regulation reduce the magnitude of the incumbent's competency signal leading voters to discount the importance of the economy in their voting decision. Paradoxically, countries that respond to increased exposure to global economic forces by privatizing their economies and adopting liberal economic policies are more likely to compensate for the decline in economic voting associated with globalization.

Second, the empirical work in Chapter 9 explores how various indicators of policy making responsibility impact economic voting. We assume that voters believe policymaking responsibility to be less concentrated when cabinet seats are more equally divided among parties, when minority cabinets rule, and when oppositions are empowered. Our empirical evidence for each of these claims is strong. In addition to these general hypotheses about the concentration of responsibility, we also assume that voters attribute more responsibility to prime ministers than their partners, more to parties holding more seats in cabinet than to those holding less, more to parties holding the finance ministry than to those who do not, more to cabinet parties than to opposition parties, more to single-party prime ministers than to prime ministers of coalition cabinets, and (most controversially) more to single party prime ministers than to presidents. Again, our empirical evidence supports each of these predicted relationships strongly. Indeed, there is simply no doubt that the current distribution of policy making responsibility strongly conditions the economic vote in Western democracies.

Finally, in Chapter 10 we explore how voters' beliefs about the pattern of contention for policymaking responsibility impact the economic vote. Though our specific hypotheses rely on various aspects of contention – e.g., whether a party is in many competitive alternative cabinets or how competitive is the incumbent cabinet as a whole – each ultimately requires that we say something about the voters' beliefs about which parties or coalitions of parties are likely to be in competition to obtain executive authority. We used two strategies (or auxiliary hypotheses) for doing this. First, we assumed the voters' beliefs about the competitiveness of different parties or coalitions resulted either from the historical record – i.e., which parties or coalitions had policy making responsibility in the past. Second, we assumed that these beliefs stemmed from the rational voter's "model" of how current observable characteristics of parties (e.g., their size or ideology) get translated into policymaking responsibility. Thus, we tested hypotheses linking a party's history of cabinet participation to the economic vote of opposition parties, as well as ones linking these parties' predicted probabilities of entering a cabinet to their economic vote – where the predicted probabilities come from a statistical model of cabinet formation. Likewise, for incumbents, we examine the connection between economic voting and the chance that cabinets similar to the current cabinet come to power. Finally, we show that the economic vote for

incumbent parties that are members of competitive alternative cabinets is muted relative to their partners.

Taken together, these results provide strong support for the idea that the pattern of contention for policymaking authority has an important, though previously under appreciated, impact on the economic vote. Further, when we examine this impact alongside measures of the current distribution of responsibility, we find that the pattern of contention has a strong independent impact on economic voting. This is seen most clearly in the case of perennial prime ministers. According to both our theory and the clarity of responsibility argument mentioned above, these parties should, based on their large share of policy making responsibility, have large economic votes. However, when one adds to this the implications of the pattern of contention for the economic vote (which our theory does and the clarity argument does not), we predict that perennial prime ministers – who will be in all the cabinets that form – will actually have a depressed economic vote relative to their smaller partners. This is exactly what we find in countries like Belgium, the Netherlands, and Italy – all cases in which the critical role of the dominant Christian parties in policymaking could not have been lost on most voters, but where, nevertheless, economic voting for these parties is smaller relative to their partners than we would otherwise expect.

What Remains to be done?

In this section we ask what remains to be done both in extending the approach to economic voting we have developed in this book and developing the study of comparative economic voting more generally. Specifically, we think there are at least three fruitful areas in which the work begun in this book can and should be extended: (1) explorations of the assumptions of our model with the goal of improving its theoretical foundations, (2) development of alternative theoretical models so that they are both precise enough and general enough to be usefully contrasted to ours, and (3) continued empirical testing of the implications of our model that more completely capture the complex interactions implied by the theory. We discuss each of these possible areas of future work below.

Exploring our assumptions and their implications for differences among individuals

While we think that our theory produces plausible empirical hypotheses about the impact of political and economic institutions on the average level of economic voting for a party or in a system, these hypotheses rest on a foundation of assumptions about the knowledge and behavior of individuals that, with some exceptions – Chapter 6 and the beginning of Chapter 10 – we have ignored empirically.

In this brief section we highlight several places in which our theory makes strong assumptions about what individuals do and do not know about their economies and political systems, but that are quite amenable to future empirical exploration. Further, while our focus has been on the impact of political and economic institutions on economic voting, our theory is ultimately about individuals and so we point out places in which its direct implications could be tested by looking at differences amongst individuals.

The first set of contextual implications that we produced from our theory rested on micro-level assumptions regarding what individuals actually know about electorally versus non-electorally accountable decision making in different national contexts have not been empirically explored in this book. Is it reasonable to think that large government contexts have a smaller ratio of non-electorally accountable to politically accountable decision makers than limited government contexts? Moreover, do voters perceive these differences? And is it reasonable to think that voters in open economies recognize that their economic outcomes are more affected by non-electorally accountable decision makers? In general, are voters sensitive to the electoral accountability of decision makers and do they make the kind of inferences from this information that we attribute to them? We do not have definitive answers to this question. Clearly though our theoretical claims regarding the impact of electorally versus non-electorally decision making on the economic vote are contingent on the answers to these empirical questions. The importance of addressing these empirical issues goes well beyond explaining the economic vote. These empirical issues have broader implications for our understanding of democratic governance because we are proposing here a model of how citizens learn about the competency of governing elites that we believe is critical to understanding the extent to which they hold elected officials accountable for policy outcomes.

Second, economic voters, according to our theory, use the pattern of contention for policy making responsibility in their vote choice. Moreover, the pattern of contention may concern not just electoral contention between single parties, but also the pattern of contention amongst potential *distributions of responsibility*. We assume that voters have knowledge about which parties or coalitions of parties are likely to form the government (or participate in policy making). They are also sensitive to how voting for a party affects the likelihood of different coalitions forming the government. Understanding the extent to which voters make these assessments, how they obtain the information necessary to do so, and how this affects their vote choice, will represent an important extension of our theoretical and empirical work on the strategic economic vote reported in this book. We present, in Chapter 10, some preliminary evidence from three elections that support our assumption that voters understand which parties are likely to participate in a governing coalition and their likely roles in the cabinet. But we have not explored this across a variety of different contexts; nor have we investigated how voters actually obtain this information. It might be that media messages regarding

contention are very informative in multi-party contexts and that in these contexts voters pay close attention to them. Alternatively, the strategic economic vote may be mediated by elite activity in a fashion similar to the argument developed by Cox and Munger (1989) to explain the relationship between contention and turnout in U.S. congressional elections. Elite efforts on behalf of parties in contention for administrative responsibility may be the mechanism that mobilizes the economic vote for parties in contention for administrative responsibility. Developing and empirically evaluating these precise mechanisms will be a critical step toward confirming (or maybe challenging) the strategic economic voting results reported in Chapter 10.

Finally, since each of the aggregate level hypotheses explored in this book stems from a more direct hypothesis about individuals, it would be worthwhile for scholars to investigate whether differences in beliefs among individuals in the same electoral context lead to the difference in economic voting at the individual level that our theory would predict. So for example, it would be worthwhile to develop a set of survey questions that could tap the extent to which voters believe economic outcomes are the result of decisions by electorally accountable individuals or non-electorally accountable ones. With such a measure one could examine directly whether individuals with different beliefs about this were more or less likely to cast economic votes. Similarly, in Chapter 10 we saw one way that survey questions can be used to discover what voters believe about the competitiveness of different potential coalitions. Again, with such a measure, surveys that also included economic perceptions and vote choice questions could test directly for the impact of difference in such beliefs on individual level economic voting.

Alternative Models

We see no reason in principal why an alternative theoretical approach building on different micro-foundations might not follow the approach we have taken in this book and so derive alternative explanations for contextual variation in the economic vote. Alternative theoretical approaches building on conventional sanctioning models are an obvious candidate. As we noted in the introduction, Hibbs (2006) argues for a contextual theory of the economic vote that builds on the conventional retrospective punishment model and that incorporates Kramer's (1983) "error-in-variables" conception of economic evaluations as a measure of the signal-to-noise ratio regarding incumbent economic performance. Hibbs (2006) suggests that contexts in which the politically relevant portion of macro-economic outcomes is small – reflected in relatively high measurement error when overall economic evaluations are used – will be those that have a low correlation between economic evaluations and vote choice. While a potentially promising theoretical approach, to our knowledge no one has developed a fully-specified contextual theory of economic voting that builds on these insights.

There are other possible micro-foundations for a contextual theory of the economic vote. A theoretical approach that builds on emotional responses to the macro-economy might be another candidate. One example is Lodge's model in which people's evaluations of candidates are "read" from a long-term tally that tracks the totality of the emotional responses that people have had to messages they have received about the candidates. Some of these messages concern the economy and if one assumes that people have a negative emotional response to the incumbent when bad economic news is received, then we would expect an association between their perceptions of the economy and their evaluations of the incumbent candidate, mediated by their tally of negative emotional responses. In Zaller's model (1990) voters also receive messages, but in his case their evaluations of candidates are based on a random selection from the pool of messages they have received and accepted. If a voter receives and accepts negative economic messages and understands these as negative incumbent messages, then they will be more likely to sample a negative opinion of the incumbent. And Zaller (2004) has explored the application of this model to the economic vote by exploring empirically which groups in the population actually accept these negative messages regarding the economy.

Instead of thinking of sanctioning as a way that rational voters induce politicians not to shirk their responsibilities (i.e., as a moral hazard problem), we could ask how context would impact economic voting in a sanction model that invokes emotional response. This enterprise, however, would require that we have a well-specified theory of emotional response and how these responses connect to individual perceptions of aggregate economic performance. While there has been an effort among political psychologists to develop general models of the impact of emotions on political judgments (for example the *Affective Intelligence Theory* proposed by Marcus, Neuman and Mackuen (2000)), we simply know too little about what triggers specific emotional responses to predict with any confidence how variations in political and economic context are likely to effect emotional responses at the individual level (Ladd and Lenz 2004).

Any one of these alternative modeling approaches might serve as the basis for developing explanations for the contextual variation described in Chapter 3. But, at least to our knowledge, no one has taken these alternative models of vote choice as a point of departure and then derived rigorous explanations for how the vote decisions are conditioned by context. Much of the empirical literature that examines economic voting in different political contexts (e.g., the clarity of responsibility literature) is motivated from a loose discussion of elections as devices of electoral sanction (particularly the moral hazard version of the argument). And hence it remains to be seen whether alternative models of vote choice might provide more powerful explanations for the contextual variation in the economic vote.

Complex Causation and Alternative ways to Explore the Implications of our Theory

At any point in time context will of course provide multiple cues to the voter regarding strategic incentives in addition to the competency signals that are central to our theory. Exploring in greater detail precisely how voters respond to these multiple, sometimes contradictory, signals will be an important extension of the research we've conducted to date. Indeed we used the fact that voters receive different contextual signals as a critical test of our contention argument: This critical condition occurs when the voter is evaluating an incumbent party that has a high degree of administrative responsibility but one that is not in "contention", i.e., either certain to enter the governing coalition or certain not to participate in government. Our theory clearly predicts that there should be no economic vote in this situation while conventional explanations, such as clarity of responsibility, suggest exactly the opposite: there should be a significant economic vote.

In fact, these juxtapositions of contextual features are likely to be most revealing about how context shapes the economic vote. So, for example, we predict little economic voting in contexts with very open economies and with an extensive state sector *even* for cases in which administrative responsibility is highly concentrated, say in the case of single party majority governments (which, all things being equal, contributes to a high economic vote). A more likely scenario though is that each of these measures – our trade/state sector index and our responsibility measure – assumes values that either moderate or exaggerate the importance of the economy in vote choice. The interesting question in this case is identifying the unique contribution of each of these contextual variables. These examples illustrate the two estimation issues that we do not address in any depth in this book but clearly are of central importance to future research in this area: determining the independent effects of each of these three contextual variables and sorting out the causal complexity of our hypothesized relationships.

First, do each of these contextual effects have independent effects on the economic vote and what are their relative magnitudes? Clearly there will be economic votes that are exercised in contexts where the ratio of electorally to non-electorally accountability decision makers is high, where a party has high administrative responsibility and where a party is also very much in contention. Readers will note that in many of the bi-variate plots presented throughout the book, there are countries that seem to have values on all of the contextual measures that are driving the economic vote in the same direction. The Netherlands, for example, is highly integrated into the global economy; has a history of a perennial incumbent PM party; and administrative responsibility is shared by a number of parties. Hence the values of all three contextual variables for The Netherlands predict a low economic vote. In fact it turns out that this collinearity is relatively

significant in our data set. In addition, as we saw in Chapter 7, some of the country characteristics, such as trade openness, simply do not vary much over the sample. These features of the data plus the fact that we typically have only 163 observations of the contextual variables for analysis make it difficult to estimate these independent effects.

As the number of these voter preference studies increases over time, we will undoubtedly get better estimate of these independent effects. In the mean time we can obtain at least some order of magnitude estimates of the relative importance of these three contextual variables for the economic vote. Our goal here is simply to demonstrate that each of the three contextual variables exercises some independent effect on the economic vote. Accordingly we present in the first three columns of Table 11.1 model specifications that pair each contextual variable with one of the other contextual variables. Combining the interaction effects of the different context variables reduces some of the interactions that we saw in the models estimated in Chapters 7, 9 and 10 where each context effect was estimated on its own. The interaction of concentration of responsibility with positive economic evaluations is attenuated when it is included in models with the open economy/extensive state sector context interaction terms. The PM contender interacted with negative economic evaluations is attenuated whenever these effects are paired with another contextual effect. But really what is important here is that all three of these contextual variables have the hypothesized effect – even when paired with each other – of exaggerating the difference in vote probabilities of individuals with positive versus negative economic evaluations.

Table 11.1 Multi-level Logistic Regression Model of Chief Executive Vote

	Model	Model 2	Model 3	Model 4
Left-Right Self Placement (dev. from mean)	0.48 (0.01)	0.48 (0.01)	0.49 (0.01)	0.48 (0.01)
Left-Right Self Placement X Leftist PM (Leftist PM is indicator variable)	-0.88 (0.01)	-0.88 (0.01)	-0.75 (0.01)	-0.88 (0.01)
Voter Perceives Economy got Better (indicator variable)	0.45 (0.03)	0.45 (0.03)	0.45 (0.03)	0.44 (0.03)
Voter Perceives Economy got Worse (Indicator variable)	-0.59 (0.03)	-0.56 (0.04)	-0.57 (0.03)	-0.59 (0.03)
Open Economy/Extensive State Sector	0.07 (0.04)	-0.06 (.04)		0.04 (0.04)
Voter Perceives Economy got Better X Open Economy/Extensive State Sector	-0.22 (0.05)	-0.18 (0.04)		-0.18 (0.05)
Voter Perceives Economy got Worse X Open Economy/Extensive State Sector	0.09 (0.05)	0.19 (0.05)		0.09 (0.05)
Concentration of Executive Responsibility (deviation from mean)	0.89 (0.20)		0.88 (0.15)	0.91 (.19)
Voter Perceives Economy got Better X Concentration of Executive Responsibility	0.01 (0.22)		0.47 (0.15)	-0.07 (0.21)
Voter Perceives Economy got Worse X Concentration of Executive Responsibility	-0.78 (0.21)		-0.92 (0.15)	-0.78 (0.22)
PM Contenders		-0.23 (0.09)	-.11 (.07)	-0.23 (0.08)
Voter Perceives Economy got Better X PM Contenders		0.26 (0.09)	.29 (.08)	0.26 (0.09)
Voter Perceives Economy got Worse X PM Contenders		-0.05 (0.11)	-.11 (.08)	-0.01 (0.10)
Size of chief executive's Party	5.49 (0.32)	6.30 (0.29)	4.61 (0.26)	5.66 (0.32)
Constant	-2.48 (0.10)	-2.74 (0.09)	-2.21 (0.08)	-2.53 (0.10)
$\sigma^2_{\omega 1}$	0.05	0.06	0.07	0.05
$\sigma^2_{\omega 2}$	0.06	0.06	0.07	0.05
$\sigma^2_{\omega 3}$	0.06	0.08	0.07	0.06
$\sigma_{\omega 1, \omega 2}$	-0.02	-0.02	-0.03	-0.02
$\sigma_{\omega 1, \omega 3}$	-0.01	-0.02	-0.02	-0.01
$\sigma_{\omega 2, \omega 3}$	-0.03	-0.04	-0.03	-0.03
N	201,87	201,876	201,876	201,876

Standard Errors are listed below the logit coefficients. All coefficients are statistically significant at $p < .1$ except the shaded cells.

These effects are confirmed in Model 4 that includes the three context variables and their interactions with the two economic evaluation variables. In spite of the relatively high degree of collinearity amongst these variables, the three basic contextual effects are confirmed in this combined model: The difference in the economic vote of those with positive versus negative evaluations is significantly greater, in contexts with an open economy and an expansive state sector; when administrative responsibility is concentrated; and when there are more PM contenders. This results from the fact that in the case of each context interaction at least one of

the interaction terms is significant in the predicted direction (and none of the interaction effects are significant in the wrong direction). The most important consequence of combining the three context effects (compared to the estimating separate context effects) is that the interaction effects for each contextual interaction term tend to be significant in only one direction: the interaction effects for the *open economy/extensive state sector* and *PM contenders* variables are significant in interaction with the positive economic evaluation variable; while, the *concentration of administrative responsibility* variable is significant in interaction with negative economic evaluations. Given more observations and variation on the contextual variables we would likely see the same symmetric effects we saw in the individual contextual models.

Figure 11.1 illustrates the change in predicted probabilities of voting for the chief executive party associated with different values of the three contextual variables – in each case we have plotted the direction of the economic evaluation change that matches the significance of the interaction effects in Table 11.1. The first graph illustrates how the political economic context conditions the impact of an improving economic evaluation on the vote for the chief executive party: For countries scoring very low on the *open economy/extensive state sector* measure – that is to say a closed economy with a limited government sector – a unit improvement in economic perceptions results in an improvement of .15 in the parties vote probabilities. At the average reading on this scale, zero, the economic vote is about 9 percent, dropping to around 4 percent for countries that have the most open economies and a large state sector. For the variable measuring the *concentration of administrative responsibility* note that at zero, the mean value for the responsibility variable, the economic vote is 10 percent, while at the extreme value of concentration of responsibility – about 0.2 – the economic vote increases to 15 percent. The economic vote drops to zero for those cases that score very low on the measure (in the neighborhood of -0.4). In the case of our summary measure of contention – *PM contenders* – the economic vote is about 9 percent in contexts with the average value of zero and it rises to 12 percent in contexts with high *PM contenders* score and falls to 6 percent in those with very low score. We do not want to make too much of these multivariate results for the reasons we pointed out earlier. Nevertheless, in spite of the data limitations here, we think the results summarized in Table 11.1 and illustrated in Figure 11.1 confirm that each of our three theoretical variables make an important independent contribution to explaining contextual variation in the economic vote. With time of course we will have a much larger sample of voter preferences studies that will provide the added contextual variation that we suspect will provide even stronger confirmation of the independent effect of these three contextual variables.

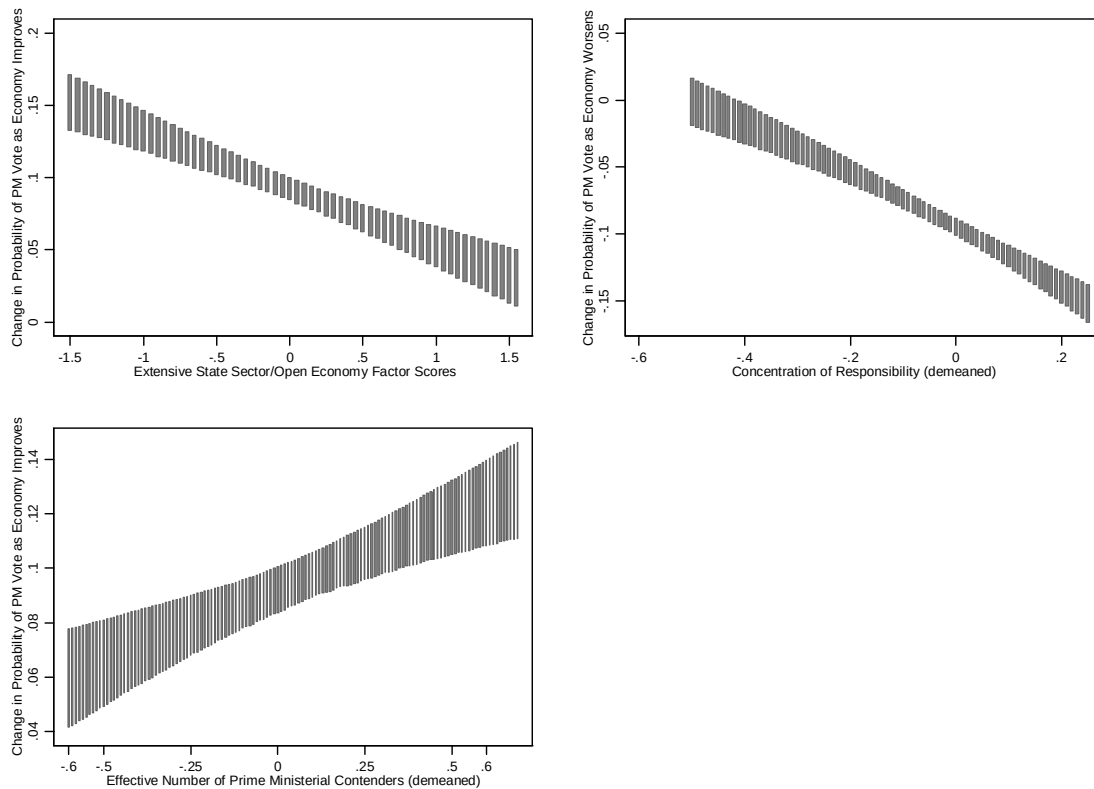


Figure 11.1 Multivariate estimates of the impact of extensive state sector/open economy, concentration of responsibility and effective number of primeministerial contenders on the economic vote.

In presenting our empirical results we have alluded to a second reason why the independent effects of our contextual variables may be attenuated in linear models: Our theory implies that the impact of these contexts on the economic vote should, in fact, not be independent of one another. Recall that in the equations describing the voter's utility of voting for a party, the β 's, α 's, P 's, λ 's and σ^2 's all end up in the voter's value function as multiplicative coefficients on the change in the economy. If any one of these is zero, the previous economy will fall out of the voter's value function for a party and the other contexts can have no further impact on economic voting, whatever their value. More generally, for example, this occurs when three conditions, C_1 and C_2 and C_3 , jointly produce Y , so that the impact of changes in one condition on Y depends on the others. There are in effect multiple paths of non-occurrence of the dependent variable such that the absence of any one condition nullifies the impact of the others. For example, with respect to our model, attribution of responsibility to a party **and** a belief that the party is in contention for the prime ministry **and** the possibility of discerning a competency signal from the background

noise of economic fluctuations **produce** an economic vote for a party. If any one of these conditions is not met then we do not expect any economic voting.²⁹¹ Further, as any one variable moves toward zero it will diminish the impact of the other three. The competency model of rational retrospective economic voting suggests that economic voting will be contingent on a number of conditions being met - any of which, if absent, will sap the incentive of voters to use the retrospective economy in their vote choice.

This effect results in a triangular or wedge shape that is found in many of our graphs. We see this because the theory states that when C_1 is absent then Y (economic voting) cannot occur. However, when C_1 is large, it may occur, but whether it does or not depends on the values of C_2 and C_3 . Clearly, a linear approximation to this relationship will often reveal a generally positive relationship, but if the data have this pattern, it will tend to be flattened relative to a true linear relationship because there will be cases in which C_1 is large but there is nevertheless no economic voting. We think that our bivariate graphs of context and economic voting reflect this kind of wedge shaped structure. Hence it is important, when we use linear models to approximate this relationship, that we do not simply run statistical models and look for statistical significance; rather we need to examine graphs to evaluate whether we see the expected wedge shapes. Accordingly, we relied extensively on the graphical presentation of relationships throughout the book.

The hypothesis tests that we report in our empirical chapters can only be approximate because they do not account for the multiple conjunctural nature of the hypotheses developed from our theory. While it would be ideal to test our hypotheses using techniques that do capture this causal structure, the statistical literature on multiple conjunctural causation does not yet provide appropriate techniques for doing this with the kind of data we have.²⁹² For this reason we did not abandon traditional techniques of examining the data. Ultimately though the validity of the theoretical arguments we developed here will depend on estimating the multiple conjunctural nature of our hypotheses. Accordingly, solving the technical challenges associated with estimating multiple conjunctural causation will be critical for making progress in our understanding of how context shapes vote choice.

²⁹¹ This is an estimation problem that Ragin (1987) initially identified as being particularly problematic in qualitative studies in comparative research.

²⁹² See Braumoeller (2003) for a discussion of some of the estimation issues associated with the multiple conjunctural causation we described here.

Implications for Comparative Studies of Political Behavior

In spite of these challenges that remain to be addressed, we have made significant advances in understanding how context conditions the vote choice in general even though our focus here has been on the economic vote. Two very broad insights regarding models of vote choice emerge: 1) modeling efforts that do not incorporate contextual variation are very likely wrong; 2) theories of contextual variation in vote choice need to address the full range of contextual variation in vote choice which is rarely the case in current research.

In estimating individual-level vote choice models, where the political or economic context is of consequence for the voting decision (and it is difficult to imagine when this might not be the case), contextual variation is not an option, or an interesting extension of a single case study, it is an essential part of the research design. The empirical ramifications of ignoring this maximum for the analysis of individual survey data are rarely demonstrated primarily because of small N problems. Our project is unique in analyzing a large number of voter preference surveys that allow us to get some insight into the consequences of ignoring context in empirical estimations. This is nicely illustrated by our empirical analysis presented in Part I of the book where we demonstrate that the economic vote for any particular party varies quite significantly, even from one election to the next. Our density plot of economic vote magnitudes in Chapter 3 suggests that the estimated economic vote in about a quarter of the surveys in our sample is essentially zero. This of course is the logical consequence of context being important – there will be contexts in which the estimated effects will be very low or zero. Hence, to say there is “no economic vote” in a particular context is not the same thing as saying there is no economic voting. And it certainly does not imply that voters are not rational. In fact, the absence of an economic vote in some contexts, as we demonstrate, is a testament to instrumentally rational voting. This implies, and the empirical results confirm, that there will be political parties – including the Prime Minister’s party – that receive no economic vote with some regularity. Many – although certainly not all – of these observed zero economic votes are predicted to have this value because of the contextual factors identified in our theory.

And even more importantly it suggests that drawing conclusions about vote choice (or political behavior more broadly) based on estimates from a single context (either in time or cross-sectionally) could very well lead to the wrong conclusion. So, for example, estimating a vote choice model from surveys conducted in 1997 in the UK and finding an insignificant coefficient for the economic evaluations variable might lead some to conclude that there is no economic vote in the UK and that this might constitute evidence that voters are not behaving “rationally”. As our theory and empirical evidence in this book strongly suggest, this would be patently wrong.

But of course this applies to any model of vote choice – not including contextual variables, including the wrong ones, or simply drawing conclusions based on analysis from a single context, all could very likely lead to the wrong conclusions regarding the behavior of rational voters. A properly specified vote choice model presumes you have properly specified contextual effects. At least in the case of individual vote-choice model this is difficult to accomplish without samples extending over time and across national contexts.²⁹³

We believe that incorporating this contextual variation in individual-level models is both essential and currently feasible. As a result of almost fifty years of conducting voter preference surveys throughout the developed world, we now have surveys from a large number of political and economic contexts. The strategy we describe in Chapter 4 for estimating contextual effects on the economic vote can easily be adapted for estimating the impact of context on other variables in the vote choice calculus. Moreover, there are a number of factors that are widely considered to play an important role in vote choice which means there are measures of these concepts in most voter preference surveys – for example, ideology, postmaterialism, democratic satisfaction, partisanship, class, etc. Hence in our view political scientists are well equipped with the data and the methods necessary to estimate the conditional effects of context on the factors shaping vote choice. The challenge is theoretical: developing contextual theories, as we have done in this book, that explain how these diverse factors shaping vote choice are conditioned by context.

Our findings also provide an important insight into the nature of this theoretical challenge. In particular, our results indicate that any contextual theory of vote choice that informs a well specified model of vote choice needs to account for contextual effects that are very diverse or multidimensional in character. The analysis of variance summarized in Chapter 3 provides an important clue as to the nature of the variation in vote choice that needs to be accounted for by contextual variables. This analysis demonstrated that the economic vote can vary significantly from one voter preference survey to the next. While the median economic vote is around 7 percent, in any survey it could range between -20 and 8 percent. We have little doubt that this is also the case for other vote choice models – there will be contextual variation in the “ideological vote” (in fact we document the extent of this in Chapter 3) in the “partisan vote”, in the “spatial vote”, etc. What is particularly interesting about the economic vote variation we described in Chapter 3 is that much of it is survey-to-survey rather than a function of a temporal trend or cross-national effects. But to the extent that the literature addresses context at all it tends

²⁹³ One could think of experimental research designs that might represent another approach to incorporating context in empirical models of vote choice.

to favor explanations grounded in temporal trends – think of “the end of class voting” or “the end of ideology” or “the decline of partisanship”. Or, alternatively, the literature focuses on cross national effects such as institutional characteristics or features of the economy that are quite static. In contrast, our examination of variation in the economic vote suggests that much of its variation is unique to each voter preference survey. In other words a lot of the interesting action in the data is distinct variations in the economic vote from one voter preference study to the next within each country. Accordingly we need theories that not only address cross-national variations in vote choice models and temporal trends in these models but also that explain variation in vote choice models that is unique to each voter preference survey (or unique to each election, as the case might be). For example, we need theory that explains why ideology might be important in a Dutch vote choice model in 2000 but much less important in 2004. Theories that are unable to address this dimension of contextual variation in vote choice models are likely to fair quite poorly in terms of accounting for how voters factor context into their voting calculus.

This book was initially motivated by a simple concern with understanding contextual variation in the economic vote. It became clear to us – probably not as quickly as it should have – that this required that we think about a more general puzzle: How is the *vote calculus* conditioned by context? In particular, how does context condition the importance that voters accord their evaluations of government performance or their issue positions or maybe candidate personality when they make a vote decision? We have focused here on economic evaluations in the voting calculus because we believe (and have conclusively demonstrated) that this is one of the most important considerations shaping vote choice. And while it is true that our theory of context explicitly concerns economic evaluations and vote choice we think the general theoretical and empirical approaches can inform efforts to understand how contexts conditions the importance of most other important factors that shape the vote decision.

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Appendix A

Define a *vote total* for party $j \in \mathbf{J}$ as $v_j = \sum_{i=1}^n v_{ij}$, where i indexes n voters and v_{ij} is 1 if voter i votes for party j and 0 otherwise. We will denote the vector of vote totals for the J parties in $\mathbf{J}$ as $\mathbf{v}_J = \{v_1, v_2, \dots, v_J\}$. We will consider two different characterizations of the distribution of policy-making responsibility over parties. First, define $\lambda_k = \{\lambda_{k1}, \lambda_{k2}, \lambda_{kj}, \dots, \lambda_{kJ}\}$ as a *distribution of responsibility* over the J parties, where $\lambda_k \in \Lambda$ is the finite set of all such distributions (of which there are K). Further, we assume that $\lambda_{kj} \geq 0$ for all λ_{kj} and that $\sum_{j \in \mathbf{J}} \lambda_{kj} = 1$.²⁹⁴ Next, define

$\mathbf{g}_m = \{g_{m1}, g_{m2}, g_{mj}, \dots, g_{mJ}\}$, where $g_{mj} = 1$ if party j is a member of cabinet (or proposed cabinet) $\mathbf{g}_m$ and 0 otherwise. $\mathbf{g}_m \in \mathbf{G}$ is the finite set of all possible cabinets (of which there are M).

Define the *electoral support* associated with any distribution of responsibility λ_k as $v_{\lambda_k} = \sum_{j \in \mathbf{J}} \lambda_{kj} v_j$ and $\mathbf{v}_\Lambda = \{v_{\lambda_1}, v_{\lambda_2}, v_{\lambda_k}, \dots, v_{\lambda_K}\}$. The impact of an increase in the vote total for party j on the electoral support for a particular distribution of responsibility is thus proportional to party j 's responsibility in the proposed distribution. Thus, assuming a large enough electorate that the difference quotients for the selection function that are associated with a change in a single vote can be well approximated by derivatives, we have $\frac{\partial v_{\lambda_k}}{\partial v_j} = \lambda_{kj}$ for all J parties. Further, define the *electoral support* associated with any cabinet $\mathbf{g}_m$ as $v_{\mathbf{g}_m} = \sum_{j \in \mathbf{J}} g_{mj} v_j$ and $\mathbf{v}_\mathbf{G} = \{v_{\mathbf{g}_1}, v_{\mathbf{g}_2}, v_{\mathbf{g}_m}, \dots, v_{\mathbf{g}_M}\}$. The impact of an increase in the vote total for party j on the electoral support for a particular cabinet is thus g_{mj} , which is just 1 if the party is in the cabinet and zero otherwise.

Now define $\mathbf{s}_\Lambda = \{s_1, s_2, s_k, \dots, s_K\}$, where $s_k = s(v_{\lambda_1}, v_{\lambda_2}, v_{\lambda_k}, \dots, v_{\lambda_K}, \mathbf{z}_{\lambda_1}, \mathbf{z}_{\lambda_2}, \mathbf{z}_{\lambda_k}, \dots, \mathbf{z}_{\lambda_K})$ is a *selection function* defined over the

²⁹⁴ We assume that Λ is finite to avoid having to work with matrices that have an infinite number of rows and columns. Given Λ can be as large as we like, this is not a substantive restriction.

K elements of Λ . The $\mathbf{z}_{\lambda_k}$'s represent any factors associated with λ_k other than electoral support that impact the value of s_k . More compactly, we can write this selection function as: $s_k = s(\mathbf{v}_\Lambda, \mathbf{Z}_\Lambda)$, where $\mathbf{Z}_\Lambda = \{\mathbf{z}_{\lambda_1}, \mathbf{z}_{\lambda_2}, \mathbf{z}_{\lambda_k}, \dots, \mathbf{z}_{\lambda_K}\}$.²⁹⁵ We assume that s_k is everywhere differentiable with respect to $\mathbf{v}_\Lambda$, again assuming a large enough electorate that we can approximate difference quotients with derivatives.

Similarly, we can define a version of the selection function that maps electoral support for cabinets (and other factors defined over cabinets) to selection values. This is just $\mathbf{s}_G = \{s_1, s_2, s_m, \dots, s_M\}$ with typical element $s_m = s(\mathbf{v}_G, \mathbf{Z}_G)$, where the $\mathbf{z}$'s are defined analogously to those above and we make the same assumption of linearity between integer values of v_j .

We want to use these definitions to provide general characterizations of how a vote for party j impacts the chance that each possible cabinet or distribution of responsibility is selected to govern. To do that, we will first define the derivatives of the selection functions with respect to party vote totals and then define a rule for choosing either a cabinet or a distribution of responsibility given values of the selection functions for each possible cabinet or distribution of responsibility.²⁹⁶

We want to derive $\frac{\partial \mathbf{s}_\Lambda}{\partial \mathbf{v}_j}$. Since $\mathbf{s}_\Lambda$ and $\mathbf{v}_j$ are both vectors, this is the matrix:

$$(A.1) \quad \frac{\partial \mathbf{s}_\Lambda}{\partial \mathbf{v}_j} = \begin{bmatrix} \frac{\partial s_1}{\partial v_1} & \frac{\partial s_1}{\partial v_2} & \frac{\partial s_1}{\partial v_j} & \dots & \frac{\partial s_1}{\partial v_J} \\ \frac{\partial s_2}{\partial v_1} & \frac{\partial s_2}{\partial v_2} & & & \vdots \\ \frac{\partial s_k}{\partial v_1} & & \frac{\partial s_k}{\partial v_j} & & \\ & & & \ddots & \\ \frac{\partial s_K}{\partial v_1} & \dots & & & \frac{\partial s_K}{\partial v_J} \end{bmatrix}$$

²⁹⁵ $\mathbf{Z}_\Lambda$ is not strictly necessary but is included to remind the reader that the selection function does not only depend on electoral support.

²⁹⁶ Since each v_j is just the sum of the v_{ij} over n , we need only consider the derivative of $\mathbf{s}$ with respect to the vote totals v_j instead of individual votes.

By definition, votes play no role in $\mathbf{Z}_\Lambda$ and we assume for convenience that $\mathbf{v}_\Lambda$ and $\mathbf{Z}_\Lambda$ appear in separable parts of the selection function and ignore $\mathbf{Z}_\Lambda$ in the discussion that follows. To find expressions for the elements of the matrix in (A.1), we can use the relationship between the electoral support for distributions of responsibility and party vote totals. Specifically, from the definitions above, we can write the relationship between $\mathbf{v}_\Lambda$ and $\mathbf{v}_J$ as:

$$(A.2) \quad \mathbf{v}_\Lambda = \mathbf{A}\mathbf{v}_J$$

where $\mathbf{A}$ is the Jacobean of the transformation. Consequently, using standard results on vector derivatives, we have for some possible distribution of responsibility $\lambda_{\tilde{k}}$:

$$(A.3)$$

$$\begin{aligned} \frac{\partial s_{\tilde{k}}}{\partial \mathbf{v}_J} &= \mathbf{A} \frac{\partial s_{\tilde{k}}}{\partial \mathbf{v}_\Lambda} \\ \left[\frac{\partial s_{\tilde{k}}}{\partial v_1}, \frac{\partial s_{\tilde{k}}}{\partial v_2}, \frac{\partial s_{\tilde{k}}}{\partial v_j}, \dots, \frac{\partial s_{\tilde{k}}}{\partial v_J} \right]^T &= \begin{bmatrix} \lambda_{11} & \lambda_{12} & \lambda_{1j} & \dots & \lambda_{1J} \\ \lambda_{21} & \lambda_{22} & & & \vdots \\ \lambda_{k1} & & \lambda_{kj} & & \\ \vdots & & & \ddots & \\ \lambda_{K1} & \dots & & & \lambda_{KJ} \end{bmatrix}^T \left[\frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_1}}, \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_2}}, \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}}, \dots, \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_K}} \right]^T \\ &= \left[\sum_{\lambda_k \in \Lambda} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} \lambda_{k1}, \sum_{\lambda_k \in \Lambda} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} \lambda_{k2}, \sum_{\lambda_k \in \Lambda} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} \lambda_{kj}, \dots, \sum_{\lambda_k \in \Lambda} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} \lambda_{kJ} \right]^T \end{aligned}$$

which are the rows of equation (A.1). The typical element of $\mathbf{A}$, λ_{kj} , is simply the responsibility of party j in distribution of authority λ_k (where, again, there are K such distributions). If we use $\tilde{k}$ and k as separate indexes over the set Λ , the whole matrix is as follows:

$$(A.4) \quad \frac{\partial \mathbf{s}_\Lambda}{\partial \mathbf{v}_J} = \begin{bmatrix} \sum_{\lambda_k \in \Lambda} \frac{\partial s_1}{\partial v_{\lambda_k}} \lambda_{k1} & \sum_{\lambda_k \in \Lambda} \frac{\partial s_1}{\partial v_{\lambda_k}} \lambda_{k2} & \sum_{\lambda_k \in \Lambda} \frac{\partial s_1}{\partial v_{\lambda_k}} \lambda_{kj} & \dots & \sum_{\lambda_k \in \Lambda} \frac{\partial s_1}{\partial v_{\lambda_k}} \lambda_{kJ} \\ \sum_{\lambda_k \in \Lambda} \frac{\partial s_2}{\partial v_{\lambda_k}} \lambda_{k1} & \sum_{\lambda_k \in \Lambda} \frac{\partial s_2}{\partial v_{\lambda_k}} \lambda_{k2} & & & \vdots \\ \sum_{\lambda_k \in \Lambda} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} \lambda_{k1} & & \sum_{\lambda_k \in \Lambda} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} \lambda_{kj} & & \\ \vdots & & & \ddots & \\ \sum_{\lambda_k \in \Lambda} \frac{\partial s_K}{\partial v_{\lambda_k}} \lambda_{k1} & \dots & & & \sum_{\lambda_k \in \Lambda} \frac{\partial s_K}{\partial v_{\lambda_k}} \lambda_{kJ} \end{bmatrix}$$

The columns of this matrix are of the most interest for our purposes. Each column indicates how a vote for party j impacts the selection values of each possible distribution of responsibility among the parties. We will assume that the selection function for λ_k never decreases when v_{λ_k} increases – so more electoral support for a particular distribution of responsibility never hurts its selection value. Since λ_{kj} is always positive, the elements of (A.4) are all non-negative.

For cabinets, an exactly parallel argument produces:

(A.5)

$$\frac{\partial \mathbf{s}_G}{\partial \mathbf{v}_J} = \begin{bmatrix} \sum_{\mathbf{g}_m \in G} \frac{\partial s_1}{\partial v_{\mathbf{g}_m}} g_{m1} & \sum_{\mathbf{g}_m \in G} \frac{\partial s_1}{\partial v_{\mathbf{g}_m}} g_{m2} & \sum_{\mathbf{g}_m \in G} \frac{\partial s_1}{\partial v_{\mathbf{g}_m}} g_{mj} & \dots & \sum_{\mathbf{g}_m \in G} \frac{\partial s_1}{\partial v_{\mathbf{g}_m}} g_{mJ} \\ \sum_{\mathbf{g}_m \in G} \frac{\partial s_2}{\partial v_{\mathbf{g}_m}} g_{m1} & \sum_{\mathbf{g}_m \in G} \frac{\partial s_2}{\partial v_{\mathbf{g}_m}} g_{m2} & & & \vdots \\ \sum_{\mathbf{g}_m \in G} \frac{\partial s_{\tilde{m}}}{\partial v_{\mathbf{g}_m}} g_{m1} & & \sum_{\mathbf{g}_m \in G} \frac{\partial s_{\tilde{m}}}{\partial v_{\mathbf{g}_m}} g_{mj} & & \\ \vdots & & & \ddots & \\ \sum_{\mathbf{g}_m \in G} \frac{\partial s_M}{\partial v_{\mathbf{g}_m}} g_{m1} & \dots & & & \sum_{\mathbf{g}_m \in G} \frac{\partial s_M}{\partial v_{\mathbf{g}_m}} g_{mJ} \end{bmatrix}$$

where we use $\tilde{m}$ and m as separate indexes over the set $\mathbf{G}$.

These two matrices describe how votes for parties are connected to selection values for each cabinet or distribution of executive authority. When we make assumptions about the form of these expressions, these are translatable into assumptions about the process through which cabinets get formed or, more generally, how policy-making responsibility gets distributed among parties. One assumption that we frequently use in the text to simplify the examples and explanation is that the selection function for any particular cabinet or distribution of executive authority only responds to changes in the electoral support of that cabinet or distribution of executive authority, respectively. Formally, *assumption #1* is:

$$\frac{\partial s_{\tilde{m}}}{\partial v_{\mathbf{g}_m}} = \begin{cases} \frac{\partial s_{\tilde{m}}}{\partial v_{\mathbf{g}_{\tilde{m}}}} & \text{if } m = \tilde{m} \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_k}} = \begin{cases} \frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_{\tilde{k}}}} & \text{if } k = \tilde{k} \\ 0 & \text{otherwise} \end{cases}$$

This means that the elements of A.4 and A.5 can be written as $\frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_{\tilde{k}}}} \lambda_{\tilde{k}j}$ and $\frac{\partial s_{\tilde{m}}}{\partial v_{\mathbf{g}_{\tilde{m}}}} g_{\tilde{m}}$,

respectively.

A second assumption that we will use less often but that is sometimes convenient is that the derivatives of the selection function with respect to each distribution of authority (or cabinet) are equal.²⁹⁷ Specifically, we assume:

²⁹⁷ For convenience this is stated globally, but is really much less restrictive than it appears, since we only ever require that this equality hold for two distributions of responsibility that are in a tie for selection (i.e., the two derivatives need only be equal when they are evaluated for a vector of electoral support in which all but one voter has voted the resulting values of the selection function are equal or can be made equal by a vote)

$$\frac{\partial s_{\tilde{k}}}{\partial v_{\lambda_{\tilde{k}}}} = \frac{\partial s_{\tilde{k}'}}{\partial v_{\lambda_{\tilde{k}'}}} \quad \forall \tilde{k}, \tilde{k}'$$

$$\frac{\partial s_{\tilde{m}}}{\partial v_{\mathbf{g}_{\tilde{m}}}} = \frac{\partial s_{\tilde{m}'}}{\partial v_{\mathbf{g}_{\tilde{m}'}}} \quad \forall \tilde{m}, \tilde{m}'$$

This assumption, which says, for example, that a small change in the *electoral support* associated with one cabinet has the same impact on the selection value of that cabinet as the same change in the *electoral support* of another cabinet has on the selection value of that cabinet. It does not say anything about the impact of support for individual parties on selection values.

Now, continuing with our main discussion, suppose that we adopt the following simple *selection rule*: the distribution of responsibility or the cabinet with the largest selection value is always chosen to be the actual distribution of responsibility. Further, we assume that the probability that more than two distributions of executive authority are tied for selection is zero. With this, we can use (A.4) and (A.5) to characterize how a vote for party j will impact the choice of a distribution of responsibility or cabinet, respectively.

Denote voter i 's utility for the distribution of responsibility λ_k as $u(\lambda_k)$ and for cabinets similarly (i.e., $u(\mathbf{g}_m)$). Also, denote the vote totals for each party when all voters except voter i have cast their ballots as $\mathbf{v}_{J/i} = \{v_{1/i}, v_{2/i}, v_{j/i}, \dots, v_{J/i}\}$. We will first focus our attention on the two distributions of responsibility, $\lambda_{\tilde{k}}$ and $\lambda_{\tilde{k}'}$, with the highest selection values given all votes other than voter i 's vote have been cast. We will also assume that the selection value of the second largest of these distributions is sufficiently large relative to that of the other $K-2$ possible distributions that, given (A.4), no single vote for any of the j parties could place a different distribution of responsibility in one of the top two positions.

To simplify notation, we will use $D_{\tilde{k}j}$ and $D_{\tilde{k}'j}$ to indicate the derivatives of $s_{\tilde{k}}$ and $s_{\tilde{k}'}$, with respect to v_j , evaluated at $\mathbf{v}_{J/i}$. Similarly, we will use $s_{\tilde{k}, \sim i}$ and $s_{\tilde{k}', \sim i}$ to refer to the value of $s_{\tilde{k}}$ and $s_{\tilde{k}'}$, when electoral support is $\mathbf{v}_{J/i}$. Let $D_{\tilde{k}j} - D_{\tilde{k}'j}$ be δ_j .

Consider the case in which δ_j is positive so $D_{kj} > D_{\tilde{k}'j}$. If we assume that ties for selection are broken randomly, the difference in voter i 's expected utility for a vote for party $j \in \mathbf{J}$ rather than abstaining (v_0) is:

$$(A.6) \quad u(v_{ij} | \mathbf{v}_{\mathbf{J}/i}, \delta_j > 0) - u(v_0 | \mathbf{v}_{\mathbf{J}/i}, \delta_j > 0) = \begin{cases} u(\lambda_{\tilde{k}'} - u(\lambda_{\tilde{k}'}) & \text{if } s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} > \delta_j \\ \frac{u(\lambda_{\tilde{k}}) + u(\lambda_{\tilde{k}'})}{2} - u(\lambda_{\tilde{k}'}) & \text{if } s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} = \delta_j \\ u(\lambda_{\tilde{k}}) - u(\lambda_{\tilde{k}'}) & \text{if } 0 < s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} < \delta_j \\ u(\lambda_{\tilde{k}}) - \frac{u(\lambda_{\tilde{k}}) + u(\lambda_{\tilde{k}'})}{2} & \text{if } s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} = 0 \\ u(\lambda_{\tilde{k}}) - u(\lambda_{\tilde{k}'}) & \text{if } s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} < 0 \end{cases}$$

If we denote the probability of various electoral outcomes (before voter i casts her vote) with $p(\cdot)$ then we can write the expected utility difference of voting for party j rather than abstaining when $D_{kj} > D_{\tilde{k}'j}$ as:

$$(A.7) \quad u(v_{ij} | \mathbf{v}_{\mathbf{J}/i}, \delta_j > 0) - u(v_0 | \mathbf{v}_{\mathbf{J}/i}, \delta_j > 0) = p(s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} = \delta_j) \left[\frac{u(\lambda_{\tilde{k}}) - u(\lambda_{\tilde{k}'})}{2} \right] \\ + p(0 < s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} < \delta_j) [u(\lambda_{\tilde{k}}) - u(\lambda_{\tilde{k}'})] + p(s_{\tilde{k}', \sim i} - s_{\tilde{k}, \sim i} = 0) \left[\frac{u(\lambda_{\tilde{k}}) - u(\lambda_{\tilde{k}'})}{2} \right]$$

In large electorates in which the impact of a single vote on the selection function is likely to be quite small, we can assume that the probability of events very close to a tie for selection are the same as for a tie itself. Thus, we can rewrite equation (A.7) as:

$$(A.8) \quad u(v_{ij} | \mathbf{v}_{\mathbf{J}/i}, \delta_j > 0) - u(v_0 | \mathbf{v}_{\mathbf{J}/i}, \delta_j > 0) = 2p(s_{\tilde{k}', \sim i} = s_{\tilde{k}, \sim i}) [u(\lambda_{\tilde{k}}) - u(\lambda_{\tilde{k}'})]$$

We can do a similar derivation for the case in which $D_{\tilde{k}j} < D_{\tilde{k}'j}$. This produces:

$$(A.9) \quad u(v_{ij} | \mathbf{v}_{J/i}, \delta_j > 0) - u(v_0 | \mathbf{v}_{J/i}, \delta_j > 0) = 2p(s_{\tilde{k}', \sim i} = s_{\tilde{k}, \sim i}) [u(\lambda_{\tilde{k}'}) - u(\lambda_{\tilde{k}})]$$

Finally, when $D_{\tilde{k}j} = D_{\tilde{k}'j}$, a vote for j will necessarily leave the ordering of the selection functions for $\lambda_{\tilde{k}}$ and $\lambda_{\tilde{k}'}$ the same and so the utility difference of voting for party j or abstaining will always be zero. Expressions for cabinets rather than distributions of responsibility are exactly the same as (A.8) and (A.9), except replacing the λ 's with $\mathbf{g}$'s. These expressions follow from exactly the same argument.

Now, we are finally in a position to formally define $P_{j,\lambda\lambda'}$ and $P_{j,\mathbf{g}\mathbf{g}'}$ from the text and to derive equations (5.10) and (5.24). Specifically, we define $P_{j,\lambda\lambda'}$ as the probability that the selection values of $\lambda_{\tilde{k}}$ and $\lambda_{\tilde{k}'}$ are tied for first place among all the potential distributions of responsibility and $D_{\tilde{k}j} > D_{\tilde{k}'j}$. Likewise, $P_{j,\lambda'\lambda}$ is the probability that the selection values of $\lambda_{\tilde{k}}$ and $\lambda_{\tilde{k}'}$ are tied for first place among all the potential distributions of responsibility and $D_{\tilde{k}j} < D_{\tilde{k}'j}$. For cabinets rather than distributions of responsibility,, we define $P_{j,\mathbf{g}\mathbf{g}'}$ as the probability that the selection values of $\mathbf{g}_{\tilde{m}}$ and $\mathbf{g}_{\tilde{m}'}$ are tied for first place among all the potential cabinets and $D_{\tilde{m}j} > D_{\tilde{m}'j}$. Likewise, $P_{j,\mathbf{g}'\mathbf{g}}$ is the probability that the selection values of $\mathbf{g}_{\tilde{m}}$ and $\mathbf{g}_{\tilde{m}'}$ are tied for first place among all the potential cabinets and $D_{\tilde{m}j} < D_{\tilde{m}'j}$.

Before using these definitions to derive the difference in expected utility of casting a vote for some party versus abstaining, we need to be somewhat more precise about what we mean when we say, for example, “the probability that the selection values of $\mathbf{g}_{\tilde{m}}$ and $\mathbf{g}_{\tilde{m}'}$ are tied for first place among all the potential cabinets and $D_{\tilde{m}j} < D_{\tilde{m}'j}$ ”.

Specifically, we invoke the probability that $D_{\tilde{m}j} < D_{\tilde{m}'j}$ not because we think the cabinet formation process that defines these derivatives is random, but instead as a way to represent the voter’s subjective beliefs about this aspect of the cabinet formation process. Substantive assumptions about these derivatives are thus assumptions about what voters

believe to be true about cabinet formation. The notation $P_{j,gg'}$ and the like is used only as a shorthand for the joint probability $P(s_{\mathbf{g}_m, \sim i} = s_{\mathbf{g}_{m'}, \sim i})P(D_{\mathbf{g}_m j} > D_{\mathbf{g}_{m'} j} | s_{\mathbf{g}_m, \sim i} = s_{\mathbf{g}_{m'}, \sim i})$ which becomes $P(s_{\mathbf{g}_m, \sim i} = s_{\mathbf{g}_{m'}, \sim i})P(D_{\mathbf{g}_m j} > D_{\mathbf{g}_{m'} j})$ under the assumption that the chance of tie for selection between two cabinets is independent of their selection functions (which can be thought of as part of a pre-existing process of cabinet selection). Another shorthand that we will use when we need to make the two pieces $P_{j,gg'}$ explicit is $P_{gg'} = P(s_{\mathbf{g}_m, \sim i} = s_{\mathbf{g}_{m'}, \sim i})$.

With these definitions, we can use A.8 and A.9 to write the difference in expected utility of casting a vote for any party versus abstaining as the sum of all the ways this vote might result in a non-zero utility difference between any two possible distributions of responsibility (or cabinets). Specifically,

$$(A.10) \quad 2 \sum_{\lambda \in \Lambda} \sum_{\lambda' \in \Lambda} P_{j, \lambda \lambda'} [u(\lambda_k) - u(\lambda_{k'})]$$

and

$$(A.11) \quad 2 \sum_{\mathbf{g} \in G} \sum_{\mathbf{g}' \in G} P_{j, \mathbf{g} \mathbf{g}'} [u(\mathbf{g}_m) - u(\mathbf{g}_{m'})]$$

Now define $\mathbf{g}_m$ as the incumbent cabinet and $\mathbf{g}_{m'}$ and $\mathbf{g}_{m''}$ as generic non-incumbent cabinets. Likewise define λ_k as the “incumbent” or status quo distribution of responsibility and $\lambda_{k'}$ and $\lambda_{k''}$ as generic non-status quo distributions of responsibility. If we let $\mathbf{A}$ do double duty as the set of these alternative (i.e., non-incumbent) cabinets and the set of non status-quo distributions of responsibility, ignore the multiplication by two (which is inconsequential), and modify some of the notational conventions to match the simpler (but less precise) forms used in the text, we can rewrite the above results exactly as equation (5.14) and the first lines of (5.24), respectively.

$$\begin{aligned}
(A.12) \quad E[u \mid v_j] - E[u \mid v_0] &= \sum_{\mathbf{g}' \in \Lambda} P_{j, \mathbf{g} \mathbf{g}'} (E[u_{\mathbf{g}, t+1}] - E[u_{\mathbf{g}', t+1}]) \\
&+ \sum_{\mathbf{g}' \in \Lambda} P_{j, \mathbf{g}' \mathbf{g}} (E[u_{\mathbf{g}', t+1}] - E[u_{\mathbf{g}, t+1}]) \\
&+ \sum_{\mathbf{g}' \in \Lambda} \sum_{\mathbf{g}'' \in \Lambda} P_{j, \mathbf{g}' \mathbf{g}''} (E[u_{\mathbf{g}', t+1}] - E[u_{\mathbf{g}'', t+1}])
\end{aligned}$$

$$\begin{aligned}
(A.13) \quad E[u \mid v_j] - E[u \mid v_0] &= \sum_{\lambda' \in \Lambda'} P_{j, \lambda \lambda'} (E[u_{\lambda, t+1}] - E[u_{\lambda', t+1}]) \\
&+ \sum_{\lambda' \in \Lambda'} P_{j, \lambda' \lambda} (E[u_{\lambda', t+1}] - E[u_{\lambda, t+1}]) \\
&+ \sum_{\lambda' \in \Lambda'} \sum_{\lambda'' \in \Lambda'} P_{j, \lambda' \lambda''} (E[u_{\lambda', t+1}] - E[u_{\lambda'', t+1}])
\end{aligned}$$

Appendix B

Proof that the minimum of the responsibility augmented competency signal is an equal division of responsibility.

(B.1)

$$\begin{aligned}
 \frac{\partial}{\partial \lambda_i} \left(\frac{\sigma_\mu^2 \sum_j \lambda_{j,t}^2}{\sigma_\xi^2 + \sigma_\mu^2 \sum_j \lambda_{j,t}^2} \right) &= \frac{\partial}{\partial \lambda_i} \left(\sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right) \left(\sigma_\xi^2 + \sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right)^{-1} - \frac{\partial}{\partial \lambda_i} \left(\sigma_\xi^2 + \sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right) \left(\sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right)^{-1} \\
 &= \frac{\partial}{\partial \lambda_i} \left(\sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right) \sigma_\xi^2 + \frac{\partial}{\partial \lambda_i} \left(\sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right) \sigma_\mu^2 \sum_j \lambda_{j,t}^2 - \frac{\partial}{\partial \lambda_i} \left(\sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right) \sigma_\mu^2 \sum_j \lambda_{j,t}^2 \\
 &= \frac{\partial}{\partial \lambda_i} \left(\sigma_\mu^2 \sum_j \lambda_{j,t}^2 \right) \sigma_\xi^2
 \end{aligned}$$

Since we have $\sum_j \lambda_{j,t} = 1$, we can rewrite $\sum_j \lambda_{j,t}$ as:

$$\sum_j \lambda_{j,t} = \sum_{j=1}^{J-1} \lambda_{j,t} + \left(1 - \sum_{j=1}^{J-1} \lambda_{j,t} \right)$$

(B.2) and

$$\sum_j \lambda_{j,t}^2 = \sum_{j=1}^{J-1} \lambda_{j,t}^2 + \left(1 - \sum_{j=1}^{J-1} \lambda_{j,t} \right)^2$$

We can now write:

$$(B.3) \quad \left(\frac{\partial \sigma_\mu^2 \sum_j \lambda_{j,t}^2}{\partial \lambda_i} \right) \sigma_\xi^2 = \left(\frac{\partial \left(\sigma_\mu^2 \sum_{j=1}^{J-1} \lambda_{j,t}^2 + \sigma_\mu^2 \left(1 - \sum_{j=1}^{J-1} \lambda_{j,t} \right)^2 \right)}{\partial \lambda_i} \right) \sigma_\xi^2$$

By definition we have:

$$\sum_{j=1}^{J-1} \lambda_{j,t} = \lambda_{i,t} + \sum_{j \neq i}^{J-1} \lambda_{j,t}$$

where the sum on the right hand side is the sum over the J-2 free λ_j 's, not including λ_i .
Replacing this in equation (B.3) and expanding gives:

(B.4)

$$\begin{aligned} \left(\frac{\partial \sigma_\mu^2 \sum_j \lambda_{j,t}^2}{\partial \lambda_i} \right) \sigma_\xi^2 &= \left(\frac{\partial \sigma_\mu^2 \left(\lambda_{i,t}^2 + \sum_{j=1}^{J-1} \lambda_{j,t}^2 \right) + \sigma_\mu^2 - 2\sigma_\mu^2 \left(\lambda_{i,t} + \sum_{j \neq i}^{J-1} \lambda_{j,t} \right) + \sigma_\mu^2 \left(\lambda_{i,t} + \sum_{j \neq i}^{J-1} \lambda_{j,t} \right)^2}{\partial \lambda_i} \right) \sigma_\xi^2 \\ &= \left(2\sigma_\mu^2 \lambda_{i,t} - 2\sigma_\mu^2 + \frac{\partial \sigma_\mu^2 \left(\lambda_{i,t} + \sum_{j \neq i}^{J-1} \lambda_{j,t} \right)^2}{\partial \lambda_i} \right) \sigma_\xi^2 \\ &= \left(2\sigma_\mu^2 \lambda_{i,t} - 2\sigma_\mu^2 + \frac{\partial \sigma_\mu^2 \left(\lambda_{i,t}^2 + 2\lambda_{i,t} \sum_{j \neq i}^{J-1} \lambda_{j,t} + \left(\sum_{j \neq i}^{J-1} \lambda_{j,t} \right)^2 \right)}{\partial \lambda_i} \right) \sigma_\xi^2 \\ &= \left(2\sigma_\mu^2 \lambda_{i,t} - 2\sigma_\mu^2 + 2\sigma_\mu^2 \lambda_{i,t} + 2\sigma_\mu^2 \sum_{j \neq i}^{J-1} \lambda_{j,t} \right) \sigma_\xi^2 \\ &= \left(4\sigma_\mu^2 \lambda_{i,t} - 2\sigma_\mu^2 + 2\sigma_\mu^2 \sum_{j \neq i}^{J-1} \lambda_{j,t} \right) \sigma_\xi^2 \\ &= 2\sigma_\mu^2 \sigma_\xi^2 \left(2\lambda_{i,t} - 1 + \sum_{j \neq i}^{J-1} \lambda_{j,t} \right) \end{aligned}$$

Setting the last line of equation (B.4) to zero for each i defines a system of $J-1$ equations, each of which has the form:

$$(B.5) \quad \lambda_{i,t} + \frac{1}{2} \sum_{j \neq i}^{J-1} \lambda_{j,t} = \frac{1}{2}$$

In matrix notation this system of equations is just:

$$(B.6) \quad \mathbf{A}\boldsymbol{\lambda} = \mathbf{i} \frac{1}{2}$$

where $\mathbf{i}$ is a $J-1$ vector of ones and:

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \lambda_1 \\ \vdots \\ \lambda_{J-1} \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 1 & \frac{1}{2} & \cdots & \frac{1}{2} \\ \frac{1}{2} & 1 & \vdots & \vdots \\ \vdots & \cdots & \ddots & \vdots \\ \frac{1}{2} & \cdots & \cdots & 1 \end{bmatrix}$$

The solution to equation (B.6) is:

$$(B.7) \quad \boldsymbol{\lambda} = \mathbf{A}^{-1} \mathbf{i} \frac{1}{2}$$

Let $k = J-1$ and define,

$$\mathbf{A}^{-1} = \begin{bmatrix} \frac{2k}{k+1} & -\frac{2}{k+1} & \dots & -\frac{2}{k+1} \\ -\frac{2}{k+1} & \frac{2k}{k+1} & \vdots & \vdots \\ \vdots & \dots & \ddots & \vdots \\ -\frac{2}{k+1} & \dots & \dots & -\frac{2k}{k+1} \end{bmatrix}$$

It follows that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$, so $\mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$ and, thus, solving for the λ that minimizes this we get:

$$\mathbf{A}^{-1} \frac{1}{2} = \begin{bmatrix} \frac{2k}{k+1} & -\frac{2}{k+1} & \dots & -\frac{2}{k+1} \\ -\frac{2}{k+1} & \frac{2k}{k+1} & \vdots & \vdots \\ \vdots & \dots & \ddots & \vdots \\ -\frac{2}{k+1} & \dots & \dots & -\frac{2k}{k+1} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \vdots \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{k+1} \\ \vdots \\ \vdots \\ \frac{1}{k+1} \end{bmatrix}$$

since $k = J-1$, and we have the sum of all shares equal to one, this implies that for all J parties, the vector of shares that minimize the competency signal allocates all parties share $1/J$.

Appendix C

Derivation of Equation 8.34

Growth in the current period under distribution of responsibility λ is:

$$(C.1) \quad y_{\lambda,t} = \bar{y} + \eta_{\lambda,t} \\ = \bar{y} + \sum_{j \in \mathbf{J}} \{ \lambda_{j,t} \mu_{j,t} + \lambda_{j,t-1} \mu_{j,t-1} \} + \xi_t$$

Rearranging this gives:

$$(C.2) \quad \sum_{j \in \mathbf{J}} \{ \lambda_{j,t} \mu_{j,t} \} + \xi_t = y_{\lambda,t} - \bar{y} - \sum_{j \in \mathbf{J}} \{ \lambda_{j,t-1} \mu_{j,t-1} \} \\ = y_{\lambda,t} - \bar{y} - \mu_{\lambda,t-1}$$

Where everything on the right hand side of this equality is observed and so the sum of the terms on the left is also observed, though not the individual components. Denote the sum on the left hand side as $k_{\lambda,t} = \sum_{j \in \mathbf{J}} \{ \lambda_{j,t} \mu_{j,t} \} + \xi_t$. Since $k_{\lambda,t}$ is observed, the voter can

compute her expectation about any $\mu_{j,t}$ given $k_{\lambda,t}$. To calculate this conditional expectation, we need to know the distribution of both $k_{\lambda,t}$ and $\mu_{j,t}$. $\mu_{j,t}$ is a normally distributed random variable with zero mean and variance σ_{μ}^2 . Since

$$k_{\lambda,t} = \mu_{\lambda,t} + \xi_t = \sum_{j \in \mathbf{I}} \{ \lambda_{j,t} \mu_{j,t} \} + \xi_t, \text{ its distribution depends on the distribution of } \mu_{\lambda,t},$$

which is just:

$$(C.3) \quad \mu_{\lambda,t} = \sum_{j \in \mathbf{I}} \lambda_{j,t} \mu_{j,t} \sim N \left(0, \sigma_{\mu}^2 \sum_{j \in \mathbf{I}} \lambda_{j,t}^2 \right) \\ \sim N \left(0, \sigma_{\mu_{\lambda}}^2 \right)$$

Thus $k_{\lambda,t}$ is the un-weighted sum of two normally distributed random variables, both with zero means and variances $\sigma_{\mu_{\lambda}}^2$ and σ_{ξ}^2 , respectively. The distribution of $k_{\lambda,t}$ is thus:

$$(C.4) \quad k_{\lambda,t} = (\mu_{\lambda} + \xi_t) \sim N(0, \sigma_{\mu_{\lambda}}^2 + \sigma_{\xi}^2)$$

Given that both, $k_{\lambda,t}$ and $\mu_{j,t}$ are distributed normally, their joint distribution is bivariate normal and the optimal forecast of $\mu_{j,t}$ given $k_{\lambda,t}$ is just this conditional expectation, which is computed from the appropriate conditional distribution of the bivariate normal. Using standard results on the bivariate normal distribution:

$$(C.5) \quad E[\mu_{i,t} | k_{\lambda,t}] = E[\mu_{i,t}] + \frac{\sigma_{\mu, k_{\lambda,t}}}{\sigma_{k_{\lambda}}^2} (y_{\lambda,t} - \bar{y} - \mu_{\lambda,t-1}) - E[k_{\lambda,t}]$$

$$= \frac{\sigma_{\mu}^2 \lambda_{i,t}}{\sigma_{\mu_{\lambda}}^2 + \sigma_{\xi}^2} (y_{\lambda,t} - \bar{y} - \mu_{\lambda,t-1})$$

Where the second line uses the fact that the covariance between $k_{\lambda,t}$ and $\mu_{i,t}$ is equal to $\sigma_{\mu}^2 \lambda_{i,t}$. To see this:

$$\begin{aligned} \sigma_{\mu, k_{\lambda,t}} &= E[(\mu_{i,t} - E[\mu_{i,t}])(k_{\lambda,t} - E[k_{\lambda,t}])] \\ &= E[\mu_{i,t} k_{\lambda,t}] \\ &= E[\mu_{i,t} (\mu_{\lambda,t} + \xi_t)] \\ &= E\left[\mu_{i,t} \left(\sum_{j \in \mathbf{J}} \{\lambda_{j,t} \mu_{j,t}\} + \xi_t\right)\right] \\ &= E\left[\mu_{i,t}^2 \lambda_{i,t} + \mu_{i,t} \sum_{j \neq i \in \mathbf{J}} \{\lambda_{j,t} \mu_{j,t}\} + \mu_{i,t} \xi_t\right] \\ &= \sigma_{\mu}^2 \lambda_{i,t} \end{aligned}$$

Where the last line follows because we assumed independence between competency draws. Finally, substituting for $\sigma_{\mu_{\lambda}}^2$ and noting that $E[\mu_{i,t} | k_{\lambda,t}] = E[\mu_{i,t} | y_{\lambda,t}]$, we have equation 8.34 in the text:

$$(C.6) \quad E\left[\mu_{i,t} \mid y_{\lambda,t}\right] = \frac{\sigma_{\mu}^2 \lambda_{i,t}}{\sigma_{\xi}^2 + \sigma_{\mu}^2 \sum_{j \in \mathbf{J}} \lambda_{j,t}^2} (y_{\lambda,t} - \bar{y} - \mu_{\lambda,t-1})$$